

RADIANT Worked Examples & Validation

Case Studies, Scenario Digests, and Validation Evidence

RADIANT Project

v0.4.0

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Running the Examples

RADIANT ships its worked examples in two forms, and this volume uses both. Some examples are **configurations you open** — a YAML file that the desktop GUI loads, evaluates, and lets you drive by hand. Others are **programs you run** — a Python script that builds a sensor, sweeps it, and prints its results to the terminal. The two modalities describe the same physics through the same public API; they differ only in who is holding the controls.

This chapter is the setup instruction for every chapter that follows: what to install, where the files live, and the two commands that open an example.

1.1 What ships

Kind	Where	Opened with
Reference configs	examples/*.yaml	GUI, CLI, or <code>Sensor.from_yaml</code>
Example programs	examples/scripts/*.py	<code>python examples/scripts/<name>.py</code>
Scenario exercise baselines	scenarios/NN_persona/N.M_slug/inputs/<slug>.gui.yaml	GUI (File → Open YAML)
Scenario run scripts	scenarios/NN_persona/N.M_slug/scripts/run_<slug>.py	<code>python run_<slug>.py</code>
Flagship-mission configs	scenarios/09_flagship_missions/9.N_*/**.yaml	GUI, CLI, or the API

Two of those rows deserve a note.

The exercise baselines are generated, not hand-written. Each scenario was built as a *backend* test case: a run script that assembles a Sensor in Python, sweeps it, and writes a workbook. A Python script is not something File → Open can consume, so `scenarios/tools/emit_gui_yaml.py` imports each runner's validated config factory and serializes `Sensor.to_yaml()` into `inputs/<slug>.gui.yaml`, alongside a headline-metric snapshot in `inputs/<slug>.gui.expected.json`. Opening a baseline in the GUI therefore reproduces the same numbers the backend scenario validated — that equality is a gate (`scenarios/tools/verify_gui_yaml.py`), not a hope.

The baselines pick one portable point of each trade. A scenario's run script may sweep aperture from 0.15 m to 0.60 m against a staged MODTRAN tape7; its baseline freezes a mid-range aperture and, where the tape7 is not committed, the simple parametric atmosphere. The baseline is for *driving the GUI*; the full sweep still lives in the run script.

1.2 Installing what the examples need

RADIANT needs **Python 3.11 or 3.12**. From a clone:

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\Activate.ps1
pip install -e ".[dev]"
```

The optional extras each unlock a different part of this volume:

Extra	Installs	Needed for
gui	PySide6, matplotlib, qtconsole, openpyxl	the GUI chapters (chapter 2)
scenarios	openpyxl, matplotlib	scenario run scripts (vendor-format workbooks, figures)
dev	pytest, mypy, ruff, import-linter, hypothesis, h5py	running the test suite

```
pip install -e ".[gui,scenarios]"
```

gui is a back-compatibility alias: the Qt stack moved into the base dependencies, so a plain `pip install radiant` already ships the desktop application. Installing the extra explicitly still works and costs nothing.

The six programs in `examples/scripts/` need **no extra at all** except `aperture_sweep.py`, which saves a figure and therefore needs `matplotlib` (present in both `gui` and `scenarios`).

Verify the install, and check *which* RADIANT you just got — a stale editable install pointing at another checkout is the single most common source of “my edit did nothing”:

```
radiant --version
```

1.3 Where inputs and outputs live

Inputs are committed; results generally are not.

- **Committed:** every `.yaml` config, every `.csv/.xlsx` vendor-format input, every figure a `walkthrough.md` actually references (each with a provenance line in the folder’s `outputs/MANIFEST.md`).
- **Not committed:** `*_results.xlsx` workbooks, ad-hoc plots, and the built PDF manuals. These are regenerate-on-demand and gitignored.

A scenario folder has a fixed shape:

```
NN_persona/
  N.M_scenario_slug/
    inputs/      # vendor-format data, the .gui.yaml baseline, the metric snapshot
    scripts/     # run_<slug>.py (the trade study), gui_console_<slug>.py
    outputs/     # committed figures + MANIFEST.md; workbooks are regenerated
    walkthrough.md # what was run and what the numbers mean
    gaps.md      # every RADIANT limitation the scenario hit
    gui_workflow.md # what the GUI must provide to support this workflow
```

examples/ is flatter and deliberately so: its scripts carry their explanation in the module docstring and in their own printed output, not in a companion walkthrough. When chapter 3 quotes a script’s commentary, it is quoting text the script itself prints.

1.4 Modality one — open a baseline in the GUI

```
radiant gui                                # empty, pick a file from inside
radiant gui examples/mwir_leo_minimal.yaml # straight onto a config
```

or, once the window is up, File → Open YAML...

The GUI reads **every** document through one reader. A plain config opens as a single session; a file carrying a configurations: section opens as the whole study, with a configuration selector across the top and one metric column per configuration. You do not choose between two open commands — the file’s content decides, and the status bar says which happened (Opened oli2_all_bands_study.yaml – 9 configurations (B1_CA, B2_Blue, ...); displaying B1_CA).

Loading evaluates immediately. Editing any parameter marks the view stale and re-evaluates after a short debounce; Evaluate (F5) forces it. Everything the window shows — metric cards, MTF curves, noise tables — comes from the last completed run, so a figure in this manual always shows an *evaluated* window.

To confirm a baseline landed correctly, compare the headline cards against the scenario’s inputs/<slug>.gui.expected.json. A mismatch is a GUI wiring bug, not a physics result.

1.5 Modality two — run a script

From the repository root:

```
python examples/scripts/basic_evaluation.py
```

If you are working in a git worktree, or in any checkout that is not the one pip install -e . registered, put the checkout’s sources first so you exercise *those* edits:

```
PYTHONPATH=src python examples/scripts/basic_evaluation.py
```

The same config runs from the command line, which is the fastest way to get one number without writing anything:

```
radiant run examples/mwir_leo_minimal.yaml
radiant run examples/mwir_leo_minimal.yaml --set optics.aperture_diameter_m=0.50
radiant run scenarios/09_flagship_missions/9.4_landsat_oli2_snr/oli2_all_bands_study.yaml \
    --configuration B4_Red
```

Scenario run scripts are launched from inside their own folder, because they resolve their vendor inputs relative to themselves:

```
cd scenarios/01_sarah_systems_engineer/1.1_mwir_maritime_surveillance/scripts
python run_mwir_maritime_surveillance.py
```

1.6 Three things that will otherwise surprise you

Warnings are results. RADIANT raises a `UserWarning` when the physics stops meaning what you think it means — most often full-well saturation. Several examples in this volume print such warnings on purpose, and the saturated points are part of the lesson. A clipped pixel's SNR no longer responds to the scene, so a saturated configuration reads as *insensitive*, not as high-performing. Read the warning before reading the number it decorates.

The spectral grid is an input. Chain results depend on how finely the band is sampled. `Sensor.from_yaml(path, wavelength_points=300)` and `radiant run --wavelength-points 300` set it; the GUI's status bar reports what was used (Evaluated – 500 wavelength points). Two runs that disagree in the fourth digit usually disagree about this, not about physics.

The regime is decided for you, once. Whether a target is treated as extended, sub-pixel, or a point source is classified in the optics stage and read unchanged by every stage after it. You influence it through the scene you describe — target extent, range, aperture — never by asserting it downstream. Each worked example names the regime it lands in, because almost every surprising number in this volume traces back to that one classification.

1.7 What the next two chapters do

Chapter 2 drives the GUI: six short task-oriented walkthroughs, each with numbered steps and screenshots of the real application, from building a sensor out of nothing to comparing nine configurations side by side.

Chapter 3 drives the API: the six programs in `examples/scripts/`, each with the listing excerpt that matters, the output it actually produced, and the physics that output is showing you.

GUI Worked Examples

Six short walkthroughs, each a task rather than a tour: build a sensor from nothing, read a validated flagship baseline metric group by metric group, adopt a detector from the part library, edit an optical train, sweep a parameter, and compare nine configurations at once. Each one is a few minutes of clicking and ends with numbers you can check against this page.

Every figure in this chapter is a capture of the real application, generated by a generator script driving the real main window offscreen on a committed config. Regenerate the whole set with one command:

```
python scripts/gen_gui_screenshots.py --all
```

The offscreen renderer paints platform-neutral chrome, so the figures look the same whoever produced them and on whichever operating system. Your own window will carry your platform’s title bar and, if you have chosen the dark theme, your own colors; nothing else differs.

A convention used throughout. The window has three permanent columns: the **Parameters** dock on the left (the full schema, as a tree, with a provenance badge on every value), the **stage workspace** in the middle (whichever of the ten chain stages is selected in the strip along the top), and the **right rail** (pinned metric cards, the Edit Config (YAML) button, and the Messages panel). Stages whose content does not fit one pane are tabbed — Geometry is *Inputs* | *Schematic*, Optics is *Inputs* | *Transmission* | *MTF* | *PSF* + *Pupil*, Detector is *Inputs* | *Noise* | *Detector* + *PSF*. Walkthroughs below name the tab whenever it matters.

2.1 Walkthrough 1 — Build a sensor from scratch

Goal: go from an empty window to a first evaluated result, understanding at each step which stage you are describing.

The chain runs geometry first (where is the sensor, where is the target, how far apart), then source, atmosphere, optics, platform, spectral integration, detector, readout, calibration, and finally performance. Filling the stages in strip order is therefore also filling them in the order the physics consumes them.

1. **Start empty.** `radiant gui` with no path, or File → New. The window opens on the welcome screen — “Start a scenario” — whose first tile is **Blank config** (“Start from schema defaults and set everything yourself”). Take it and you land in the Geometry workspace with every parameter at its schema default.
2. **Place the sensor and the target.** Select stage **1 Geometry**, tab **Inputs**. The center pane leads with a derived scene-class label — Scene: `air` → `ground` (`air_to_ground`) – derived — and then offers one card per geometry *family*, each asking you to pick one mode. Set `sensor_altitude_m` to 8000 m and leave `target_altitude_m` at 0 m and `path_zenith_rad` at 0 deg; that is a nadir view from 8 km.

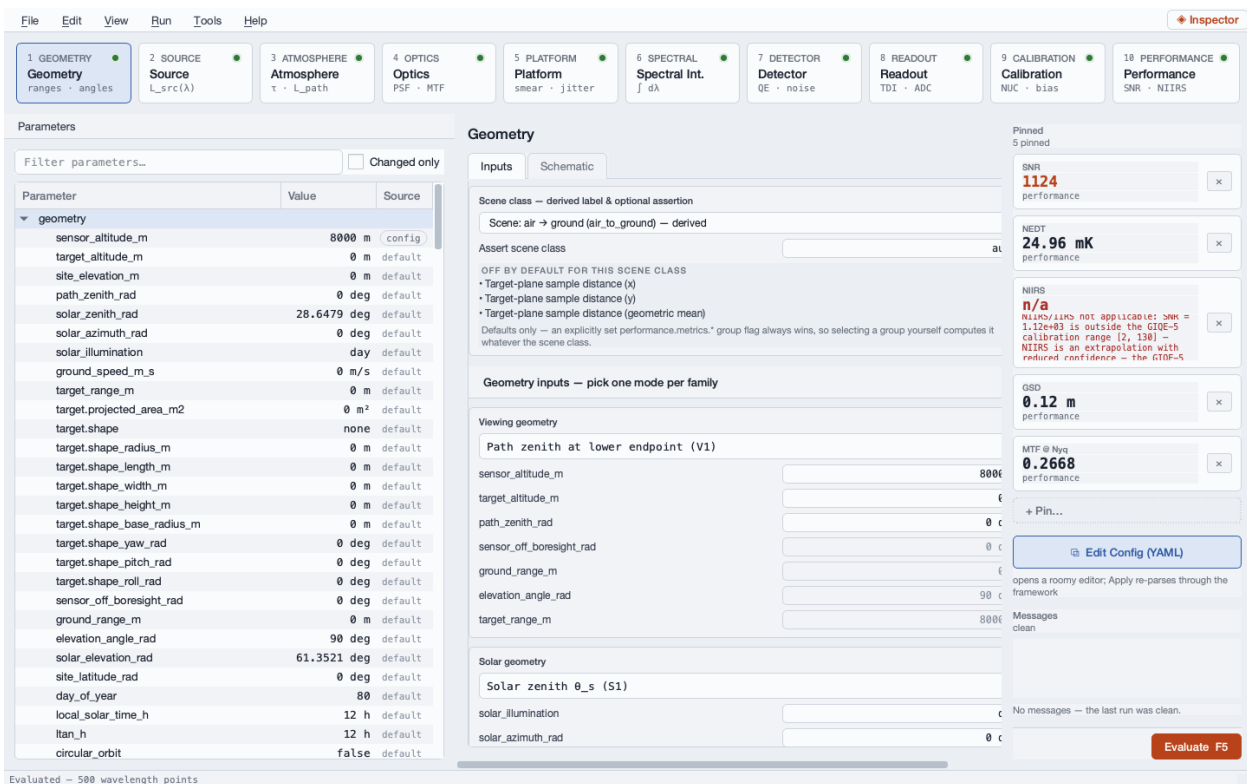


Figure 2.1: The whole window on the geometry step: the Parameters dock on the left, the mode cards in the center, the pinned cards on the right. This is one of the few figures kept at full width, because the next two paragraphs read the dock and the form together.

Note the third column of the Parameters dock. `sensor_altitude_m` reads `config` because you set it; everything else reads `default`. That badge is how you tell what you have actually specified from what the schema supplied — and it is the same distinction the YAML round-trip preserves.

The grayed fields are not disabled features. Geometry accepts one mode per family, and the fields of the modes you did not pick are shown inert rather than hidden, so the alternatives stay visible. Here `path_zenith_rad` is the chosen viewing mode, so `sensor_off_boresight_rad`, `ground_range_m`, `elevation_angle_rad`, and `target_range_m` are all derived rather than entered.

3. **Check the geometry you just described, not the numbers you typed.** Switch to the **Schematic** tab.

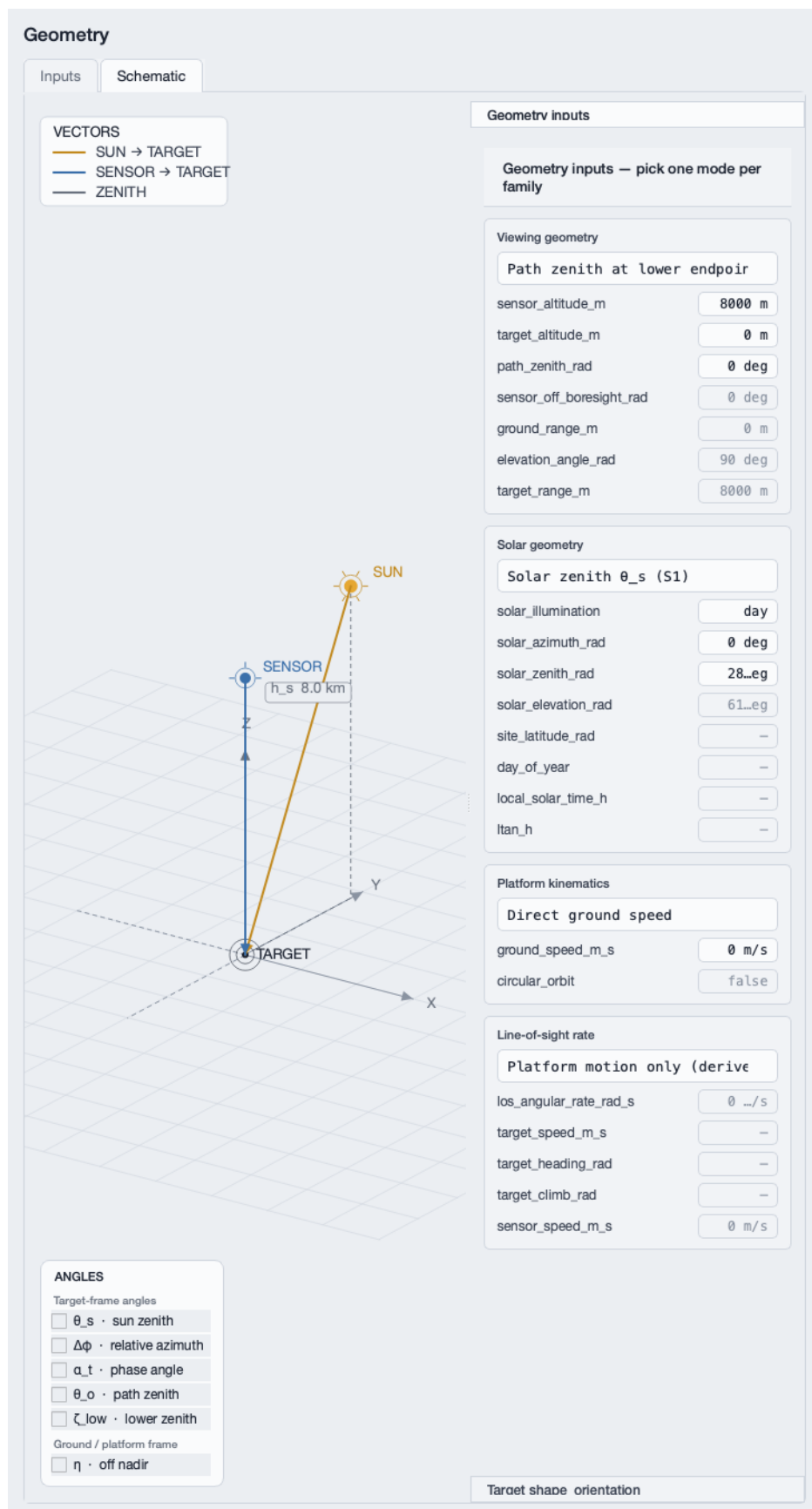


Figure 2.2: Geometry workspace, Schematic tab — the 2D viewing-triangle schematic with the sensor, target, sun vector, and the labeled angle set.

The schematic is deliberately **not to scale**: an 8 km altitude drawn to scale against a 6371 km Earth radius would be invisible. Altitudes are carried by leader labels (h_s 8.0 km) while the geometry itself stays legible. The sun vector (orange) and the sensor line of sight (blue) are drawn from the solar geometry you set — here a 28.65° solar zenith at day-of-year 80, local solar time 12 h.

4. **Give it a telescope.** Select stage **4 Optics**, tab **Inputs**.

Optics

Inputs Transmission MTF PSF + Pupil

Optics — aperture and wavefront error (transmission: Transmission tab)

Aperture diameter	0.3 m
Focal length	1.2 m
f-number	4
Obscuration ratio	0
Spider arms	0
WFE RMS (scalar)	0 waves
WFE reference wavelength	0.633 μm
Zernike file (Zemax export)	
Defocus	0 μm

Import Zemax Zernike...

Outputs

A collect	0.0706858 m ²	
Omega pixel	2.25e-10 sr	
D eff	0.3 m	
F number eff	4	
Obscuration eff	0	
Omega cone	0.0485195 sr	
Transmission input mode	scalar	
Pupil plane extent	0.3	
Pupil wavelength μm	4.2515	
Stray includes thermal	false	
Regime	extended	

Figure 2.3: Optics workspace, Inputs tab — aperture and wavefront-error entry on the left, the derived optics outputs beneath.

Set `aperture_diameter_m` to 0.3 m and `focal_length_m` to 1.2 m. The f-number field immediately reads 4 and carries a lightning bolt with a derived badge — it is computed, not stored, and typing into it is not how you change it. Beneath the form, the Outputs readout shows what the stage handed down-

stream: collecting area 0.0706858 m^2 , pixel solid angle $2.25 \times 10^{-10} \text{ sr}$, cone solid angle 0.0485195 sr , and — the important one — Regime: extended.

That last line is the radiometric regime, classified here in the optics stage and never re-decided later. No target extent was specified, so the 300 K scene fills the pixel footprint: the target signal is a radiance integrated over the pixel solid angle, and the ensquared-energy fraction is reported as a spatial metric rather than applied to the signal.

5. **Evaluate.** Press Evaluate (F5), or simply keep editing — the window re-evaluates on a debounce after every committed change. The status bar reports Evaluated – 500 wavelength points.

6. **Read the result.** Select stage **10 Performance**.

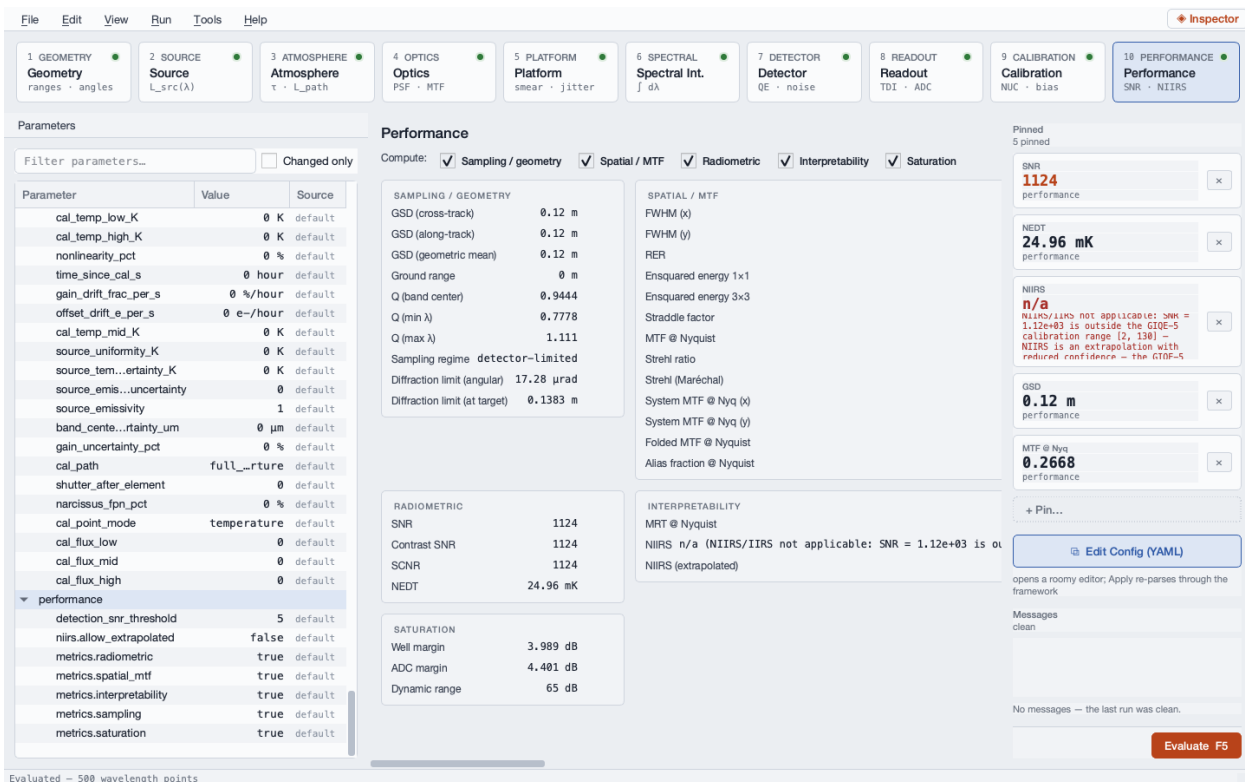


Figure 2.4: Performance workspace on the minimal MWIR example after evaluation.

For this configuration the headline numbers are SNR 1124, NEDT 24.96 mK, GSD 0.12 m, and system MTF at Nyquist 0.2668. The pinned cards in the right rail carry the same four plus NIIRS, which reads n/a with its reason — *NIIRS/IIRS not applicable: SNR = 1.12e+03 is outside the GIQE-5 calibration range [2, 130]* — the tool says so rather than extrapolating quietly.

7. **Save it.** File → Save As... writes exactly what you specified — the config rows, not the defaults — so the file stays readable and the schema stays free to move underneath it.

2.2 Walkthrough 2 — Open a flagship baseline and read every metric group

Goal: understand what each metric group is telling you, on a configuration whose answers are already known.

We use `scenarios/09_flagship_missions/9.2_landsat_tirs_nedt/tirs_b10_nedt_300k.yaml`: Landsat 9 TIRS band 10, a 705 km sun-synchronous f/1.64 refractive LWIR pushbroom on a 25 μm QWIP focal plane, viewing a 300 K unit-emissivity blackbody through a vacuum path — the same view the published on-orbit NEdT is measured against.

1. **Open it.** File → Open YAML..., or `radiant gui scenarios/09_flagship_missions/9.2_landsat_tirs_nedt/tirs_b10_nedt_300k.yaml`. The window evaluates on load.
2. **Go to Performance.**

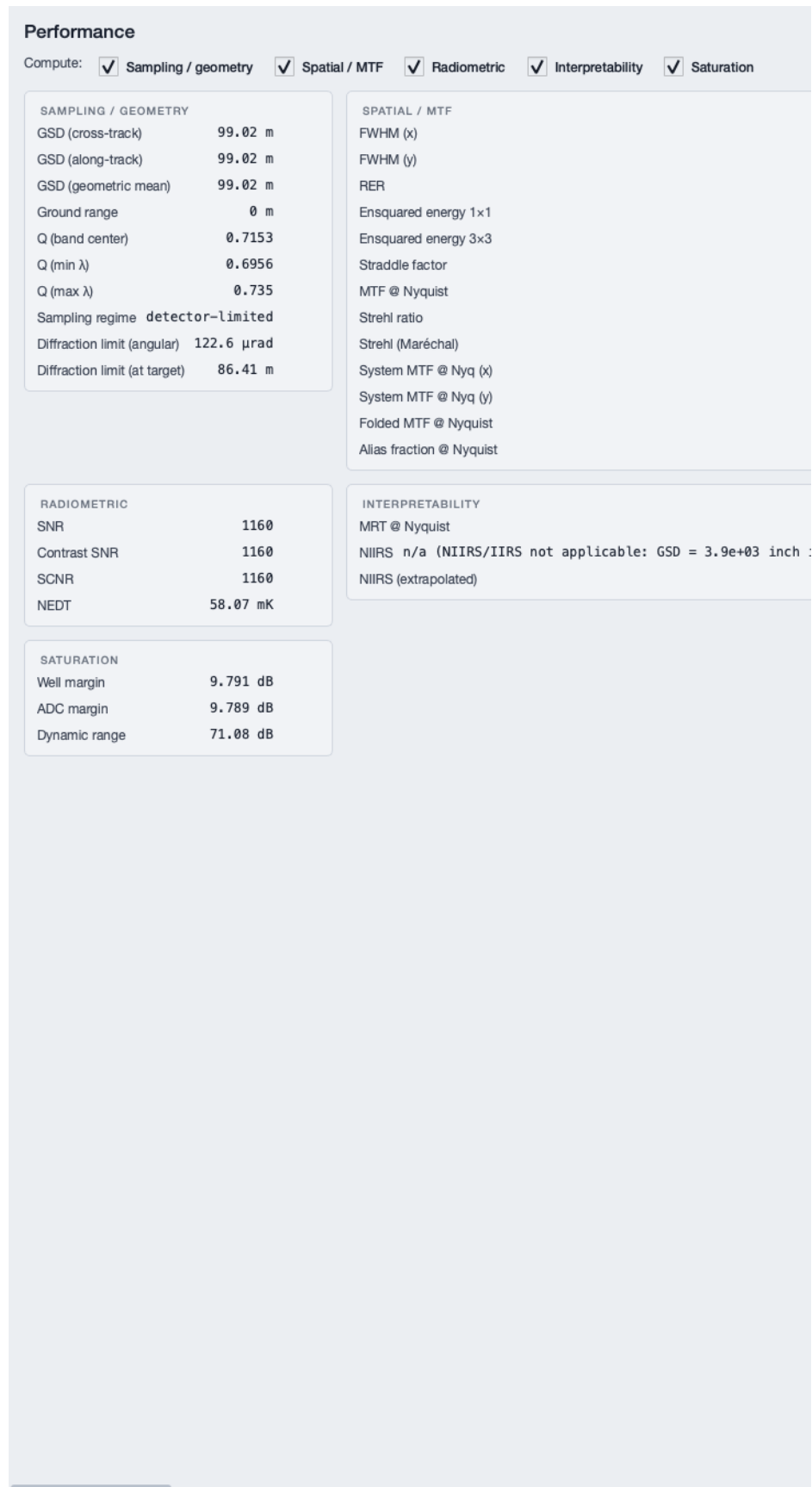


Figure 2.5: Performance workspace on the TIRS band-10 baseline. All five metric groups are enabled and populated.

3. **Read the groups in order.** The Compute: row of checkboxes above the cards is the metric-group selector; each ticked group is a family of metrics the chain was asked to produce. Untick one and its inputs stop being computed, which is how you buy back time on an expensive configuration.

- **Sampling / geometry** — GSD 99.02 m in both axes (against the published 100 m thermal product), ground range 0 m (nadir), $Q = 0.7153$ at band center, and the resulting classification detector-limited. $Q = \lambda F / \# / p$ compares the optical blur to the pixel: below 1 the optics pass more detail than the 25 μm pixels can sample. The diffraction limit is 122.6 μrad , which projects to 86.41 m at the target — smaller than the 99.02 m sample, which is the same statement again.
- **Spatial / MTF** — FWHM 26.28 μm in both axes, RER 0.6791, ensquared energy 0.508 in the central pixel and 0.9081 over 3×3, MTF at Nyquist 0.3533, folded MTF 0.7066, alias fraction 0.5. The folded value is exactly twice the unfolded one and the alias fraction is exactly 0.5: at Nyquist the first sampling replica lands back on Nyquist, so half the apparent contrast there is folded-in above-Nyquist scene content. Strehl reads 1 because this config carries no wavefront error.
- **Radiometric** — SNR 1160, contrast SNR 1160, SCNR 1160, NEDT 58.07 mK. The three SNR flavors coincide because the scene is a uniform blackbody: there is no separate clutter or contrast term to divide them.
- **Interpretability** — minimum resolvable temperature difference (MRT) at Nyquist 0.3698 K, and **no NIIRS value** — the card reports the configuration as outside the GIQE-5 calibration envelope rather than printing a rating. The MRT is the NEDT divided by the system MTF at that frequency; it is the temperature difference a human observer needs to resolve a bar pattern at the sampling limit.
- **Saturation** — well margin 9.791 dB, ADC margin 9.789 dB, dynamic range 71.08 dB. Margins are the headroom between the filled well and its capacity; they go to 0 dB exactly when the pixel clips.

The result to check: RADIANT predicts NEDT = 58.07 mK where the flight-measured value is 49 mK and the requirement is ≤ 400 mK. The prediction sits inside the 58–124 mK bracket the scenario derives from the two published conversion-efficiency bounds, and it clears the spec by almost an order of magnitude.

4. **Ask where the spatial number came from.** Select stage **4 Optics**, tab **MTF**.

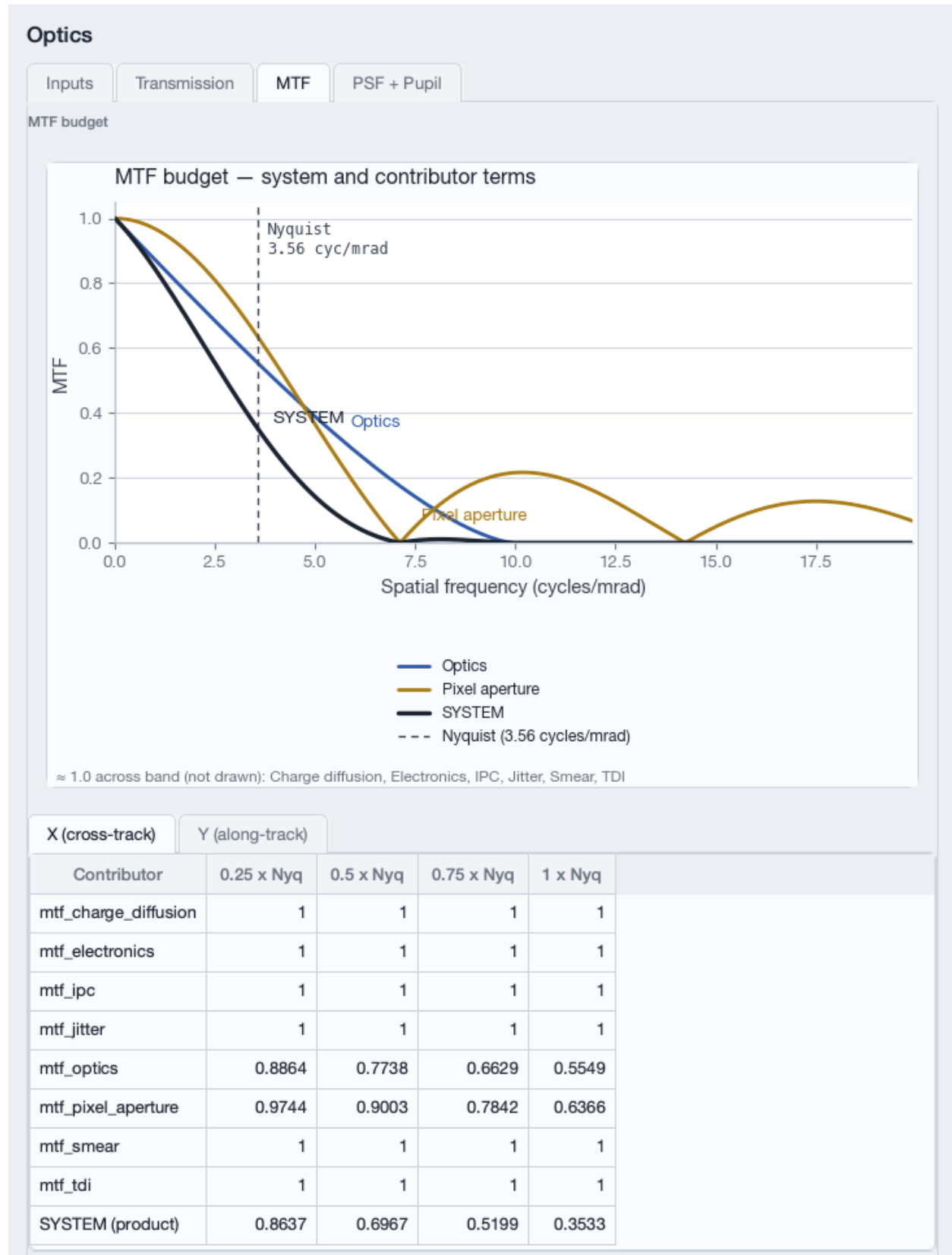


Figure 2.6: Optics workspace, MTF tab — the system MTF with its contributor terms, and the full per-term budget table beneath.

The system curve (black) is the product of every contributor. Only two contributors bite here: the optics term (blue, from the autocorrelation of the complex pupil) and the pixel-aperture term (gold, the sinc of the 25 μm detector footprint). The table below reads them at fractions of Nyquist — at $1 \times \text{Nyquist}$, $\text{mtf_optics} = 0.5549$ against $\text{mtf_pixel_aperture} = 0.6366$, while charge diffusion, electronics, IPC, jitter, smear, and TDI all sit at 1 and are noted as “ ≈ 1.0 across band (not drawn)”. The dashed vertical marks Nyquist at 3.56 cycles/mrad. The pixel term’s first zero is out past 7 cycles/mrad and it rings afterwards — that gold lobe beyond 10 cycles/mrad is the sinc sidelobe, above Nyquist and therefore aliased rather than resolved.

5. Ask where the radiometric number came from. Select stage **7 Detector**, tab **Noise**.

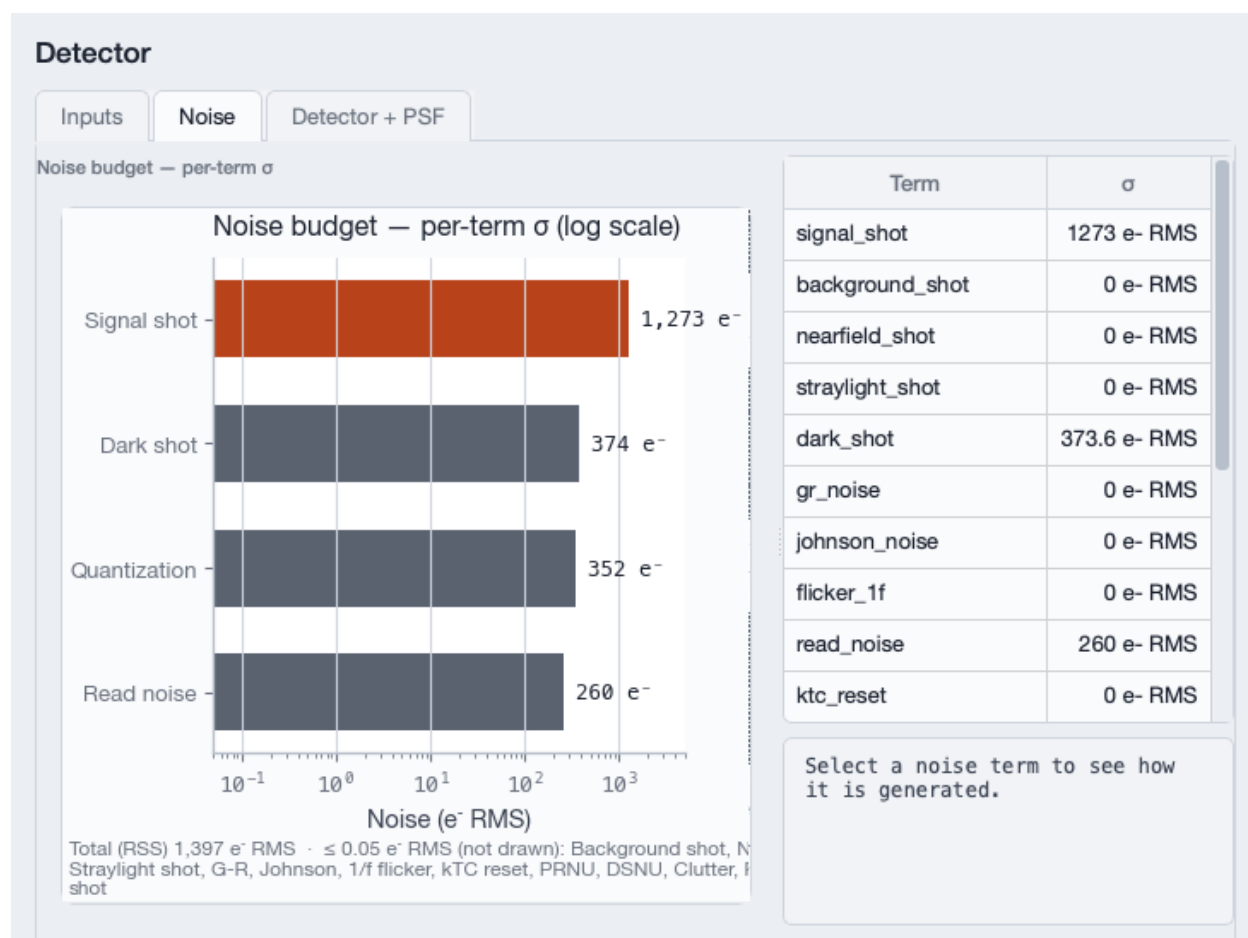


Figure 2.7: Detector workspace, Noise tab — the per-term noise budget on a log axis beside the same terms as a table.

Sixteen noise terms are always computed; four are non-zero here. Signal shot dominates at 1273 e- RMS, then dark shot at 373.6 e- RMS, quantization at 352.4 e- RMS, and read noise at 260 e- RMS. The terms combine in quadrature to a total of 1397 e- RMS, and the caption lists the twelve terms below 0.05 e- RMS that were computed and not drawn. This is a shot-noise-limited detector, which is what “photon-limited” means in practice: the 260 e- RMS read noise, large as it is in absolute terms, contributes under 4 % of the total in quadrature.

2.3 Walkthrough 3 — Pick an FPA from the part library

Goal: stop typing detector parameters one at a time and adopt a real part, with its provenance attached.

RADIANT ships a library of focal-plane presets — real parts, each traced to a datasheet line in a manifest. Applying one writes a *set* of detector parameters at once and marks each of them as preset-supplied, so a later reader can tell which numbers came from the part and which you chose yourself.

1. **Select stage 7 Detector, tab Inputs.** The top strip of the pane is the FPA part library row. On a config with no part applied it offers Choose part & apply... alone.
2. **Choose a part.** Choose part & apply... opens the **FPA part library** table — one row per preset, with columns *Part*, *Kind*, *Class*, *Band*, and *Values (basis census)*, plus a details pane for the selected row. Select a part and confirm to apply it.

The picker is a modal dialog, and the figure set for this manual is captured offscreen where modal dialogs cannot be photographed cleanly; the figure below therefore shows the state you are left in *after* applying, which is the state that matters.

3. **Read what the preset did.**

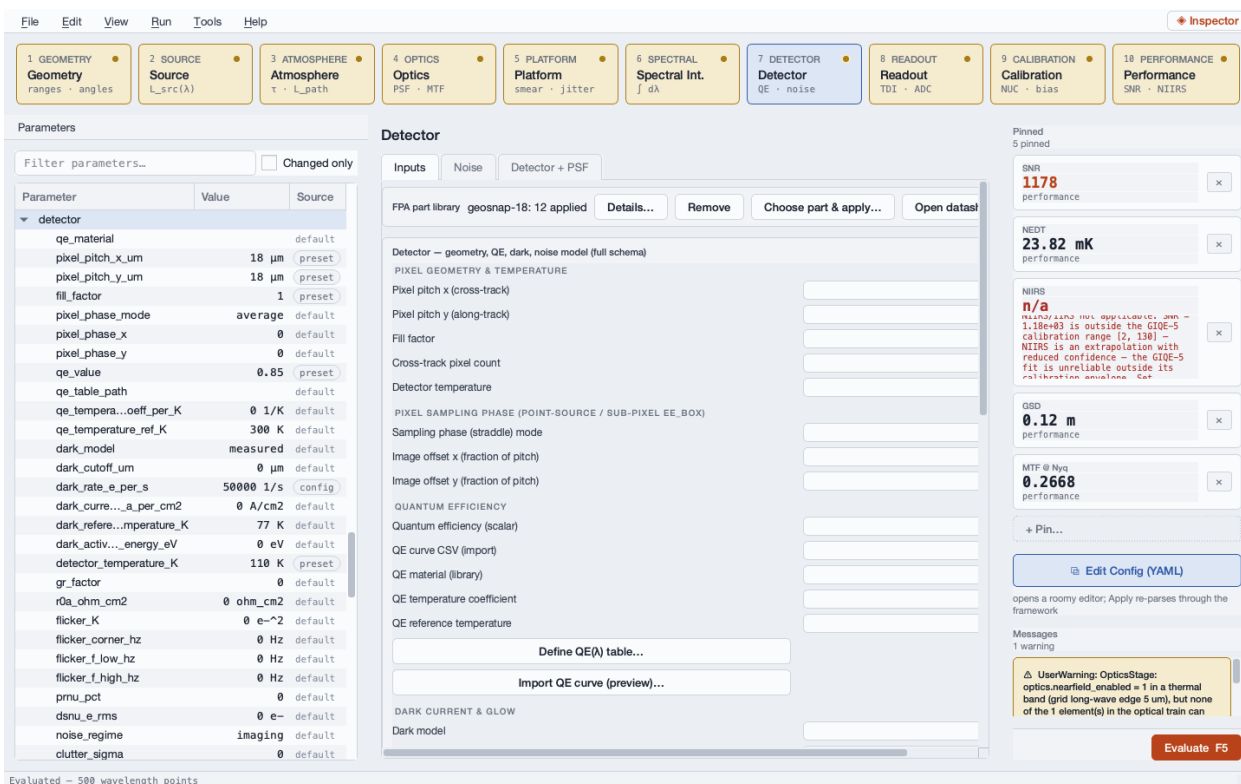


Figure 2.8: Detector workspace, Inputs tab, on a config built from the GeoSnap-18 preset. The library row reports the part and how many parameters it supplied; the Parameters dock badges each of them preset.

The library row now reads FPA part library geosnap-18: 12 applied, with Details... and Remove beside it. In the Parameters dock the twelve preset-supplied values carry a preset badge instead of default or config: `pixel_pitch_x_um = 18 μm`, `pixel_pitch_y_um = 18 μm`, `fill_factor = 1`, `qe_value = 0.85`, `detector_temperature_K = 110 K`, and the rest. Every one of them traces `fpa:geosnap-18/<source>` in the run's provenance record.

4. **Override what the part does not know.** `dark_rate_e_per_s` reads 50000 1/s with a config badge, not a preset one — it was set in the config *after* the preset was applied, and that is deliberate: a datasheet dark current is quoted at the vendor's operating point, not at yours, so the analyst owns it. A preset supplies a starting point; it never locks a parameter.
5. **Evaluate.** For this MWIR LEO configuration the result is SNR 1178, NEDT 23.82 mK, GSD 0.12 m, MTF at Nyquist 0.2668.
6. **Remove it cleanly.** Remove lifts the preset and returns each of the twelve parameters to whatever it was before — it does not leave the values behind as orphaned config entries.

2.4 Walkthrough 4 — Edit an optical element train

Goal: replace a single scalar throughput with a real train of coated surfaces, and see both the transmission and the self-emission that follow from it.

A scalar `transmission_scalar` is a fine first cut. A thermal system needs more: every warm optical surface both attenuates the scene and emits into the detector, and the emission is set by Kirchhoff's law from the surface's own reflectance and transmittance. RADIANT will not accept an emissivity you type in for an optical element; it derives it.

We use `scenarios/09_flagship_missions/9.4_landsat_oli2_snr/oli2_b04_snr_ltyp.yaml` — Landsat 9 OLI-2 band 4 (red), a config that carries the published four-mirror silver train plus a refractive window and a per-band interference filter.

1. **Select stage 4 Optics, tab Transmission.**

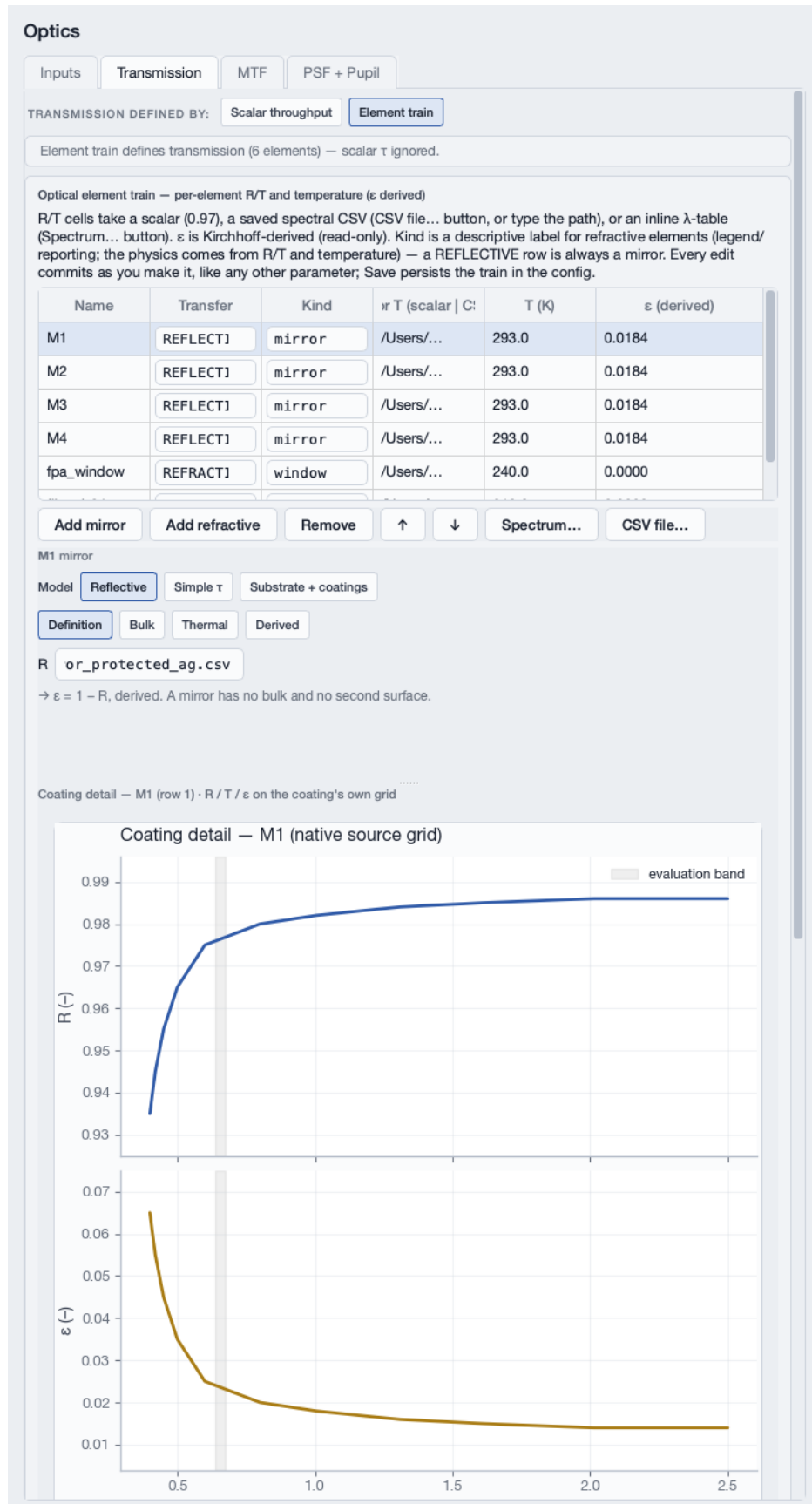


Figure 2.9: Optics workspace, Transmission tab — the element train as editable rows, with the selected element's coating curve beneath.

2. **Read the mode switch.** The TRANSMISSION DEFINED BY: selector at the top has two positions, Scalar throughput and Element train, and here the train is selected. The note under it is explicit about the consequence: Element train defines transmission (6 elements) – scalar τ ignored. The scalar is still in the config; it is simply not in the calculation. Nothing is silently merged.
3. **Read a row.** Each row is one surface: a name, a transfer mode, a descriptive kind, the R or T value (a scalar, a CSV path, or an inline table), the element’s own temperature in kelvin, and the **derived** emissivity. The four silver mirrors M1–M4 are REFLECTIVE / mirror at 293.0 K with $\varepsilon = 0.0184$ each; fpa_window is REFRACTIVE / window at 240.0 K with $\varepsilon = 0.0000$.

The ε column is read-only. For a mirror it is $\varepsilon = 1 - R$; for a transmissive element it is $\varepsilon = 1 - T - R$. Specifying both a reflectance and an emissivity for the same surface over-specifies the energy balance, so the tool derives rather than accepts. $\varepsilon = 0.0184$ on these mirrors is $1 - R$ for $R = 0.9816$ — the value the silver coating’s own curve takes where the derivation is evaluated.

4. **Edit it.** Add mirror, Add refractive, Remove, and the up/down arrows reorder the train; Spectrum... opens an inline wavelength/value table and CSV file... points a row at a measured coating curve. Every edit commits as you make it — there is no Apply button to forget — and each is undoable from Edit → Undo. The chain re-evaluates on the debounce, so the transmission plot and the downstream metrics move as you type.
5. **Inspect one coating.** Selecting a row draws its curve on its own native source grid beneath the table — here Coating detail – M1 (native source grid), showing reflectance rising from about 0.94 at the short-wavelength edge to above 0.98 across the rest of the plotted range. The curve is drawn on the grid the data came in on, not on the chain’s resampled grid, so you are checking the input rather than the interpolation.
6. **Read the effect.** Band 4 evaluates to SNR 340.6, NEDT 12.02 mK, GSD 28.65 m, MTF at Nyquist 0.5937. The published requirement for OLI-2 band 4 is $\text{SNR} \geq 90$ at typical radiance and the recalled Landsat 8 flight value is about 225, so 341 is a consistent over-prediction of the kind this model produces where band-average QE and filter peak are generous and where PRNU, striping, and calibration-transfer noise are not modeled.

2.5 Walkthrough 5 — Run a sweep

Goal: turn one evaluated point into a trade curve without writing a script.

1. **Choose the axis.** Open examples/mwir_leo_minimal.yaml and select stage **4 Optics**. The Parameters dock shows the optics branch; the parameter we are about to sweep, optics.aperture_diameter_m, currently reads 0.3 m with a config badge.

☐ Changed only

Parameter	Value	Source
▼ optics		
zernike_file		default
aperture_diameter_m	0.3 m	config
obscuration_ratio	0	default
n_spiders	0	default
spider_width_m	0 m	default
spider_angle_deg	0 deg	default
focal_length_m	1.2 m	config
f_number	⚡ 4	derived
cold_stop_undersize_frac	0	default
cold_stop_obscuration_ratio	0	default
transmission_scalar	0.7	config
transmission_input_mode	scalar	default
defocus_um	0 μm	default
wfe_mode	scalar_rms	default
wfe_rms_waves	0 waves	default
wfe_reference_wavelength_um	0.633 μm	default
field_position_x	0 deg	default
field_position_y	0 deg	default
surface_roughness_nm	0 nm	default
scatter_halo_sigma_um	100 μm	default
nearfield_enabled	0	config
stray.input_mode	veili...glare	default
stray.veiling_glare_fraction	0	default
stray.absolute_irradiance_W_m2	0 W/m^2	default
stray.includes_thermal	0	default
stray.veiling_glare_mtf	0	default
stray.halo_sigma_um	50 μm	default
psf_n_wavelengths	1	default

Figure 2.10: The Parameters dock, optics branch. `aperture_diameter_m` is the sweep axis; `f_number` below it is derived and moves with it.

Note `f_number` two rows down: 4, lightning bolt, derived. Sweeping the aperture at fixed focal length sweeps the f-number too. A sweep axis drags its dependants with it, which is usually what you want and occasionally a trap — if you meant to hold $f/\#$ constant you must sweep focal length alongside.

2. **Open the sweep.** Run → Run Sweep.... The dialog asks for a parameter, a start, a stop, a number of points, and a metric; a second parameter block turns the run into a 2-D surface. The metric list is populated from the metrics the last evaluation actually produced, so you cannot sweep against a metric this configuration does not compute.
3. **Fill it in.** Parameter `optics.aperture_diameter_m`, start 0.15 m, stop 0.60 m, 10 points, metric `snr`. The run evaluates on a worker thread, so the window stays responsive and the run can be canceled.
4. **Read the curve — and the place it stops being a curve.** The result for this configuration:

Aperture diameter [m]	SNR [-]
0.150	561.94
0.200	749.31
0.250	936.67
0.300	1124.03
0.350	1311.38
0.400	1414.17
0.450	1414.17
0.500	1414.17
0.550	1414.17
0.600	1414.17

Up to 0.35 m, SNR is exactly linear in diameter: $561.94 / 0.150 = 3746$ and $1311.38 / 0.350 = 3747$ per meter. That is the signature of a shot-noise-limited extended-source system. Collected signal goes as D^2 , shot noise goes as $\sqrt{S} \propto D$, so $\text{SNR} \propto D$ — not D^2 .

At 0.40 m the curve goes flat at 1414.17 and stays there. This is not a physical asymptote; it is the full well. The Messages panel fills with saturation warnings (“full well saturated — signal + dark + glow + near-field + stray = 2.246×10^6 e- exceeds full_well_capacity_e = 2×10^6 e-, fill fraction 1.12. Signal clipped ... Downstream SNR/NEDT/NIIRS reflect the CLIPPED signal”). Clipped signal divided by clipped shot noise is $\sqrt{2 \times 10^6} = 1414.2$ — a constant, and a number that no longer responds to the scene at all. **The flat part of this curve is the tool refusing to lie, not the sensor plateauing.** The fix is a shorter integration time, a deeper well, or less throughput; not a bigger telescope.

5. **Keep the numbers.** The finished sweep is retained on the window; File → Export Sweep CSV... writes it out. The same run from a script is `sensor.sweep("optics.aperture_diameter_m", apertures, metric="snr")` — one GUI action, one API call, and chapter 3 runs exactly that.

2.6 Walkthrough 6 — Compare configurations

Goal: hold one instrument fixed and vary only what genuinely differs between several versions of it, then read them side by side.

RADIANT’s multi-configuration model works like CODE V zoom configurations. A parameter is **shared** — one value for every configuration — until you explicitly configure it, at which point it carries one value per configuration, densely. A study therefore states what differs, not what is repeated.

We use `scenarios/09_flagship_missions/9.4_landsat_oli2_snr/oli2_all_bands_study.yaml`: all nine Landsat 9 OLI-2 bands as one study on one instrument.

1. **Open it.** File → Open YAML.... Because the file carries a `configurations:` section, the same Open action that loads a plain config loads the whole study; the status bar says so.

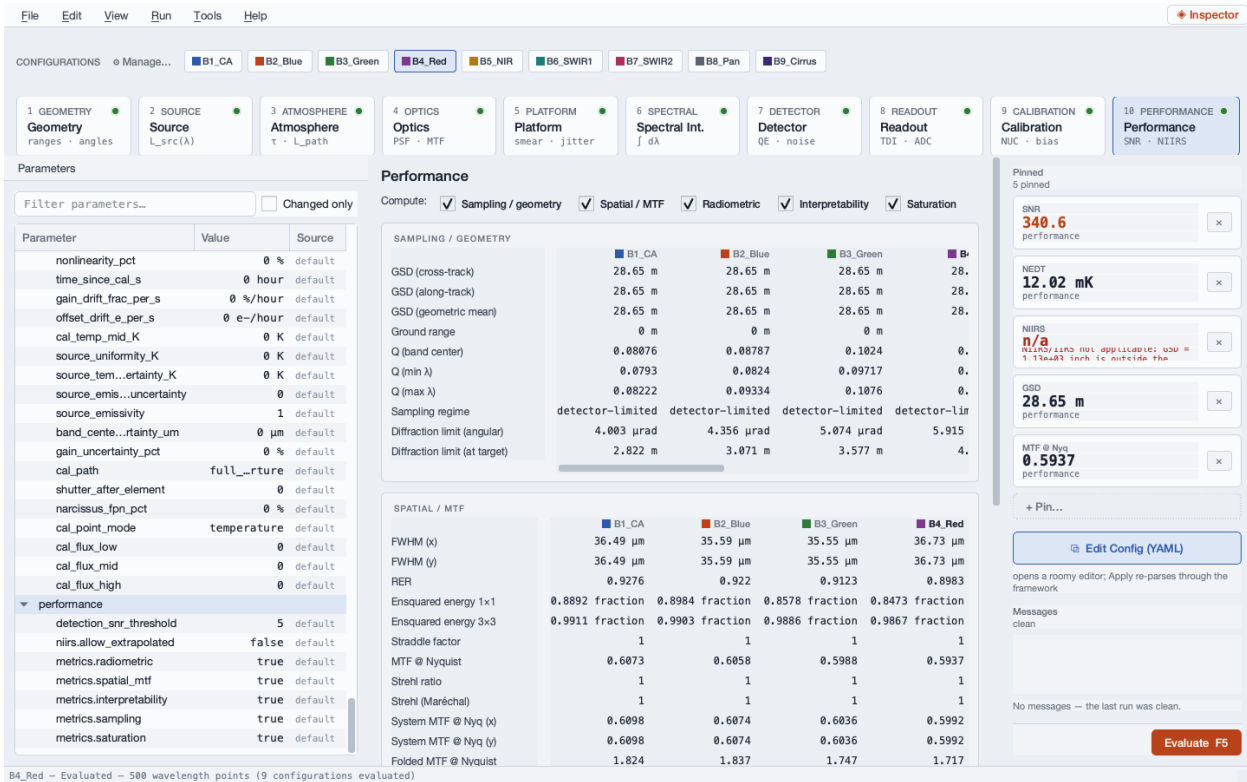


Figure 2.11: Performance workspace on the nine-band OLI-2 study. The configuration selector runs across the top; every metric group shows one column per configuration.

2. **Read the selector.** The CONFIGURATIONS bar lists B1_CA through B9_Cirrus, each with a color chip, and B4_Red is the displayed one. Selecting a different tab which configuration the stage workspaces and the pinned cards describe; it does not re-run anything, because all nine were evaluated in one pass on load. The status bar confirms it: B4_Red - Evaluated - 500 wavelength points (9 configurations evaluated).
3. **Read the matrix.** Every metric group card grows one column per configuration, color-keyed to the selector. In Sampling / geometry, GSD is 28.65 m in every column — pitch, focal length, and altitude are shared, so the ground sample cannot differ. What does differ is everything wavelength-dependent: the diffraction limit runs 4.003 μrad in B1 (443 nm) to 5.915 μrad in B4 (655 nm), and at the target 2.822 m to 4.17 m. *Q* runs 0.08076 to 0.1193 — deeply detector-limited in every band, as a 0.135 m aperture at 705 km must be.

In Spatial / MTF the same wavelength dependence shows up as blur: FWHM 35.55 μm in B3 against 36.73 μm in B4, MTF at Nyquist 0.6073 in B1 falling to 0.5937 in B4, ensquared energy 0.8892 in B1

against 0.8473 in B4. One telescope, one focal plane, and a monotone degradation with wavelength that nobody had to enter.

4. **Check the study against flight.** The nine SNR predictions are 177, 496, 472, 341, 238, 328, 367, 227, and 139 for B1 through B9 respectively, against requirements of 130, 130, 100, 90, 90, 100, 100, 80, and 50. Every band clears its requirement; the ratio to recalled Landsat 8 flight values sits between 0.77 and 2.04.
 5. **Edit once, move everything.** Change a shared parameter — `aperture_diameter_m`, say — and all nine configurations move together on the next evaluation. Change a configured one and only its own column moves. `Edit → Configurations...` is where a shared parameter becomes configured (and back); the Parameters dock marks a configured parameter with a badge so you can always tell which kind you are about to edit.
 6. **Compare files instead of configurations.** When the versions you want to compare are separate files rather than configurations of one study — a vendor’s proposal against your own model, say — use `Tools → Compare Config Files...` instead. It takes the current config plus any number of files added with `Add config file...`, evaluates them all on a worker thread with a progress bar, and lays the metrics out as one table. Use the configuration set when the instrument is one instrument; use the file comparison when it is genuinely several.
-

2.7 Where to go next

Every walkthrough above has a scripted counterpart. The same sweep is `Sensor.sweep`, the same comparison is `ConfigurationSet.compare`, and the same element train is a list in the YAML. Chapter 3 runs six programs that do exactly that, and prints what they produced.

For the full per-scenario campaign — which GUI surface each of the scenarios is meant to stress, and the automated checks that confirm the baselines still open and still reproduce their metrics — see the repository’s scenario exercise index.

Scripting Examples

Six programs ship in `examples/scripts/`. Between them they cover the whole scripting surface an analyst needs: load and evaluate, sweep, clone-and-diff, hand-rolled loops with derived metrics, Monte Carlo tolerancing, and the multi-configuration study. They are deliberately small — the largest is under sixty lines apart from the configuration-set study — and each one is self-documenting, with its explanation in the module docstring and in the output it prints.

Every output block below is the program’s **actual** output, produced by running it from the repository root:

```
PYTHONPATH=src python examples/scripts/<name>.py
```

Long tables are trimmed where the trim is marked; nothing is retyped or idealized.

All six run against `examples/mwir_leo_minimal.yaml`: a 0.30 m f/4 telescope at 8 km altitude looking straight down at a 300 K, $\epsilon = 0.95$ extended scene through a mid-latitude-summer atmosphere, on an 18 μm -pitch focal plane with a 2 Me- well. Because no target extent is given, the scene fills the pixel footprint and every one of these examples runs in the **extended** radiometric regime. That single fact explains most of what follows.

3.1 `basic_evaluation.py` — load, evaluate, read

Demonstrates: the three-line core of the API, plus the two result surfaces that are not metrics — the noise budget and the stage history.

```
from radiant.api.sensor import Sensor

CONFIG = Path(__file__).resolve().parent.parent / "mwir_leo_minimal.yaml"

sensor = Sensor.from_yaml(CONFIG)
result = sensor.evaluate()

for name, value in sorted(result.metrics.items()):
    print(f" {name:20s} = {value:.6g}")

for nt in result.noise_terms:
    print(f" {nt.name:25s} {nt.value_e:.4f} e- RMS")

print(f"Stages executed: {' → '.join(result.history)}")

Output (metric list trimmed to the rows discussed):

=== RADIANT Basic Evaluation ===
```

Config: mwir_leo_minimal.yaml

```

adc_margin_dB           = 4.40063 dB
alias_fraction_at_nyquist = 0.500007 fraction
contrast_snr            = 1124.03
diffraction_limit_angular_urad = 17.2833  $\mu$ rad
diffraction_limit_ground_m = 0.138267 m
diffraction_limit_target_plane_m = 0.138267 m
dynamic_range_dB        = 65.0043 dB
ee_1x1                  = 0.414137 fraction
ee_3x3                  = 0.878017 fraction
fwhm_x_m                = 2.12972e-05 m
fwhm_y_m                = 2.12972e-05 m
ground_range_m          = 0 m
gsd_along_track_m       = 0.12 m
gsd_cross_track_m       = 0.12 m
gsd_geometric_mean_m    = 0.12 m
mrt_at_nyquist_K        = 0.210501 K
mtf_at_nyquist          = 0.266771
mtf_folded_at_nyquist   = 0.53355
mtf_system_at_nyquist_x = 0.26884
mtf_system_at_nyquist_y = 0.26884
nedt_K                  = 0.024958 K
niirs_extrapolated      = 1 0/1 flag
q_center                = 0.944444
q_max                   = 1.11111
q_min                   = 0.777778
rer                     = 0.610069
sampling_regime_code    = 0 code
scnr                    = 1124.03
snr                     = 1124.03
straddle_factor         = 1
strehl                  = 1
strehl_marechal         = 1
well_margin_dB          = 3.98876 dB

```

Noise budget:

```

signal_shot            1124.0766 e- RMS
background_shot        0.0000 e- RMS
nearfield_shot         0.0000 e- RMS
straylight_shot        0.0000 e- RMS
dark_shot              0.7071 e- RMS
gr_noise               0.0000 e- RMS
johnson_noise          0.0000 e- RMS
flicker_1f            0.0000 e- RMS
read_noise             5.0000 e- RMS
ktc_reset              0.0000 e- RMS
quantization           9.2376 e- RMS
prnu                   0.0000 e- RMS
dsnu                   0.0000 e- RMS
clutter                0.0000 e- RMS
persistence_noise      0.0000 e- RMS

```

```
glow_shot          0.0000 e- RMS
```

Stages executed: geometry → source → atmosphere → optics → platform → spectral_integration
→ detector → readout → calibration → performance

What to notice.

The noise budget always has sixteen terms. Twelve of them are exactly zero here because this minimal config does not enable the physics that produces them — no 1/f corner, no PRNU or DSNU, no persistence, no clutter, no cold-shield glow. They are printed anyway. A term that is zero because it was not configured and a term that is zero because it is negligible look identical in a result object; they are distinguished by the configuration, and printing the full taxonomy is what makes that checkable.

SNR is essentially the signal-shot term. $SNR = 1124.03$ against a signal shot noise of 1124.0766 e-RMS. In a shot-limited extended-source system the signal in electrons is the square of the shot noise, so $SNR = S/\sqrt{S} = \sqrt{S}$ — the SNR and the shot-noise term in electrons are numerically the same to within the contribution of everything else. Quadrature-summing the four non-zero terms gives $\sqrt{1124.08^2 + 0.71^2 + 5.00^2 + 9.24^2} = 1124.13$ e- RMS, which is why the 5 e- RMS read noise and the 9.24 e- RMS quantization noise are visible in the budget and invisible in the answer.

The history is the architecture. `result.history` is the ten stages in the order they ran, geometry first. Nothing downstream re-decides what an earlier stage concluded — the radiometric regime, for instance, is classified in the optics stage and read unchanged by the six stages after it.

3.2 aperture_sweep.py — one axis, one metric, one figure

Demonstrates: `Sensor.sweep`, and the plotting helper that turns a `SweepResult` into a figure.

```
import numpy as np
from radiant.api.plot import plot_sweep
from radiant.api.sensor import Sensor

sensor = Sensor.from_yaml(CONFIG)

apertures = np.linspace(0.15, 0.60, 10)
result = sensor.sweep(
    "optics.aperture_diameter_m",
    apertures,
    metric="snr",
)

for d, snr in zip(result.values, result.metric_values, strict=True):
    print(f"{d:10.3f}  {snr:12.2f}")

fig = plot_sweep(result)
fig.savefig(OUTPUT, dpi=150)
```

Output (five saturation warnings elided — they are discussed below):

```
=== Aperture Sweep: SNR vs Aperture Diameter ===
```

D (m)	SNR
0.150	561.94
0.200	749.31
0.250	936.67
0.300	1124.03
0.350	1311.38
0.400	1414.17
0.450	1414.17
0.500	1414.17
0.550	1414.17
0.600	1414.17

Plot saved to: examples/scripts/aperture_sweep_snr.png

What to notice.

The first five points are a straight line through the origin. $561.94 / 0.150 \text{ m} = 3746$ per meter; $1311.38 / 0.350 \text{ m} = 3747$ per meter. For a shot-limited extended source the collected signal scales as the collecting area, $S \propto D^2$, while the dominant noise is its own square root, $\sqrt{S} \propto D$. So $\text{SNR} \propto D$ — linear in diameter, not quadratic. Doubling the aperture doubles the SNR and costs four times the glass.

The last five points are not physics. From 0.40 m onward the answer is pinned at 1414.17 and the run emits a warning per point:

```
UserWarning: ReadoutStage: full well saturated – signal + dark + glow + near-field
+ stray = 2.246e+06 e- exceeds full_well_capacity_e = 2e+06 e- (fill fraction 1.12).
Signal clipped to 2e+06 e-. Downstream SNR/NEDT/NIIRS reflect the CLIPPED signal and
will not respond to scene/atmosphere changes. ...
```

(Quoted with its remedy clause elided. The same points emit a second, distinct warning from the ADC — ReadoutStage: ADC saturated – ... Signal clipped to 16383 DN — because a well that overfills also overfills the digitizer.)

1414.17 is $\sqrt{2 \times 10^6}$ — the shot-limited SNR of a pixel filled exactly to its 2 Me- capacity, and therefore a constant independent of the scene. A trade study that reads this plateau as “diminishing returns above 0.4 m” has drawn a conclusion about an arithmetic clamp. The warning is the result; the number beside it is not. The fix is a shorter integration time, a deeper well, or less throughput.

This is also the reason RADIANT raises rather than clips silently. A clipped chain still produces a plausible-looking float, and a plausible-looking float in a spreadsheet is indistinguishable from a real answer a week later.

3.3 compare_configs.py — clone, change, diff

Demonstrates: `Sensor.clone()` and `Sensor.set()`, and the fact that a configuration change moves many metrics at once in directions that are not all the same.

```
baseline = Sensor.from_yaml(CONFIG)
baseline_result = baseline.evaluate()
```

```

modified = baseline.clone()
modified.set("optics.aperture_diameter_m", 0.45)
modified.set("spectral_integration.integration_time_s", 0.010)
modified_result = modified.evaluate()

all_metrics = sorted(set(baseline_result.metrics) | set(modified_result.metrics))
for name in all_metrics:
    v_base = baseline_result.metrics.get(name, float("nan"))
    v_mod = modified_result.metrics.get(name, float("nan"))
    delta = v_mod - v_base
    pct = (delta / v_base * 100.0) if v_base != 0.0 else float("nan")
    print(f"{name:>20s} {v_base:12.4f} {v_mod:12.4f} {delta:+12.4f} {pct:+9.1f}%")

```

Output (trimmed to the informative rows; the full table is 33 metrics):

=== Configuration Comparison ===

	Metric	Unit	Baseline	Modified	Delta	%Change
	adc_margin_dB	dB	4.4006	0.4119	-3.9888	-90.6%
	alias_fraction_at_nyquist	fraction	0.5000	0.5043	+0.0043	+0.9%
	contrast_snr	dimensionless	1124.0273	4020.4751	+2896.4478	+257.7%
	diffraction_limit_angular_urad	μrad	17.2833	11.5222	-5.7611	-33.3%
	diffraction_limit_ground_m	m	0.1383	0.0922	-0.0461	-33.3%
	diffraction_limit_target_plane_m	m	0.1383	0.0922	-0.0461	-33.3%
	dynamic_range_dB	dB	65.0043	63.0101	-1.9942	-3.1%
	ee_1x1	fraction	0.4141	0.5444	+0.1302	+31.4%
	ee_3x3	fraction	0.8780	0.9199	+0.0419	+4.8%
	fwhm_x_m	m	0.0000	0.0000	-0.0000	-12.9%
	fwhm_y_m	m	0.0000	0.0000	-0.0000	-12.9%
	ground_range_m	m	0.0000	0.0000	+0.0000	n/a
	gsd_along_track_m	m	0.1200	0.1200	+0.0000	+0.0%
	gsd_cross_track_m	m	0.1200	0.1200	+0.0000	+0.0%
	gsd_geometric_mean_m	m	0.1200	0.1200	+0.0000	+0.0%
	mrt_at_nyquist_K	K	0.2105	0.1158	-0.0947	-45.0%
	mtf_at_nyquist	dimensionless	0.2668	0.3854	+0.1186	+44.5%
	mtf_folded_at_nyquist	dimensionless	0.5335	0.7774	+0.2439	+45.7%
	mtf_system_at_nyquist_x	dimensionless	0.2688	0.3859	+0.1170	+43.5%
	mtf_system_at_nyquist_y	dimensionless	0.2688	0.3859	+0.1170	+43.5%
	nedt_K	K	0.0250	0.0198	-0.0051	-20.5%
	niirs_extrapolated	0/1 flag	1.0000	1.0000	+0.0000	+0.0%
	q_center	dimensionless	0.9444	0.6296	-0.3148	-33.3%
	q_max	dimensionless	1.1111	0.7407	-0.3704	-33.3%
	q_min	dimensionless	0.7778	0.5185	-0.2593	-33.3%
	rer	dimensionless	0.6101	0.7065	+0.0964	+15.8%
	sampling_regime_code	code	0.0000	0.0000	+0.0000	n/a
	scnr	dimensionless	1124.0273	4020.4751	+2896.4478	+257.7%
	snr	dimensionless	1124.0273	1414.1738	+290.1465	+25.8%
	straddle_factor	dimensionless	1.0000	1.0000	+0.0000	+0.0%
	strehl	dimensionless	1.0000	1.0000	+0.0000	+0.0%

strehl_marechal	dimensionless	1.0000	1.0000	+0.0000	+0.0%
well_margin_dB	dB	3.9888	0.0000	-3.9888	-100.0%

Changes applied:

optics.aperture_diameter_m: 0.30 → 0.45 m

spectral_integration.integration_time_s: 0.005 → 0.010 s

What to notice.

The spatial metrics are unambiguously better. A 1.5× aperture shrinks the diffraction limit at the ground from 0.1383 m to 0.0922 m — exactly 1/1.5 — pushes Q from 0.944 to 0.630 (deeper into detector-limited), lifts MTF at Nyquist from 0.2668 to 0.3854, RER from 0.6101 to 0.7065, and ensquared energy in the central pixel from 0.4141 to 0.5444. GSD does not move at all, because GSD is pitch times range over focal length and none of those changed. Sharper, not finer-sampled.

The radiometric metrics are a trap. SNR rises only 25.8 %, and well_margin_dB falls to exactly 0.000. A 1.5× aperture and a 2× integration time together deliver 4.5× the electrons into a well that was already 63 % full, so the modified configuration is clipped — and 1414.17 is once again $\sqrt{2 \times 10^6}$. The +257.7 % on contrast_snr and the -20.5 % on NEDT are computed from a clipped signal. well_margin_dB = 0.000 is the tell: zero headroom, by definition, is a full well.

Read well_margin_dB and adc_margin_dB before reading anything in the radiometric group. They are cheap and they are the honesty check on everything else.

Units are in the names, not in the columns. This script prints a bare numeric table whose only unit information is what the metric names carry (nedt_K, gsd_along_track_m, well_margin_dB). That is enough to read it, but it is thinner than the rest of the repository's convention: compare the configuration-set study of dual_band_configuration_set.py below, which pulls each metric's unit from the metric registry and prints it in its own column.

3.4 custom_loop.py — when the built-in sweep is not the shape you want

Demonstrates: that clone() + set() + evaluate() in a plain Python loop is a first-class way to work, and that derived quantities the metric registry does not define are yours to compute.

```
base = Sensor.from_yaml(CONFIG)
```

```
t_int_values = [0.001, 0.002, 0.005, 0.010, 0.020, 0.050]
```

```
for t_int in t_int_values:
```

```
    s = base.clone()
```

```
    s.set("spectral_integration.integration_time_s", t_int)
```

```
    result = s.evaluate()
```

```
    snr = result.metrics["snr"]
```

```
    nedt_val = result.metrics.get("nedt")
```

```
    nedt = float(nedt_val) if nedt_val is not None else float("nan")
```

```
    # Time-normalized SNR: SNR / sqrt(t_int) — useful for comparing
```

```
    # detector performance independent of integration time.
```

```
snr_norm = snr / math.sqrt(t_int)

print(f"{t_int * 1000:12.1f} {snr:10.2f} {snr_norm:10.2f} {nedt:10.6f}")
```

Output (three saturation warnings elided):

```
=== Custom Loop: Integration Time Analysis ===
```

t_int (ms)	SNR	SNR/ $\sqrt{t}$	NEDT (K)
1.0	502.59	15893.37	0.055817
2.0	710.85	15895.11	0.039465
5.0	1124.03	15896.15	0.024958
10.0	1414.17	14141.74	0.019837
20.0	1414.17	9999.71	0.019837
50.0	1414.17	6324.37	0.019837

SNR/ $\sqrt{t}$ should be approximately constant if the system is photon-noise-limited (shot noise $\propto \sqrt{\text{signal}} \propto \sqrt{t_{\text{int}}}$).

What to notice.

The script's own hypothesis is confirmed, then broken, and the break is the point. For the first three rows SNR/ $\sqrt{t}$ is constant to four significant figures — 15893, 15895, 15896 per $\sqrt{s}$. Signal scales with integration time, shot noise with its square root, so SNR scales as $\sqrt{t}$: this system is photon-noise-limited. From 10 ms the ratio collapses (14142, 9999.7, 6324.4) because the SNR column has stopped moving. The well fills at about 8 ms, and past that point longer integration buys nothing but a clipped pixel. A constant SNR/ $\sqrt{t}$ is a diagnostic for “photon-limited”; a *falling* one is a diagnostic for “saturated or read-noise-limited”, and telling those two apart is exactly what this column is for.

The NEDT column reads in kelvin, from the registered key. The metric is registered as nedt_K, carrying its unit in its name, and the script reads result.metrics["nedt_K"] — 0.024958 K for the 5 ms row, the same value basic_evaluation.py prints. (An earlier revision asked for nedt with .get(), which returned None and printed nan down the whole column: result.metrics is a plain mapping, .get() on a misremembered key fails silently, and result.metric_records() is the surface that tells you the registered names and their units.) Past the saturation knee the NEDT column freezes with the SNR column, for the same reason.

3.5 tolerance_analysis.py — Monte Carlo over manufacturing spread

Demonstrates: set_tolerance, monte_carlo, and the statistics and correlation surface on the result — the answer to “which of my tolerances actually matters”.

```
sensor = Sensor.from_yaml(CONFIG)

sensor.set_tolerance("optics.aperture_diameter_m", "gaussian", std_fraction=0.02)
sensor.set_tolerance("optics.transmission_scalar", "gaussian", std_fraction=0.05)
sensor.set_tolerance("detector.qe_value", "gaussian", std_fraction=0.03)

mc_result = sensor.monte_carlo(n_trials=50, seed=42)
```

```

units = {rec.name: rec.unit for rec in sensor.evaluate().metric_records()}
for metric in mc_result.metric_names:
    unit = units.get(metric, "")
    print(f"{metric} [{unit}]:")
    print(f"  Mean:  {mc_result.mean(metric):.4f} {unit}")
    print(f"  Std:    {mc_result.std(metric):.4f} {unit}")
    print(f"  5th %: {mc_result.percentile(metric, 5.0):.4f} {unit}")
    print(f"  95th %: {mc_result.percentile(metric, 95.0):.4f} {unit}")

corr = mc_result.correlation("snr")
for param, r in sorted(corr.items(), key=lambda kv: abs(kv[1]), reverse=True):
    print(f"  {param:40s}  r = {r:+.4f}")

```

Output (per-metric blocks trimmed to four of the thirty-three):

```

=== Monte Carlo Tolerance Analysis ===
Trials: 50
Seed:   42

```

```

adc_margin_dB [dB]:
  Mean:  4.4759 dB
  Std:   0.5916 dB
  5th %:  3.7031 dB
  95th %:  5.3286 dB

```

```

alias_fraction_at_nyquist [fraction]:
  Mean:  0.5000 fraction
  Std:   0.0000 fraction
  5th %:  0.5000 fraction
  95th %:  0.5000 fraction

```

```

contrast_snr [dimensionless]:
  Mean: 1119.8083 dimensionless
  Std:  38.5357 dimensionless
  5th %: 1065.5523 dimensionless
  95th %: 1170.0833 dimensionless

```

```

diffraction_limit_angular_urad [μrad]:
  Mean: 17.2882 μrad
  Std:  0.3000 μrad
  5th %: 16.7862 μrad
  95th %: 17.6788 μrad

```

```

diffraction_limit_ground_m [m]:
  Mean: 0.1383 m
  Std:  0.0024 m
  5th %: 0.1343 m
  95th %: 0.1414 m

```

```

diffraction_limit_target_plane_m [m]:
  Mean: 0.1383 m
  Std:  0.0024 m

```

```
5th %: 0.1343 m
95th %: 0.1414 m

dynamic_range_dB [dB]:
  Mean: 65.0419 dB
  Std: 0.2958 dB
  5th %: 64.6556 dB
  95th %: 65.4683 dB

ee_1x1 [fraction]:
  Mean: 0.4142 fraction
  Std: 0.0073 fraction
  5th %: 0.4049 fraction
  95th %: 0.4268 fraction

ee_3x3 [fraction]:
  Mean: 0.8780 fraction
  Std: 0.0020 fraction
  5th %: 0.8754 fraction
  95th %: 0.8813 fraction

fwhm_x_m [m]:
  Mean: 0.0000 m
  Std: 0.0000 m
  5th %: 0.0000 m
  95th %: 0.0000 m

fwhm_y_m [m]:
  Mean: 0.0000 m
  Std: 0.0000 m
  5th %: 0.0000 m
  95th %: 0.0000 m

ground_range_m [m]:
  Mean: 0.0000 m
  Std: 0.0000 m
  5th %: 0.0000 m
  95th %: 0.0000 m

gsd_along_track_m [m]:
  Mean: 0.1200 m
  Std: 0.0000 m
  5th %: 0.1200 m
  95th %: 0.1200 m

gsd_cross_track_m [m]:
  Mean: 0.1200 m
  Std: 0.0000 m
  5th %: 0.1200 m
  95th %: 0.1200 m

gsd_geometric_mean_m [m]:
```

```
Mean: 0.1200 m
Std: 0.0000 m
5th %: 0.1200 m
95th %: 0.1200 m

mrt_at_nyquist_K [K]:
Mean: 0.2117 K
Std: 0.0106 K
5th %: 0.1956 K
95th %: 0.2281 K

mtf_at_nyquist [dimensionless]:
Mean: 0.2668 dimensionless
Std: 0.0065 dimensionless
5th %: 0.2584 dimensionless
95th %: 0.2779 dimensionless

mtf_folded_at_nyquist [dimensionless]:
Mean: 0.5336 dimensionless
Std: 0.0131 dimensionless
5th %: 0.5168 dimensionless
95th %: 0.5558 dimensionless

mtf_system_at_nyquist_x [dimensionless]:
Mean: 0.2688 dimensionless
Std: 0.0058 dimensionless
5th %: 0.2612 dimensionless
95th %: 0.2786 dimensionless

mtf_system_at_nyquist_y [dimensionless]:
Mean: 0.2688 dimensionless
Std: 0.0058 dimensionless
5th %: 0.2612 dimensionless
95th %: 0.2786 dimensionless

nedt_K [K]:
Mean: 0.0251 K
Std: 0.0008 K
5th %: 0.0240 K
95th %: 0.0263 K

niirs_extrapolated [0/1 flag]:
Mean: 1.0000 0/1 flag
Std: 0.0000 0/1 flag
5th %: 1.0000 0/1 flag
95th %: 1.0000 0/1 flag

q_center [dimensionless]:
Mean: 0.9447 dimensionless
Std: 0.0164 dimensionless
5th %: 0.9173 dimensionless
95th %: 0.9661 dimensionless
```

```
q_max [dimensionless]:
  Mean:  1.1114 dimensionless
  Std:   0.0193 dimensionless
  5th %: 1.0792 dimensionless
  95th %: 1.1365 dimensionless

q_min [dimensionless]:
  Mean:  0.7780 dimensionless
  Std:   0.0135 dimensionless
  5th %: 0.7554 dimensionless
  95th %: 0.7956 dimensionless

rer [dimensionless]:
  Mean:  0.6101 dimensionless
  Std:   0.0047 dimensionless
  5th %: 0.6042 dimensionless
  95th %: 0.6182 dimensionless

sampling_regime_code [code]:
  Mean:  0.0000 code
  Std:   0.0000 code
  5th %: 0.0000 code
  95th %: 0.0000 code

scnr [dimensionless]:
  Mean:  1119.8083 dimensionless
  Std:   38.5357 dimensionless
  5th %: 1065.5523 dimensionless
  95th %: 1170.0833 dimensionless

snr [dimensionless]:
  Mean:  1119.8083 dimensionless
  Std:   38.5357 dimensionless
  5th %: 1065.5523 dimensionless
  95th %: 1170.0833 dimensionless

straddle_factor [dimensionless]:
  Mean:  1.0000 dimensionless
  Std:   0.0000 dimensionless
  5th %: 1.0000 dimensionless
  95th %: 1.0000 dimensionless

strehl [dimensionless]:
  Mean:  1.0000 dimensionless
  Std:   0.0000 dimensionless
  5th %: 1.0000 dimensionless
  95th %: 1.0000 dimensionless

strehl_marechal [dimensionless]:
  Mean:  1.0000 dimensionless
  Std:   0.0000 dimensionless
```

```

5th %: 1.0000 dimensionless
95th %: 1.0000 dimensionless

well_margin_dB [dB]:
  Mean:  4.0640 dB
  Std:   0.5916 dB
  5th %: 3.2912 dB
  95th %: 4.9168 dB

Correlation with SNR:
  optics.transmission_scalar      r = +0.8184
  optics.aperture_diameter_m     r = +0.5109
  detector.qe_value              r = +0.4983

```

What to notice.

The correlations rank the tolerances, and the ranking is not the obvious one. Transmission dominates at $r = +0.82$ despite being only a 5 % 1σ spread, because it enters the signal linearly. Aperture is next at $+0.51$ even though signal goes as D^2 — because its spread is only 2 %, so its contribution to the *variance* is $2 \times 2 \% = 4 \%$ against transmission's 5 %. QE at 3 % lands at $+0.50$, essentially tied with aperture. Read the correlations together with the spreads you assigned: they measure the tolerance you set, not the sensitivity of the physics alone. If tightening one tolerance is cheap, this table says which one to tighten.

The metric means are not the nominal values. Nominal SNR is 1124.03; the Monte Carlo mean is 1119.81 with $\sigma = 38.54$. The shortfall is about 0.1σ — sampling noise at 50 trials, not a bias — but it is the kind of difference worth tracking: a nonlinear response to symmetric input spread genuinely does move the mean, and with only 50 samples you cannot yet tell which you are looking at. Raise `n_trials` before reading anything into a sub- σ shift.

The run is reproducible. `seed=42` is echoed in the output, and the same seed gives the same 50 draws and the same statistics on any machine.

Every statistic carries its unit. The script reads each metric's registered unit from `metric_records()` once and prints it beside the mean, the spread and both percentiles, so a line such as `Std: 0.0007 K` is a statement, not a bare number.

3.6 dual_band_configuration_set.py — one telescope, three ways to operate it

Demonstrates: the multi-configuration API — `ConfigurationSet`, `configure()`, `evaluate_all()`, `compare()`, `save()/load()` — and per-configuration warning attribution. At 470 lines it is the largest example, and most of those lines are the printed explanation rather than the modeling.

The study is a single 0.30 m f/4 telescope at 8 km altitude on one 18 μm -pitch focal plane, operated three ways:

Configuration	Band	Integration	QE	Well
MWIR	3.5–5.0 μm	5.0 ms	0.70	2.0 Me-
LWIR	8.0–12.0 μm	0.5 ms	0.55	6.0 Me-
LWIR_long	8.0–12.0 μm	2.0 ms	0.55	6.0 Me-

```
from radiant.api import ConfigSetRunResult, ConfigurationSet, Sensor
```

```
base = Sensor.from_yaml(BASE_CONFIG, wavelength_points=300)
study = ConfigurationSet(base, names=["MWIR", "LWIR", "LWIR_long"])
```

```
study.configure("spectral_integration.filter_min_um", [3.5, 8.0, 8.0])
study.configure("spectral_integration.filter_max_um", [5.0, 12.0, 12.0])
study.configure("spectral_integration.integration_time_s", [0.005, 0.0005, 0.002])
study.configure("detector.qe_value", [0.70, 0.55, 0.55])
study.configure("detector.dark_rate_e_per_s", [1.0e5, 2.0e6, 2.0e6])
study.configure("readout.full_well_capacity_e", [2.0e6, 6.0e6, 6.0e6])
study.configure("readout.gain_e_per_dn", [32.0, 92.0, 92.0])
```

```
study.baseline = "MWIR" # deltas in compare() are measured against this
study.active = "MWIR" # evaluated first (this is the GUI's displayed one)
```

```
run = study.evaluate_all()
comparison = study.compare(run)
```

Seven parameters are configured; eleven stay shared — aperture, focal length, altitude, pixel pitch, read noise, scene temperature, emissivity, and the atmosphere. A change to any shared value moves all three configurations at once. That is the whole point of the model: the study states what differs, not what is repeated.

Output (heavily trimmed — the program prints about 310 lines; the focus-metric table is re-laid out to fit the page, with each metric's unit moved into its name column, and every value is the program's own):

Summary (one line per configuration, evaluation order = active first):

```
MWIR      * ok  snr = 1124 [dimensionless]; nedt_K = 0.02496 [K];
           gsd_geometric_mean_m = 0.12 [m]
LWIR      ok  snr = 2248 [dimensionless]; nedt_K = 0.02761 [K];
           gsd_geometric_mean_m = 0.12 [m]
LWIR_long ok  snr = 2448 [dimensionless]; nedt_K = 0.02536 [K];
           gsd_geometric_mean_m = 0.12 [m] (2 warnings)
(* = baseline: 'MWIR'; 0 of 3 configuration(s) failed)
```

Warning attribution — each warning belongs to exactly one configuration:

```
MWIR      no warnings
LWIR      no warnings
LWIR_long 2 warning(s): ...
```

Focus metrics (value [unit], then delta vs baseline):

metric [unit]	MWIR	LWIR	LWIR_long
---------------	------	------	-----------

```

-----
snr [dimensionless]          1123.79  2248.25 (+1.12e+03)  2447.71 (+1.32e+03)
nedt_K [K]                   0.0249632 0.0276141 (+0.00265) 0.0253638 (+0.000401)
gsd_geometric_mean_m [m]     0.12                0.12                0.12
diffraction_limit_ground_m [m] 0.138267    0.325333 (+0.187)    0.325333 (+0.187)
q_center [dimensionless]     0.944444    2.22222 (+1.28)    2.22222 (+1.28)
mtf_at_nyquist [dimensionless] 0.266683 5.42079e-17 (-0.267) 5.42079e-17 (-0.267)
ee_1x1 [fraction]           0.414038    0.135344 (-0.279)    0.135344 (-0.279)
well_margin_dB [dB]         3.98901    1.48629 (-2.5)    0.00579252 (-3.98)

```

and, later, the program's own reading of those numbers:

```

MWIR      GSD = 0.12 m  diffraction blur = 0.1383 m  Q = 0.944 [dimensionless]
LWIR      GSD = 0.12 m  diffraction blur = 0.3253 m  Q = 2.22 [dimensionless]
LWIR_long GSD = 0.12 m  diffraction blur = 0.3253 m  Q = 2.22 [dimensionless]

MWIR      t_int = 5 ms  SNR = 1124  NEDT = 24.96 mK  well fill = 63.2 %
LWIR      t_int = 0.5 ms  SNR = 2248  NEDT = 27.61 mK  well fill = 84.3 %
LWIR_long t_int = 2 ms  SNR = 2448  NEDT = 25.36 mK  well fill = 337.2 %

LWIR      well 5.057e+06 e- of 6e+06 e- (84.3 %) -> ok
LWIR_long well 2.023e+07 e- of 6e+06 e- (337.2 %) -> clipped

```

What to notice.

Same GSD, different imaging regime. All three configurations sample the ground at 0.12 m, because GSD is set by pitch, focal length, and range — all shared. The *optical* blur is not shared: diffraction scales with wavelength, so the LWIR spot on the ground is 0.3253 m against the MWIR's 0.1383 m, a factor of 2.35 from the very same 0.30 m aperture. $Q = \lambda F / \# / p$ therefore runs 0.944 in the MWIR (undersampled, detector-limited, MTF at Nyquist a healthy 0.2667) and 2.22 in the LWIR (oversampled, optics-limited, MTF at Nyquist 5.4×10^{-17} — numerically zero, because diffraction cuts off *below* Nyquist). One telescope, one focal plane, two genuinely different imaging regimes, chosen only by the band.

A metric can diverge without being wrong. Because MRT at Nyquist scales as $1/\text{MTF}$, the LWIR configurations report `mrt_at_nyquist_K` of order 10^{15} K. That is the metric correctly reporting “this system resolves nothing at Nyquist in this band”, not a temperature. Evaluate LWIR resolution below the optical cutoff instead.

Aliasing behaves as the sampling theorem says it must. The MWIR folded MTF at Nyquist is $0.5334 = 2 \times 0.2667$, and the alias fraction is 0.5000: at Nyquist the first sampling replica lands back on Nyquist, so exactly half the apparent contrast there is folded-in above-Nyquist content. The LWIR folded value is 9.7×10^{-16} — zero — and its alias fraction reads exactly 0. The optics cut off below Nyquist, so there is nothing to alias. That zero is *reported*, not computed: the ratio is evaluated only where the folded MTF exceeds 10^{-9} of its DC value, below which both terms are round-off.

The interesting sensitivity result is not the smaller NEDT. The LWIR configuration reaches 27.61 mK — within about 11 % of the MWIR's 24.96 mK — while integrating ten times shorter, and still fills a three-times-deeper well to 84.3 %. At 300 K the scene's spectral radiance peaks near $9.7 \mu\text{m}$, and dL/dT — which is what NEDT actually measures — is far larger in absolute terms in the LWIR. The band is not photon-starved; it is well-depth-limited. At equal integration time the LWIR configuration would be decisively more sensitive, and it cannot use equal integration time on this focal plane.

LWIR_long demonstrates the trap it was built to demonstrate. Four times the LWIR integration time overfills a 6 Me- well to 337.2 %. Its SNR of 2448 is the highest in the study and it collects the * best-value mark on several metrics, and it has won nothing: its signal was clipped, so its SNR and NEDT no longer respond to the scene. The two warnings that say so are attributed to LWIR_long alone — not re-raised into the caller's warning filters, not attached to the study as a whole — so a noisy configuration cannot be mistaken for a property of the run. Trade integration time against well depth, not against SNR alone.

The whole study is one file. `study.save(path)` writes an ordinary RADIANT config plus one configurations: section carrying the names, the baseline, the active configuration, and the dense per-configuration value lists. A file without that section is byte-for-byte today's format and still loads as the degenerate one-configuration set. The script round-trips its own output and checks that the names, the configured table, and the shared parameters all come back identical.

3.7 What these six do not cover

Batch matrices over more than two axes, measured-data import and reconciliation, Zernike wavefront import, and the flagship-mission validation comparisons all live in the scenario suite rather than in `examples/scripts/`. The repository's scenario catalog indexes them, and later chapters of this volume work several of them end to end.

Reference — Every Way to Define Optical Transmission

Example: `transmission_doors · ships in the wheel as radiant/data/examples/transmission_doors.yaml · modality: config-led, GUI-readable`

4.1 What this example is

A syntax reference that happens to run. The other six bundled examples are curated snapshots of real scenarios; this one is a deliberately unrealistic six-element train whose only job is to use **every way of defining optical transmission at once**, so you can read the forms side by side and diff your own config against it.

Open it, copy the element whose shape matches the datasheet in front of you, and delete the rest.

It evaluates to SNR 1181.00 [-] with a band-mean system throughput of 0.80595 [-]. Those numbers are not a result anyone should cite; they exist so the file stays honest.

4.2 The first fork: scalar τ , or an element train

RADIANT accepts optical transmission in exactly two mutually exclusive forms.

	How you say it	When it is right
Scalar	<code>optics.transmission_scalar: 0.70</code>	You know the net throughput and nothing else — a vendor’s single number, an early trade.
Element train	<code>an optical_elements: list</code>	You have a prescription: surfaces, coatings, temperatures.

A config carrying both is refused — there is no rule for which should win. This example uses the train, so it has **no** `transmission_scalar` line at all.

The choice is not only bookkeeping. A scalar τ is a lumped number with no surfaces, so it has **no Kirchhoff emissivity** and contributes no warm-optics self-emission. In a thermal band that omission is not small: declaring the same net τ as a real train of 290 K surfaces is what puts the optics’ own glow into the background. If you are working in the MWIR or LWIR and you care about the noise floor, the train is not a refinement — it is the model.

4.3 The second fork: three forms, per quantity

Every optical quantity — a mirror’s reflectance, a lens surface’s coating, a bulk absorption coefficient — takes any of three forms, independently of every other quantity in the file.

Scalar. One number, flat across the band.

```
reflectance: 0.980
```

Spectral CSV. A path, resolved relative to *the config file*, not the working directory. Two columns, wavelength_um then value. Lines starting with # are comments, and a single column-header row is accepted — which is what Excel and pandas write, so you do not have to strip it.

```
reflectance: transmission/protected_silver_R.csv
```

Inline λ -table. The same meaning as a CSV, carried in the config itself. Right for a handful of points you do not want to manage as a separate file.

```
reflectance:
  wavelength_um: [3.0, 4.0, 5.0]
  values: [0.975, 0.978, 0.974]
```

NaN and inf are refused in a CSV rather than read as data: one of them interpolates across the whole grid, and a silently-NaN emissivity is worse than a stopped run.

4.4 The third fork: three element models

Model	What it is	ϵ
Reflective	A mirror: one surface, one reflectance.	$1 - R$, derived
Simple refractive	One net transmittance and nothing else.	0, by construction
Cavity	Two coated surfaces around an absorbing bulk.	derived from $\alpha \cdot d$ and the coatings

The middle row is the one that surprises people. A simple refractive element takes the remaining $1 - \tau$ to be **reflection, not absorption**, so its emissivity is zero and it emits nothing at any temperature. That is the correct model for a cold window whose losses you only know in aggregate — and the wrong one if you need its thermal emission. In this example dewar_window sits at 80 K where it does not matter; put the same element at 290 K and you would be silently omitting its glow.

Use the cavity model when you need an element’s emission. It is also the only model a substrate can act on, which is why naming one on a mirror or on a simple refractive row is refused rather than quietly ignored.

4.5 The fourth fork: where the bulk absorption comes from

A cavity element needs the bulk absorption coefficient α . There are two doors, and they reach the identical physics.

Named substrate — the bundled material library supplies $\alpha(\lambda)$:

```
substrate: germanium
thickness_m: 0.008
R1: 0.010
R2: 0.010
```

Explicit α — the custom-material path, for a measured coupon or a material the library does not carry. It takes all three forms, and a measured coupon is a spectrum:

```
alpha: transmission/chalcogenide_alpha.csv
thickness_m: 0.006
```

Naming a substrate **and** giving α over-specifies the element and is refused: a substrate supplies exactly that quantity, so there is no rule for which should win. `thickness_m` is required either way — thickness belongs to the lens, never to the material.

The library carries six materials in two confidence tiers. ZnSe, CaF₂ and multispectral ZnS are built from five or six published laser-line α anchors; Ge, Si and BaF₂ are flagged as class-typical, because an analyst cannot avoid germanium and silicon whatever the data density. The tier is shown wherever you pick a material, and a substrate used outside its published window is refused rather than extrapolated — past an absorption edge, an extrapolated α is not an approximation, it is a different material.

There is no refractive index input. It entered no formula.

4.6 The fifth fork: the two surfaces are independent

A cavity's faces may differ in **value and in form**, which is what a real lens looks like — a measured coating on one face and a nominal number on the other. `imager_custom` in this example does exactly that:

```
R1: transmission/ar_coating_ge_R.csv # measured curve
T2: 0.988 # nominal scalar
```

Note also that surface 1 is given as **R** and surface 2 as **T**. Either is accepted per surface, and the other is derived: coatings are lossless by model rule, so $R + T = 1$. Give whichever the datasheet quotes.

4.7 What each element in the file demonstrates

#	Element	Door	Through-value [-]	ϵ [-]
1	primary	REFLECTIVE, scalar	0.98000	0.02000
2	secondary	REFLECTIVE, spectral CSV	0.98000	0.02000
3	fold	REFLECTIVE, inline λ -table	0.97567	0.02433
4	dewar_window	simple REFRACTIVE, scalar τ	0.92000	0.00000
5	collimator_ge	cavity, named substrate	0.96600	0.01438
6	imager_custom	cavity, explicit α from CSV, mixed-form surfaces	0.96781	0.01066

Row 4's zero emissivity is the simple-refractive model doing what it says. Rows 5 and 6 differ in where α came from and in nothing else.

4.8 Reading it in the GUI

Open the example and go to **Optics ► Transmission**. The train table is the overview; selecting a row opens that element in the detail editor beside its figure.

- **Model** names which of the three models the element uses, and switches between them. A switch states what it will cost before it applies — converting a simple element to a cavity has to *seed* coatings, because one net transmittance cannot determine two coatings plus a bulk.
- **Definition** shows the two surfaces independently, each with its own R-or-T choice and its own scalar / table / CSV source.
- **Bulk** offers the named substrate or the explicit α as one choice, never two fields you can both fill — so the over-specification above cannot be constructed. On rows 1–4 the same tab explains why there is no bulk.
- **View $\alpha(\lambda)$...** draws the chosen material's absorption on a log axis across its whole published window with the run band shaded. α spans orders of magnitude inside a transparency window — CaF_2 's rises $645\times$ across its own — so a linear axis would collapse the transparent region onto zero.

4.9 See also

- docs/guides/examples_running.md — running a bundled example
- docs/architecture/RADIANT_Optics.md §6 — the cavity model and the first-order one-interaction-per-surface rule
- docs/guides/parameter_reference.md — every parameter, generated from the schema

Case Study — MWIR Maritime Surveillance

Scenario 1.1 · persona: Sarah, systems engineer on a proposal team · modality: GUI-led · scenario folder: scenarios/01_sarah_systems_engineer/1.1_mwir_maritime_surveillance/

5.1 The question

Sarah's team is bidding a sun-synchronous MWIR ship-detection payload. The procurement asks for detection of a small surface vessel from a 500 km orbit, and the proposal has to defend an aperture: somewhere between 15 cm and 45 cm at a fixed $f/2.5$, on a 640×512 InSb focal plane with $15 \mu\text{m}$ pixels. Two questions have to be answered before the aperture can be argued:

1. How do SNR, NEDT, NIIRS and detection range move as the aperture grows?
2. A colleague has handed her a MODTRAN 6 tape7 for a maritime column. How much does using that real transmittance instead of RADIANT's own parametric model change the answer?

The second question is the one that used to have no answer. Before RADIANT could read a tape7, an analyst holding a colleague's MODTRAN run still had to re-derive the atmosphere from a parametric model — which is to say, ignore the data she had.

This chapter walks the GUI half of that study on the scenario's committed baseline, `inputs/1.1_mwir_maritime_surveillance.gui.yaml`, which is the 30 cm mid-sweep point with the parametric atmosphere. The MODTRAN comparison and the full aperture sweep are the scripted half; the closing section points at them.

5.2 The inputs, and why they are what they are

Quantity	Value	Why
Sensor altitude	500,000 m	The procurement's sun-synchronous reference orbit.
Path zenith angle	20°	A representative off-nadir look, not a nadir best case.
Slant range to target	532,089 m	Carried in the config file; the altitude and look angle above imply it to within 0.5 %.
Target projected area	240 m^2	A $30 \text{ m} \times 8 \text{ m}$ hull, presented broadside.
Target temperature	288 K	Sea-surface temperature; the hull is in thermal equilibrium with it.

Quantity	Value	Why
Target emissivity	steel curve, band-average $\epsilon = 0.266$	RADIANT's library curve over 3.5–5.0 μm ; the catalog wants a rust-specific curve, which does not exist.
Background	water_calm, 288 K, $\epsilon = 0.985$	Calm-ocean emissivity from the same library.
Aperture diameter	0.30 m	Mid-point of the 0.15–0.45 m trade.
Focal length	0.75 m	Holds $f/2.5$ at the mid-point.
Optical transmission	0.85	Scalar lump — this study does not model a coated train.
Spectral band	3.5–5.0 μm	MWIR cold-filter passband.
Integration time	0.005 s	5 ms, a long dwell for a pushbroom but plausible for a staring frame.
Pixel pitch	15 $\mu\text{m} \times 15 \mu\text{m}$	The vendor's InSb array.
Quantum efficiency	0.773476	Band-average of the vendor's InSb QE curve over 3.5–5.0 μm .
Dark rate	50,000 e-/s at 77 K	Vendor figure at the cryogenic set point.
Read noise	30 e- RMS	Vendor ROIC.
Full well / gain / ADC	8×10^6 e- / 500 e-/DN / 14 bit	Vendor ROIC.
Atmosphere	parametric, midlat_summer, maritime aerosol, 23 km visibility	Matched to the profile the MODTRAN column was run against.

Two of those deserve a second look. The **target and the background are at the same temperature**: a steel hull floating in 288 K water is not a hot target. Everything this sensor detects comes from the *material* difference between painted steel and calm water — the library's steel curve band-averages to $\epsilon = 0.266$ over 3.5–5.0 μm against the sea's 0.985, and by Kirchhoff the hull's complementary reflectance $\rho = 0.734$ bounces sky and (in daylight) sun back into the aperture. RADIANT models the hull as a mixed emit-plus-reflect target for exactly that reason. The two effects run opposite ways, and how nearly they cancel is the whole detection problem.

And the **quantum efficiency is a scalar**, band-averaged from the vendor's curve rather than carried spectrally — the same simplification scenario 1.2 makes, and the reason this baseline and the scenario's spectral-QE runs agree only to about a percent.

5.3 The GUI walk

5.3.1 Step 1 — The atmosphere the colleague supplied

The scenario's own GUI workflow opens with File → Import Atmosphere → MODTRAN tape7. That is a modal file dialog, and the figures in this manual are captured offscreen where modal dialogs cannot be photographed cleanly, so the figure below shows the state that matters instead: the Atmosphere workspace as the baseline config leaves it, on the **parametric** side of the comparison.



Figure 5.1: Atmosphere workspace on the scenario 1.1 baseline — the parametric model’s four inputs above, the target-path transmittance and path radiance they produce below.

The model selector reads *simple*, and the four parametric inputs beneath it are the whole specification: standard atmosphere *midlat_summer*, aerosol type *maritime*, visibility 23 km, precipitable water 1.4 cm. The Fried parameter is 0 m, which is how this workspace says *turbulence off* — appropriate for a down-looking space sensor where the turbulent layer is at the far end of a 532 km path.

The plots underneath are the stage’s output, drawn on the chain’s 500-point wavelength grid. The **target path** panel carries two curves: τ_{up} , flat at a little under half scale from 3.5 μm to about 4.9 μm and then falling off the 5.0 μm filter edge, and L_{path} , the up-welling path radiance, climbing steadily across the band. The scenario’s runner reports the in-band mean as $\bar{\tau} = 0.4594$ — a maritime MWIR column at 20° off nadir passes under half the target’s emission. The **background path** panel below repeats the exercise for the ground-to-sensor leg.

Where the tape7 goes. Switching the model selector to modtran exposes atmosphere. `modtran.tape7_path`; RADIANT parses and unit-converts the deck before the chain runs, with no temporary-CSV side door. The scenario’s script runs both ways on one otherwise-identical config, which is what makes the comparison a clean A/B on the atmosphere term alone.

5.3.2 Step 2 — Place the sensor and the target

Select stage 1 **Geometry**, tab **Inputs**.

The screenshot displays the RADIANT software interface for the Geometry stage, specifically the Inputs tab. The top navigation bar includes tabs for 1 GEOMETRY, 2 SOURCE, 3 ATMOSPHERE, 4 OPTICS, 5 PLATFORM, 6 SPECTRAL, 7 DETECTOR, 8 READOUT, 9 CALIBRATION, and 10 PERFORMANCE. The left sidebar shows a list of parameters for the geometry stage, including sensor altitude, target altitude, site elevation, path zenith, solar zenith, solar azimuth, solar illumination, ground speed, target range, target projected area, target shape, target shape radius, target shape length, target shape width, target shape height, target shape yaw, target shape pitch, target shape roll, sensor off-boresight, ground range, elevation angle, solar elevation, site latitude, day of year, local solar time, ltan, and circular orbit. The main workspace is divided into two sections: 'Inputs' and 'Schematic'. The 'Inputs' section contains a 'Scene class' dropdown set to 'space → ground (space_to_ground) → derived', an 'Assert scene class' dropdown set to 'auto', and a list of 'Geometry inputs' with values for sensor altitude (500000 m), target altitude (0 m), path zenith (20 deg), sensor off-boresight (18.4895 deg), ground range (167956 m), elevation angle (70 deg), and target range (532089 m). The 'Schematic' section shows a 'Viewing geometry' section with 'Path zenith at lower endpoint (V1)' and a 'Solar geometry' section with 'Solar zenith 0_s (S1)' set to 'day' and 'solar illumination' set to '0 deg'. The right sidebar displays performance metrics: SNR (965.7), NEDT (26.04 mK), NIIRS (4.582), GSD (10.93 m), and MTF @ Nyq (0.3553). At the bottom right, there is an 'Evaluate F5' button.

Figure 5.2: Geometry workspace, Inputs tab — the derived scene class, the off-by-default metric list, and the V1 path-zenith viewing mode.

The card at the top reads Scene: space → ground (space_to_ground) — derived. Nobody typed that; it follows from a 500 km sensor looking at a 0 m target, and it is what decides which metrics are relevant. For this scene class only three metrics default off — the three target-plane sample distances — because a ground scene *has* a ground plane and every ground-projection metric applies.

The viewing family sits in mode **V1**, Path zenith at lower endpoint. The three entered fields are `sensor_altitude_m = 500,000 m`, `target_altitude_m = 0 m` and `path_zenith_rad = 20 deg`; `sensor_off_boresight_rad`, `ground_range_m`, `elevation_angle_rad` and `target_range_m` are grayed because they belong to *other* doors of the same family, not because they are unavailable. The slant range reads 532,089 m — 6 % longer than the 500 km altitude, which is the cost of a 20° look.

In the Parameters dock on the left, five rows carry a config badge — `sensor_altitude_m`, `path_zenith_rad`, `target_range_m`, `target.projected_area_m2` at 240 m², and the solar setting — and everything else reads default. That column is how you tell what the analyst specified from what the schema supplied.

`target_range_m` appearing in both lists is not a contradiction, and it is worth understanding once. The *mode form* greys it because mode V1 does not read it — the slant range is solved from the altitude and the zenith angle. The *dock* badges it config because the baseline file carries a value for it anyway: this file was written by `Sensor.to_yaml()`, which records the range it resolved. The form is telling you what this mode takes as input; the dock is telling you where a value came from. They answer different questions about the same row.

5.3.3 Step 3 — Declare the scene type, and let the tool argue with you

Select stage **2 Source**, tab **Scene & regime**.

Source

Scene & regime | Target — thermal | Target — point source | Target — reflective | Background & co

SCENE TYPE & REGIME

Scene type (declared)	sub_pixel
Regime override (force)	sub_pixel
Fill fraction (sub-pixel)	1

Outputs

Regime tentative	sub_pixel	
Projected area	240 m ²	
Range	532089 m	
Fill fraction	1	
Angular extent	2.91153e-05 rad	
Regime override	sub_pixel	
Scene type declared	sub_pixel	

Figure 5.3: Source workspace, Scene & regime tab — the declared sub-pixel scene and the angular extent that justifies it.

Scene type (declared) and Regime override (force) both read `sub_pixel`, and the Outputs list beneath reports `Regime tentative = sub_pixel`, projected area 240 m², range 532,089 m, fill fraction 1, and — the number the declaration rests on — an **angular extent of 2.91153×10^{-5} rad**, i.e. 29.1 μ rad.

That is the number to reason from. The diffraction-limited angular blur at this aperture is $17.28\ \mu\text{rad}$ (the Performance workspace reports it in a moment), so the ship subtends about 1.7 times the diffraction spot. A point-source treatment requires the target to be *small compared with* the PSF — conventionally under about a tenth of it — and $1.7\times$ is not that. RADIANT’s own point-source consistency check rejects the point-source classification here, and the scenario declares `sub_pixel` instead. That is not a workaround: at 30 cm of aperture this ship is genuinely on the edge of being resolved, and `sub_pixel` is the physically correct description of a target that fills part of a pixel against a background that fills the rest.

The scenario’s GUI workflow asks for this refusal to be surfaced inline, with the numeric ratio shown, rather than as a raw traceback. That advisory is still a gap; the numbers it would quote are the two on this page.

5.3.4 Step 4 — Give it a telescope, and watch the regime be finalized

Select stage **4 Optics**, tab **Inputs**.

Optics

Optics — aperture and wavefront error (transmission: Transmission tab)

Aperture diameter	0.3 m
Focal length	0.75 m
f-number	2.5
Obscuration ratio	0
Spider arms	0
WFE RMS (scalar)	0 waves
WFE reference wavelength	0.633 μm
Zernike file (Zemax export)	
Defocus	0 μm

Outputs

A collect	0.0706858 m ²	
Omega pixel	4e-10 sr	
D eff	0.3 m	
F number eff	2.5	
Obscuration eff	0	
Omega cone	0.122015 sr	
Transmission input mode	full_prescription	
Pupil plane extent	0.3	
Pupil wavelength um	4.2515	
Stray includes thermal	false	
Regime	sub_pixel	

Figure 5.4: Optics workspace, Inputs tab — the 30 cm f/2.5 head and the stage outputs, ending in the final radiometric regime.

Aperture diameter 0.3 m, focal length 0.75 m, and the f-number field immediately reads 2.5 with a lightning bolt and a derived badge. WFE RMS is 0 waves against a 0.633 μm reference wavelength — this study treats the optics as diffraction-limited, which is why the Strehl ratio will come out at exactly 1.

The Outputs block underneath is what the stage handed downstream: collecting area 0.0706858 m², pixel solid angle 4×10^{-10} sr, effective diameter 0.3 m, effective f-number 2.5, cone solid angle 0.122015 sr, pupil evaluated at 4.2515 μm — and Regime: sub_pixel.

That last line is the architectural point. The source stage made a *tentative* classification; the optics stage, which is the first stage that knows the PSF, makes the **final** one. Every downstream stage reads this value and

none re-decides it. The practical consequence here is that the ensquared-energy fraction will be applied to the target term and not to the background term, because in a sub-pixel scene the background genuinely fills the pixel while the target does not.

5.3.5 Step 5 — Read the noise budget

Select stage **7 Detector**, tab **Noise**.

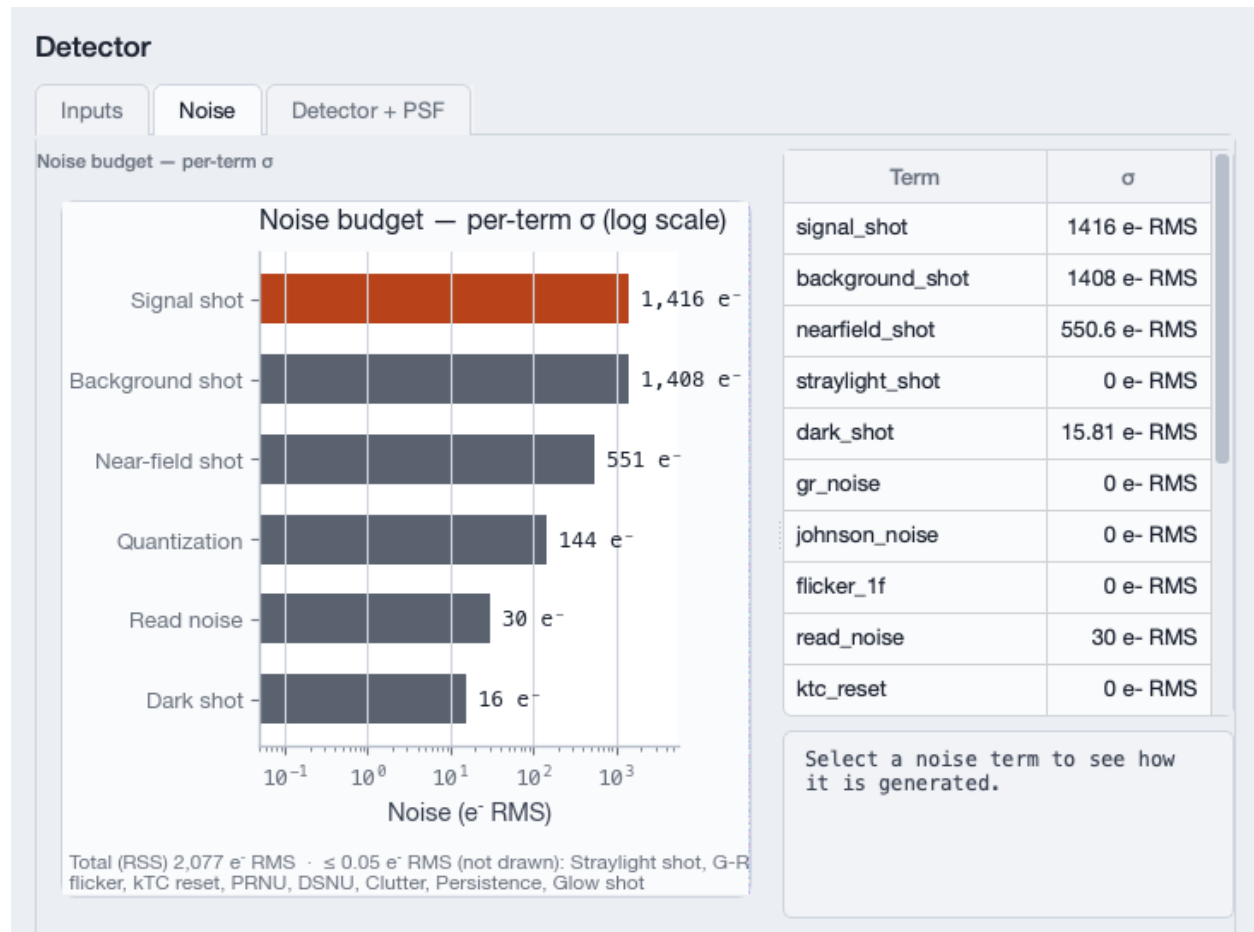


Figure 5.5: Detector workspace, Noise tab — the maritime noise budget, where background shot nearly matches signal shot.

Sixteen noise terms are computed; six are non-zero. In order:

Term	σ [e- RMS]	Note
signal_shot	1416	The hull's own emission.
background_shot	1408	The sea filling the rest of the pixel.
nearfield_shot	550.6	The telescope's own two 280 K mirrors (CU-380).
quantization	144.3	$g/\sqrt{12}$ at 500 e-/DN.
read_noise	30	Vendor ROIC.
dark_shot	15.81	50,000 e-/s over 5 ms.

Term	σ [e- RMS]	Note
Total (RSS)	2076.9	

Signal shot and background shot are the same size, and the figure lets you read why. Shot noise is $\sqrt{S}$, so the two terms square straight back to collected charge: $1416^2 = 2.005 \times 10^6$ e- from the target arm against $1408^2 = 1.982 \times 10^6$ e- from the background arm. The difference is 2.3×10^4 e-, about **1.1 % of either one**. The sea is not a dark backdrop — it is a 288 K surface at $\epsilon = 0.985$ filling the pixel, and the hull is a 288 K surface at $\epsilon = 0.266$ that makes up most of the shortfall by reflecting sky and sun. Nothing about this scene is signal-versus-dark.

Quantization at 144.3 e- RMS is nearly five times the read noise, which is what a 500 e-/DN gain buys: the ADC, not the detector, sets the electronic floor. On this bright scene it costs nothing — 144 e- against 2003 e- total is 0.5 % in quadrature — but it is worth remembering before this ROIC is flown against a fainter one.

The third term is the instrument itself. `nearfield_shot` at 550.6 e- RMS is the telescope's own emission — two mirrors at $R = \sqrt{0.85}$ and 280 K, each radiating at its Kirchhoff $\epsilon = 1 - R = 0.078$ — and it contributes 3.03×10^5 e- of charge, 13.3 % of this pixel's background. Before CU-380 (2026-10-03) this scenario declared its train as a scalar transmittance, which has no Kirchhoff emissivity, so the term was identically zero and absent from this table. The figure above is generated from the shipped GUI on this scenario's committed config, so it shows the near-field bar and the 2,077 e- RMS total alongside the others.

5.3.6 Step 6 — Read the result, and pick the right SNR

Select stage **10 Performance**.

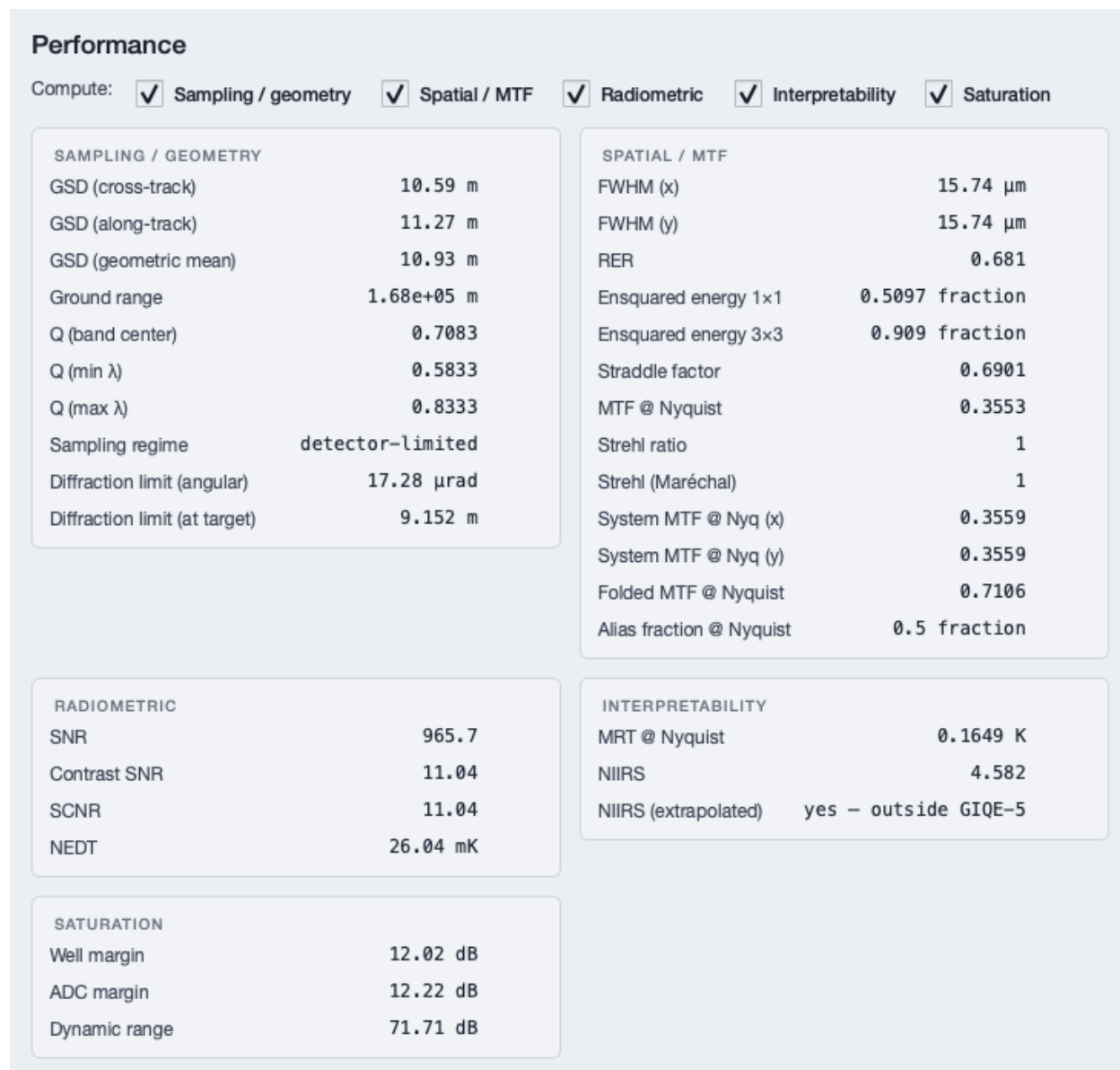


Figure 5.6: Performance workspace on the scenario 1.1 baseline — all five metric groups after evaluation.

Sampling / geometry. GSD 10.59 m cross-track and 11.27 m along-track, geometric mean 10.93 m; ground range 1.68×10^5 m; $Q = 0.7083$ at band center (0.5833 at the short edge, 0.8333 at the long one), classified detector-limited; diffraction limit 17.28 μ rad, projecting to 9.152 m at the target. The two GSD axes are not equal, and the ratio is exactly the obliquity: $11.27/10.59 = 1.064 = 1/\cos 20^\circ$. A 20° look stretches the ground footprint in the plane of the tilt and leaves the perpendicular axis alone.

Spatial / MTF. FWHM 15.74 μ m in both axes, RER 0.681, ensquared energy 0.5097 in the central pixel and 0.909 over 3×3 , straddle factor 0.6901, MTF at Nyquist 0.3553, folded MTF 0.7106, alias fraction 0.5. Strehl is exactly 1 in both the degraded-PSF and the Maréchal form, because the config carries no wavefront error.

Radiometric — read this group carefully.

Metric	Value
SNR	966
Contrast SNR	11.04
SCNR	11.04
NEDT	26.04 mK

SNR and contrast SNR differ by a factor of 87. SNR is the collected signal over the total noise — it says the pixel is well exposed. **Contrast SNR is the target-minus-background difference over that same noise, and it is the number that decides whether the ship is detectable.** The 1.1 % radiance difference read off the noise budget a moment ago, divided by 2076.9 e- RMS, *is* 11.04. That is comfortably above a detection threshold of 5, so the answer is still *yes, detectable* — but an aperture trade argued on SNR = 966 is arguing about the wrong number. The margin here is 11.04, and it is set by how nearly the hull’s reflected sky makes up for its missing emission.

Interpretability. MRT at Nyquist 0.165 K, NIIRS 4.582, flagged yes – outside GIQE-5. The flag is not a footnote to ignore: a 10.93 m GSD thermal scene sits outside the GIQE-5 regression’s calibration range, and the config opts into the extrapolated value explicitly via `performance.niirs.allow_extrapolated`. Read NIIRS here as a relative trend across the aperture sweep, not as an absolute rating.

Saturation. Well margin 12.02 dB, ADC margin 12.22 dB, dynamic range 71.71 dB. The margin is $20 \log_{10}$ of capacity over filled charge, so 12.02 dB is a factor of 4: the 5 ms dwell fills about a quarter of the 8 Me-well. There is room to integrate longer if the frame rate allows it, though not a great deal.

5.4 What the study concludes

At the 30 cm mid-point, with the parametric maritime atmosphere, the GUI reports SNR 966, contrast SNR 11.04, NEDT 26.04 mK, NIIRS 4.582 and GSD 10.93 m. The scenario’s runner, which evaluates the same configuration headless, reports SNR 965.71, NEDT 0.0260 K and NIIRS 4.58 — the same numbers to the precision the cards display.

Three findings survive out of the aperture sweep the script runs:

- **SNR is flat across 15 → 45 cm of aperture.** The sweep holds $f/2.5$ fixed, so focal length scales with diameter and the per-pixel étendue — and therefore the photon flux, for both the sub-pixel hull and the extended sea — is invariant. The aperture buys *resolution*, not signal: NIIRS climbs from 3.50 to 5.10 across the sweep while SNR does not move. An aperture trade that wants an SNR benefit has to sweep aperture at **fixed focal length**, letting $f/\#$ vary.
- **The parametric atmosphere and the real MODTRAN column agree to about 7 % on band transmittance** here — $\bar{\tau}$ 0.4594 parametric against 0.4277 measured, with the parametric model marginally the more transparent. SNR follows at about 10 % high (965.71 against 875.45) and detection range at 6.2 % (2326.8 km against 2191.6 km). The spread widened from 9 % when CU-380 gave the telescope its real warm mirrors: the same warm-optics term lands on both columns, which is the larger fractional insult to whichever has the weaker scene signal — the MODTRAN one. Note that τ , SNR and range do not move together; quote the one the decision needs.
- **The regime is sub_pixel, and the metric that matters is contrast SNR.** Both follow from the same fact: this is a modest target against a bright, isothermal background at the edge of being resolved.

One fact worth holding onto when reading the atmosphere-column reasoning. The config — the scenario's runner and this GUI baseline alike — points `source.target.emissivity_path` at the library's steel curve, which band-averages to $\epsilon = 0.266$ over 3.5–5.0 μm , so the hull is $\rho = 0.734$ and strongly reflective in this band. That is why every change to the modelled downwelling sky moves the SNR: the reflected-sky term is the larger of the hull's two signal contributions here.

One caveat carries into the proposal. The detection ranges above come from `detection_range_beer_lambert`, which extrapolates the reference-range SNR outward on an extinction coefficient measured at 532 km. It is not a full chain re-evaluation at each range, and it assumes that coefficient holds out past 2000 km, which curvature, refraction and the real atmospheric profile all say it does not. The GUI baseline does not publish a `detection_range_m` metric for this scene at all — the scenario's runner composes it. Treat those kilometers as a *relative* parametric-versus-MODTRAN sensitivity, not as an operational range.

5.5 The same study from a script

Everything above is one evaluation of one config; the study is a sweep of both. The scenario's runner does the whole thing:

```
cd scenarios/01_sarah_systems_engineer/1.1_mwir_maritime_surveillance
python scripts/run_mwir_maritime_surveillance.py
```

It builds the two atmosphere configurations — `SimpleAtmosphere` and the imported `tape7` — on one otherwise-identical sensor, sweeps the aperture from 0.15 m to 0.45 m at fixed $f/2.5$, composes detection range through `radiant.performance.detection_range_beer_lambert`, and writes `outputs/fig1_snr_and_range_vs_aperture.png` plus the comparison table this chapter quotes. The MODTRAN column needs the real-run `tape7` staged; without it the script falls back to the synthetic deck behind a loud banner and says so. `scripts/gui_console_1.1_mwir_maritime_surveillance.py` is the same work as a paste-in for the GUI's scripting console, if you would rather stay in the window.

The full narrative, the atmosphere-provenance caveats, and the gap list are in the scenario's own `walkthrough.md`.

Case Study — InSb versus HgCdTe on One Bench

Scenario 2.1 · persona: Mike, detector engineer · modality: GUI-led · scenario folder: scenarios/02_mike_detector_engineer/2.1_insb_vs_hgcdte_noise_budget/

6.1 The question

Mike has to pick a focal plane for a space MWIR instrument, and two vendors have sent him data. Not comparable data: one QE curve arrives as `wavelength_nm`, `QE_pct`, the other as `lambda_um`, `quantum_efficiency`; both send measured dark current against temperature as `T_K`, `Jdark_A_cm2`; and both quote read noise against the *same* 15 μm , 33 fF, CDS-mode ROIC — 18 e- RMS for the InSb part, 12 e- RMS for the HgCdTe part.

He wants a like-for-like noise budget on a common bench — a flat 300 K blackbody filling the aperture, a 3.5–5.0 μm cold filter, no atmosphere, no scene — and the three numbers that actually size the cryocooler:

- the **dark-current crossover temperature**, where dark shot noise reaches the read noise floor;
- the **BLIP temperature**, where the dark rate reaches the photon rate;
- the **NEI**, the noise-equivalent irradiance.

This chapter walks the bench in the GUI on the scenario’s committed baseline, `inputs/2.1_insb_vs_hgcdte_noise_budget.gui.yaml`, which carries the **InSb** branch. The HgCdTe branch and the cooler trade are the scripted half; the closing section points at them, and the “Running the second detector” box below says what the GUI action is.

6.2 The bench, and why it is what it is

Quantity	Value	Why
Blackbody temperature	300 K	A flat plate filling the aperture — the standard flat-field source.
Blackbody emissivity	0.995	Cavity source, effectively unity.
Bench ambient	295 K, $\epsilon = 0.95$	The room the plate sits in.
Collimator aperture	0.025 m	2.5 cm.
Collimator focal length	0.0575 m	5.75 cm, giving $f/2.3$.
Optical transmission	0.90	Collimator plus window, as a scalar lump.
Cold filter passband	3.5–5.0 μm	The common band both parts are traded in.

Quantity	Value	Why
Pixel pitch	$15\ \mu\text{m} \times 15\ \mu\text{m}$	The shared ROIC.
Fill factor	1	Full-fill.
Node capacitance	$3.3 \times 10^{-14}\ \text{F}$	33 fF.
CDS	on	kTC reset noise suppressed.
ROIC glow	5 e-/s	Vendor figure.
Full well / gain / ADC	$5 \times 10^6\ \text{e-} / 305\ \text{e-/DN} / 14\ \text{bit}$	Shared ROIC.
Integration time	0.001 s	1 ms frame.
Operating temperature	77 K	The nominal set point the trade starts from.
Quantum efficiency (InSb)	0.86	Band-average of the vendor curve over 3.5–5.0 μm .
Dark rate (InSb)	280,868 e-/s at 77 K	$J(77\ \text{K}) \cdot A_{pix}/q$ from the vendor table.
Read noise (InSb)	18 e- RMS	Vendor CDS figure.
Atmosphere	exo	There is no atmosphere between a plate and a collimator 5 cm away.
Sensor altitude	1.0 m	A bench placeholder — the chain needs a geometry, and this one is not a real one.

The HgCdTe branch changes exactly three of those: QE 0.782, dark rate 7022 e-/s at 77 K, read noise 12 e- RMS. Everything else — optics, filter, ROIC, well, gain, frame time — is held. That is what makes the comparison a detector comparison.

6.3 The GUI walk

6.3.1 Step 1 — Get the vendor curves in

The scenario's workflow opens with File → Import → Detector QE Curve and File → Import → Dark Current Curve. Both are modal dialogs, outside what these offscreen figures can show, so the figure below is the state they leave behind: the Detector workspace on the InSb branch.

Detector

Inputs Noise Detector + PSF

FPA part library no part applied Choose part & apply... Open datasheet/paper

Detector — geometry, QE, dark, noise model (full schema)

PIXEL GEOMETRY & TEMPERATURE

Pixel pitch x (cross-track) 15 μm

Pixel pitch y (along-track) 15 μm

Fill factor 1

Cross-track pixel count 0

Detector temperature 77 K

PIXEL SAMPLING PHASE (POINT-SOURCE / SUB-PIXEL EE_BOX)

Sampling phase (straddle) mode average

Image offset x (fraction of pitch) 0

Image offset y (fraction of pitch) 0

QUANTUM EFFICIENCY

Quantum efficiency (scalar) 0.86

QE curve CSV (import)

QE material (library)

QE temperature coefficient 0 1/K

QE reference temperature 300 K

Define QE(λ) table...

Import QE curve (preview)...

DARK CURRENT & GLOW

Dark model measured

Dark cutoff wavelength 0 μm

Dark rate 280868 1/s

Dark current density 0 A/cm²

Dark reference temperature 77 K

Dark activation energy 0 eV

ROIC glow 5 1/s

1/F NOISE

1/f coefficient K 0 e⁻²

1/f corner frequency 0 Hz

1/f band low edge (override) 0 Hz

1/f band high edge (override) 0 Hz

G-R & JOHNSON NOISE

G-R factor (HgCdTe) 0

R₀A product (Johnson) 0 ohm_cm²

FIXED-PATTERN NOISE & REGIME

PRNU 0

DSNU 0 e⁻

Clutter σ (SCNR) 0

Noise regime imaging

PERSISTENCE

Persistence fraction 0

Persistence time constant 1 s

Prior-frame signal 0 e⁻

Figure 6.1: Detector workspace, Inputs tab, on the InSb bench branch — the FPA part-library row, the QE entry group with its import affordances, and the vendor dark rate.

Three things on this page are worth naming.

The FPA part library row across the top reads FPA part library no part applied, with Choose part & apply... live and Open datasheet/paper grayed out. RADIANT ships a library of real focal-plane presets, each traced to a datasheet line in a manifest, and applying one writes a whole set of detector parameters at once with a preset badge on each. Mike is not using one here — and that is the right call for this study, because the whole point is to model *these two vendors' parts*, from *their* data, not a library part that resembles them. The row is worth knowing about for the moment he wants a sanity reference: apply a comparable preset, read the deltas, remove it again.

The quantum-efficiency group offers four ways in: a scalar (Quantum efficiency (scalar) = 0.86 here), a QE curve CSV (import) path, a QE material (library) name, and two buttons — Define QE(λ) table... for an inline wavelength/value table and Import QE curve (preview)... for the vendor CSV. The importer behind that second button resolves units from the header tokens, so wavelength_nm / QE_pct and lambda_um / quantum_efficiency both land in canonical μm and fraction from the same call, and a QE above 1 after conversion is a hard error naming the override rather than a silently unphysical curve.

This baseline carries the **scalar**, 0.86 — the band-average of the InSb curve. That choice is visible in the results and is discussed under step 4.

The dark current reads 280,868 1/s at a 77 K reference temperature. That is not a datasheet number typed in; it is $J(77\text{ K}) A_{\text{pix}}/q$ evaluated from the vendor's measured $J_{\text{dark}}(T)$ table, with the $\text{A}/\text{cm}^2 \rightarrow \text{e-}/\text{s}$ conversion done once in the loader. The loader's interpolation is Arrhenius-faithful — $\ln J$ linear in $1/T$, exact between nodes for $J \propto e^{-E_a/kT}$ — and it refuses to extrapolate outside the measured range. That refusal is a feature: the first run of this scenario tripped it, because the BLIP temperatures lie above 110 K and the vendor tables stopped at 110. The tables were extended to 130 K rather than extrapolated through.

6.3.2 Step 2 — Check the shared ROIC

Select stage **8 Readout**.

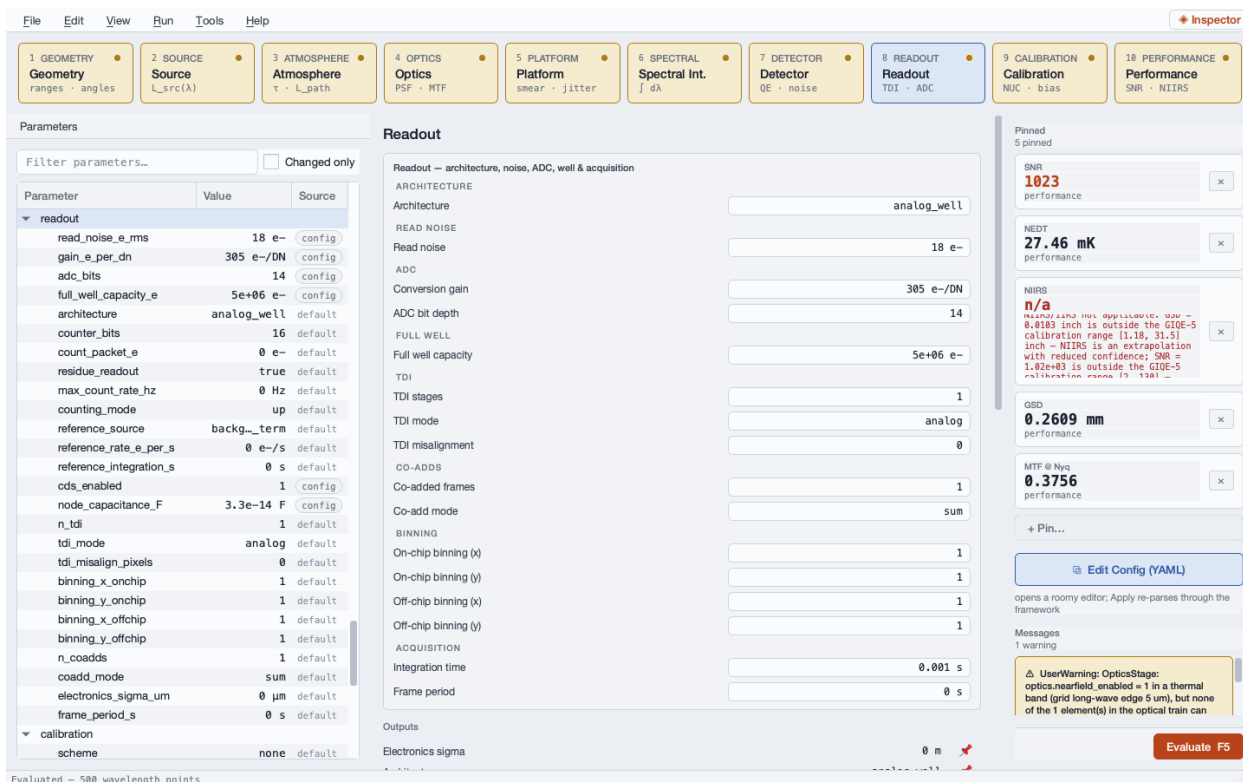


Figure 6.2: Readout workspace — the shared ROIC both detectors are traded on.

Architecture `analog_well`; read noise 18 e-; conversion gain 305 e-/DN over 14 bits; full well 5×10^6 e-; TDI stages 1 in `analog` mode with zero misalignment; one co-added frame; no binning on or off chip; integration time 0.001 s. In the Parameters dock, `cds_enabled` = 1 and `node_capacitance_F` = 3.3×10^{-14} both carry config badges.

Two of those numbers are about to matter. **305 e-/DN is 5 Me- spread over 14 bits**, and a uniform quantizer contributes $g/\sqrt{12} = 88$ e- RMS regardless of how quiet the detector is. **CDS on** means the reset noise $\sqrt{k_B T C}/q - 37.0$ e- RMS at 33 fF and 77 K by hand — is suppressed, and the noise budget will report `ktc_reset` = 0 rather than quietly omitting it.

6.3.3 Step 3 — Read the noise budget

Select stage **7 Detector**, tab **Noise**.

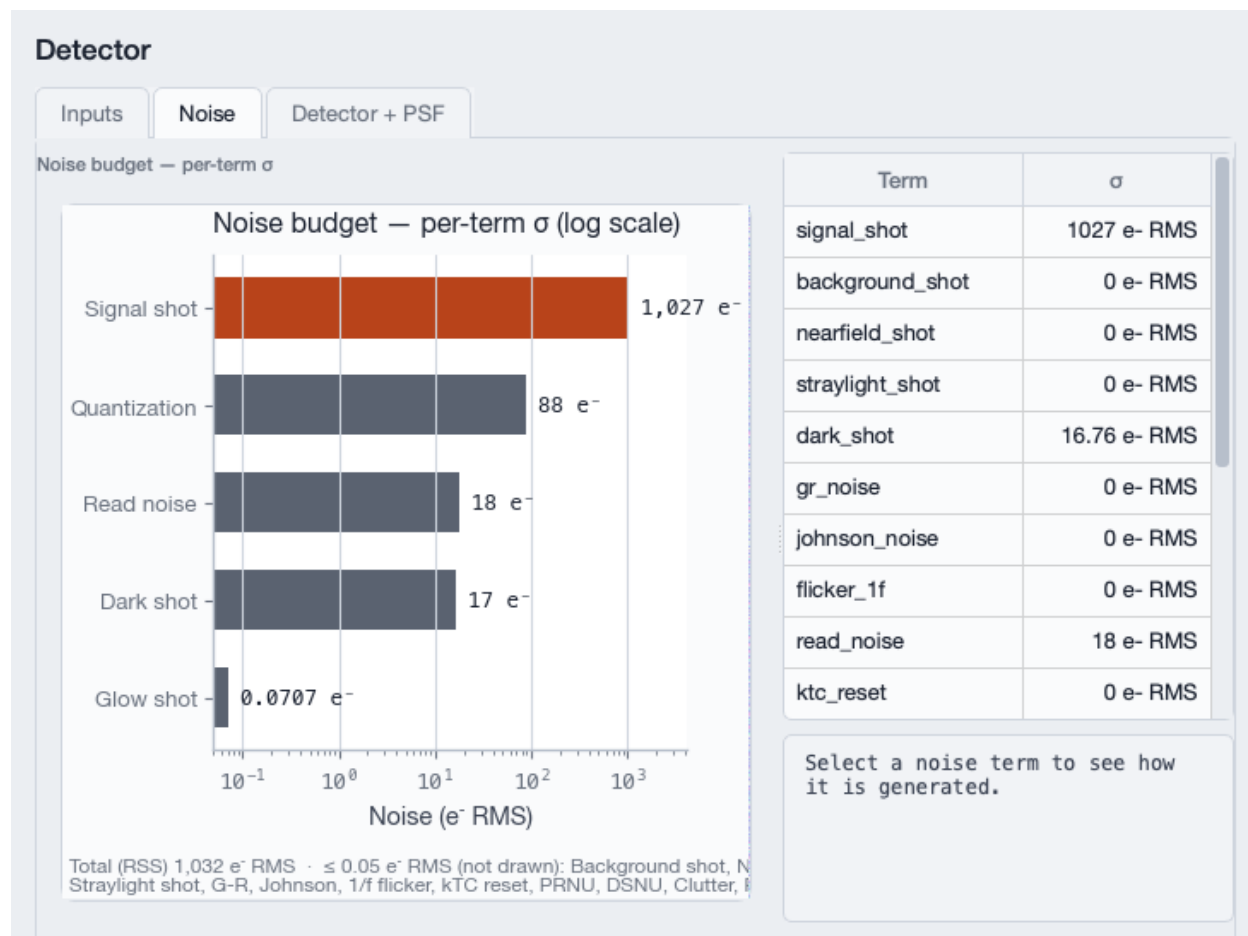


Figure 6.3: Detector workspace, Noise tab — the InSb bench budget, photon-limited, with quantization as the second term.

Term	σ [e- RMS]	Comment
signal_shot	1027	$\sqrt{S}$ — dominant by a factor of twelve.
quantization	88.05	$g/\sqrt{12}$ at 305 e-/DN.
read_noise	18	Vendor CDS figure.
dark_shot	16.76	$\sqrt{280\,868 \times 0.001}$.
glow_shot	0.0707	5 e-/s of ROIC glow over 1 ms.
ktc_reset	0	Suppressed by CDS, as configured.
background_shot	0	Zero by design — see below.
Total (RSS)	1032	

`background_shot = 0` is the entry that looks like a bug and is not. In the extended regime the scene *is* the source — a 300 K plate filling the aperture — so there is no separate background photon population to add on top of it. The bench-ambient temperature in the config feeds the contrast scene, not a second shot term. The signal-integration stage's own outputs say the same thing in numbers: `signal_e = 1,055,678 e-`, `background_e = 0`, `contrast_e = 1,055,678 e-`, which is why SNR, contrast SNR and SCNR will all read the same value on the Performance page.

The headline for the cooler budget: at 77 K the dark current is irrelevant. Dark shot noise is 16.76 e- RMS against 1027 e- RMS of photon noise. The 40× dark-current advantage HgCdTe holds over InSb — 7022 e-/s against 280,868 e-/s — buys essentially nothing *at this set point on this bright bench*. The trade at 77 K is a QE trade, and InSb wins it.

The second-largest term is the ADC, not the detector. Quantization at 88.05 e- RMS is five times the read noise and five times the dark shot noise. On this bench it is invisible — 88 e- against 1032 e- is 0.4 % in quadrature — but on a low-background scene it becomes the floor, and no detector choice can move it. That is a question for the ROIC vendor (a low-gain or dual-gain mode), not for the detector vendor.

6.3.4 Step 4 — Read the result

Select stage **10 Performance**.

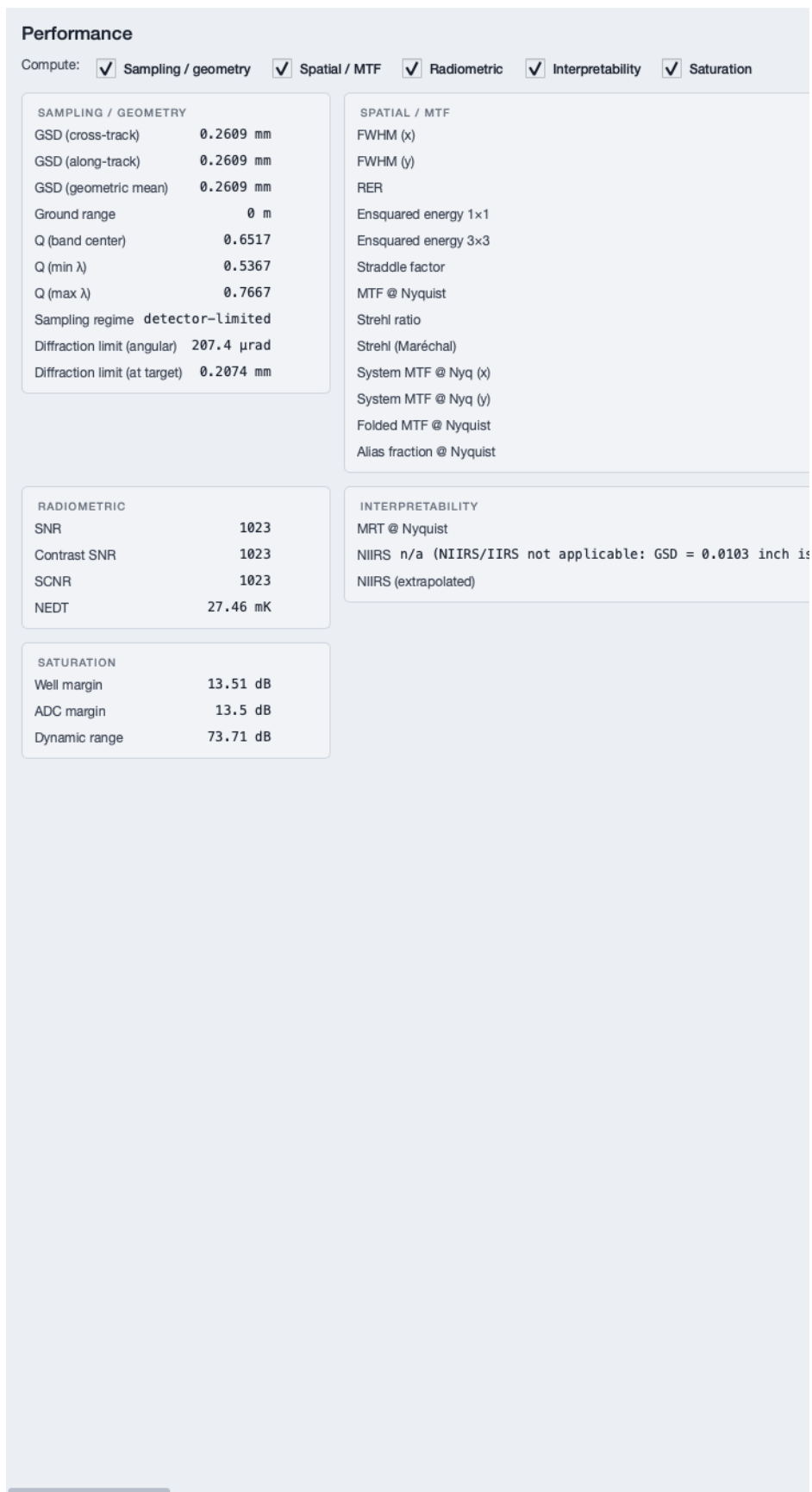


Figure 6.4: Performance workspace on the InSb bench branch.

Radiometric. SNR 1023, contrast SNR 1023, SCNR 1023, NEDT 27.46 mK. The three SNR flavors coincide because `contrast_e` equals `signal_e`, as step 3 explained.

Spatial / MTF. FWHM 15.54 μm , RER 0.6992, ensquared energy 0.5316 in the central pixel and 0.917 over 3×3 , MTF at Nyquist 0.3756, folded MTF 0.7529, alias fraction 0.5012, Strehl 1. $Q = 0.6517$ at band center, classified detector-limited — an f/2.3 collimator at 4.25 μm blurs to well under a 15 μm pixel, which is what a bench is supposed to do.

Sampling / geometry — do not read this group. GSD reads 0.2609 mm in both axes and the diffraction limit “at target” reads 0.2074 mm. Those are honest arithmetic on a placeholder: $ph/f = 15 \mu\text{m} \times 1.0 \text{ m} / 0.0575 \text{ m} = 0.2609 \text{ mm}$, where the 1.0 m is the bench stand-in for a sensor altitude. There is no ground plane in a laboratory, and the ground-projection family means nothing here. The tool does refuse the one metric it can refuse: the pinned NIIRS card reads n/a with its reason — *NIIRS/IIRS not applicable: GSD = 0.0103 inch is outside the GIQE-5 calibration range [1.18, 31.5] inch* — and the Interpretability group carries only MRT at Nyquist (0.1645 K) plus the yes – outside GIQE-5 flag. A dedicated detector-only bench mode, which would switch the whole family off the way an air target does, is a tracked gap.

Saturation. Well margin 13.51 dB, ADC margin 13.50 dB, dynamic range 73.71 dB. The margin is $20 \log_{10}$ of capacity over filled charge, so 13.51 dB is a factor of 4.74: the 1 ms frame fills about a fifth of the 5 Me-well — 1,055,678 e- of 5×10^6 e-, which is the signal-integration output step 3 quoted.

Running the second detector. The committed GUI baseline is the InSb branch. In the window, the HgCdTe branch is three edits in the Parameters dock — `detector.qe_value` 0.86 $\rightarrow$ 0.782, `detector.dark_rate_e_per_s` 280,868 $\rightarrow$ 7022, `readout.read_noise_e_rms` 18 $\rightarrow$ 12 — after which the window re-evaluates on its debounce and the Noise tab redraws. To see both at once rather than one after the other, save the edited version and use **Tools $\rightarrow$ Compare Config Files...**, which evaluates the current config plus any number of added files on a worker thread and lays the metrics out as one table. (The multi-configuration surface — **Edit $\rightarrow$ Configurations...** — is the other option, and the right one when the two versions are genuinely one instrument; here they are two different focal planes in one dewar, which is a file comparison.)

6.4 What the study concludes

The scenario’s runner evaluates both detectors with their **spectral** QE curves rather than a band-averaged scalar, and reports:

Term	InSb [e- RMS]	HgCdTe [e- RMS]
signal_shot	1033.69	985.31
quantization	88.05	88.05
read_noise	18.00	12.00
dark_shot	16.76	2.65
glow_shot	0.07	0.07
Total (RSS)	1037.73	989.32
Signal [e-]	1,068,522	970,841
SNR	1029.7	981.3

and, from the exact Arrhenius inversion of the vendor tables rather than a temperature sweep:

Quantity	InSb	HgCdTe
Dark rate at 77 K [e-/s]	280,868	7,022
Crossover temperature [K]	77.3	84.1
BLIP temperature [K]	101.7	115.4
NEI [photons/s/cm ²]	5.363×10^{11}	5.620×10^{11}

The GUI baseline is the InSb scalar-QE branch, and it reproduces the runner’s own scalar column exactly. The runner reports scalar-QE signal 1,055,678 e- against spectral 1,068,522 e- (+1.22 %), and $\sqrt{1\,055\,678} = 1027.5$ — the 1027 e- RMS on the figure above. Carried through, that is SNR 1023 in the window against SNR 1029.7 in the spectral run. The gap is not a disagreement: it is the price of a band-averaged QE. A flat average cannot know that the in-band photons concentrate at the long-wavelength end, where both vendors’ curves are near their peaks, so it under-counts by about a percent here — and by much more for a steeper QE slope or a wider band.

The verdict the trade produces:

- **At 77 K both parts are photon-noise-dominated**, so the SNR ratio is just the $\sqrt{\text{signal}}$ ratio, which tracks the QE ratio. InSb’s higher, flatter QE (0.860 against 0.782 band-averaged) buys about 10 % more signal and about 5 % more SNR. The 40× dark-current difference does not appear.
- **The parts separate at warmer set points.** HgCdTe’s dark shot noise reaches the read-noise floor at 84.1 K against 77.3 K for InSb, and it stays background-limited to 115.4 K against 101.7 K. That 7–14 K of set-point margin is cryocooler mass, power and lifetime at the system level — the classic InSb (QE and uniformity) versus HgCdTe (operability margin) trade, with vendor-derived numbers under it.
- **Quantization is the number-two noise term on both**, and it belongs to the ROIC. Fix it in the gain plan, not in the detector choice.

6.5 The same study from a script

```
cd scenarios/02_mike_detector_engineer/2.1_insb_vs_hgcdte_noise_budget
python scripts/run_detector_shootout.py
```

The runner loads all four vendor CSVs through `radiant.io.qe_csv.load_qe_csv` and `radiant.io.dark_current_csv.load_dark_current_csv`, runs each detector twice — once with the spectral QE curve injected onto the chain grid, once with the band-averaged scalar — inverts the dark-current curves for the crossover and BLIP temperatures, and writes the noise-budget and cooler-trade sheets plus four figures into `outputs/. scripts/gui_console_2.1_insb_vs_hgcdte_noise_budget.py` is the same work as a paste-in for the GUI’s scripting console.

The vendor-import details, the kTC hand cross-check, and the gap list are in the scenario’s own `walkthrough.md`.

Case Study — ISR Pass Planning

Scenario 3.1 · persona: Raj, collection planner · modality: GUI-led · scenario folder: scenarios/03_raj_mission_planner/3.1_isr_pass_planning/

7.1 The question

Raj tasks a sun-synchronous imager and has to defend a collection plan in a tasking review. The spacecraft flies at 600 km, can slew 45° off nadir, and the customer will not accept imagery below NIIRS 6.0. Three questions come out of that:

1. What are the orbit kinematics — period, ground speed, revisit rate — that set how often the target is reachable at all?
2. How far off nadir can he point before image quality falls below the floor, and how wide is the resulting access corridor?
3. What area can the spacecraft cover per unit time?

The third one is the reason this scenario exists as a *composition* rather than a single chain run. The RADIANT signal chain has no concept of platform velocity, so it cannot produce a coverage rate on its own. The orbit model supplies the ground-track speed; the chain supplies the swath; `performance.access_rate` multiplies them.

This chapter walks the image-quality half in the GUI on the scenario's committed baseline, `inputs/3.1_isr_pass_planning.gui.yaml`, which is the 30° off-nadir nominal look. The angle sweep and the orbit arithmetic are the scripted half.

7.2 The system, and why it is what it is

Quantity	Value	Why
Sensor altitude	600,000 m	The mission orbit.
Path zenith angle	30°	The nominal look this baseline evaluates; the study sweeps 0–45°.
Solar zenith angle	35°	A mid-morning sun-synchronous descending pass.
Aperture diameter	0.50 m	The instrument.
Focal length	6.0 m	$f/12$ — a long-focal-length imager, as a sub-meter GSD needs.
Optical transmission	0.85	Scalar lump.
Spectral band	0.45–0.70 μm	Panchromatic visible.

Quantity	Value	Why
Integration time	0.0005 s	0.5 ms — a fast pushbroom line time.
Pixel pitch	6.5 μm $\times$ 6.5 μm	Silicon detector.
Cross-track pixel count	8000	The array width — this is what sets the swath.
Quantum efficiency	0.85	Silicon in the visible.
Target reflectance	0.30	A generic land scene.
Dark rate	50 e-/s at 280 K	Room-temperature silicon; negligible at 0.5 ms.
Full well / gain / ADC	30,000 e- / 8 e-/DN / 12 bit	A small-well visible ROIC.
Read noise	20 e- RMS	Typical CMOS.
Atmosphere	parametric, us_standard	No specific site; the generic column.
Collection constraints	slew $\leq 45^\circ$, NIIRS ≥ 6.0	The tasking rules the plan has to satisfy.

The two collection constraints are not RADIANT parameters — they are the customer’s rules. What RADIANT supplies is the curve they are applied to.

7.3 The GUI walk

7.3.1 Step 1 — Describe the look

Select stage **1 Geometry**, tab **Inputs**.

Geometry

Scene class — derived label & optional assertion

Scene: space → ground (space_to_ground) — derived

Assert scene class auto

OFF BY DEFAULT FOR THIS SCENE CLASS

- Target-plane sample distance (x)
- Target-plane sample distance (y)
- Target-plane sample distance (geometric mean)

Defaults only — an explicitly set performance.metrics.* group flag always wins, so selecting a group yourself computes it whatever the scene class.

Geometry inputs — pick one mode per family

Viewing geometry

Path zenith at lower endpoint (V1)

sensor_altitude_m	600000 m
target_altitude_m	0 m
path_zenith_rad	30 deg
sensor_off_boresight_rad	27.1914 deg
ground_range_m	312301 m
elevation_angle_rad	60 deg
target_range_m	683151 m

Solar geometry

Solar zenith θ_s (S1)

solar_illumination	day
solar_azimuth_rad	0 deg
solar_zenith_rad	35 deg
solar_elevation_rad	55 deg
site_latitude_rad	—
day_of_year	—
local_solar_time_h	—
ltan_h	—

Platform kinematics

Direct ground speed

ground_speed_m_s	0 m/s
circular_orbit	false

Line-of-sight rate

Platform motion only (derived)

los_angular_rate_rad_s	0 rad/s
target_speed_m_s	—
target_heading_rad	—
target_climb_rad	—
sensor_speed_m_s	0 m/s

Site elevation — terrain under the line of sight

Site elevation m

Figure 7.1: Geometry workspace, Inputs tab — the 600 km orbit looking 30° off nadir through the V1 path-zenith mode.

The scene-class card derives Scene: space $\rightarrow$ ground (space_to_ground), and only the three target-plane sample distances default off — a ground scene wants every ground-projection metric there is, which is exactly what a collection planner is after.

The viewing family is in mode **V1**, Path zenith at lower endpoint. Raj entered sensor_altitude_m = 600,000 m and path_zenith_rad = 30 deg; ground_range_m, target_range_m, elevation_angle_rad and sensor_off_boresight_rad are grayed because this mode derives all four. That grayed-not-hidden idiom matters for a planner: the alternatives stay visible, so it is obvious that a pointing plan expressed as a *ground range* rather than an angle is available in another mode of the same family, not in another tool.

Solar geometry sits in mode **S1**, Solar zenith θ_s , at 35 deg with solar_illumination = day. For a reflective-band scene that is not decoration: it sets the irradiance the 0.30-reflectance surface is lit by, and it moves SNR directly.

7.3.2 Step 2 — Look at the geometry, not the numbers

Switch to the **Schematic** tab.

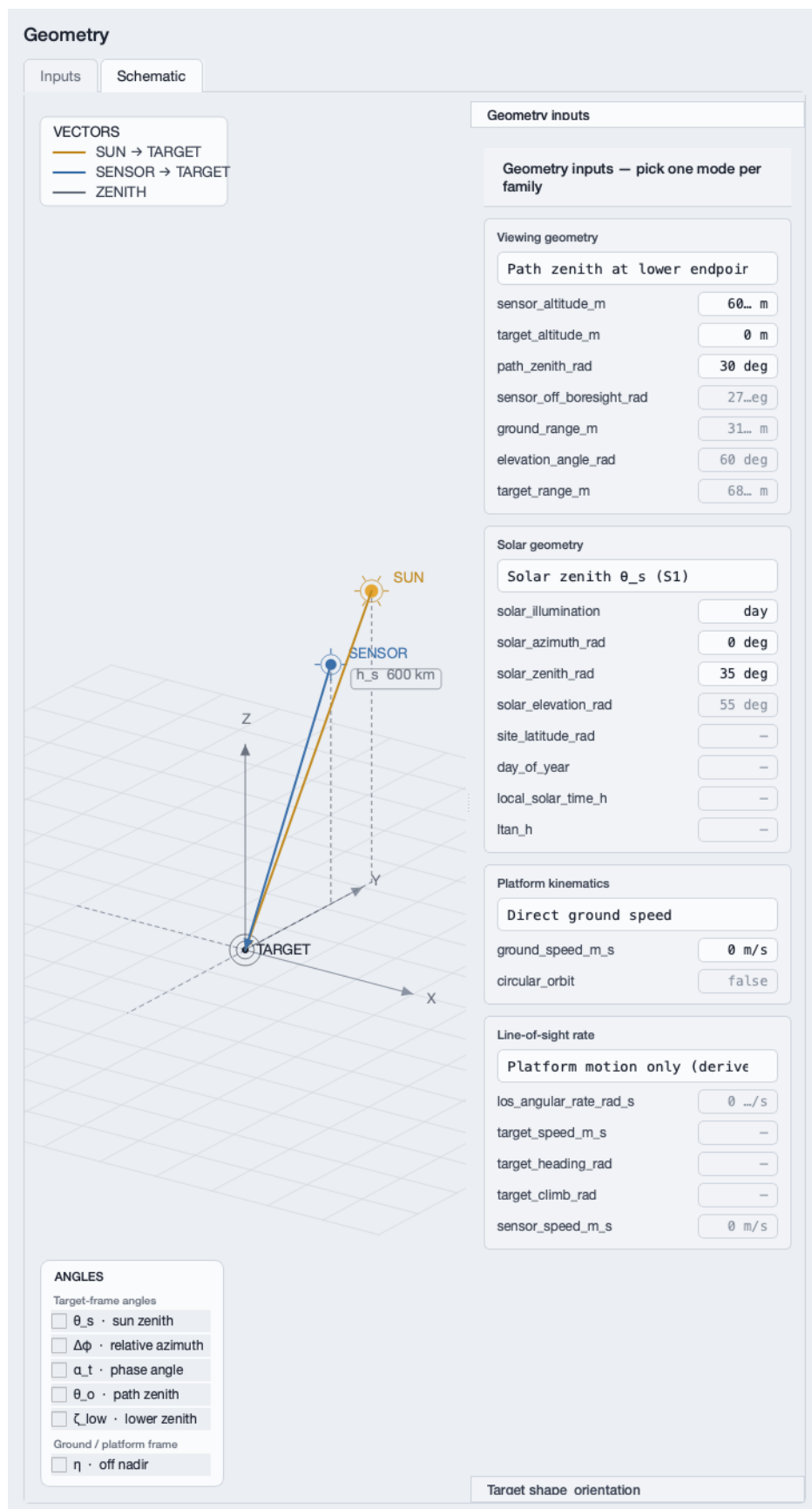


Figure 7.2: Geometry workspace, Schematic tab — the off-nadir look with the sun vector and the not-to-scale altitude leader pill.

The orthographic schematic draws the sensor above and to one side of the target, the sun vector in orange, and the sensor line of sight in blue. It is deliberately **not to scale**: 600 km against a 6371 km Earth radius drawn honestly would put the sensor inside the target marker. Altitude is therefore *told*, via the `h_s 600 km` leader pill, while the geometry stays legible.

The **ANGLES** panel in the lower left is the part a planner uses. Each of θ_s (sun zenith), $\Delta\phi$ (relative azimuth), α_t (phase angle), θ_o (path zenith), ζ_{low} (lower-endpoint zenith) and η (off-nadir) is an individually revealable arc, unchecked here. They are separate toggles rather than one “show angles” switch because the angles differ from each other by fractions of a degree at LEO, and drawing all six at once is unreadable. Ticking θ_o and η together is the quickest way to see that the 30° the config names at the *target* is not the 30° the spacecraft slews through at the *sensor* — they differ by the Earth-center central angle.

The mode form is repeated on the right of this tab, so a pointing change can be made without leaving the picture: edit `path_zenith_rad` there and the schematic and the metrics both move on the next debounce.

7.3.3 Step 3 — Read the four numbers the plan is argued from

Select stage **10 Performance**.

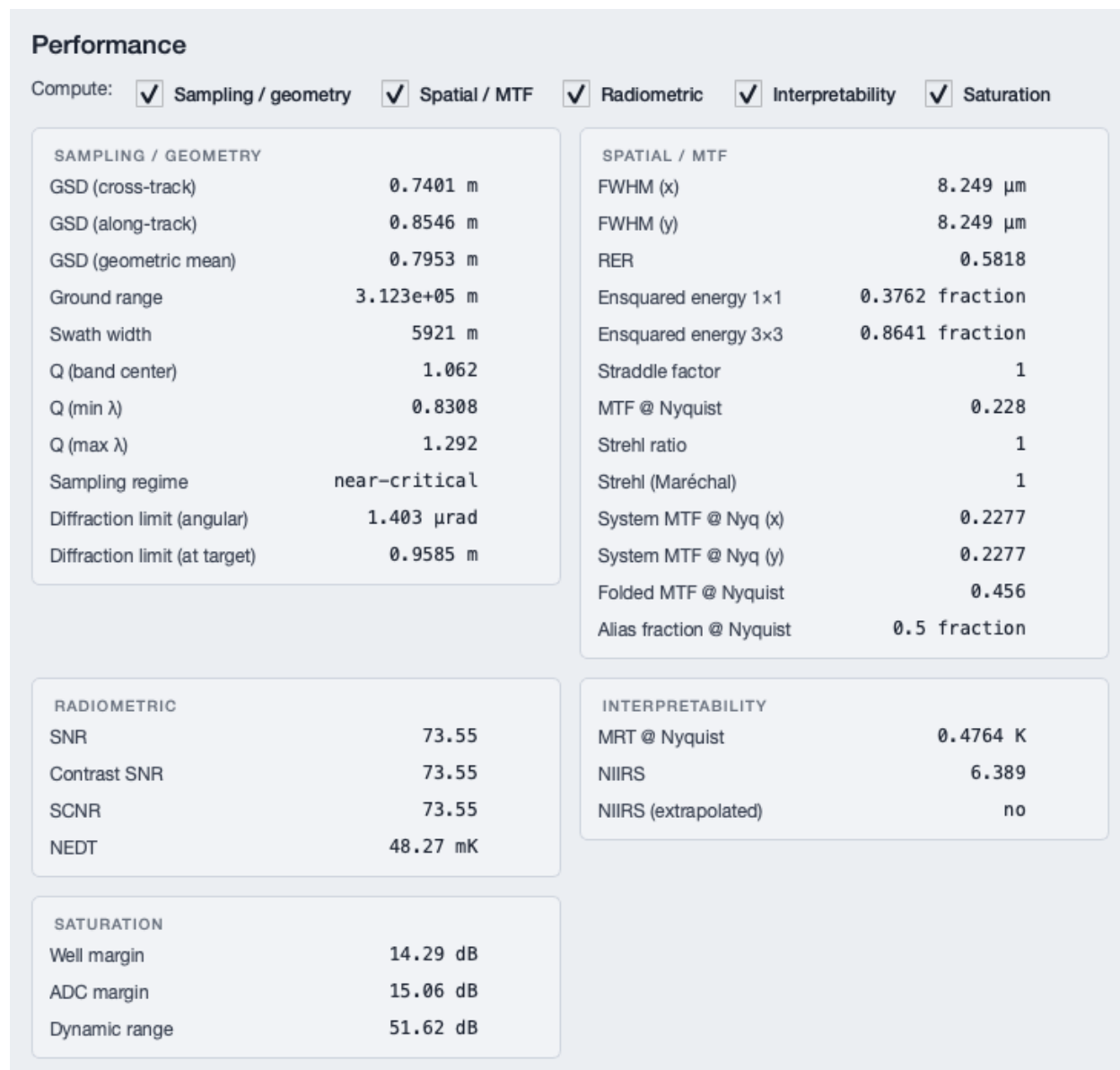


Figure 7.3: Performance workspace at 30° off nadir.

Sampling / geometry is the group a collection planner lives in:

Metric	Value
GSD (cross-track)	0.7401 m
GSD (along-track)	0.8546 m
GSD (geometric mean)	0.7953 m
Ground range	3.123×10^5 m
Swath width	5921 m
Q (band center)	1.062
Sampling regime	near-critical
Diffraction limit (angular)	1.403 μ rad

Metric	Value
Diffraction limit (at target)	0.9585 m

Three of those carry the whole argument. **Ground range 312 km** is how far cross-track this one look reaches — that is the corridor coordinate. **Swath width 5921 m** is 8000 pixels of 0.7401 m cross-track GSD, and it is what will multiply the ground-track speed into a coverage rate. And the **GSD ratio**, $0.8546/0.7401 = 1.155 = 1/\cos 30^\circ$, is the obliquity stretch: a 30° look lengthens the footprint in the plane of the tilt by exactly the secant, and leaves the perpendicular axis alone.

$Q = 1.062$ with the near-critical label is a well-matched instrument. $Q = \lambda F/\# / p$ compares optical blur to pixel; at 1 the optics and the sampling are balanced, below 1 the pixels throw away resolution the optics delivered, above 1 the optics blur below what the pixels could sample. A 0.5 m aperture at f/12 on $6.5\ \mu\text{m}$ pixels lands at 1.06 — the diffraction limit projects to 0.9585 m at the target against a 0.7953 m sample, which is the same statement.

Spatial / MTF. FWHM $8.249\ \mu\text{m}$, RER 0.5818, ensquared energy 0.3762 in the central pixel and 0.8641 over 3×3 , MTF at Nyquist 0.228, folded MTF 0.456, alias fraction 0.5, Strehl 1. The folded value is exactly twice the unfolded one and the alias fraction is exactly 0.5: at Nyquist the first sampling replica lands back on Nyquist, so half the apparent contrast there is folded-in above-Nyquist scene content.

Radiometric. SNR 72.69, contrast SNR 72.69, SCNR 72.69, NEDT 48.83 mK. The three coincide because the scene is a uniform reflective surface with no separate clutter or contrast term.

Interpretability. MRT at Nyquist 0.4819 K, **NIIRS 6.381**, and NIIRS (extrapolated): no — this GSD is inside the GIQE-5 calibration range, so the rating is a real one rather than an extrapolated trend. At 30° the look clears the 6.0 floor by 0.38.

Saturation. Well margin 14.48 dB, ADC margin 15.25 dB, dynamic range 51.71 dB. The margin is $20 \log_{10}$ of capacity over filled charge, so 14.48 dB is a factor of 5.3 — the 30,000 e⁻ well is about a fifth full on a 0.30-reflectance scene. Enough here, and worth watching over bright desert or cloud.

7.3.4 Step 4 — Sweep the pointing angle

The single look above answers one question. The plan needs the curve.

☐ Changed only

Parameter	Value	Source
▼ geometry		
sensor_altitude_m	600000 m	config
target_altitude_m	0 m	default
site_elevation_m	0 m	default
path_zenith_rad	30 deg	config
solar_zenith_rad	35 deg	config
solar_azimuth_rad	0 deg	default
solar_illumination	day	default
ground_speed_m_s	0 m/s	default
target_range_m	0 m	default
target.projected_area_m2	0 m ²	default
target.shape	none	default
target.shape_radius_m	0 m	default
target.shape_length_m	0 m	default
target.shape_width_m	0 m	default
target.shape_height_m	0 m	default
target.shape_base_radius_m	0 m	default
target.shape_yaw_rad	0 deg	default
target.shape_pitch_rad	0 deg	default
target.shape_roll_rad	0 deg	default
sensor_off_boresight_rad	0 deg	default
ground_range_m	0 m	default
elevation_angle_rad	90 deg	default
solar_elevation_rad	61.3521 deg	default
site_latitude_rad	0 deg	default
day_of_year	80	default
local_solar_time_h	12 h	default
ltan_h	12 h	default
circular_orbit	false	default

Figure 7.4: Parameters dock (panel grab) — the geometry branch, with path_zenith_rad as the sweep axis.

The dock, widened so the full dot-paths are legible, shows what will be swept and what will follow it. geometry.path_zenith_rad reads 30 deg with a config badge; sensor_altitude_m 600,000 m and solar_zenith_rad 35 deg are the other two config rows and are held. Everything grayed — target.

`projected_area_m2`, the whole `target.shape.*` family — belongs to target-extent modes this extended scene does not use.

`Run → Run Sweep...` takes a parameter, a start, a stop, a point count and a metric, and evaluates on a worker thread so the window stays responsive. Parameter `geometry.path_zenith_rad`, 0 to 45 deg, 16 points, metric `niirs` reproduces the scenario's own trade. The metric list is populated from what the last evaluation actually produced, so a metric this configuration does not compute cannot be selected by accident.

The result, from the scenario's runner over the same range:

Off-nadir [deg]	GSD [m]	NIIRS	SNR	Ground range [km]	Swath [km]
0	0.65	6.67	72.0	0	5.2
15	0.68	6.60	72.8	146	5.4
30	0.80	6.38	72.7	312	5.9
45	1.05	5.97	71.3	527	7.1

The 30° row is the baseline this chapter evaluated, and the GUI's own numbers — GSD 0.7953 m, NIIRS 6.381, SNR 72.69, ground range 3.123×10^5 m, swath 5921 m — are that row at full precision.

7.4 What the study concludes

Orbit kinematics, from `radiant.core.orbit` on a 600 km circular orbit: period 5792 s (96.5 min), inertial velocity 7.56 km/s, sub-satellite ground-track speed 6.91 km/s, 14.9 orbits per day. The ground-track speed is below the orbital speed by the factor R_E/a — the nadir point traces a circle of radius R_E while the spacecraft traces one of radius $a = R_E + h$. Earth rotation is neglected, which is a few-percent direction-dependent cross-term at LEO and adequate for coverage sizing.

Coverage, composed from the two halves: nadir swath 5.2 km $\times$ 6.91 km/s gives an area-coverage rate of **35.9 km²/s**, or roughly 104,000 km² per daylight pass.

The corridor, and the planning verdict. NIIRS falls monotonically with off-nadir angle — 6.67 at nadir to 5.97 at 45° — and crosses the 6.0 floor at 42°. The spacecraft can *slew* to 45°, reaching 527 km cross-track, but it can only *image at spec* out to 42°, which is a 478 km half-width. **Image quality, not agility, sets the usable access corridor**, and the last 3 degrees of slew buy reach the imagery cannot use.

Two mechanisms are worth separating, because they run opposite ways:

- **GSD grows with off-nadir angle** roughly as $1/\cos^2$ — one factor from the lengthening slant range, one from the projection stretch — and drags NIIRS down with it through the GIQE-5 resolution term.
- **SNR rises slightly off nadir**, from 72.0 at nadir to 72.8 at 15°, before easing to 71.3 at 45°. The ground footprint per pixel grows faster than the slant-range path loss costs, so each pixel collects more photons from a sunlit extended scene. Image quality still degrades, because resolution and not SNR is the binding term: the GIQE-5 SNR contribution is $1.559 \log_{10}(\text{SNR})$, so a 2 % SNR change is 0.013 NIIRS against the 0.42 the slew itself costs.

A caveat for the far end of the sweep: above about 42° the GSD passes outside the GIQE-5 calibration range and NIIRS extrapolation warnings fire. The runner opts into the extrapolated trend deliberately, and those

tail values should be read as a relative trend. The floor crossing at 42° is inside the calibrated range, so the verdict does not rest on an extrapolation.

The regime throughout is **extended**: the sunlit surface fills the pixel, and the point-source and sub-pixel machinery — ensquared-energy application to a target term, angular-extent checks — is unused.

7.5 The same study from a script

```
cd scenarios/03_raj_mission_planner/3.1_isr_pass_planning
python scripts/run_pass_planning.py
```

The runner reads the mission constants transcribed from `inputs/raj_orbit_sensor.xlsx`, calls `radiant.core.orbit` for the kinematics, walks 16 pointing angles from 0 to 45° through the chain, finds the NIIRS-floor crossing, composes the coverage rate through `radiant.performance.access_rate.compute_access_rate_m2_s`, and writes `outputs/fig1_offnadir_image_quality.png` and `outputs/fig2_access_corridor.png`. `scripts/gui_console_3.1_isr_pass_planning.py` is the same work as a paste-in for the GUI's scripting console, where the orbit helpers are one-liners:

```
from radiant.core.orbit import ground_track_speed_m_s, orbital_period_s
orbital_period_s(600e3) / 60          # 96.54 min
ground_track_speed_m_s(600e3)        # 6910.9 m/s
```

The orbit-model truth anchors, the access-corridor derivation, and the gap list — a target-specific revisit model is the main one — are in the scenario's own `walkthrough.md`.

Case Study — A Target Detection Matrix

Scenario 4.1 · persona: Lisa, detection/targeting analyst · modality: script-led · scenario folder: scenarios/04_lisa_analyst/4.1_target_detection_matrix/

8.1 The question

Lisa owes a quarterly program review one slide. The program has a twelve-target library, three candidate sensors, and four atmospheric conditions it cares about, and the question the reviewers will ask is always the same: **how far off-nadir can each sensor detect each target, and which target is the hardest?**

That is $12 \times 4 \times 3 = 144$ evaluations, and each one is itself a search — detection range is not a metric RADIANT reports, it is the range at which a detection criterion stops being satisfied, which has to be found. Doing this by hand is not arithmetic-hard, it is bookkeeping-hard: 144 configurations, each with a different target temperature, emissivity and area, a different atmosphere, a different sensor, and a failure in any one of them silently poisoning a briefing chart.

This is the case for a batch runner, and for a script rather than a window. The GUI would let Lisa examine any one cell beautifully; it would not let her defend all 144.

8.2 The inputs, and why they are what they are

The target library arrives as a twelve-row workbook, `inputs/lisa_target_library.xlsx`, read by `radiant.io.target_library.load_target_library`, which derives `projected_area_m2` from `length × width` so that the analyst maintains dimensions rather than areas:

Target	A_{proj} [m ²]	T [K]	ϵ [–]	Material
MBT tank	28.4	310.0	0.90	painted steel
APC	18.2	306.0	0.90	painted steel
Cargo truck	20.0	301.0	0.92	painted steel
Technical (pickup)	10.1	303.0	0.88	painted steel
SAM TEL	35.6	305.0	0.90	painted steel
Towed artillery	26.6	296.0	0.85	painted steel
Patrol boat	145.0	299.0	0.85	painted steel
Fast attack craft	476.0	302.0	0.85	painted steel
Transport aircraft	1600.0	295.0	0.30	bare aluminum
Fighter aircraft	150.0	297.0	0.35	low-e coating
Fuel bladder farm	600.0	298.0	0.95	rubberized fabric
Small UAV (parked)	12.0	294.0	0.80	composite

Target	A_{proj} [m ²]	T [K]	ϵ [–]	Material
--------	------------------------------	-------	----------------	----------

Note the spread in emissivity, not just in size: the transport aircraft is enormous (1600 m²) but bare aluminium at $\epsilon = 0.30$ and *cold* at 295 K, while the fuel bladder farm is a third its size at $\epsilon = 0.95$. Those two facts will fight each other later.

Three sensors, all at 500 km, all read from YAML:

	A — MWIR smallsat	B — MWIR flagship	C — LWIR wide
Aperture	0.18 m	0.50 m	0.35 m
Focal length	0.45 m (f/2.5)	1.50 m (f/3.0)	0.70 m (f/2.0)
Band	3.6–4.9 μm	3.6–4.9 μm	8.0–11.5 μm
Pixel pitch	15 μm	12 μm	17 μm
GSD at nadir	16.7 m	4.0 m	12.1 m
Quantum efficiency	0.72	0.78	0.65
Dark rate	5,000 e-/s at 80 K	2,000 e-/s at 75 K	200,000 e-/s at 60 K
Read noise	35 e- RMS	25 e- RMS	40 e- RMS
Integration time	4 ms	4 ms	2 ms

Sensor C's YAML is deliberately stale — it still carries the old name `platform.h_sensor` where the current schema says `geometry.sensor_altitude_m`. RADIANT accepts it through the deprecated-alias mechanism and says so:

```
OUTDATED PARAMETER NAME absorbed: sensor C's YAML still says
'platform.h_sensor' (the superseded name). RADIANT accepted it
through the deprecated-alias mechanism (1 DeprecationWarning(s) raised) and mapped it to
geometry.sensor_altitude_m – the config still runs, loudly.
```

That is the intended behavior for an aged config: run, but warn. A config archive that silently breaks on a schema rename is a config archive nobody keeps.

Four atmospheres, each pairing a visibility with the profile that physically goes with it:

Condition	Visibility	Standard profile
clear	50 km	midlat_summer
haze	10 km	midlat_summer
tropical_haze	5 km	tropical
arctic_clear	100 km	subarctic_winter

The detection criterion is declared in the script, before any run:

```
SCNR_THRESHOLD = 5.0      # detection criterion [dimensionless]
CLUTTER_SIGMA = 0.02      # rural scene clutter, fraction of in-pixel background
ALTITUDE_M = 500_000.0    # all three sensors fly the same 500 km orbit
ZENITH_MAX_RAD = math.radians(66.0) # horizon at 500 km is 68.0° – stay clear
BISECTION_STEPS = 9       # zenith resolution ~0.13° → ~0.5% in range
```

8.3 Why SCNR and not SNR

RADIANT reports `snr` and `contrast_snr`, and this study uses neither. That decision is the analytic heart of the scenario, so it deserves the space.

Pure noise-limited SNR does not discriminate here. Every one of these sensors holds SNR above 19 across the entire swath for every target — a 500 km MWIR imager staring at a warm Earth is photon-rich. A criterion that every cell passes is not a criterion.

Real sub-pixel detection against a terrestrial background is **clutter-limited**: what defeats you is not the sensor's noise but the scene's own spatial variability, patches of ground that are as different from their neighbors as your target is. So the script models rural scene clutter at 2 % of the in-pixel background and forms a signal-to-clutter-plus-noise ratio:

```
def scnr_of(result) -> float:
    """|contrast_e| / RSS(all noise terms incl. clutter) - detection SCNR."""
    contrast_e = abs(float(result.stage_outputs["spectral_integration"]["contrast_e"]))
    total = math.sqrt(sum(nt.value_e**2 for nt in result.noise_terms))
    return contrast_e / total if total > 0 else 0.0
```

It is assembled script-side because RADIANT's own `snr` and `contrast_snr` metrics carry *temporal* noise only; the spatial clutter term is deliberately excluded from them, and this is recorded as a gap in the scenario's `gaps.md`.

The absolute value is not a convenience. In the sub-pixel regime the chain forms $\text{contrast}_e = ff \cdot (L_{\text{target}} \cdot \text{EE}_{\text{box}} - L_{\text{bg}})$: the target's compact energy is `EE_box`-weighted, because its PSF spills into neighbouring pixels, while the uniform background it **occludes** is not. A deeply sub-pixel target with a small `EE_box` is therefore detected largely by the *dip* it punches in the background, and `contrast_e` goes negative. A single hot pixel and a single occluded pixel are equally detectable against a scene, so $|\cdot|$ is the physically right criterion — but it means “detectability” here is distance from the weighted-background null in either direction, not brightness.

8.4 The scripted path

```
PYTHONPATH=src python \
    scenarios/04_lisa_analyst/4.1_target_detection_matrix/scripts/run_detection_matrix.py
```

8.4.1 Geometry: set one thing, derive the rest

```
def configure_geometry(sensor: Sensor, zenith_rad: float, area_m2: float) -> None:
    r_slant = slant_range_from_theta_o_m(zenith_rad, ALTITUDE_M, 0.0)
    sensor.set("geometry.path_zenith_rad", zenith_rad)
    footprint_m2 = (_ifov_rad(sensor) * r_slant) ** 2
    sensor.set("geometry.target.projected_area_m2", min(area_m2, footprint_m2))
    sensor.set("source.target.fill_fraction", min(1.0, area_m2 / footprint_m2))
```

Three subtleties are packed into six lines, and each one was a bug before it was a line of code.

Only θ_o is set. The target-side path zenith is the single geometric input; the chain derives the slant range from it through its own spherical viewing triangle. Setting `geometry.target_range_m` as well would over-specify the line of sight with a different (sensor-side) slant and trip RADIANT's consistency check. The

script reads the slant back through the same `slant_range_from_theta_o_m` the chain uses, so the footprint it computes and the geometry the chain runs cannot diverge.

Sensor.get returns canonical units. `detector.pixel_pitch_x_um` comes back in **meters**, despite the `_um` suffix in the name — the suffix records the *input* unit, not the storage unit. An initial version of this script multiplied by 10^{-6} again and made the pixel footprint 10^{12} times too small; the symptom was a printed GSD of 0.0 m.

Fill fraction, not projected area, drives sub-pixel weighting. This is the one that matters physically. `geometry.target.projected_area_m2` drives the *point-source* path (A_t/R^2). In the **sub-pixel** regime the target is weighted by `source.target.fill_fraction = min(1,  $A_{target}/(\text{IFOV} \cdot R)^2$ )` — the fraction of the pixel it covers. Off-nadir, the pixel footprint grows as R^2 , fill falls, and SCNR falls with it. *That* is the detection-range mechanism this whole matrix rests on, and a script that sets only the projected area would produce a matrix with no range dependence at all and no obvious sign that anything was wrong.

8.4.2 Detection range by bisection

```
if scnr_at(sensor, ZENITH_MAX_RAD, area_true) >= SCNR_THRESHOLD:
    return {... "detection_range_km": range_max_km, "horizon_limited": 1.0}

lo, hi = 0.0, ZENITH_MAX_RAD # SCNR(lo) >= threshold > SCNR(hi)
for _ in range(BISECTION_STEPS):
    mid = 0.5 * (lo + hi)
    if scnr_at(sensor, mid, area_true) >= SCNR_THRESHOLD:
        lo = mid
    else:
        hi = mid
detection_range_km = slant_range_from_theta_o_m(0.5 * (lo + hi), ALTITUDE_M, 0.0) / 1000.0
```

Nine bisection steps on the zenith angle, bracketed at nadir and at the 66° practical swath edge (the true horizon at 500 km is 68.0°). Two early exits keep the cost honest: a cell that fails at nadir is “not detectable” without any search, and a cell that still passes at the swath edge is reported as **swath-edge limited** at 1,061 km — the sensor’s access, not the atmosphere, is what ends it there. Nine steps give ~0.13° of zenith resolution, about 0.5 % in range; that figure matters when reading the tables below.

8.4.3 The matrix itself

```
for label, path in SENSOR_FILES.items():
    runner = BatchRunner(
        base_config={}, # unused - the factory loads the YAML
        axes=[("target", TARGET_AXIS), ("atmosphere", ATMOSPHERES)],
        sensor_factory=lambda cfg, p=path: _load_sensor(p),
    )
    results[label] = runner.run(evaluate_cell)
    n_failed = results[label].n_failed
    if n_failed:
        print(f"    {n_failed} cell(s) failed - recorded in the error column")
```

`radiant.api.batch.BatchRunner` takes the Cartesian product of the declared axes, applies each cell’s parameter overrides to a freshly built `Sensor`, and captures per-cell failures rather than aborting the

batch — a failed cell is recorded and reported, never dropped. `result.pivot("detection_range_km", rows="target", cols="atmosphere")` then builds the briefing tables directly.

One line inside the sensor factory is easy to miss and load-bearing:

```
s.set("performance.niirs.allow_extrapolated", True)
```

These targets are unresolved at GSDs of 4–17 m, far outside the GIQE-5 calibration envelope, so RADIANT's applicability gate would return NIIRS as N/A by default. The scenario wants the extrapolated *trend*, so it opts in explicitly. That is the intended contract: the extrapolation is available, but you have to ask for it in writing.

8.5 Real output — the three matrices

=== Running the matrix: $12 \times 4 \times 3 = 144$ cells ===

Detection criterion: $SCNR \geq 5 - |\text{contrast}|$ over $RSS(\text{noise} + \text{clutter})$,
 scene clutter = 2% of in-pixel background (rural)
 Swath edge: zenith 66° (slant 1,061 km; horizon at 68.0°)
 Running A: MWIR smallsat 18 cm (48 cells)...
 Running B: MWIR flagship 50 cm (48 cells)...
 Running C: LWIR wide 35 cm (48 cells)...

Sensor A — MWIR smallsat, 16.7 m GSD, 278 m² pixel footprint at nadir:

Target	clear	haze	tropical_haze	arctic_clear
MBT tank	not detectable	not detectable	not detectable	not detectable
APC	not detectable	not detectable	not detectable	not detectable
Cargo truck	not detectable	not detectable	not detectable	not detectable
Technical (pickup)	not detectable	not detectable	not detectable	not detectable
SAM TEL	not detectable	not detectable	not detectable	not detectable
Towed artillery	not detectable	not detectable	not detectable	not detectable
Patrol boat	not detectable	not detectable	not detectable	534
Fast attack craft	not detectable	not detectable	not detectable	788
Transport aircraft	not detectable	not detectable	not detectable	1,061*
Fighter aircraft	not detectable	not detectable	not detectable	not detectable
Fuel bladder farm	823	813	742	1,061*
Small UAV (parked)	not detectable	not detectable	not detectable	not detectable

Sensor B — MWIR flagship, 4.0 m GSD, 16 m² pixel footprint at nadir:

Target	clear	haze	tropical_haze	arctic_clear
MBT tank	696	688	560	901
APC	673	666	612	865
Cargo truck	773	763	699	1,029
Technical (pickup)	575	570	527	718
SAM TEL	870	858	782	1,061*
Towed artillery	893	883	800	1,061*
Patrol boat	1,061*	1,061*	998	1,061*
Fast attack craft	1,061*	1,061*	924	1,061*
Transport aircraft	1,047	1,022	737	1,061*
Fighter aircraft	1,059	1,033	753	1,061*

Fuel bladder farm	1,061*	1,061*	1,059	1,061*
Small UAV (parked)	686	678	621	901

Sensor C — LWIR wide, 12.1 m GSD:

Target	clear	haze	tropical_haze	arctic_clear
MBT tank	not detectable	not detectable	not detectable	not detectable
APC	not detectable	not detectable	not detectable	not detectable
Cargo truck	not detectable	not detectable	not detectable	not detectable
Technical (pickup)	not detectable	not detectable	not detectable	not detectable
SAM TEL	not detectable	not detectable	not detectable	565
Towed artillery	not detectable	not detectable	not detectable	523
Patrol boat	841	830	711	1,061*
Fast attack craft	1,061*	1,061*	981	1,061*
Transport aircraft	1,061*	1,061*	1,061*	1,061*
Fighter aircraft	938	926	782	1,061*
Fuel bladder farm	1,061*	1,061*	1,029	1,061*
Small UAV (parked)	not detectable	not detectable	not detectable	not detectable

=== Worst-case target ===

Hardest: Technical (pickup) – mean detection range 199 km
across all 12 sensor×atmosphere cells

Easiest: Fuel bladder farm – mean 991 km

=====

DONE – 144 cells evaluated (0 failures recorded)

=====

All ranges are **slant range in kilometers** at SCNR = 5; * marks a cell that is swath-edge limited (SCNR $\geq$ 5 all the way out to the 66° practical edge, slant 1,061 km). The full run takes 5 min 15 s — about 1,300 chain evaluations.

8.6 What the matrix says

Regime: sub-pixel, forced, in every cell. The script sets both `source.scene_type` and `source.regime_override` to `sub_pixel`. The override is deliberate and worth understanding: left to itself, RADIANT's classifier picks `point_source` for the smaller targets, and the point-source path drops the in-pixel background photons entirely. Detection against a bright Earth background *needs* those photons — they are the thing the target must be distinguished from, and they carry the clutter term the criterion is limited by. Forcing `sub_pixel` keeps the background and clutter terms in every cell, which is the physically correct description of a target that fills part of a pixel while the scene fills the rest.

8.6.1 Aperture buys targets; size buys range

The single clearest read on the slide. Sensor B's 4.0 m GSD gives a 16 m² pixel footprint; sensor A's 16.7 m GSD gives 278 m² — seventeen times larger. Fill fraction is target area over footprint, so B fills a pixel with targets that vanish into A's. **B detects all twelve targets in all four conditions. A detects exactly one — the fuel bladder farm — in the three temperate columns.**

Within a sensor, larger targets reach further, because fill stays near unity out to longer slant ranges before the growing footprint dilutes it: on B, the fuel bladder farm runs to the swath edge in clear air while the Technical pickup stops at 575 km.

For the program review this is the constellation-versus-exquisite decision reduced to one table. Many small apertures buy revisit; one large aperture buys the target set.

8.6.2 The hardest target is not the smallest or the coldest

Technical (pickup) — 10.1 m^2 , $\epsilon = 0.88$, 303 K — has a mean detection range of 199 km across all twelve sensor $\times$ atmosphere cells and falls out entirely on the LWIR sensor. The easiest is the **fuel bladder farm** at a mean of 991 km.

But the ordering is *not* a size ranking and *not* a temperature ranking, and the matrix shows why. On sensor B in clear air, the cool 12 m^2 Small UAV at 294 K reaches 686 km and the hot 28.4 m^2 MBT tank at 310 K reaches 696 km — within 1.5 % of each other, despite a $2.4\times$ difference in area and 16 K in temperature. Both beat the *smaller, cooler* Technical pickup.

The mechanism is the EE_box-weighted contrast introduced earlier. Detectability is $ff \cdot |L_{\text{target}} \cdot \text{EE}_{\text{box}} - L_{\text{bg}}|$: how far the target pixel departs from the background **in either direction**, times fill. The Technical pickup is simply the target whose EE_box-weighted radiance lands closest to the background's after weighting, so its pixel departs least and it separates worst. It is nearest the null.

Two consequences follow, and Lisa should carry both into the room:

- **Which target sits at the null is model-dependent, not a property of the target library.** It has moved before, under changes to the EE_box weighting and to the down-looking path emission, and it will move again. “The Technical pickup is our hardest target” is a statement about the current model, not about the pickup.
- **A multi-pixel matched filter would recover the hardest targets.** EE_box spreads the target's energy into neighbouring pixels and the single-pixel criterion throws that energy away. Summing it back is a real detection-model refinement, and it is exactly the targets marked “not detectable” here that it would recover. This is recorded in the scenario's `gaps.md`.

8.6.3 The atmosphere sets how far, not what

Every sensor's `arctic_clear` column leads its temperate columns, by 25–35 % on sensor B, and on sensor A the arctic column is the *only* one where three extra targets — patrol boat, fast attack craft, transport aircraft — clear threshold at all.

The mechanism is not target-path absorption. That axis was recalibrated away years ago, and real MOD-TRAN columns span only about $1.3\times$ in band-mean MWIR transmittance across these profiles. What drives the spread now is the **atmosphere's own thermal emission**: the down-looking path carries a $(1 - \tau)B(\lambda, T_{\text{eff}})$ term, so a warm, moist column puts more radiance into the in-pixel background — and the criterion is limited by 2 % of that background. A cold, dry subarctic column emits least, so its clutter floor is lowest and every range lengthens. The `tropical_haze` column, warmest and wettest, is uniformly the worst.

This is a case where the correct qualitative conclusion (“climate matters”) was reached for the wrong reason for a long time, corrected to the opposite conclusion (“climate barely matters”) when the spurious absorption

response was removed, and then restored on genuinely different physics when the emission term landed. The scenario's `walkthrough.md` keeps a full drift archaeology of that history; it is worth reading before quoting any number in this matrix as settled.

8.6.4 NIIRS is present and should be read as nothing

Every cell also reports NIIRS at nadir, and every value is a **GIQE-5 extrapolation** far outside the regression's calibration envelope: these GSDs are 4–17 m against a calibration range that stops below about 0.8 m. The script says so in its own physics notes. It is reported because the briefing template asks for it. Treat it as a relative ordering at best, and never as a rating.

8.7 Numbers: this run versus the committed walkthrough

The runner was re-executed unmodified for this chapter. **142 of the 144 cells reproduce the scenario's committed `walkthrough.md` matrices exactly**, including every detectable / not-detectable verdict, every swath-edge flag, the hardest-target identity and mean (Technical pickup, 199 km) and the easiest (fuel bladder farm, 991 km).

Two cells differ, both on sensor B, and both are reported here with both values rather than reconciled away:

Cell	Walkthrough, as committed	This run	Δ
Fighter aircraft, haze	1,029 km	1,033 km	+4 km (+0.39 %)
Small UAV (parked), clear	685 km	686 km	+1 km (+0.15 %)

Both shifts are below the bisection's own stated resolution — nine steps give $\sim 0.13^\circ$ in zenith, about 0.5 % in range — so a sub-resolution change in the underlying physics is enough to move a bisection midpoint across a reporting boundary. Neither changes a verdict, a ranking, or a conclusion in this chapter. They are recorded because a detection matrix that is re-run and silently re-quoted is how two results-affecting landings previously passed through this scenario unremarked; nothing in the machine baselines covers a script-side SCNR bisection.

A related inconsistency, noted in passing and not corrected here: the walkthrough's own refresh note attributes its single moving cell to "sensor A's small UAV, 686 $\rightarrow$ 685 km", but sensor A's Small UAV is not detectable in every column in both the walkthrough's own table and this run. The cell it describes is sensor B's, which this run reports back at 686 km. The walkthrough's physics discussion also quotes "688 vs 695 km" for the Small UAV and MBT tank on sensor B where its own table says 685 and 696; the prose rounds, the table is the record.

8.8 The takeaway

1. **Sensor B detects all twelve targets in all four conditions; sensor A detects one.** The difference is GSD, hence pixel footprint, hence fill fraction — a $17\times$ ratio in footprint area.
2. **The hardest target is the one nearest the weighted-background null**, not the smallest or the coldest. Right now that is the Technical pickup at a 199 km mean. It is a model-dependent verdict.
3. **Climate sets range, not target set.** `arctic_clear` leads by 25–35 % because a cold dry column emits less thermal path radiance and therefore carries less clutter.

4. **Detection here is clutter-limited, not noise-limited.** Noise-limited SNR exceeds 19 in every cell of the matrix and discriminates nothing.
5. **The recoverable margin is in the detection model, not the sensor.** A multi-pixel matched filter would bring back the targets the single-pixel SCNR marks undetectable.

8.9 The same study in the GUI

The scenario ships a GUI-openable baseline of one representative cell, `inputs/4.1_target_detection_matrix.gui.yaml`, with headline metrics snapshotted in `.gui.expected.json` — enough to open a single configuration in the window and inspect what the batch does 144 times in a loop: the forced sub-pixel regime, the fill fraction, the in-pixel background, and the clutter term the criterion is limited by. The scenario's `gui_workflow.md` specifies what a full matrix workflow would need: a **Target Library** import that shows the derived `projected_area_m2` column and flags duplicate or non-numeric rows before accepting; a deprecation *banner* rather than a console warning when sensor C's aged `platform.h_sensor` is loaded; a matrix builder with target, atmosphere and sensor axes backed by the same `BatchRunner`; a progress grid that fills in cell by cell with failures shown in red rather than dropped; per-sensor result heatmaps carrying the “not detectable” and swath-edge states and a GIQE-extrapolation caveat chip on the NIIRS column; and a worst-case panel that explains the EE_box occlusion mechanism in place rather than leaving the reader to infer it. The two committed figures, `outputs/fig1_detection_range_matrix.png` and `outputs/fig2_nadir_scnr_by_target.png`, are what that panel would show.

Case Study — Allocating a Wavefront-Error Budget

Scenario 5.1 · persona: Tom, optical designer · modality: script-led · scenario folder: scenarios/05_tom_optical_designer/5.1_wfe_budget_allocation/

9.1 The question

Tom has a 40 cm Cassegrain on his screen in Zemax: f/10, 35 % linear central obscuration, operating in the VNIR from 500 to 800 nm on a 500 km orbit. The design closes. What he does not yet have is a **budget** — a defensible statement of how much total wavefront error the telescope may accumulate, across design residual, assembly tolerance and on-orbit thermal distortion, before the imagery stops being worth flying.

Two versions of that question have to be answered separately, and conflating them is the classic error:

1. **The trade.** How do Strehl, MTF at Nyquist, ensquared energy, RER and NIIRS move as total WFE RMS grows from 0 to 0.25 waves? This defines where the cliff is.
2. **The as-built.** Tom's current design has a *specific* Zernike decomposition — 12 terms exported from Zemax. Where does that particular prescription sit against the budget, and does its *shape* matter, or only its RMS?

The second question is the one a scalar sweep cannot answer on its own, and the one this scenario exists to demonstrate. A single RMS number is a summary statistic of a pupil; two pupils with identical RMS and different modal content produce different point-spread functions.

9.2 The inputs, and why they are what they are

Quantity	Value	Why
Aperture diameter	40 cm (= 0.400 m)	Tom's design.
Focal length	400 cm (= 4.000 m)	Holds f/10.
Central obscuration	35 % linear	Cassegrain secondary plus its baffle.
Optical transmission	75 %	Two-mirror train plus filter, lumped.
Optics temperature	20 °C (= 293.15 K)	Not load-bearing in the VNIR; the scene is solar-reflective.
WFE reference wavelength	633 nm (= 0.633 μm)	HeNe. Every interferometric measurement Tom will ever take is referenced here.
Spectral band	500–800 nm (= 0.500–0.800 μm)	The VNIR passband; band center 650 nm.

Quantity	Value	Why
Pixel pitch	10.0 μm	The CCD.
Quantum efficiency	85 %	Detector, band-average.
Dark current	3.0 e-/s	Cooled CCD.
Read noise	5.0 e- RMS	Detector.
Full well	100,000 e-	Detector.
Integration time	2.0 ms	A short VNIR dwell.
Orbit altitude	500 km	Sets GSD 1.25 m at 2.5 μrad IFOV.
Target / background reflectance	0.15 / 0.10	A low-contrast land scene.
Solar zenith angle	30°	A good-illumination reference case.
Atmosphere	simple, visibility 23 km, PWV 20 mm	Standard clear column.

The derived numbers the script prints before it starts are the ones to hold onto: **GSD 1.25 m, IFOV 2.5 μrad , Airy disk 15.9 μm (1.59 pixels), and $Q = 0.650$.**

Q — the ratio of optical cutoff to detector sampling — is 0.650, which means this system is **undersampled**. The pixel is larger than the Airy core. That single fact determines the shape of everything below: the detector aperture MTF, not the optics, is what limits MTF at Nyquist, and adding wavefront error eats into a budget the detector has already spent most of.

Tom’s Zernike prescription arrives as `inputs/tom_zernike_zemax.txt`, the Zemax “Zernike Standard Coefficients” text export, in Noll indexing at the 633 nm reference:

Index	Name	Coefficient [waves]
Z4	Defocus	0.020
Z5	Astigmatism 0	0.015
Z6	Astigmatism 45	0.010
Z7	Coma Y	0.025
Z8	Coma X	0.018
Z9	Trefoil Y	0.005
Z10	Trefoil X	0.004
Z11	Spherical	0.030
Z12	2nd Astigmatism 0	0.003
Z13	2nd Astigmatism 45	0.002
Z14	2nd Coma Y	0.002
Z15	2nd Coma X	0.001
RSS total		0.0513

Spherical, coma-Y and defocus dominate. Because Noll-normalized coefficients *are* per-mode RMS contributions, the total RMS is their root-sum-square — which is also what makes the allocation below a clean RSS budget rather than a linear one.

9.3 The scripted path

One script does the whole study:

```
S=scenarios/05_tom_optical_designer/5.1_wfe_budget_allocation
PYTHONPATH=src python $S/scripts/run_wfe_budget_trade.py
```

9.3.1 Step 1 — Parse the Zemax export, and refuse to run on a mismatch

```
zemax = load_zemax_zernike(ZEMAX_FILE)

# Cross-check: workbook sheet vs Zemax export must agree.
sheet_coeffs = {idx: coeff for idx, _, coeff in zernike_data}
if sheet_coeffs != zemax.zernike_coeffs:
    raise ValueError(
        "Workbook 'Zernike Coefficients' sheet disagrees with the Zemax "
        f"export: sheet={sheet_coeffs}, zemax={zemax.zernike_coeffs}"
    )
```

`radiant.io.zemax_zernike.load_zemax_zernike` handles the parts of a real Zemax export that break naive parsers: UTF-16 and UTF-8 encodings, Noll-index validation, and capture of the reference wavelength from the file’s own header rather than from a hard-coded assumption.

The cross-check against the workbook sheet is the more interesting line, and it is worth stealing. Tom has the same twelve numbers in two places — the optical export and the systems-engineering workbook. The script compares them and raises rather than proceeding on whichever it happened to read second. Two sources of truth for one quantity is a latent defect; making the script assert their agreement converts it into a loud one.

9.3.2 Step 2 — Express the allocation as an error budget

```
WFE_ALLOCATION_WAVES = 1.0 / 14.0

wfe_budget = ErrorBudget(
    name="wfe",
    unit="waves @ 633 nm",
    contributors=tuple(
        BudgetContributor(name=f"Z{idx} {zern_names.get(idx, '')}".strip(), value=coeff)
        for idx, coeff in sorted(zemax.zernike_coeffs.items())
    ),
    allocation=WFE_ALLOCATION_WAVES,
)
```

The allocation is $\lambda/14 = 0.0714$ waves, the Maréchal criterion for “diffraction-limited” (Strehl ≈ 0.80). `radiant.api.ErrorBudget` does the RSS, the margin, the per-contributor variance shares, and — the number Tom actually needs — `remaining_allocation()`.

9.3.3 Step 3 — Sweep the scalar RMS

```
for wfe_rms in wfe_values:
    config["optics"] = {**base_config["optics"], "wfe_rms_waves": wfe_rms}
    sensor = Sensor.from_dict(config)
    r = sensor.evaluate()

    strehl = r.metrics.get("strehl", 0.0)
    mtf_nyq = r.metrics.get("mtf_at_nyquist", 0.0)
    ee_1x1 = r.metrics.get("ee_1x1", 0.0)
```

```
rer = r.metrics.get("rer", 0.0)
niirs = r.metrics.get("niirs", 0.0)
```

Thirteen sweep points from 0 to 0.25 waves. RADIANT expands the scalar RMS over a fixed, deterministic equal-RMS low-order Zernike set (Noll Z4–Z11) to build the aberrated pupil.

9.3.4 Step 4 — Run the actual prescription, in Zernike mode

The scalar sweep is the *trade*. The as-built *truth* is a run of Tom's real coefficients:

```
wfe_zernike = zemax.to_wavefront_error()

config_zern["optics"] = {**base_config["optics"], "wfe_rms_waves": 0.0}
session_zern = RadiantSession(wavelength_um=wl_grid)
params_zern = session_zern.default_params()
load_config(config_zern, params_zern)
params_zern.resolve()
r_zern = session_zern.run(
    params_zern,
    extra_stage_outputs={"optics_config": {"wavefront_error": wfe_zernike}},
)

# Scalar-RMS run at exactly the same total RMS for a like-for-like pair.
config_scal["optics"] = {**base_config["optics"], "wfe_rms_waves": total_rms}
r_scal = Sensor.from_dict(config_scal).evaluate()
```

This is the architectural pattern for any file-derived object in RADIANT. Stages do not read files; the IO layer builds the object and the API injects it into the chain before execution, here through `RadiantSession.run(extra_stage_outputs=...)` into `optics_config`. Note also that `wfe_rms_waves` is set to **0.0** on the Zernike run — the scalar parameter and the injected prescription are two routes to the same pupil phase, and stacking them would double-count the aberration.

The paired scalar run at exactly `total_rms` is what makes the comparison meaningful: same RMS, different modal mix, everything else identical.

9.3.5 Real output — the sweep

WFE [waves]	Strehl [--]	MTF@Nyq [--]	EE(1x1) [--]	EE(3x3) [--]	RER [--]	SNR [--]	NIIRS [--]
0.000	1.0000	0.2546	0.4288	0.8833	0.6114	120.2	6.14
0.020	0.9879	0.2488	0.4242	0.8805	0.6079	120.2	6.13
0.040	0.9534	0.2327	0.4107	0.8720	0.5966	120.2	6.11
0.060	0.9011	0.2101	0.3903	0.8581	0.5786	120.2	6.06
0.071	0.8671	0.1968	0.3770	0.8484	0.5665	120.2	6.03
0.080	0.8402	0.1862	0.3654	0.8394	0.5556	120.2	6.00
0.100	0.7794	0.1663	0.3384	0.8167	0.5295	120.2	5.93
0.120	0.7210	0.1530	0.3117	0.7912	0.5022	120.2	5.86
0.140	0.6750	0.1450	0.2866	0.7640	0.4751	120.2	5.78
0.160	0.6334	0.1383	0.2637	0.7367	0.4492	120.2	5.70
0.180	0.5953	0.1296	0.2428	0.7101	0.4248	120.2	5.62
0.200	0.5589	0.1178	0.2234	0.6852	0.4018	120.2	5.54

0.250 0.4681 0.0863 0.1783 0.6300 0.3490 120.2 5.33

9.3.6 Real output — the budget

```

=== WFE Error Budget (ErrorBudget, Gaps 23+28) ===
Error budget: wfe [waves @ 633 nm]
Contributor          RMS      Share
Z11 Spherical         0.03    34.2%
Z7 Coma Y             0.025   23.7%
Z4 Defocus            0.02    15.2%
Z8 Coma X             0.018   12.3%
Z5 Astigmatism 0      0.015    8.5%
Z6 Astigmatism 45     0.01     3.8%
Z9 Trefoil Y          0.005    0.9%
Z10 Trefoil X         0.004    0.6%
Z12 2nd Astigmatism 0 0.003    0.3%
Z13 2nd Astigmatism 45 0.002    0.2%
Z14 2nd Coma Y        0.002    0.2%
Z15 2nd Coma X        0.001    0.0%
RSS total             0.05131
Allocation            0.07143
Margin (linear)       0.02012  within budget
RSS headroom          0.04969

```

9.3.7 Real output — the prescription against its scalar twin

Both runs: total WFE RMS = 0.0513 [waves at 633 nm]

Metric	Zernike (actual)	Scalar screen	Δ
Strehl [--]	0.9174	0.9256	-0.0082
MTF@Nyquist [--]	0.2246	0.2204	+0.0041
EE(1x1) [--]	0.3953	0.3999	-0.0046
RER [--]	0.5812	0.5872	-0.0060
NIIRS [--]	6.07	6.08	-0.0149
SNR [--]	120.2	120.2	+0.0000

(The run also prints a Maréchal reference table, a degradation table, a Strehl comparison, a noise budget, and four figures. The whole thing takes 2.6 s.)

9.4 What the numbers say

Regime: extended. The scene is a uniform land surface filling the pixel footprint completely, so the optics stage classifies it extended and every downstream stage reads that. In the extended regime there is no separable in-pixel background term — which is why the noise budget below has no background_shot row.

9.4.1 The budget closes, with real headroom

Quantity	Value [waves at 633 nm]
RSS total (Tom's 12 terms)	0.0513
Allocation ($\lambda/14$)	0.0714
Linear margin	+0.0201
RSS headroom	0.0497

The two margins are not the same number and the difference is not bookkeeping. The **linear margin**, 0.0201 waves, is what is left if the next contributor adds *arithmetically*. The **RSS headroom**, 0.0497 waves, is what is left if it adds *in quadrature* — which is what independent error sources do: $\sqrt{0.0714^2 - 0.0513^2} = 0.0497$. Tom can absorb an assembly-plus-thermal contributor of nearly 0.05 waves RMS, two and a half times what the linear reading would let him book. Quoting the linear margin to a mechanical engineer under-sells the design by a factor of 2.5.

The variance-share column tells him where reduction effort pays: spherical alone is 34.2 % of the variance, coma-Y 23.7 %, defocus 15.2 %. The bottom six terms together are under 2 %. Polishing trefoil is wasted money.

9.4.2 Tom's design sits comfortably inside diffraction-limited territory

At his actual 0.0513-wave prescription, run in Zernike mode: **Strehl 0.9174**, MTF at Nyquist 0.2246, RER 0.5812, and Δ NIIRS of -0.07 against perfect optics. The conventional diffraction-limited threshold is Strehl 0.80; he is well clear of it.

9.4.3 Shape versus RMS — a smaller effect than it used to be, and worth understanding why

The prescription and its equal-RMS scalar twin now agree closely: $|\Delta|$ of 0.008 in Strehl, 0.004 in MTF at Nyquist, and 0.01 in NIIRS. That is the *expected* answer, and it is expected because both pupils are now **low-order**. RADIANT's scalar expansion is a deterministic equal-RMS spread over Noll Z4–Z11, so comparing it with Tom's spherical-and-coma-dominated set is comparing two low-order pupils at one RMS. Two such pupils should be near-twins at Strehl 0.92, and they are.

The signs are informative even so: Tom's actual mix sits *slightly below* the equal-weight expansion in Strehl, EE and RER, and *slightly above* it in MTF at Nyquist. Concentrating the variance in spherical and coma pulls a little more energy out of the core than an even spread does, while leaving marginally more mid-frequency contrast. The gap grows with WFE, and only the Zernike route reproduces aberration-specific PSF structure — coma's asymmetric flare, spherical's rings — that no single RMS number can encode. **Use the prescription whenever one exists.**

9.4.4 SNR does not move, and that is the design insight

SNR is 120.2 at every one of the thirteen sweep points. Wavefront error redistributes energy within the point-spread function; it does not remove photons from the aperture. The noise budget is likewise constant:

Noise term	σ [e- RMS]	Fraction
signal_shot	120.3	99.8 %
read_noise	5.0	0.2 %

Noise term	σ [e- RMS]	Fraction
quantization	0.3	< 0.1 %
dark_shot	0.1	< 0.1 %
Total (RSS)	120.4	

Signal 14,468 e-, and $\sqrt{14\,468} = 120.3$: shot-limited, as a bright VNIR scene at 2 ms should be. NIIRS therefore changes with WFE almost entirely through the GIQE-5 RER term ($3.32 \log_{10} \text{RER}$), with the SNR term held fixed. That is the cleanest possible separation of a spatial effect from a radiometric one, and it means every NIIRS movement in the sweep table above is attributable to image quality alone.

9.4.5 Where the cliff is

Degradation	First sweep point past it	Interpolated crossing
−0.25 NIIRS	0.120 waves	≈ 0.110 waves
−0.50 NIIRS	0.180 waves	≈ 0.175 waves
−1.00 NIIRS	not reached (−0.81 at 0.250)	—

The script reports the first *sweep grid point* whose NIIRS has fallen past each threshold; the scenario’s `walkthrough.md` quotes the linear interpolation between bracketing points. Both readings describe the same thirteen rows, and the difference between them is sweep resolution, not disagreement.

The practical reading for Tom: the $\lambda/14$ allocation of 0.0714 waves costs him 0.11 NIIRS, which is nothing. He could relax the allocation to roughly 0.10 waves and still be inside a quarter-NIIRS degradation. Whether he *should* is a separate argument — Strehl at 0.10 waves is 0.78, just under the diffraction-limited line, and “diffraction-limited” is a phrase that appears in requirements documents.

Note also that the spatial columns now **decouple**. From 0 to 0.25 waves, MTF at Nyquist falls 66.1 % while RER falls only 42.9 %. A low-order aberration attacks mid-frequency contrast harder than it attacks edge slope. A model that degraded every spatial metric by one common factor could not produce that spread, and the fact that RADIANT does is a check on the pupil-domain implementation: all five spatial metrics come off the same EffectivePSF, so their *relative* behavior is physics rather than parameterization.

9.5 Two caveats the run itself raises

Annular Zernikes are not implemented. The very first line of output is a warning:

```
UserWarning: Obscuration ratio 0.35 > 0.30: standard Zernike polynomials are NOT
orthogonal on an annular pupil. Individual coefficient interpretations may be
misleading. Annular Zernikes (Mahajan 1981) are not implemented.
```

This is honest and it matters for the *interpretation* of individual coefficients, not for the total. On a 35 %-obscured pupil the standard circular Zernike basis is no longer orthogonal, so the variance shares in the budget table above are approximate attributions rather than an exact decomposition. The RSS total and the resulting pupil phase are unaffected; what is affected is the claim “spherical is 34.2 % of the variance”, which on an annulus mixes slightly with defocus and the higher spherical terms. Tom should read the ranking as a priority ordering, not as an exact accounting.

The Maréchal approximation is a reference, not the model. The run prints a side-by-side Maréchal-versus-RADIANT Strehl table. They diverge past about 0.1 waves — at 0.25 waves, Maréchal gives 0.085 against RADIANT’s 0.468. Two separate effects produce that. Maréchal’s $S = \exp[-(2\pi \text{OPD}_{\text{rms}}/\lambda)^2]$ is a small-aberration expansion that is simply invalid below $\text{Strehl} \approx 0.3$; and the table’s Maréchal column is evaluated at the 633 nm reference while RADIANT works at the 650 nm band center, where the same physical OPD is a smaller fraction of a wave. RADIANT’s reported Strehl is the degraded-PSF peak over the diffraction-limited reference peak — a computed ratio, not a closed-form approximation — and it is the one to quote.

9.6 The takeaway

Tom’s 40 cm Cassegrain carries 0.0513 waves RMS of design wavefront error against a $\lambda/14$ (0.0714-wave) allocation. It closes with **0.0497 waves of RSS headroom** for assembly and thermal contributors — the number to hand the mechanical team, not the 0.0201-wave linear margin. At that prescription the telescope delivers Strehl 0.9174, MTF at Nyquist 0.2246 and a NIIRS cost of 0.07, which is negligible.

The budget’s cliff is much further out than the allocation: a quarter-NIIRS degradation does not arrive until roughly 0.11 waves. And the modal *shape* of the error, at this RMS, is worth about 0.01 NIIRS — small, but the only way to know that was to run the actual prescription rather than trust the scalar summary.

Numbers in this chapter come from a re-run of the committed runner and reproduce the scenario’s `walkthrough.md` tables exactly, to every printed digit.

One documentation defect surfaced in the re-run and is recorded rather than quietly fixed: the script’s closing “Limitations” block still describes the scalar sweep as using “a random phase screen”. It does not — the scalar expansion is now the deterministic equal-RMS Z4–Z11 set, which is why the prescription and its twin agree as closely as they do. The walkthrough’s tables and discussion are current; the runner’s printed prose is one revision behind.

9.7 The same study in the GUI

The scenario ships a GUI-openable baseline of this telescope, `inputs/5.1_wfe_budget_allocation.gui.yaml`, with its headline metrics snapshot in `.gui.expected.json`. Opening it puts the 40 cm f/10 Cassegrain in the window with `optics.wfe_rms_waves` as an editable field, so the sweep above can be walked by hand one point at a time with the Performance workspace’s Spatial/MTF group updating as you go — which is a genuinely better way to build intuition for the Strehl-versus-MTF decoupling than reading the table. The scenario’s `gui_workflow.md` specifies what a complete WFE workspace would add: a WFE input-mode selector (Scalar RMS / Zernike / OPD map), an **Import from Zemax** button calling the same `load_zemax_zernike` the script uses, a live table of the parsed Z4–Z15 terms with variance shares, and an `ErrorBudget` panel showing allocation, linear margin and RSS headroom with a tooltip explaining why the quadrature headroom exceeds the linear margin. Zernike mode is functional through API-level injection today, as the script shows; what it lacks is a configuration-surface route, so the GUI would perform the injection on the analyst’s behalf.

Case Study — Benchmarking Against a Published Datasheet

Scenario 6.1 · persona: Dr. Chen, researcher validating a model against the literature · modality: script-led
 · scenario folder: `scenarios/06_dr_chen_researcher/6.1_published_snr_benchmark/`

10.1 The question

Dr. Chen has a published cooled-LWIR HgCdTe focal-plane datasheet on her desk. It quotes two numbers and the conditions they were measured under: a specific detectivity $D^* = 2.00 \times 10^{11}$ Jones and a NETD of 25.0 mK, at $f/2$, against a 300 K scene, over 8–12 μm , on a 30 μm pixel. She wants to know whether RADIANT, configured to those same conditions, reproduces them.

This is not a trade study. There is no sweep, no aperture axis, no design decision at the end of it. It is the question a researcher asks before she is willing to publish anything *else* the tool says: **does the noise model land on a number somebody else measured?**

The obstacle is that the datasheet and the chain speak different languages. A datasheet quotes **optical-power** figures of merit — watts of noise-equivalent power, and the detectivity derived from them. RADIANT’s chain computes noise in **electrons RMS**, because that is where the physics lives: shot noise is the square root of a collected charge, read noise is a ROIC property in electrons, dark shot is a rate times a time. Getting from one to the other is a unit bridge, and it is the bridge this scenario was written to exercise.

10.2 The inputs, and why they are what they are

Everything below is transcribed from `inputs/chen_lwir_ddatasheet.xlsx` (sheet `Datasheet`) into the run script as module constants, so that the run is self-contained and reproducible without the workbook. `inputs/create_spreadsheet.py` regenerates the workbook from the same values.

Quantity	Value	Why
Published D^*	2.00×10^{11} Jones	The figure under test. “Jones” is $\text{cm}\cdot\text{Hz}^{1/2}/\text{W}$.
Published NETD	25.0 mK	The second figure under test.
Spectral band	8.0–12.0 μm	The datasheet’s stated LWIR passband.
Pixel pitch	30 $\mu\text{m} \times 30 \mu\text{m}$	Sets the detector area $A_d = 9.00 \times 10^{-6} \text{ cm}^2$.
f-number	2.0	The datasheet’s reference cold-stop $f/\#$; realized as a 0.05 m aperture at 0.10 m focal length.
Scene temperature	300 K, $\epsilon = 0.98$	The reference scene the NETD is quoted against.

Quantity	Value	Why
Integration time	30 μ s	The datasheet’s reference dwell. It is short <i>because</i> the LWIR flux is intense.
Quantum efficiency	0.75	Datasheet, band-average.
Dark rate	1.0×10^5 e-/s at 77 K	Datasheet, at the cryogenic set point.
Read noise	60 e- RMS	Datasheet ROIC.
Full well	1.0×10^7 e-	Datasheet ROIC.
Optical transmission	0.85	Scalar lump — this benchmark is not a coated-train study.
Atmosphere	simple, us_standard, nadir from 1000 m	A short, nearly transparent column; the benchmark is about the detector, not the air.
Tolerance	± 15 %	Dr. Chen’s declared pass band, set before the run.

Two entries carry more weight than they look. The **30 μ s integration time** is the whole reason this configuration is plausible: an 8–12 μ m staring array pointed at a room-temperature scene collects charge extraordinarily fast, and 30 μ s is what keeps the well from filling. LWIR staring FPAs are integration-time-limited, and the datasheet’s reference conditions quietly encode that.

The **± 15 % tolerance is declared in advance**, in the script, as TOLERANCE_PCT. A benchmark whose tolerance is chosen after the residual is known is not a benchmark.

10.3 The converter bridge

Three relations link the two languages. All three ship as RADIANT functions with their own Level-0 tests, and this scenario was the first consumer of them:

$$\text{NEP} = \sigma_e \cdot \frac{hc}{\eta \lambda t_{int}} \quad D^* = \frac{\sqrt{A_d \Delta f}}{\text{NEP}} \quad \Delta f = \frac{1}{2 t_{int}}$$

The first converts a noise in electrons to a noise-equivalent optical power in watts: divide by the quantum efficiency to get from electrons to absorbed photons, multiply by the photon energy hc/λ to get joules, divide by the integration time to get watts. The second is the definition of specific detectivity — detector area and noise bandwidth normalized out, so that arrays of different size and frame rate can be compared.

The third is the one worth pausing on. $\Delta f = 1/(2t_{int})$ is the **equivalent noise bandwidth of an integrating detector**: a device that accumulates charge for t_{int} and then reads it out is, to a noise calculation, a filter of that bandwidth. It is a convention, not a measurement, and it is what makes a per-frame electron count commensurable with a per- $\sqrt{\text{Hz}}$ detectivity. At the datasheet’s 30 μ s it gives $\Delta f = 16.7$ kHz.

10.4 The scripted path

One script does the whole study:

```
S=scenarios/06_dr_chen_researcher/6.1_published_snr_benchmark
PYTHONPATH=src python $S/scripts/run_datasheet_benchmark.py
```

10.4.1 The sensor, built in code rather than loaded from YAML

Because every parameter is a datasheet figure, the script constructs the sensor directly instead of maintaining a parallel config file:

```
def build_sensor() -> Sensor:
    """LWIR FPA configured to the datasheet reference conditions."""
    aperture = 0.05
    focal = aperture * FNUM # f/# = focal / aperture
    s = Sensor()
    s.set("source.scene_type", "extended")
    s.set("source.target.temperature", SCENE_TEMP_K, unit="K")
    s.set("source.target.emissivity", 0.98)
    s.set("optics.aperture_diameter_m", aperture)
    s.set("optics.focal_length_m", focal)
    s.set("optics.transmission_scalar", TRANSMISSION)
    s.set("detector.pixel_pitch_x_um", PITCH_UM)
    s.set("detector.qe_value", QE)
    s.set("detector.dark_rate_e_per_s", DARK_E_PER_S)
    s.set("spectral_integration.filter_min_um", BAND[0])
    s.set("spectral_integration.filter_max_um", BAND[1])
    s.set("spectral_integration.integration_time_s", T_INT_S)
    s.set("readout.read_noise_e_rms", READ_NOISE_E)
    s.set("readout.full_well_capacity_e", FULL_WELL_E)
    return s
```

Note the f-number: it is not a settable input. RADIANT derives it from aperture and focal length, so the script inverts the datasheet's $f/2$ into a focal length and lets the chain re-derive it. That is the general pattern for any derived quantity in RADIANT — you specify the independent inputs and the tool computes the rest.

10.4.2 The two directions of the bridge

The script crosses the bridge twice, in opposite directions, and that is the whole design of the benchmark. First it takes the **datasheet's** D^* down to an electron count, so that Dr. Chen can see what total noise the published figure implies:

```
nep_spec = nep_from_dstar(DSTAR_SPEC, AREA_CM2, DELTA_F_HZ)
sigma_e_spec = noise_electrons_from_nep(nep_spec, QE, LAMBDA_C_UM, T_INT_S)
```

Then it runs the chain and takes the **chain's** noise up to a detectivity:

```
r = build_sensor().evaluate()
signal_e = r.stage_outputs["spectral_integration"]["signal_e"]
snr = r.metrics["snr"]
noise_e = signal_e / snr

nep_chain = nep_from_noise_electrons(noise_e, QE, LAMBDA_C_UM, T_INT_S)
dstar_chain = dstar_from_nep(nep_chain, AREA_CM2, DELTA_F_HZ)
nedt_chain_mk = r.metrics["nedt_K"] * 1e3
```

Two details are worth copying into your own benchmark scripts. The total noise is recovered as `signal_e / snr` rather than by quadrature-summing the sixteen noise terms by hand — the chain already did that sum, and

re-doing it is an opportunity to disagree with it. And NETD is read straight out of `result.metrics["nedt_K"]`, which is the chain's band-integrated $\partial S/\partial T$ value, not a single-wavelength approximation.

10.4.3 Real output

```
=====
SCENARIO 6.1 – PUBLISHED-DATASHEET BENCHMARK (D*/NETD)
=====
Datasheet: cooled LWIR HgCdTe, D* = 2.00e+11 Jones, NETD = 25 mK @ f/2, 300 K,
8–12 μm, 30 μm pixel.
Pixel area 9.00e-06 cm², noise bandwidth Δf = 1/(2·t_int) = 16.7 kHz.

-----
DATASHEET D* → NEP → equivalent noise electrons (converter chain)
-----
NEP(D*)           = 1.936e-12 W
σ_e equivalent     = 2,193 e⁻ (total noise the D* implies)

-----
CHAIN RESULT → D* / NETD (via the converters)
-----
Signal            = 6.335e+06 e⁻ (well 63%)
Total noise σ_e    = 2,518 e⁻ (√signal = 2,517 → BLIP)
NEP(chain)         = 2.223e-12 W
D*(chain)          = 1.742e+11 Jones
NETD(chain)        = 24.62 mK

-----
BENCHMARK (tolerance ±15%)
-----
Metric            Datasheet      Chain      Residual    Verdict
D* [Jones]        2.00e+11    1.74e+11   -12.9%      PASS
NETD [mK]         25.0        24.6       -1.5%      PASS
```

Wrote `fig1_datasheet_benchmark.png`

(One interpretive paragraph is trimmed from the block above; it is quoted in full in the next section. The run takes 0.83 s.)

10.5 The comparison

Metric	Datasheet	Chain	Residual	Verdict (±15 %)
D^* [Jones]	2.00×10^{11}	1.74×10^{11}	-12.9 %	PASS
NETD [mK]	25.0	24.6	-1.5 %	PASS

Both pass. But they pass by very different margins — 1.5 % on NETD against 12.9 % on D^* — and the asymmetry is the interesting part of the result, not a blemish on it.

The chain is background-limited, and that is why D^* comes out low. The run reports a signal of 6.335×10^6 e- and a total noise of 2518 e- RMS, against $\sqrt{6.335 \times 10^6} = 2517$ e-. The two agree to one part in 2500: every other noise term in the budget — the 60 e- read noise, the 3 e- of dark shot over 30 μ s — is invisible in quadrature against photon shot noise. The system is **BLIP**, background-limited in performance.

A datasheet's peak D^* is a near-intrinsic figure: it characterizes the *detector*, measured under conditions chosen to expose the detector's own noise floor. A real system staring at a 300 K scene through an f/2 cold stop is not in that condition — it is drowning in scene photons — so its *system* detectivity is necessarily lower than the detector's intrinsic one. The -12.9 % residual is that effect. It is in the expected direction, and a residual of the *opposite* sign would have been the alarming result, because it would mean the chain had found less noise than the physics of the scene allows.

The script's own summary of this, printed verbatim with the table:

Interpretation: the chain is photon-shot- (BLIP-) limited ($\sigma_e \approx \sqrt{\text{signal}}$), so its system D^* reflects the background-limited detectivity at these conditions. Agreement within tolerance validates RADIANT's noise model against the published figure; a residual would flag a background-flux or QE mismatch versus the datasheet reference.

The electron-domain cross-check makes the same point in the other units. The datasheet's D^* , converted down, implies 2193 e- of total noise; the chain finds 2518 e-. That is 14.8 % more noise, which is the same statement as the -12.9 % detectivity residual seen from the other side of the reciprocal. Dr. Chen gets to check the bridge's arithmetic against itself without leaving the output block.

NETD agrees far better because NETD is a ratio in which the background cancels. $\text{NETD} = \sigma_{\text{total}}/(\partial S/\partial T)$. In the BLIP limit the numerator is $\sqrt{S}$ and the denominator is proportional to S times the Planck derivative's logarithmic slope, so the background flux largely divides out and what survives is the thermal contrast of the scene and the band. The datasheet and the chain are describing the same physics there, and they agree to 1.5 %.

Regime: extended. The 300 K scene fills the pixel footprint completely, so RADIANT classifies the scene extended in the optics stage and every downstream stage reads that classification. In the extended regime there is no separable background term inside the target pixel — the pixel sees one radiance field — which is why the noise budget here is signal shot plus electronics and nothing else. The well fills to 63 % in 30 μ s, confirming that the datasheet's short dwell is a saturation constraint rather than a stylistic choice.

10.6 Truth anchors behind the converters

The benchmark above tests the *model*. The converters it runs through were tested first, on their own, in `src/radiant/performance/tests/test_noise_spec_converters.py` (13 Level-0 tests), before this scenario was allowed to consume them:

Anchor	Setup	Expected
D^* definition	$A_d = 1 \times 10^{-4}$ cm ² , $\Delta f = 1$ Hz, $D^* = 1 \times 10^{10}$ Jones	NEP = 1×10^{-12} W (hand calculation)
NEP $\leftrightarrow \sigma_e$	NEP = $\sigma_e hc / (\eta \lambda t_{\text{int}})$, both directions	Exact round-trip
NEP $\leftrightarrow$ NETD	NEP 1×10^{-12} W, $\partial P / \partial T = 1 \times 10^{-11}$ W/K	NETD = 0.1 K
Full loop	$D^* \rightarrow \text{NEP} \rightarrow \text{NETD} \rightarrow \text{NEP} \rightarrow D^*$	Closes

That ordering is the point. A unit bridge that has not been closed against a hand calculation is not evidence about the model — it is a second, untested model sitting between you and the answer.

10.7 The takeaway

RADIANT reproduces a published cooled-LWIR FPA’s NETD to 1.5 % and its D^* to 12.9 %, both inside a ± 15 % tolerance declared before the run, with the D^* residual explained in the right direction by the system’s background-limited operating point. For a researcher, the usable conclusion is narrower and more valuable than “the tool works”: **the electron-domain noise model, propagated from component specifications plus scene photon shot noise, lands on published optical-power figures of merit** — and when it does not, the converters give you the residual in whichever of the two unit systems makes the diagnosis obvious.

The verification-of-numbers reflex here is worth keeping. The run was re-executed for this chapter and reproduces the scenario’s committed `walkthrough.md` table digit for digit: $\sigma_e = 2518\text{e-}$, $D^* = 1.742 \times 10^{11}$ Jones, NETD = 24.62 mK, residuals -12.9 % and -1.5 %.

10.8 The same study in the GUI

The scenario ships a GUI-openable baseline of this configuration, `inputs/6.1_published_snr_benchmark.gui.yaml`, with its headline metrics snapshot in the neighbouring `.gui.expected.json`. Opening it puts the same LWIR FPA in the window, where the Detector workspace’s Noise tab shows the sixteen-term budget the `signal_e / snr` shortcut above summarizes, and the Performance workspace reports the NETD directly. The scenario’s `gui_workflow.md` specifies what a full benchmark workflow would add on top: a `datasheet-import` step, a side-by-side benchmark panel with the tolerance band shaded and a PASS/FAIL verdict — the content of `outputs/fig1_datasheet_benchmark.png` — and a one-call “`chain → D*/NETD`” helper callable from the GUI’s scripting window, so that the converter chain is available interactively rather than only in a script. The converter functions themselves are already importable there.

Case Study — Reconciling Predicted and Measured NEDT

Scenario 7.1 · persona: Karen, test engineer · modality: script-led · scenario folder: scenarios/07_karen_test_engineer/7.1_nedt_reconciliation/

11.1 The question

Karen has just finished TVAC testing an as-built MWIR sensor. She put a calibrated blackbody in front of it at seven temperatures from 15 °C to 50 °C, took 500 frames at each, and computed NEDT. At her primary test point, 25 °C, she measured **127.0 mK ± 6.0 mK**.

She then configured RADIANT with the as-built parameters — the *measured* QE, the *measured* dark current, the *measured* read noise, the *as-built* f-number, not the nominal design values — and it predicted **74.17 mK**.

That is a 52.83 mK gap, and the prediction is the optimistic one. This is the canonical test-engineer question and it has no comfortable answer: **does the model match the measurement, and if not, what is the missing noise?** A gap that cannot be attributed is a gap that will be re-argued at every design review until somebody attributes it.

11.2 The inputs, and why they are what they are

Everything arrives in one three-sheet workbook, inputs/karen_nedt_lab_data.xlsx, which is Karen's lab notebook in vendor format. The script reads it with openpyxl and converts at the boundary.

System configuration (sheet 1) — the optical bench as built:

Quantity	Lab value	Canonical	Why it differs from nominal
Aperture diameter	30.0 cm	0.300 m	As designed.
Effective focal length	121.5 cm	1.215 m	Gives f/4.05, not the f/4.00 nominal — 1.25 % slow.
Optical transmission	71.0 %	0.710	73 % nominal; 2 points lost in the as-built coatings.
Optics temperature	22.0 °C	295.15 K	Lab ambient. The optics are <i>warm</i> .
Spectral band	3500–5000 nm	3.50–5.00 μm	MWIR cold filter at 77 K.
Blackbody emissivity	0.995	0.995	CI Systems SR-800R, ±0.02 °C stability.
Shroud	22.0 °C, $\epsilon = 0.95$	295.15 K	The TVAC chamber wall.

As-built detector (sheet 2) — the FPA as measured, with the nominal value for comparison:

Quantity	As-built	Nominal	Deviation
Pixel pitch	18 μm	18 μm	—
Quantum efficiency	68.0 %	72.0 %	−4 points
Dark current	135 e-/s	100 e-/s	+35 %
Read noise (post-CDS)	14.2 e- RMS	12.0 e- RMS	+18 %
Full well	500,000 e-	—	—
System gain / ADC	12 e-/DN / 14 bit	—	—
Integration time	0.5 ms	—	Lab test value; 8 ms in orbit.
IPC coupling	1.5 %	—	—
ROIC glow	5 e-/s	—	Measured, and not modeled — see below.

NEDT measurements (sheet 3) — seven temperatures, 500 frames each, with the measurement's own σ so that the comparison carries error bars.

The single most consequential configuration choice is not in the workbook at all:

```
"atmosphere": {
    "model": "exo", # Vacuum – no atmospheric effects
},
```

The sensor is inside a TVAC chamber. There is no atmosphere. `exo` sets $\tau_{atm} = 1$, $L_{path} = 0$ and $L_{atm,down} = 0$, so the only thermal sources in the model are the blackbody, the shroud and the warm optics. Getting this wrong — by leaving a default terrestrial column in place — would put an entire atmosphere between a blackbody and a sensor sitting 30 cm apart on a bench, and no amount of gap analysis downstream would recover from it.

11.3 The physics being tested

$$\text{NEDT} = \frac{\sigma_{total}}{\partial S / \partial T}$$

A noise in electrons over a responsivity in electrons per kelvin. Both halves have to be right, and they fail in different ways: an error in σ_{total} is a noise-model error, an error in $\partial S / \partial T$ is a radiometry error. The decomposition that follows is what separates them.

Each noise term carries its own share:

$$\text{NEDT}_i = \frac{\sigma_i}{\partial S / \partial T}, \quad \text{NEDT}_{total} = \sqrt{\sum_i \text{NEDT}_i^2}$$

11.4 The scripted path

```
S=scenarios/07_karen_test_engineer/7.1_nedt_reconciliation
PYTHONPATH=src python $S/scripts/run_nedt_reconciliation.py
```

11.4.1 Importing measured data

```
INPUT_FILE = Path(__file__).parent.parent / "inputs" / "karen_nedt_lab_data.xlsx"
```

```

wb = openpyxl.load_workbook(INPUT_FILE)

ws_sys = wb["System Configuration"]
sys_specs: dict[str, object] = {}
for row in ws_sys.iter_rows(min_row=5, max_col=4, values_only=False):
    name, value = row[0].value, row[1].value
    if name and value is not None:
        try:
            sys_specs[name] = float(value)
        except (ValueError, TypeError):
            sys_specs[name] = value

```

Unremarkable code doing the essential thing: the lab's own workbook is the input, in the lab's own units, and the conversion to canonical units happens exactly once, here at the boundary, before anything touches the chain. The script prints the conversion table so Karen can audit it — every row of it appears in the output below.

11.4.2 Computing $\partial S/\partial T$ by finite difference

RADIANT does not expose its internal thermal derivative, so the script builds one:

```

DELTA_T = 0.5 # K - finite difference step for dS/dT

result_center = Sensor.from_dict(make_config(T)).evaluate()
result_plus   = Sensor.from_dict(make_config(T + DELTA_T)).evaluate()
result_minus  = Sensor.from_dict(make_config(T - DELTA_T)).evaluate()

signal_e      = result_center.stage_outputs["readout"]["signal_e_final"]
signal_plus   = result_plus.stage_outputs["readout"]["signal_e_final"]
signal_minus  = result_minus.stage_outputs["readout"]["signal_e_final"]
ds_dt = (signal_plus - signal_minus) / (2.0 * DELTA_T) # e-/K

```

Three full chain evaluations per measurement temperature, twenty-one in all. A central difference at $\delta T = 0.5$ K is accurate to better than 0.1 % against the analytic Planck derivative over this range — the Planck function's curvature in the MWIR at 288–323 K is gentle enough that the second-order error term is negligible.

11.4.3 The per-term decomposition and the gap

```

noise_dict = {nt.name: nt.value_e for nt in result_center.noise_terms}
total_noise = math.sqrt(sum(v**2 for v in noise_dict.values()))
nedt_pred_mK = total_noise / ds_dt * 1000.0

nedt_per_term = {name: sigma / ds_dt * 1000.0 for name, sigma in noise_dict.items()}

```

and then the reconciliation itself, which is three lines of arithmetic and the whole point of the exercise:

```

sigma_meas = primary["meas_nedt_mK"] / 1000.0 * ds_dt # measured NEDT → electrons
sigma_missing_sq = sigma_meas**2 - sigma_pred**2
sigma_missing = math.sqrt(sigma_missing_sq)

```

Run the measured NEDT *backwards* through the same $\partial S/\partial T$ to get the noise the measurement implies, then subtract the model's noise in quadrature. What is left is the unmodeled noise, in electrons, which is a far more actionable quantity than a millikelvin discrepancy.

11.4.4 Real output — the comparison

=== Predicted vs. Measured NEDT ===

BB T [°C]	BB T [K]	Meas [mK]	Pred [mK]	Δ [mK]	Signal [e ⁻]	dS/dT [e ⁻ /K]	σ_{total} [e ⁻]
15.0	288.15	160.0	83.86	76.14	99,144	3758.9	315.21
20.0	293.15	142.0	78.76	63.24	119,491	4392.9	345.98
25.0	298.15	127.0	74.17	52.83	143,203	5105.7	378.70
30.0	303.15	114.0	70.03	43.97	170,688	5903.0	413.40
35.0	308.15	104.0	66.28	37.72	202,383	6790.9	450.11
40.0	313.15	95.0	62.87	32.13	238,756	7775.2	488.84
50.0	323.15	81.0	56.93	24.07	327,553	10056.9	572.51

11.4.5 Real output — the noise budget at the primary test point

=== NEDT Breakdown at Primary Test Point (25°C / 298.15 K) ===

Signal: 143,203 e⁻
dS/dT: 5105.7 e⁻/K
Total noise: 378.70 e⁻ RMS

Noise Term	σ [e ⁻ RMS]	NEDT_i [mK]	Fraction [%]
signal_shot	378.42	74.117	99.9
read_noise	14.20	2.781	0.1
quantization	3.46	0.678	0.0
dark_shot	0.26	0.051	0.0
background_shot	0.00	0.000	0.0
nearfield_shot	0.00	0.000	0.0
...	(ten further terms, all 0.00)		
RSS TOTAL	378.70	74.173	100.0

Measured NEDT: 127.0 mK
Predicted NEDT: 74.17 mK
Gap: 52.83 mK

11.4.6 Real output — the gap analysis

=== Gap Analysis ===

$\sigma_{\text{predicted}}$: 378.70 e⁻ RMS
 σ_{measured} : 648.42 e⁻ RMS (from NEDT = 127.0 mK)
 σ_{missing} : 526.34 e⁻ RMS (RSS gap)
NEDT_missing: 103.09 mK

Noise Term	Current σ [e ⁻]	Required σ [e ⁻]	Increase [%]	Plausible?
signal_shot	378.42	648.26	71.3	Unlikely alone

read_noise	14.20	526.54	3608.0	Cannot explain alone
quantization	3.46	526.36	15094.6	Cannot explain alone
dark_shot	0.26	526.34	202489.9	Cannot explain alone
...	(twelve further terms, all "Cannot explain alone")			

11.4.7 Real output — sensitivity, and nominal versus as-built

Ranked by |sensitivity| (most impactful first):

1. f-number (via focal length): 0.7428 mK per 1% change
2. Optical transmission: 0.3714 mK per 1% change
3. QE: 0.3714 mK per 1% change
4. Integration time: 0.3714 mK per 1% change
5. Read noise: 0.0010 mK per 1% change
6. Dark current: 0.0000 mK per 1% change

=== Nominal vs. As-Built NEDT ===

NEDT (nominal params):	70.19 mK
NEDT (as-built params):	74.17 mK
NEDT (measured):	127.0 mK

Nominal → As-built degradation:	3.98 mK
As-built → Measured gap:	52.83 mK

(The run takes 7.2 s and also writes outputs/nedt_reconciliation_results.xlsx.)

11.5 What the numbers say

Regime: extended. The blackbody overfills the field of view — every pixel sees blackbody radiance directly. The optics stage classifies the scene extended, and in that regime the pixel is a single radiance field with no separable in-pixel background term.

11.5.1 The model is signal-shot-limited, and nothing else is close

99.9 % of the predicted noise is `signal_shot`. Read noise contributes 2.78 mK of the 74.17 mK, quantization 0.68 mK, dark shot 0.05 mK. On a 143,203 e⁻ signal the electronics are simply not in the conversation.

This is worth stating plainly because it forecloses an entire class of corrective action. If the *model* is right, no better ROIC and no colder detector improves this sensor: the remedy for a shot-limited system is more signal or less background — cold optics, a tighter cold shield, spectral narrowing — not quieter electronics.

11.5.2 The shroud background is *not* counted, and that is correct

An older vintage of this scenario carried a `background_shot` term of roughly 349 e⁻ from the 295 K shroud, about 46 % of the budget. It is now exactly zero, and the reason is the extended regime: during a NEDT measurement the large-area blackbody **fills the target pixel**. The shroud is not in that pixel. Adding its shot noise to the target pixel's budget was a double count.

Removing it dropped the prediction from about 100 mK to 74 mK — and therefore **widened** the gap against the 127 mK measurement. That is the right outcome and the uncomfortable one. The old prediction was not closer to the measurement because it was better; it was closer because a spurious term happened to push it

that way. Anyone reading an older baseline of this scenario should know that the smaller gap it reports is not a better model.

11.5.3 526 e- of noise is missing, and no single modeled term can supply it

The reconciliation: $\sigma_{meas} = 0.127 \times 5105.7 = 648.42$ e- RMS against $\sigma_{pred} = 378.70$ e- RMS, leaving $\sqrt{648.42^2 - 378.70^2} = 526.34$ e- RMS unaccounted — **139 % of the predicted noise**, worth 103.09 mK on its own.

The “what would it take” table is the script’s most useful output and its verdict is uniform: read noise would have to rise 3608 %, dark shot 202,490 %, quantization 15,095 %. Even signal shot — the one term large enough to be in the running — would need to rise 71.3 %, which would require the collected signal to be 2.9 times what the radiometry says it is. **No single modeled term explains the gap**, which is precisely the finding that sends Karen to look outside the model.

Four candidates, in the order Karen should test them:

1. **Unmodeled mirror self-emission.** `nearfield_shot` is 0.00 e- and should not be. In scalar-transmission mode the lumped optical element is treated as refractive: by Kirchhoff, $T + R = 1$ so $\varepsilon = 1 - T - R = 0$, and a zero-emissivity element emits nothing. But these are **mirrors at 295 K in the MWIR**, and real mirrors with $\varepsilon = 1 - R$ emit. Capturing that requires the `key_elements` or `full_prescription` optical mode rather than a scalar throughput. This is the leading physical candidate and the one that would change the model’s answer.
2. **The measurement is spatial, the prediction is temporal.** Karen computes NEDT as the σ across a 100×100 pixel ROI. That statistic contains pixel-to-pixel responsivity and offset variation — PRNU and DSNU — which are *not* temporal noise. RADIANT’s `prnu` and `dsnu` terms are both 0.00 e- here because the config does not enable them. A spatial σ compared against a temporal σ is not the same measurement, and on an uncorrected FPA the difference is easily this large.
3. **Unmodeled ROIC glow.** The workbook records 5 e-/s of ROIC glow. RADIANT’s `glow_shot` term is always zero — the physics is not implemented. At 0.5 ms this is a small contributor, but it is a known-missing term rather than a negligible one.
4. **Blackbody calibration and chamber reflections.** The SR-800R’s ± 0.02 °C stability contributes roughly $0.02 \times 5106 = 102$ e- of apparent noise, and the $\varepsilon = 0.95$ shroud reflects 5 % of the blackbody’s emission back toward the sensor.

11.5.4 The as-built deviations explain almost nothing, and that is good news

Nominal 70.19 mK against as-built 74.17 mK: the QE loss, the 35 % higher dark current, the 18 % higher read noise and the 1.25 % slower optics together cost **3.98 mK**, 7.5 % of the 52.83 mK gap. The sensitivity table explains why. Only four parameters move NEDT at all, and the largest — f-number — is worth 0.74 mK per 1 % change; read noise is 0.0010 mK per 1 % and dark current is 0.0000 mK per 1 %. In a shot-limited system, $NEDT \propto 1/\sqrt{S}$ and $S \propto \tau/(f/\#)^2$, so f-number enters squared and the electronics do not enter at all.

The message to Karen is therefore quite specific: **the sensor is performing close to its as-built specification. The discrepancy is in the model’s completeness and in the test method, not in the hardware.** Chasing the 35 % dark-current excess would be wasted effort.

11.5.5 NEDT improves at higher blackbody temperature

From 15 °C to 50 °C, measured NEDT falls 160 → 81 mK and predicted 83.86 → 56.93 mK. Both curves fall because $\partial S/\partial T$ grows faster than the noise does: across that range $\partial S/\partial T$ rises 3759 → 10,057 e-/K (2.68×) while σ_{total} rises only 315 → 573 e- (1.82×), because shot noise grows as $\sqrt{S}$. This is the Wien-side behavior of the Planck function in the MWIR, and the fact that both the measurement and the model show it is a check that the *responsivity* half of the NEDT ratio is right even though the *noise* half is not.

Note that the gap narrows with temperature too, 76.14 mK at 15 °C to 24.07 mK at 50 °C. The missing noise is therefore not a fixed electron count — it scales with the scene. That is itself a clue, and it points away from a constant additive term like ROIC glow and toward something proportional to signal, which is what PRNU in a spatial- σ measurement looks like.

11.6 One caveat the run raises and the walkthrough does not

The three hottest test points — 35 °C, 40 °C and 50 °C, at all three finite-difference perturbations — **exceed the 14-bit ADC full scale** at the as-built 12 e-/DN gain. The run emits nine warnings of the form:

```
UserWarning: ReadoutStage: ADC saturated – signal 2.73e+04 DN exceeds full scale
16383 DN (14-bit at 12 e-/DN). Signal clipped to 16383 DN. Increase
readout.gain_e_per_dn or readout.adc_bits, or reduce the signal (integration time,
aperture). (readout.adc_status = 'clipped')
```

The NEDT column is unaffected: the signal and $\partial S/\partial T$ the script reads are electron-domain quantities, and the monotone $\partial S/\partial T$ progression in the results table confirms the clip does not propagate into them. But a *real* measurement at those three temperatures, on this ROIC at this gain, would have been digitizer-limited — 327,553 e- is 27,296 DN against a 16,383 DN full scale — and the comparison at 50 °C is therefore between a model of an unclipped sensor and a measurement of a clipped one. The three lowest test points are clean. The scenario's walkthrough.md does not mention this; the warning is the tool behaving correctly and the omission is in the narrative, not the code.

11.7 The takeaway

1. **The gap is real and it is 52.83 mK**, or equivalently 526 e- RMS of noise the model does not contain — 139 % of the noise it does.
2. **No modeled term can absorb it.** Every candidate would need an implausible increase; the reconciliation table makes that quantitative rather than rhetorical.
3. **The three leading explanations are all outside the current model:** warm-mirror self-emission that scalar transmission mode cannot produce, a spatial-versus-temporal mismatch in what “NEDT” means on each side of the comparison, and unmodeled ROIC glow.
4. **The hardware is fine.** As-built deviations cost 3.98 mK; the sensitivity analysis shows no ± 1 % parameter perturbation could close the remaining gap.
5. **The honest prediction is the wider gap.** Removing a spurious shroud term made the model better and the agreement worse. Agreement achieved through a compensating error is not agreement.

Numbers in this chapter come from a re-run of the committed runner and reproduce the scenario's walkthrough.md tables exactly — 74.17 mK predicted, 127.0 mK measured, 52.83 mK gap, 526.34 e- missing, 3.98 mK of as-built degradation, and every sensitivity coefficient.

11.8 The same study in the GUI

The scenario ships a GUI-openable baseline of the as-built sensor at the primary test point, `inputs/7.1_nedt_reconciliation.gui.yaml`, with headline metrics snapshotted in `.gui.expected.json`. Opening it puts the TVAC configuration in the window — exo atmosphere, 298.15 K blackbody, warm shroud — where the Detector workspace’s Noise tab shows the same sixteen-term budget reproduced above and the Performance workspace reports the 74.17 mK directly, so the prediction half of the reconciliation is a single click rather than a script. The scenario’s `gui_workflow.md` specifies what a complete reconciliation workspace would add: a spreadsheet-import step that maps the three sheets onto optics, detector and validation-target data with the unit conversions highlighted; a nominal-versus-as-built table with deviations color-coded; a **Lab / TVAC test mode** that recognizes the bench context and proposes `atmosphere.model = "exo"` rather than leaving the analyst to know that; a progress view over the seven temperatures $\times$ three runs; and the three comparison charts — NEDT versus temperature with measurement error bars and the gap shaded, the noise-budget breakdown, and the sensitivity ranking. The gap analysis itself has no built-in method today; the script computes $\sigma_{missing}$ by hand, and a GUI would be doing the same arithmetic behind the button.

Case Study — Air-to-Air IRST on a Level Arm

Scenario 10.2 · persona: Sarah, systems engineer sizing an airborne MWIR IRST · modality: GUI-led · scenario folder: scenarios/10_direction_general/10.2_air_to_air_level_irst/

12.1 The question

Own-ship cruises at 10 km. The target cruises at 10 km. The line of sight between them is horizontal, the target is a 500 K hot-parts source of 0.36 m² at $\epsilon = 0.90$, and the sensor is a 150 mm f/3 refractive MWIR head on a 20 μm focal plane. Sarah needs four answers:

1. How far can this IRST see the target, and how does SNR fall with range?
2. **Does RADIANT even accept a horizontal line of sight**, and what does it warn about when it does?
3. The target is manoeuvring. What does the relative line-of-sight rate do to the integration-time budget?
4. How much can she trust the atmosphere model on a 100 km horizontal path at cruise altitude?

Question 2 used to have an embarrassing answer. Equal altitudes were a geometry-free carve-out, and a path zenith at or past $\pi/2$ was rejected in three independent places. This scenario is the validation that the level arm now works end to end, and the GUI surfaces it exercises — the derived scene-class chip, the level composition of the schematic, the tangent-depression pill — are the ones built for it.

This chapter walks the nominal 50 km point in the GUI on the scenario's committed baseline, `inputs/10.2_air_to_air_level_irst.gui.yaml`. The 25–100 km range sweep, the kinematics doors and the MODTRAN cross-check are the scripted half.

12.2 The system, and why it is what it is

Vendor units on the left, canonical on the right — the vendor data arrives in mm, %, ke-, ms, km, kt and °C, and the import maps every one of them once.

Quantity	Vendor	Canonical
Entrance pupil diameter	150 mm	0.150 m
Effective focal length	450 mm	0.450 m
f-number	3.0	3.0 (derived)
Optical transmission	75 %	0.750
Central obscuration	0 %	0 — a filled refractive pupil
WFE RMS	0.05 waves	0.05 waves at 0.633 μm
Spectral band	3.50–5.00 μm	same
Pixel pitch	20 μm	20 μm

Quantity	Vendor	Canonical
Fill factor	100 %	1.00
Quantum efficiency	80 %	0.80
Dark current	50,000 e-/s	50,000 e-/s
FPA temperature	80 K	80 K
Read noise	40 e- RMS	40 e- RMS
Full well	1000 ke-	1.000×10^6 e-
System gain	61 e-/DN	61 e-/DN
ADC	14 bits	14 bits
Frame integration	0.10 ms	1.00×10^{-4} s
Own-ship / target altitude	10 km / 10 km	10,000 m / 10,000 m
Slant range	50 km	50,000 m
Target hot parts	226.85 °C, 0.36 m ² , 90 %	500.00 K, 0.36 m ² , 0.90
Own-ship TAS	480 kt	246.93 m/s
Illumination	—	night
Atmosphere	midlat_summer, PWV 2.92 cm, vis 23 km, rural	same

The derived instrument scales follow immediately: $\text{IFOV} = p/f = 44.44 \mu\text{rad}$, band center $4.250 \mu\text{m}$, $Q = \lambda F/\# / p = 0.6375$ — undersampled, which is normal for a search IRST that is trading resolution for field of view.

The atmosphere is deliberately set to the same profile, water content, visibility and aerosol as the delivered MODTRAN horizontal grid, so the scenario's cross-check is apples-to-apples rather than approximately so.

12.3 The GUI walk

12.3.1 Step 1 — Check that the tool understood the scene

Select stage **1 Geometry**, tab **Inputs**. This is the check Sarah makes before trusting a single downstream number: has the GUI read this as an air-to-air level arm, or as a space-to-ground look with a typo in it?

Geometry

Inputs
 Schematic

Scene class — derived label & optional assertion

Scene: air → air (air_to_air) — derived

Assert scene class auto

OFF BY DEFAULT FOR THIS SCENE CLASS

- NIIRS
- NIIRS (extrapolated)
- GSD (cross-track)
- GSD (along-track)
- GSD (geometric mean)
- Ground range
- Swath width
- Access rate
- Diffraction limit (at target)
- Max integration time

Defaults only — an explicitly set performance.metrics.* group flag always wins, so selecting a group yourself computes it whatever the scene class.

Geometry inputs — pick one mode per family

Viewing geometry

Direct slant range (V0)

sensor_altitude_m	10000 m
target_altitude_m	10000 m
path_zenith_rad	90.2245 deg
sensor_off_boresight_rad	90.2245 deg
ground_range_m	49921.8 m
elevation_angle_rad	-0.224479 deg
target_range_m	50000 m

Solar geometry

Solar zenith θ_s (S1)

solar_illumination	night
solar_azimuth_rad	0 deg
solar_zenith_rad	28.6479 deg
solar_elevation_rad	—
site_latitude_rad	—
day_of_year	—
local_solar_time_h	—
ltan_h	—

Platform kinematics

Direct ground speed

ground_speed_m/s	246.933 m/s
circular_orbit	false

Line-of-sight rate

Platform motion only (derived)

los_angular_rate_rad/s	0.00493866 rad/s
target_speed_m/s	—
target_heading_rad	—
target_climb_rad	—

Figure 12.1: Geometry workspace, Inputs tab — the derived air_to_air scene class, its off-by-default metric list, and the V0 direct-slant-range mode.

The card at the top reads `Scene: air → air (air_to_air) – derived`, with the two component classes spelled out. The label is read verbatim from the geometry stage’s output; the card never re-derives a class from an altitude. `Assert scene class` sits at `auto`, which is the point — the assertion is available but never required.

Underneath, **off by default for this scene class**, is a list of eleven metrics: NIIRS, NIIRS (extrapolated), the three GSD axes, ground range, swath width, access rate, two diffraction-limit-at-target entries, and max integration time. Every one of them projects something onto a ground plane, and there is no ground plane at an airborne target. The note beneath is explicit that these are *defaults only* — an explicitly set performance. `metrics.*` group flag always wins — so a missing metric is never ambiguous between “irrelevant” and “failed”.

The viewing family is in mode **V0**, `Direct slant range`. `sensor_altitude_m` and `target_altitude_m` both read 10,000 m and `target_range_m` reads 50,000 m; the four angular fields are grayed. For equal altitudes the chord fixes the Earth-center central angle directly, which is the solution that subsumed the old geometry-free carve-out.

Proving the guard works. The scenario’s workflow has Sarah set `Assert scene class = air_to_air` (nothing changes — the assertion agrees with the derivation), then change `target_altitude_m` from 10,000 m to 10 m, the classic wrong-magnitude typo. The scene now derives `air_to_ground`, contradicting the assertion, and the geometry stage raises. The GUI answers in three places at once: the actionable-error dialog carrying the what/why/action verbatim, a row in the right-rail Messages panel, and a conflict tint on the scene-class card itself with the error’s *what* line beside the chip — the contradiction shown where the assertion was made.

12.3.2 Step 2 — Look at the arm

Switch to the **Schematic** tab.

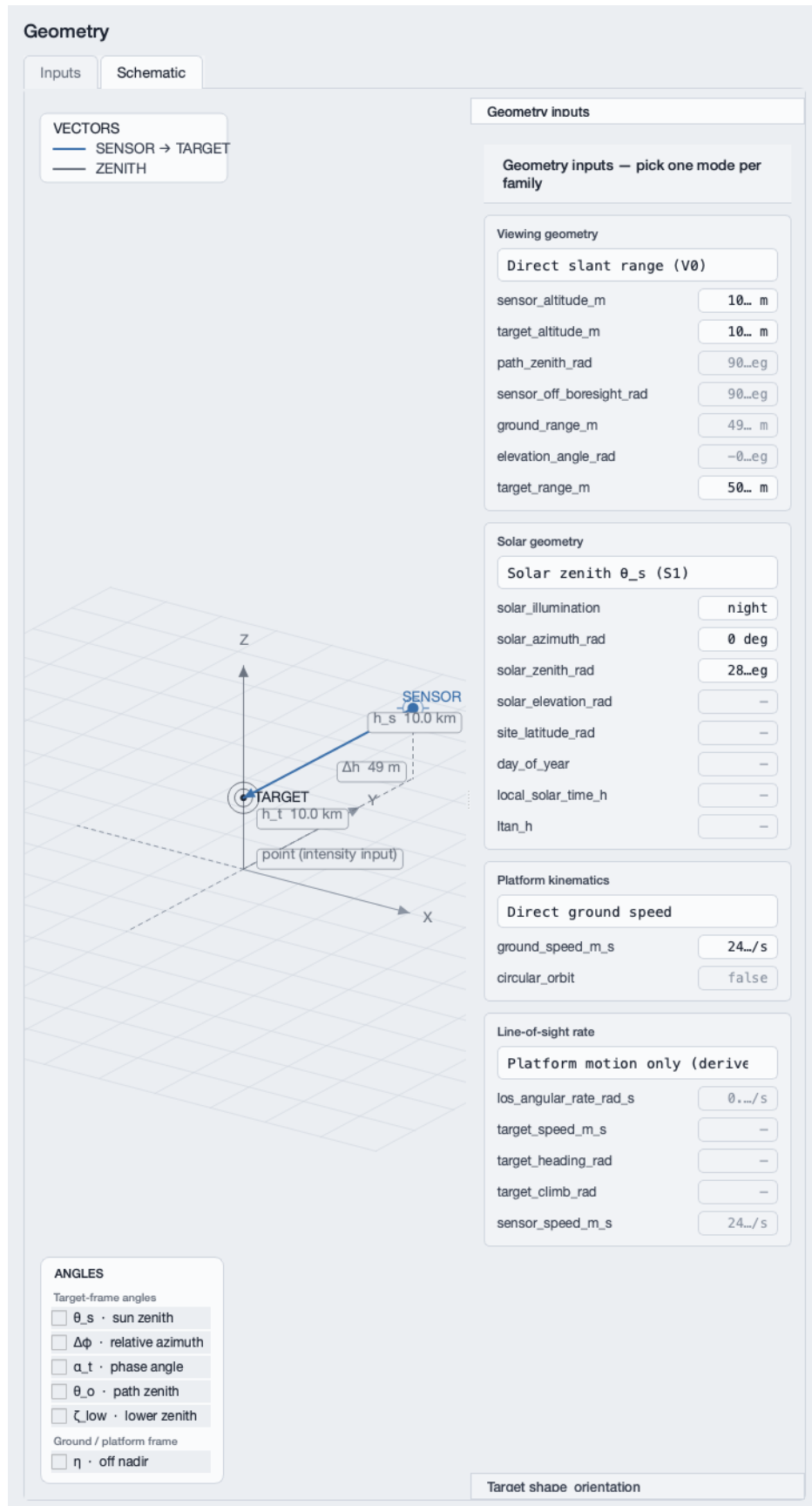


Figure 12.2: Geometry workspace, Schematic tab — the level composition, both endpoints at altitude, with the tangent-depression leader pill.

This is the figure the Phase-4 geometry work exists for. Three things to check:

- **Both endpoints are drawn apart at the same abstract height**, with the ground plane pushed below both. The composition is chosen from the derived `los_direction`, never from a user switch. If this still drew the down-looking layout — sensor above, target on the ground plane — the composition split would be broken, and that would be a regression rather than a cosmetic complaint.
- **The altitude leader pills** read `h_t 10.0 km` at the target and `h_s 10.0` at the sensor, whose unit is clipped by the right edge of the viewport; the mode form beside the picture confirms both at 10,000 m. Altitude is *told*, not drawn to scale — the not-to-scale idiom that governs every RADIANT schematic.
- **The Δh 49 m pill** on the arm itself. It has no toggle; it is drawn whenever the scene is a level arm and hidden otherwise, exactly like the altitude pills.

There is no sun vector, because `solar_illumination` is `night`.

What Δh is, and why it is a text label. Both endpoints sit on the same shell of radius $r = R_E + h$, so the straight chord between them sags *below* that shell and each endpoint looks very slightly **down** at the other. The depression is

$$\Delta h = (R_E + h) (1 - \sin \zeta_{low}) \approx \frac{L^2}{8(R_E + h)},$$

which at $L = 50$ km is 48.97 m. Forty-nine meters over a fifty-kilometer arm is invisible at any honest drawing scale, so it is annotated rather than drawn. And it matters: Δh is the variable the horizon guard classifies a level path on, so this pill is the analyst's early warning that a longer arm will trip it. The schematic calls `radiant.core.viewing_triangle.classify_horizon_topology` and formats the result rather than restating the formula, so the pill and the guard cannot disagree.

The same sag is why the path zenith at the target is *greater* than 90 degrees. The derived readout below the mode form — scroll the center pane — reports $\theta_o = 1.574714$ rad = **90.2245°** at the target and $\eta = 89.7755^\circ$ at the sensor, with ground range 49,922 m. Those two sum to exactly 180.0000° and differ by the Earth-center central angle $\varphi = 0.44896^\circ$, with $\theta_o = \pi/2 + \varphi/2$. That identity is checkable on screen by revealing the θ_o and η arcs from the ANGLES panel; on a level arm the ζ_{low} arc coincides with θ_o exactly, and seeing the two overlap is the visual proof that the down/level and up-looking branches meet continuously at equal altitudes.

12.3.3 Step 3 — Read the spatial budget of an undersampled search sensor

Select stage **4 Optics**, tab **MTF**.

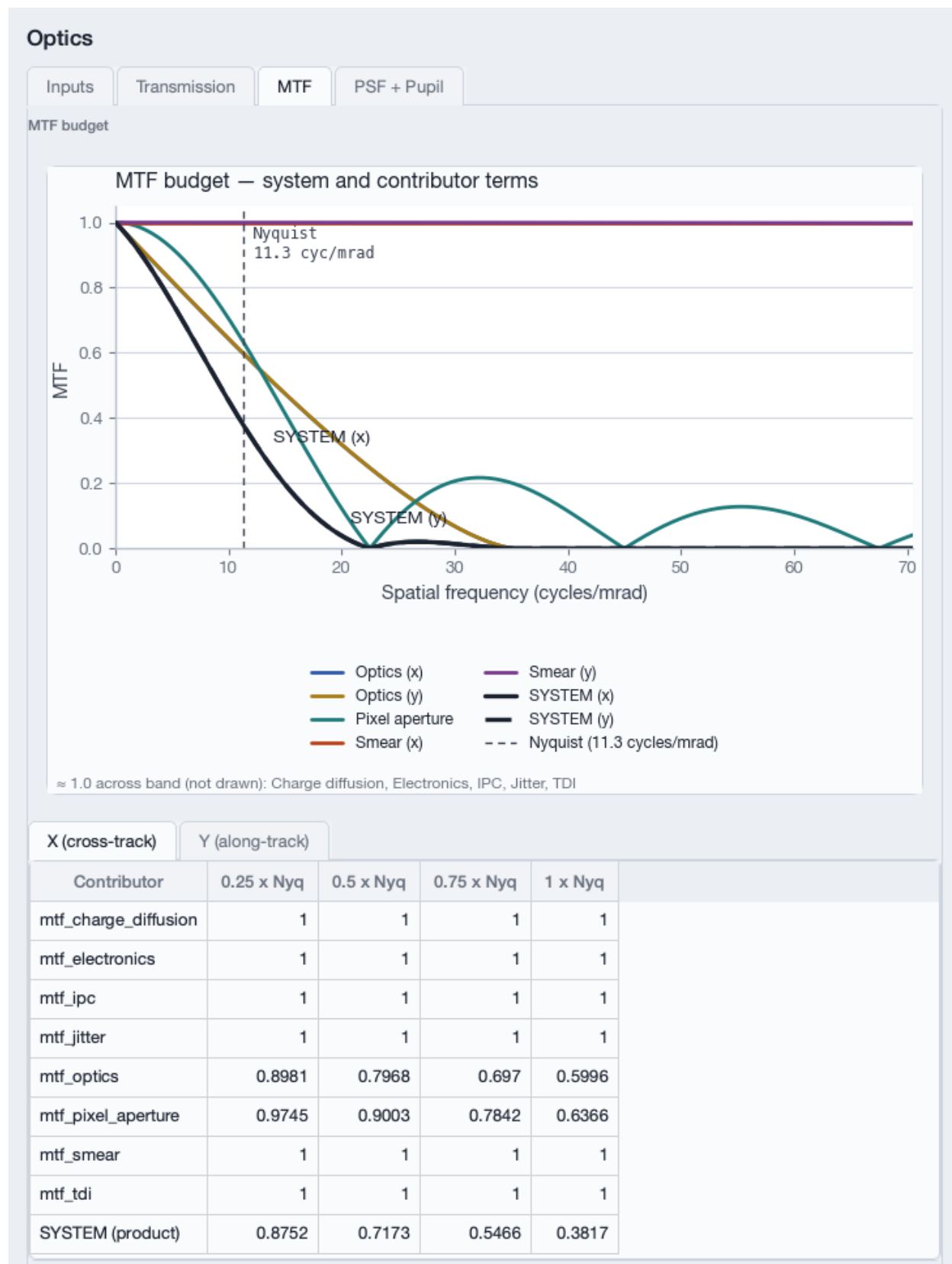


Figure 12.3: Optics workspace, MTF tab — the IRST's MTF budget, with the pixel aperture ringing past its first zero above Nyquist.

Nyquist is marked at **11.3 cycles/mrad**, which is $1/(2 \times 44.44 \mu\text{rad})$. Two contributors bite; the caption lists charge diffusion, electronics, IPC, jitter and TDI as “ ≈ 1.0 across band (not drawn)”, and the table confirms them at exactly 1 at every sampled fraction of Nyquist.

Contributor	$0.25 \times \text{Nyq}$	$0.5 \times \text{Nyq}$	$0.75 \times \text{Nyq}$	$1 \times \text{Nyq}$
mtf_optics	0.8981	0.7968	0.6970	0.5996
mtf_charge_diffusion	1	1	1	1
mtf_electronics	1	1	1	1
mtf_ipc	1	1	1	1
mtf_jitter	1	1	1	1

The optics term comes from the autocorrelation of the complex pupil — diffraction and the 0.05-wave wavefront error together, never as two factors multiplied — and the pixel aperture is the sinc of the $20 \mu\text{m}$ footprint. The shape worth noticing is the teal pixel curve: it crosses zero near 22 cycles/mrad and then *rings*, and every one of those sidelobes lies above Nyquist. In an undersampled sensor that energy is not resolution; it is aliasing. That is the price the $Q = 0.6375$ design pays for its field of view, and the Performance page reports an alias fraction of 0.503 at Nyquist to match.

12.3.4 Step 4 — Read the noise budget

Select stage **7 Detector**, tab **Noise**.

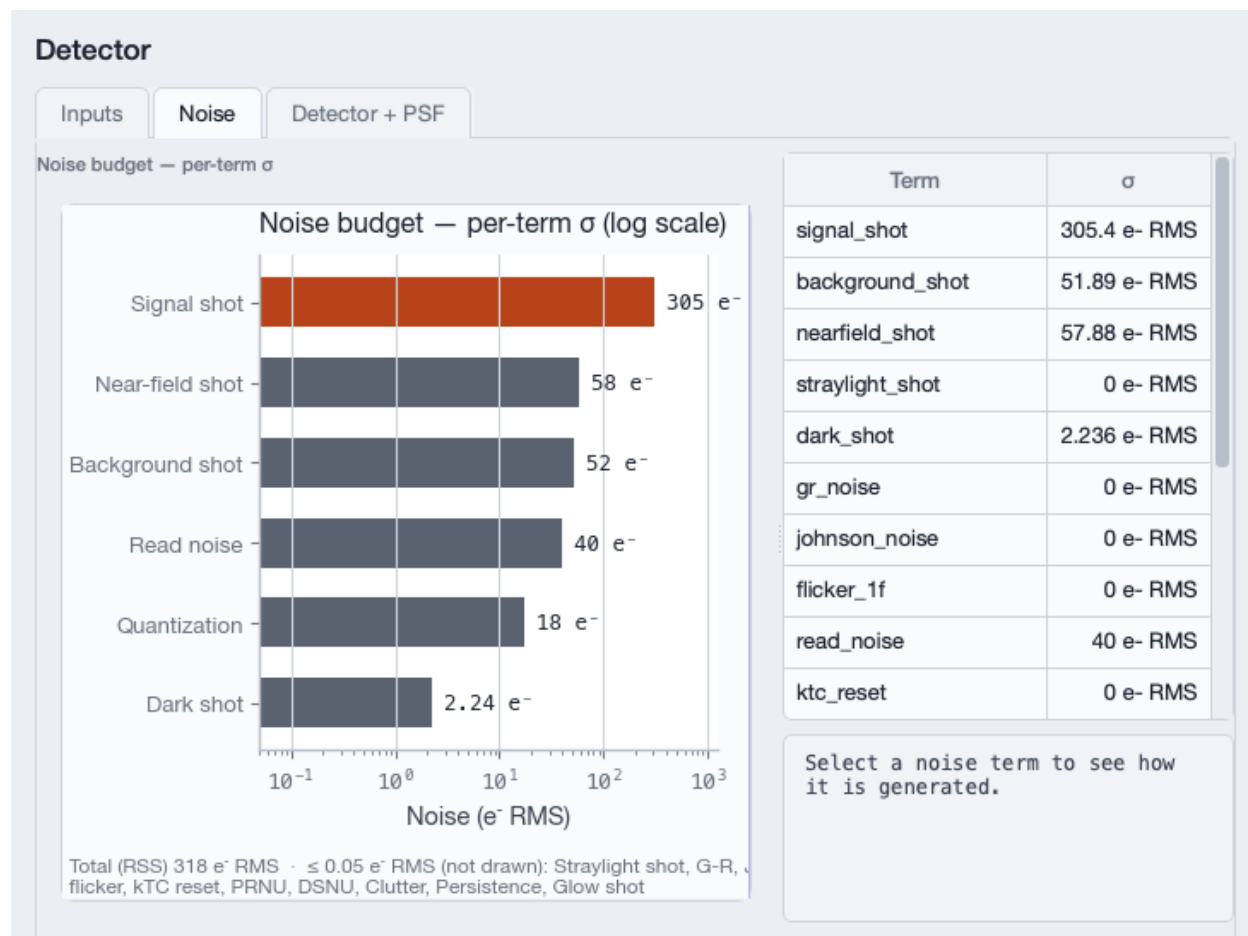


Figure 12.4: Detector workspace, Noise tab — the 50 km budget, dominated by the target’s own shot noise.

Term	σ [e- RMS]
signal_shot	305.4
background_shot	51.89
nearfield_shot	57.88
read_noise	40
quantization	17.61
dark_shot	2.236
Total (RSS)	318.2

This budget carries the most counter-intuitive result in the scenario. The target’s own shot noise is 305.4 e- RMS of a 318.2 e- RMS total — **92 % of the noise power comes from the thing being detected**. Everything that is *not* the target — sky background shot, the telescope’s own warm-optics emission, read, quantization and dark — combines to 89.2 e- RMS. (That floor was 67.9 e- RMS before CU-380, 2026-10-03, gave this uncooled head the fore-optics temperature its own datasheet states — $-23.15\text{ }^{\circ}\text{C} = 250.0\text{ K}$, “soaked at flight OAT”; nearfield_shot is the new term, and raising the floor 31 % is the whole of that change.)

That matters because a detection-range solver has to scale the noise as well as the signal. Push the target out and its own shot noise goes with it, leaving the 67.9 e- floor; freeze the total noise at its reference value

instead, as a frozen-noise solver does, and the near-field answer comes out strongly pessimistic. The shipped solver uses $\sigma^2(R) = S(R) + N_0^2$ with N_0 the target-free floor, and the consequence is visible in the next figure.

12.3.5 Step 5 — Read the result, and notice what is missing

Select stage **10 Performance**.



Figure 12.5: Performance workspace on the level arm — the ground-projection family absent by scene class, target-plane sample distance in its place.

Sampling / geometry opens with three rows that do not appear on any ground scene: target-plane sample distance 2.222 m in x, y and geometric mean. That is $p d/f = 20 \mu\text{m} \times 50 \text{ km}/0.45 \text{ m}$, and it is the right substitute for GSD when there is no ground plane to project a footprint onto. Q reads 0.6375 at band center (0.525 at the short edge, 0.75 at the long one), classified detector-limited, with a diffraction limit of 34.57 μrad . **There is no GSD row, no ground range, no swath width, no access rate** — the eleven metrics the scene-class card listed are genuinely absent from the result, not blank.

The right rail's pinned cards, outside the pane shown above, say the same thing in a second place: GSD and NIIRS both read n/a – not computed – off by default for this scene class, each naming the group to select to override it (Sampling / geometry, and Interpretability), and the Interpretability group carries only MRT at Nyquist, 0.5364 K. Nothing in the configuration asked for this; it follows from the derived scene class through one declarative relevance map.

Spatial / MTF. FWHM 20.65 μm cross-track and 20.6 μm along-track — 45.9 μrad , just over the 44.44 μrad IFOV, which is the undersampling again. RER 0.7031, ensquared energy 0.5387 in the central pixel and 0.9185 over 3×3 , MTF at Nyquist 0.3806, folded MTF 0.7659, alias fraction 0.503. Strehl ratio 0.9983 against a Maréchal value of 0.9978 — the 0.05-wave wavefront error costs essentially nothing, and the two independent Strehl routes agree to 5×10^{-4} .

Radiometric.

Metric	Value
SNR	293.1
Contrast SNR	293.1
SCNR	293.1
Detection range	$1.932 \times 10^5 \text{ m}$
NEDT	92.28 mK

Saturation. Well margin 20.61 dB, ADC margin 20.60 dB, dynamic range 69.95 dB. The margin is $20 \log_{10}$ of the ratio of capacity to filled charge, so 20.61 dB is a factor of 10.7: the 100 μs frame fills about 9 % of the 1 Me- well at 50 km. Nothing in the whole 25–100 km sweep saturates — the near end of the sweep still holds 4.6 dB.

The regime here is **POINT_SOURCE**, tentative in the source stage and final in optics. An IRST target is specified as in-band radiant intensity in W/sr rather than as a radiance, so the point-intensity door (emitter temperature 500 K, emitting area 0.36 m², emitter emissivity 0.90) is the native descriptor, and the ensquared-energy fraction is applied once, in spectral integration, to the target term only. The background is SkyBackground, selected by the line-of-sight termination classifier: the ray continues *past* the target, and a level arm at 10 km leaves the atmosphere rather than striking the ground. The same configuration aimed at a ground target would have selected a ground background. Nothing in the scenario asks for either behavior.

12.4 What the study concludes

The range sweep. Over $25 \rightarrow 100 \text{ km}$ the runner reports:

Range [km]	θ_o [deg]	Δh [m]	Guard	τ (MWIR)	Signal [e-]	Noise [e- RMS]	SNR	Det. range [km]
25	90.11224	12.2	clean	0.6337	5.8803×10^5	772.0	761.7	193.0
50	90.22448	49.0	clean	0.4023	9.3262×10^4	318.2	293.1	193.2
75	90.33672	110.2	warn	0.2556	2.6329×10^4	185.4	142.0	193.2
100	90.44896	195.9	warn	0.1625	9.4131×10^3	132.5	71.1	193.2

The 50 km row is the baseline this chapter evaluated, and the GUI's SNR 293.1, detection range 1.932×10^5 m and 20.61 dB well margin are that row. SNR falls $10.7\times$ over a $4\times$ range increase — steeper than inverse square, because the band transmittance falls from 0.634 to 0.163 across the same span. Note how unevenly CU-380's warm optics are felt down the column: -0.3% at 25 km, where the target's own shot noise still dominates, against -9.9% at 100 km, where the floor is nearly all of the noise. The range dependence of that penalty *is* the floor-limited transition, measured.

The detection range is reference-range invariant: 193.0 km solved from the 25 km point against 193.2 km solved from the 100 km point, a spread of $1.001\times$. That is the payoff of scaling the target's own shot noise along the path. Before that fix the same design returned 123.4 km from 25 km and 182.5 km from 100 km — a $1.48\times$ spread on one unchanged sensor, and the nominal 50 km answer moved from 150.9 km to about 199 km when it was corrected (193.2 km today, after CU-355 and CU-380). The residual 0.2 km of spread is the band-mean transmittance model's own reference dependence, not the noise treatment.

The horizon guard warns, and quantifies what it is warning about. Ten of the sixteen sweep arms are clean (25–70 km, Δh 12.2–96.0 m) and six sit in the warning shoulder (75–100 km, Δh 110.2–195.9 m); the analytic crossover $L = \sqrt{8r\Delta h_{clean}} = 71.45$ km falls between the 70 km and 75 km rungs. The warning text names the excluded physics — RADIANT models no atmospheric refraction — and sizes it: under the standard $k = 4/3$ effective-radius model the 100 km path would bottom out 146.9 m below its lower endpoint rather than 195.9 m, so the modeled ray samples air about 32.6 m lower on average than the real one, worth roughly **0.91 %** in band transmittance. That is the right verdict for operational work: the scene computes, the caveat is named and quantified, and the analyst decides.

Target kinematics cost more than they look like they cost. Platform motion alone gives $\omega_{LOS} = 4.939$ mrad/s; the crossing target (580 kt, heading 270° , 2° climb) raises it to **10.905 mrad/s**, a factor of 2.21. The two do *not* combine in quadrature — they are two contributions to one focal-plane translation, so they compose in the velocity domain and only then become a smear, and an RSS of two smears would have returned 7.745 mrad/s and understated the blur by 29 %. At the 100 μ s search frame the smear is 0.0111 pixel platform-only against 0.0245 pixel with the target, so it is negligible either way. The consequence is in the *budget*: the integration time that produces one pixel of smear falls from 8.999 ms to 4.076 ms, a 54.7 % cut. Target motion is free in search mode and binding in track mode.

The atmosphere is the weakest link, and it is pessimistic. Against the delivered MODTRAN horizontal decks at the same profile and altitude, the parametric model's MWIR band transmittance runs $+19.7\%$ at 5 km, -3.1% at 25 km, -30.8% at 50 km and -66.8% at 100 km. The mechanism is named: a band-averaged transmittance is not multiplicative in path length, because strong lines saturate first and flux leaks through the windows between them. The effective band extinction must therefore *fall* with path length by as much as $k(\lambda)$ varies inside the band, and the measurement is stark — MODTRAN's $\alpha(5\text{ km})/\alpha(100\text{ km})$ is $7.60\times$ in the MWIR against the model's $1.01\times$. The model's $k(\lambda)$ is essentially flat across 3.5–5.0 μ m, so it cannot reproduce saturation at all. **Usable band on that evidence: within about 5 % out to 25 km; beyond 50 km treat the MWIR numbers as a lower bound.**

Finally, the dual-path spatial consistency check runs silent on this scene: the maximum disagreement between the FFT of the convolved PSF and the MTF product is 1.0×10^{-3} against a 2×10^{-2} tolerance, twenty times inside it, with no warning records emitted. A level arm changes the radiometry and the geometry, not the spatial degradations, so there was no mechanism for the two paths to diverge — and the check confirms none did.

12.5 The same study from a script

```
cd scenarios/10_direction_general/10.2_air_to_air_level_irst
python inputs/create_spreadsheet.py
python scripts/run_air_to_air_level_irst.py
```

The runner takes about seven seconds and does 39 chain evaluations: the 16-point range sweep with both kinematics doors at each point, the horizon-guard classification, the five independent hand cross-checks, and — when the horizontal MODTRAN decks are staged — the transmittance comparison of the last section. Without those decks it prints a skipped banner naming the missing runs and every other section is unaffected. `scripts/gui_console_10.2_air_to_air_level_irst.py` is the same work as a paste-in for the GUI's scripting console, where the schematic's own pill value is one call:

```
from radiant.core.viewing_triangle import classify_horizon_topology
g = result.stage_outputs["geometry"]
res = classify_horizon_topology(g["theta_o_rad"], g["h_sensor_m"], g["h_target_m"])
print(res.topology, f"{res.dh_m:.2f} m", res.action) # interior_tangent 48.97 m clean
```

The full geometry derivation, the verbatim guard and over-specification errors, the kinematics-door matrix and the gap list are in the scenario's own `walkthrough.md`.

Scenario Digest Compendium

The eight chapters before this one walk eight scenarios at full depth — the inputs, the run, the figures, the physics argument, and the caveats. That depth costs roughly fifteen pages apiece, and there are fifty-one scenarios in the repository. Covering them all that way would produce a four-hundred-page volume in which the interesting cases were buried.

The compendium's principle is **total coverage at tiered depth**: every scenario appears, eight at full depth, four in the flagship-validation chapter, and the rest here as digests. A digest is one to two pages and answers five questions — what was the mission, what went in, what came out, which radiometric regime was in force, and what the analyst should take away — plus a pointer to the scenario folder where the full narrative, the figures, and the gap list live.

13.1 How to read a digest

Every digest carries the same six parts, in the same order:

- **Mission setup** — two or three sentences of context: who is asking, what they are looking at, from where.
- **Key inputs** — a compact table of the parameters that decide the answer, each with its unit. It is not the full configuration; the scenario folder holds that.
- **Headline results** — the numbers the scenario exists to produce, quoted from the committed walkthrough.md with their units. **No digest re-runs anything.** Where a walkthrough contradicts itself, or is missing a number a digest would want, the digest says so in one clause rather than papering over it.
- **Regime** — extended, sub_pixel, or point_source, named with one clause on why the scene classifies that way. Several scenarios never run the chain at all (pure geometry, pure sequence models); those say so.
- **Takeaway** — one or two sentences: the conclusion an engineer would carry into a design review.
- **Where to go deeper** — the scenario folder, and whether a GUI-openable baseline (inputs/<slug>.gui.yaml) ships with it. Where one does, the study can be opened through File → Open YAML and re-run in the GUI rather than from a script.

13.2 A note on vintage

RADIANT's physics has moved under these scenarios repeatedly — the atmosphere model alone has taken a dozen corrections since the first executions, each one re-baselining the affected walkthroughs. The numbers below are the **committed walkthrough values at their stated vintage**, not a fresh run. Most walkthroughs carry an italic refresh note naming the cleanup unit responsible for the last movement and its size; where that

matters to the reading of a result, the digest repeats it. A number here that disagrees with a run you do today means the scenario has been re-baselined since, and the scenario's own `walkthrough.md` is the authority.

The scenarios are grouped by persona in catalog order. Persona 8 is not a persona at all — it is a supplementary folder of tool demonstrations added outside the closed thirty-five-scenario catalog, and it is included here for the same reason as everything else: total coverage.

13.3 Persona 1 — Sarah, Systems Engineer

Sarah sizes payloads for proposals. Her scenarios are aperture-and-altitude trades where the answer has to be defensible to a customer, and the recurring theme is that the sizing basis matters more than the sizing calculation.

13.3.1 1.2 — VNIR Pan Imager: GSD vs Aperture vs Altitude

Mission setup. A sun-synchronous panchromatic imager must hold a 0.5 m ground sample distance while meeting an SNR spec. Holding GSD fixed as the orbit rises forces a longer focal length, hence a higher f-number, hence less irradiance per pixel — so aperture and altitude are coupled, and the SNR-spec contour runs diagonally across the trade space. This is the first consumer of `radiant.core.solar_geometry`, which turns the orbit's LTAN plus target latitude and date into the solar zenith angle.

Key inputs.

Quantity	Value
Held GSD	0.5 m (nadir)
Pixel pitch	6.5 μm
Band	450–700 nm (pan)
Aperture sweep	20–80 cm
Altitude sweep	400–600 km
Orbit / latitude	10:30 LTAN sun-sync, 35 °N
SNR spec	50
Detector QE	Si CCD curve, digitized from a datasheet plot, band-averaged over 450–700 nm

Headline results. At the 50 cm / 500 km reference design, winter SNR is **40.6** (solar zenith 62.2°) against a summer 70.8 (zenith 22.7°) — a **43 % seasonal swing**, and the winter case **fails** the SNR = 50 spec. The aperture ladder at 500 km reads SNR 11.8 at 20 cm, 40.6 at 50 cm, 68.7 at 80 cm; the sampling parameter $Q = \lambda (f/\#)/p$ runs 2.88, 1.15, 0.72 across the same three. Minimum aperture to hold spec climbs **50 cm at 400 km → 75 cm at 600 km**. The 20 cm design is diffraction-limited everywhere (diffraction ground spot 1.4–2.1 m against a 0.5 m pixel); the 80 cm design is detector-limited; 50 cm straddles the crossover.

Regime. extended — the sunlit surface fills the pixel, so scene radiance *is* the background, EE_box is not applied, and the point-source machinery is unused.

Takeaway. Size to the worst-case season, not the annual mean: a 50 cm aperture that looks compliant on a mean-illumination chart misses spec every winter. And buying fine GSD with a small aperture is wasted money — below about 50 cm the optics, not the pixel, set the resolution.

Where to go deeper. scenarios/01_sarah_systems_engineer/1.2_vnir_gsd_aperture_altitude/. A GUI baseline ships.

13.3.2 1.3 — Dual-Band MWIR/LWIR Wildfire Detection

Mission setup. Sarah must pick a band for a wildfire mission: a 5 m² hotspot at ~600 K against a 300 K conifer canopy, seen from a 10 km airborne platform. The vendor offers MWIR (3.5–5.0 μm , 80 K) and LWIR (8–12 μm , 60 K) HgCdTe; the canopy emissivity arrives as a JPL/NASA ASTER-library text file. The band decision is the deliverable.

Key inputs.

Quantity	Value
Fire / background	600 K hotspot, 5 m ² ; 300 K conifer canopy
Background emissivity	ASTER curve — $\epsilon = 0.9530$ (MWIR), 0.9821 (LWIR)
Platform / optics	10 km altitude, 2.5 cm f/2, 20 μm pixels
GSD / fill fraction	4.0 m $\rightarrow$ 16 m ² footprint; hotspot fills 0.31
Fire-mode integration	5 μs (MWIR), 25 μs (LWIR)
Scene clutter	3 % of background (σ)

Headline results. At 600 K: MWIR pixel signal 228,529 e- against LWIR 2,962,820 e-; contrast 220,998 e- vs 1,217,173 e-; total noise 542.7 e- RMS vs 52,418.4 e- RMS, of which clutter is 225.9 and 52,369.4 e- RMS and **warm-optics shot noise 33.7 and 609.8 e- RMS** (CU-380 — 278 K optics radiate in-band for LWIR and down the Wien flank for MWIR, a 327 $\times$ split in electrons). **SCNR including clutter: 407.2 (MWIR) vs 23.2 (LWIR)** — a 17 $\times$ advantage. NEDT runs the other way, 230.5 mK (MWIR) vs 171.8 mK (LWIR), both carrying the single-wavelength NEDT caveat. Note which metric the warm optics moved: LWIR SNR fell 3.7 % and its NEDT rose 3.9 %, while its **SCNR did not move at all** — 99.8 % of that total is scene clutter, and 610 e- RMS in quadrature with 52,369 is below the rounding. Over a 400–1200 K fire sweep, MWIR holds $P_d \approx 1$ throughout while **LWIR misses the 400 K smolder outright** (SCNR 1.1, $P_d = 0.000$ against a 4.75 σ threshold). MWIR saturates first, at ≈ 1200 K and ~ 98 % of the 4 Me- well. Band-integrated radiance contrast at 600 K is nearly equal — 373.6 W/m²/sr (MWIR) vs 382.0 W/m²/sr (LWIR).

Regime. sub_pixel — the 5 m² hotspot fills 31 % of the 16 m² pixel footprint, so the regime override keeps the in-pixel background photons and the scene clutter in the budget.

Takeaway. ΔL alone would call the two bands equivalent; the detector-level comparison separates them by a factor of seventeen, because LWIR's 3 % clutter rides on a background an order of magnitude brighter. **MWIR for fire detection, LWIR for ambient-scene mapping** — NEDT is the wrong figure of merit for the former.

Where to go deeper. scenarios/01_sarah_systems_engineer/1.3_dual_band_mwir_lwir/. A GUI baseline ships.

13.3.3 1.4 — TDI Pushbroom Optimization: Line Rate vs SNR

Mission setup. A VNIR panchromatic pushbroom imager on a 500 km sun-synchronous orbit: ground velocity caps the per-line integration at ~ 0.2 ms, so time-delay integration is what builds SNR. How many

TDI stages, and where does the well stop the gain? (The concept started as MWIR; a 300 K thermal scene saturates at $N_{TDI} = 1$, which is why TDI is fundamentally a VNIR technology.)

Key inputs.

Quantity	Value
Aperture / focal length	25 cm / 250 cm (f/10)
Band	500–850 nm
Pixel pitch / QE	7.0 μm / 80 %
Read noise / FWC	15 e- RMS / 60,000 e-
Ground velocity / line period	6954.2 m/s / 0.2013 ms
GSD / Q	1.40 m / 0.964 (undersampled)
TDI mode / misalignment	analog / 0.1 pixel per stage
N_{TDI} sweep	1–128

Headline results. Per-line signal 732 e-. SNR climbs 23.7 ($N = 1$) $\rightarrow$ 107.2 ($N = 16$) $\rightarrow$ 216.0 ($N = 64$) and **plateaus at 244.5** once the signal clips at the 60,000 e- well — because there is no separable background photon term in an extended reflective scene, the noise caps at $\sqrt{FWC} \approx 245$ e- too, so SNR flattens rather than falling. Saturation onset is $N_{TDI} = 96$ (100 % fill); $N_{TDI} = 64$ runs at 78.1 %. NIIRS rises 4.62 $\rightarrow$ 6.21 over the same span (extrapolated GIQE-5 — the configuration is outside the calibration envelope). TDI misalignment MTF falls 0.9959 $\rightarrow$ 0.5508 from $N = 1$ to $N = 128$ as registration error accumulates as $\sqrt{N}$. A calibration variant re-runs the sweep with a one-point NUC and 2 % pre-correction PRNU: the correlated floor caps SNR near **56** by $N_{TDI} \approx 16$ –32, which **changes the design answer qualitatively** from “96 stages, SNR 245.” (*The walkthrough’s Physics Discussion quotes NIIRS 6.05 at $N_{TDI} = 64$ against the sweep table’s 6.12, and twice names $N_{TDI} = 64$ as the saturation onset the refreshed table puts at 96.*)

Regime. extended — a reflective ground scene fills the pixel, which is exactly why the saturation behavior is a plateau and not a cliff.

Takeaway. Thirty-two stages is the conservative sweet spot (NIIRS 5.89, 39 % well) — but only if the calibration residual is ignored. With a realistic one-point NUC the correlated floor, not the well, sets the answer, and stages past ~16–32 buy MTF loss instead of SNR.

Where to go deeper. [scenarios/01_sarah_systems_engineer/1.4_tdi_pushbroom_optimization/](#). A GUI baseline ships.

13.3.4 1.5 — Obscured Aperture and Spider Vanes

Mission setup. Sarah is trading a Cassegrain telescope: how much do the central obscuration and the struts supporting the secondary cost, against an ideal unobstructed aperture, and how does the cost grow with strut width? This is the first consumer of RADIANT’s spider-vane pupil masking (`optics.n_spiders`, `optics.spider_width_m`, `optics.spider_angle_deg`).

Key inputs.

Quantity	Value
Aperture / f-number	50 cm / f/12

Quantity	Value
Central obscuration	$\epsilon = 0.30$ (linear)
Spider arms	4, 3 cm wide (baseline); swept 0–5 cm
Band / altitude	VNIR pan, 500 km

Headline results. Three-configuration comparison: unobstructed SNR 75.1, EE_{3×3} 0.864, RER 0.582, MTF at Nyquist 0.228; obscured-only 71.5 / 0.763 / 0.517 / 0.207; obscured plus four 3 cm spiders **66.8 / 0.657 / 0.485 / 0.221**. Going from the ideal aperture to the full Cassegrain costs **11.1 % of SNR and 24 % of the 3×3 encircled energy**. The strut-width sweep at $\epsilon = 0.30$ takes SNR 71.5 → 63.5 and EE_{3×3} 0.763 → 0.607 across 0 → 5 cm; the 3 cm baseline costs ~14 % of the encircled energy against strut-free supports. **Strehl is 1.000 in every row** — it is a wavefront-error metric measured against a reference PSF carrying the same aperture geometry, so obscuration and vanes are common-mode and cancel exactly.

Regime. extended — a sunlit surface fills the pixel; the spatial metrics, not the regime, are the point here.

Takeaway. A designer judging this telescope by Strehl alone would miss the entire obscuration and vane cost. EE and RER are the monotone degradation signals; MTF at Nyquist is non-monotonic across the three apertures (0.228 → 0.207 → 0.221) because thin high-contrast struts redistribute rather than uniformly suppress the modulation.

Where to go deeper. scenarios/01_sarah_systems_engineer/1.5_obscured_aperture_spider_vanes/. A GUI baseline ships. (The walkthrough’s aperture-comparison table carries a duplicated header row — a Markdown artifact, not a second data set.)

13.3.5 1.6 — MWIR Point-Source Space Domain Awareness

Mission setup. A space-based MWIR sensor in a 700 km orbit must detect an **unresolved** satellite at 729 km slant range. The target’s angular extent is far below one pixel IFOV, so it lands inside a single PSF: there is no surface filling a pixel, and the right radiometric quantity is radiant intensity $I(\lambda)$ [W/sr/μm], not a surface radiance times an area.

Key inputs.

Quantity	Value
Aperture / f-number	0.30 m / f/5
Band / integration	3.5–5.0 μm / 10 ms
Pixel pitch	15 μm (InSb-class)
Target intensity	graybody: 290 K, 8 m ² emitting area, $\epsilon = 0.85$
Range	geometry.target_range_m = 729,287 m (explicit)

Headline results. Signal **20,939 e-**, **SNR 19.02**, **detection range (SNR = 6) 1302.2 km**, sampling Q at band center 1.42. (CU-380, 2026-09-30: declaring the warm train this telescope always had — two mirrors at $R = \sqrt{0.85}$, 280 K LEO bus-mounted class — left the signal unchanged at 20,939 e- by construction and took SNR 20.32 → 19.02, range 1,346.7 → 1,302.2 km.) Signal scales linearly with emitting area, emissivity and intensity, and inverse-square with range — the point-source camera equation. The blackbody point-intensity input reproduces an equivalent hand-built intensity CSV exactly.

Regime. `point_source`, locked by `source.regime_override`. The chain reads range from `geometry.target_range_m` explicitly rather than deriving it from altitude and zenith, which is why the scenario sets it.

Takeaway. For an unresolved object, declare intensity. Setting a surface temperature and emissivity with zero area does *not* define a point source — it raises an actionable error steering the user toward area, which is the wrong direction. The GUI does not yet expose the point-intensity inputs.

Where to go deeper. `scenarios/01_sarah_systems_engineer/1.6_mwir_point_source_sda/`. **No GUI baseline ships** for this scenario — which is also why it was missed by one physics-refresh sweep and carried a stale SNR for a month.

13.4 Persona 2 — Mike, Detector Engineer

Mike evaluates focal planes. His scenarios are noise budgets, ROIC architecture trades, and calibration floors — the questions where a vendor datasheet number and a fielded system’s behavior diverge.

13.4.1 2.2 — 1/f Noise Corner Frequency in an LWIR Staring Array

Mission setup. A 640×512 LWIR HgCdTe staring array with a measured flicker coefficient $K = 2.5 \times 10^4$ e⁻² and a corner frequency $f_c = 200$ Hz, operated at 30, 60 and 120 Hz. How much does 1/f noise cost in NEDT, and how should the flicker band limits be set at each frame rate?

Key inputs.

Quantity	Value
Optics	15 cm Ge objective, f/2, 85 % transmission, 20 °C
Band	8.0–10.0 μm
Detector	24 μm HgCdTe, QE 55 %, dark 5×10^5 e-/s, read 350 e- RMS
FWC / integration	20 Me- / 100 μs (well-limited)
Flicker band	$f_{low} = \text{frame rate}$, $f_{high} = 1/(2t_{int}) = 5000$ Hz
Atmosphere	exo — short range, negligible path

Headline results. $\sigma_{1/f} = \sqrt{K \ln(f_{high}/f_{low})}$ gives 357.6, 332.5 and 305.4 e- RMS at 30, 60 and 120 Hz — a 4× frame-rate change moves it only 15 %, because the dependence is logarithmic. NEDT with 1/f is **27.8 / 27.7 / 27.7 mK** against 27.4 mK without: a **0.3–0.4 mK, ~1 % penalty**. The 60 Hz noise budget reads `signal_shot` 2105.0 e- RMS (92.5 % of variance), quantization 352.2, read 350.0, flicker 332.5, dark 7.1, for an RSS total of 2188.2 e- RMS. Sweeping f_{low} from 1 Hz to 500 Hz moves NEDT only 28.00 → 27.55 mK. **RADIANT overestimates $\sigma_{1/f}$ by 64–170 %** because it integrates K/f across the full band instead of capping at f_c — capped values would be 217.8 / 173.5 / 113.0 e- RMS. That is an open gap, stated plainly in the walkthrough.

Regime. `extended` — the whole-FOV LWIR radiance field is one scene, so there is no separate `background_shot` term, and `signal_shot` carries it.

Takeaway. In a background-limited LWIR system $1/f$ noise is buried: the photon shot noise from a 293 K background swamps it. Mike does not need to negotiate the flicker spec. It would matter in read-noise-limited systems, below ~ 10 Hz frame rates, or with $K > 10^6 \text{ e}^-2$ — none of which apply here.

Where to go deeper. `scenarios/02_mike_detector_engineer/2.2_1f_noise_corner_frequency/`. A GUI baseline ships.

13.4.2 2.3 — IPC Characterization: How Much MTF Loss Is Tolerable?

Mission setup. Five HgCdTe MWIR samples from one wafer lot, each with a different inter-pixel capacitance — a parasitic coupling that leaks charge to the four nearest neighbors after photon conversion. Mike has lab MTF and ensquared-energy measurements on all five and needs to know how much IPC the system can tolerate.

Key inputs.

Quantity	Value
Optics	30 cm aperture, $f/4$, 72 % transmission
Detector	18 μm pitch, QE 72 %, dark 50 e^-/s
Geometry	500 km LEO, nadir; 310 K ground scene, 295 K background
IPC sweep	0–5 % per-neighbor coupling, 51 steps
Requirements	MTF at Nyquist ≥ 0.15 , EE $1 \times 1 \geq 0.60$, SNR ≥ 100

Headline results. Baseline (no IPC): MTF at Nyquist **0.2668**, SNR **782.84**, EE 1×1 **0.4141**, GSD 7.50 m, Q 0.944, NEDT 38.1 mK, NIIRS 4.82 (extrapolated), Strehl 1.000. **The binding constraint is EE 1×1 , and it fails at baseline** (0.4141 against a 0.60 requirement) before any IPC is applied; the MTF requirement is met comfortably across the whole 0–5 % range. RADIANT’s native IPC convolution now tracks the analytic form $MTF_{IPC}(f_{Nyq}) = 1 - 4\alpha$ to within rounding — at $\alpha = 5$ %, native 0.2139 against analytic 0.2134, Δ 0.0005. Against the five lab samples, the model predicts higher MTF at low IPC ($\Delta = +0.126$ at 1.2 %, $+0.092$ at 1.8 %) and slightly lower at the highest ($\Delta = -0.020$ at 3.5 %), the gap narrowing monotonically and crossing zero near 3 % coupling.

Regime. extended — the 310 K ground scene fills the pixel, so the background temperature enters only the contrast-SNR calculation.

Takeaway. Engineers evaluating IPC reach for MTF; here the tighter requirement is ensquared energy, and the vendor’s typical IPC specification does not meet it. **IPC does not affect SNR at all** — charge is redistributed, not destroyed — so the whole cost is spatial.

Where to go deeper. `scenarios/02_mike_detector_engineer/2.3_ipc_impact_on_mtf/`. A GUI baseline ships.

13.4.3 2.4 — Persistence Characterization: Bright-Source Recovery

Mission setup. After imaging a hot 800 K calibration source, how long does a Type-II superlattice detector’s persistence ghost linger, and what does it do to the scene that follows? First consumer of the multi-frame model `radiant.detector.persistence_sequence`.

Key inputs.

Quantity	Value
Prior bright exposure	150,000 e-
Current scene	20,000 e-
Residual fraction f (frame 1)	1.5 %
Trap time constant τ	50 ms
Frame period	16.67 ms (60 Hz)
Gain (1 LSB)	100 e-/DN

Headline results. Residual decays as $\text{residual}(n) = \text{prior} \cdot f \cdot e^{-(n-1)\Delta t/\tau}$: 2250 e- (22.5 LSB) in frame 1, 828 e- (8.3 LSB) at frame 4, 218 e- (2.2 LSB) at frame 8, 11 e- (0.1 LSB) at frame 17. **Frames to clear below 1 LSB: 11**, about 183 ms of dead time. Persistence *shot* noise is small — 47.4 e- RMS in frame 1 against 300 e- read noise — so scene SNR barely moves, 59.0 $\rightarrow$ 58.4 (−1 %).

Regime. No chain regime applies: this is a closed-form temporal sequence calculation with no atmosphere, optics or chain radiometry in the loop.

Takeaway. The bias is the problem, not the noise. A 22-LSB ghost image of the calibration source is false structure a detection algorithm will flag as real, and unlike random noise it does not average away — the only remedies are waiting out the eleven frames or subtracting a modeled ghost.

Where to go deeper. `scenarios/02_mike_detector_engineer/2.4_persistence_bright_source/`. No GUI baseline ships.

13.4.4 2.5 — Well Capacity Optimization: Integration Time vs Dynamic Range

Mission setup. A MWIR HgCdTe FPA (640×512, 15 μm pitch, 2 Me- well) in an f/2 ground-based surveillance system looks at a scene containing both 200 K cold sky and 1500 K jet exhaust. Mike needs one integration time that gives $\text{SNR} \geq 10$ on the cold target without saturating on the hot one.

Key inputs.

Quantity	Value
Aperture / focal length	20 cm / 40 cm (f/2)
Band	3.5–5.0 μm
Detector	QE 72 %, dark 100 e-/s, read 20 e- RMS, FWC 2.0 Me-
ADC / gain	14 bit / 130 e-/DN
Sweep	50 log-spaced t_{int} , 1 μs – 50 ms $\times$ 10 scene temperatures (200–1500 K)
Requirements	$\text{SNR} \geq 10$ at 200 K; ≤ 70 % fill on hot targets; 90 % absolute limit

Headline results. $\text{SNR} \geq 10$ on the 200 K target requires $t_{\text{int}} \geq 103.2 \mu\text{s}$ (565 e-, SNR 11.6). At that integration the hottest scene staying under 90 % well fill is **400 K**; at 1 ms it has fallen to 300 K, barely above ambient. **1000 K and 1500 K saturate even at 1 μs .** The 200–1500 K dynamic range is therefore physically impossible in a single integration — a 1500 K blackbody radiates $\sim 389\times$ more in-band power than a 200 K one. The 1 ms noise budget at 200 K reads `signal_shot` 74.0 e- RMS (75.2 % of variance), quantization 37.5 (19.3 %), read 20.0 (5.5 %), RSS 85.3 e- RMS; at 400 K `signal_shot` is 1414.2 e- RMS and 99.9 % of the variance.

Regime. extended — one radiance field, so no separate background_shot term. The walkthrough flags nearfield_shot = 0 as a known scalar-mode limitation: warm-optics self-emission is not modeled, which under-predicts cold-target noise.

Takeaway. The requirement is not achievable and the tool says so cleanly. The cold end is where ROIC choices show up — a quarter of the 200 K noise variance is quantization plus read noise — while at 400 K the electronics are irrelevant. This scenario's dead end is what motivates 2.6.

Where to go deeper. scenarios/02_mike_detector_engineer/2.5_well_capacity_optimization/. A GUI baseline ships.

13.4.5 2.6 — DROIC vs Analog ROIC: Single-Frame HDR

Mission setup. Scenario 2.5 ended at a physical wall. A vendor offers a Senseseeker-class digital-pixel ROIC for the same MWIR HgCdTe FPA: in-pixel 16-bit counters with 4.5 ke- charge-subtraction packets and analog residue readout. What does that buy at the 1 ms cold-target working point?

Key inputs.

Quantity	Analog	DROIC
Well / effective well	2.0 Me-	$2^{16} \times 4500 = 294.9 \text{ Me-}$
Dead-time ceiling (5 MHz $\times$ 1 ms $\times$ 4500 e-)	—	22.5 Me-
Governing bound at 1 ms	charge well	dead time
ADC	14 bit at 130 e-/DN	14-bit residue, 0.275 e-/DN
Read / counting-chain noise	20 e- RMS	20 e- RMS

Shared: 20 cm f/2 optics, 3.5–5.0 μm , QE 72 %, 15 μm pitch, 1.0 ms integration.

Headline results. At 400 K the analog chain **clips** (733.0 % nominal fill, SNR 1413.6 clipped, NEDT 34.56 mK on a clipped signal) while the DROIC delivers **SNR 3828.9 at 65.2 % fill and NEDT 12.76 mK** — a 2.7 $\times$ better NEDT where the analog part has stopped measuring. In the unsaturated 200–300 K overlap the two are equivalent: SNR agrees within 0.8 %, NEDT within 0.3 mK, with the DROIC's residue quantization floor (0.079 e- RMS) actually below the analog ADC's (37.5 e- RMS). Dynamic range at the matched 200 K point goes **76.3 dB $\rightarrow$ 97.4 dB (+21.1 dB)**. Above ~450 K the DROIC clips too, with saturation_mechanism = "dead_time".

Regime. extended — an unresolved-scene radiometric comparison; the interest is entirely in the readout architecture.

Takeaway. The dead-time ceiling, not the counter, is the DROIC's real limit — 22.5 Me- is 7.6 % of the 294.9 Me- counter capacity. The 2.5 requirement is still not met end to end, but the wall moved from 300 K to 400 K; covering 1500 K needs a faster comparator ($f_{max} \gtrsim 90 \text{ MHz}$), a larger packet, or up/down mode.

Where to go deeper. scenarios/02_mike_detector_engineer/2.6_droic_vs_analog_hdr/. No GUI baseline ships.

13.4.6 2.7 — Two-Point NUC Floor in an LWIR Starer

Two distinct scenario folders in persona 02 both carry the number 2.7. They are disambiguated here by folder name.

Mission setup. The datasheet NEDT of Mike’s LWIR staring camera (~25 mK) comes from the temporal noise budget. Every such system he has fielded is limited instead by the residual fixed-pattern noise the two-point NUC leaves behind. The calibration model lets him model the calibration process itself — cal-point placement, detector nonlinearity, correction decay, and the blackbody source’s own uncertainty.

Key inputs.

Quantity	Value
Optics / band	10 cm f/1.5, 8–12 μm , 85 % transmission
Detector	12 μm pitch, QE 65 %, dark 5×10^5 e-/s at 60 K, read 350 e- RMS, 20 Me- well
Integration	120 μs (~50 % fill at 300 K)
Scheme / cal points	two-point NUC at 290 / 310 K
Nonlinearity dispersion (1σ)	1.0 % of full scale
Pre-cal PRNU / DSNU (1σ)	2.0 % / 300 e-
Drift	gain 0.005 %/hour; offset 720 e-/hour
Cal source	$\epsilon = 0.98$, $\Delta T(1\sigma) = 0.5$ K, $\Delta \epsilon(1\sigma) = 0.005$

Headline results. The achieved NEDT is **exactly the temporal floor at the two cal points** (26.13 mK at 290 K, 25.06 mK at 310 K, cal floor 0.00 mK) and degrades parabolically outside the span: 30.73 mK at 280 K, 28.84 at 320 K, 46.97 at 330 K, **79.86 mK at 340 K** — $3.3\times$ the datasheet number. Uncalibrated (raw 2 % PRNU) the same sweep reads 1086–1570 mK. Drift at 340 K takes NEDT $79.86 \rightarrow 88.90 \rightarrow 175.47 \rightarrow 475.49$ mK at 0, 6, 24 and 72 hours since calibration. Separately, at 300 K the cal-source uncertainty produces a **bias** of 0.955 % of radiance, or **591.5 mK at scene temperature**, reported beside the 26.02 mK precision figure and never RSS’d into it.

Regime. extended LWIR imaging, with `detector.noise_regime = "imaging"`. The calibration residual is added *after* readout scaling, so correlated errors do not average down.

Takeaway. Cal-point placement is now a design trade the tool can answer, and a temperature-retrieval product carries two numbers, not one: a ± 26 mK spread around a possible 0.59 K offset. The old “FPN calibrated away” assumption flatters the design everywhere outside the cal span.

Where to go deeper. `scenarios/02_mike_detector_engineer/2.7_calibration_limited_nedt/`. **No GUI baseline ships**, though a `gui_workflow.md` walks the same study through the Calibration screen.

13.4.7 2.8 — Real-Part Quick Start: GeoSnap-18 by Name

Mission setup. Every prior scenario began with Mike transcribing datasheet values by hand. For a MWIR airborne study he wants the part he is actually being offered — a Teledyne GeoSnap-18 — modeled from the FPA preset library that ships with RADIANT: one `fpa:` line, every value carrying its datasheet citation.

Key inputs.

Input	Value	Who supplies it
FPA part	geosnap - 18 (12 parameters)	library preset
Target temperature	300 K	Mike
Sensor altitude	8000 m	Mike
Aperture / focal length	0.30 m / 1.20 m (f/4)	Mike

Input	Value	Who supplies it
Band / integration	3.5–5.0 μm / 5 ms	Mike
Dark rate	5.0×10^4 e-/s	Mike (explicit, by design)

The preset supplies 18 μm pitch, 2048 cross-track pixels, 100 % fill factor, QE 85 %, 110 K detector temperature, analog_well architecture, a 2.6 Me- well, 400 e- RMS ROIC noise, a 14-bit ADC at a derived full-scale-matched 158.7 e-/DN, and an 85 Hz frame period.

Headline results. SNR 1177.9, readout architecture analog_well. The 400 e- RMS preset entry is the vendor’s ROIC-only noise for the large well; the measured 13.2 μm science device shows 360 e- RMS *system* read noise (Bowens et al. 2024), so the preset is honest to within 10 %. Quantization at the matched gain is $158.7/\sqrt{12} \approx 45.8$ e- RMS, small against 400 e- read noise, and the full well digitizes warning-free. *The digest has only this one headline metric because the walkthrough reports only one* — no NEDT or spatial metric is quoted.

Regime. extended — a 300 K scene fills the pixel and the chain is background-shot dominated; the binding capacity is the 2.6 Me- charge well.

Takeaway. The preset library is a provenance mechanism as much as a convenience: every value traces fpa : geosnap-18/<source>, the cited datasheet PDFs are kept in the repository’s validation tree with SHA-256 manifest rows, and FPALibrary refuses any preset whose attribution is incomplete. The deliberately-absent dark current is the model being honest — dark is a function of cutoff and temperature that this custom-cutoff part’s datasheet does not fix.

Where to go deeper. scenarios/02_mike_detector_engineer/2.8_fpa_part_library/. **No GUI baseline ships.**

13.4.8 2.9 — Up/Down Counting: In-Pixel Pedestal Removal

The second folder numbered 2.7; see the note above.

Mission setup. Mike’s DROIC vendor offers an up/down counting mode: the in-pixel counter increments during the scene phase and decrements during a reference phase, subtracting the background pedestal before readout. The driving case is a dim point-source target over a bright common background, where plain up-counting spends the counter range on the pedestal. What does up/down buy, and what does it cost?

Key inputs.

Quantity	Value
Aperture / focal length	15 cm / 30 cm (f/2)
Band	3.5–5.0 μm
Target	500 K, $\epsilon = 0.9$, 0.001 m ² at 10 km
Background (swept)	250–330 K, $\epsilon = 0.95$
Platform / atmosphere	8 km altitude, mid-latitude summer
Integration (up = down phase)	50 ms
Counter	14 bit $\times$ 2000 e-/count

Quantity	Value
up bound / up_down signed bound	32.77 Me- / 16.38 Me-
Counting-chain noise per phase	5 e- RMS

Headline results. Target charge over the up phase is ≈ 4.86 Me- (29.6 % of signed capacity); the pedestal grows 15.9 Me- (250 K) $\rightarrow$ 88.9 Me- (330 K). Plain up counting saturates at ~ 290 K background (113.6 % fill, usable SNR collapsing to 67.6 and then 0.0); **up_down fill is background-independent at 29.6 % across the whole sweep**, with SNR 802.2 $\rightarrow$ 359.2 over 250 $\rightarrow$ 330 K. At 250 K, where both modes are clean, the up/down penalty is visible: **802.2 vs 1066.0, a ratio of 0.75** — between 1 and $1/\sqrt{2} = 0.707$, because the target's own shot noise (2203 e- RMS) does not double, only the background terms do (reference_shot = 3984 e- RMS at 250 K). At 290 K the comparison inverts to an 8.6 $\times$ usable-SNR advantage.

Regime. A dim target over a bright common background, with the in-pixel background term explicitly retained — the noise budget carries reference_shot, a $\times\sqrt{2}$ counting read, and packet_reset over both phases.

Takeaway. Up/down moves the wall from the pedestal to the differential, at an honest $\sqrt{2}$ -class reference penalty. Any model that cancels the mean without paying the reference noise overstates the mode by up to $\sqrt{2}$ — which is exactly why the noise budget names both phases.

Where to go deeper. scenarios/02_mike_detector_engineer/2.9_updown_background_subtraction/. No GUI baseline ships.

13.5 Persona 3 — Raj, Mission Planner

Raj turns sensor performance into collection decisions. His scenarios ask where the go/no-go line sits — in weather, in look angle, in vendor selection — and they repeatedly find that the intuitive driver is not the binding one.

13.5.1 3.2 — Weather Sensitivity: How Bad Can It Get?

Mission setup. A baselined MWIR reconnaissance sensor on a 500 km sun-synchronous orbit needs a weather go/no-go threshold. At what visibility does performance drop below the NIIRS ≥ 4.0 requirement, and how much does precipitable water vapor cost?

Key inputs.

Quantity	Value
Aperture / focal length	30 cm / 120 cm (f/4)
Band / target	3.5–5.0 μm ; 300 K target, 290 K background
Detector	18 μm pitch, QE 70 %, dark 150 e-/s, read 18 e- RMS, 500 ke- well
Integration / GSD	1.0 ms / 7.5 m
Visibility sweep	2–100 km (25 points) at PWV 1.4 cm
PWV sweep	0.5–5.0 cm (15 points) at visibility 23 km
Requirement / goal	NIIRS ≥ 4.0 / ≥ 5.0

Headline results. Across the **entire** 2–100 km visibility range, band-mean τ moves $0.4738 \rightarrow 0.5580$ and NIIRS moves **4.47 $\rightarrow$ 4.49**. Across the full 0.5–5.0 cm PWV range, τ moves $0.5932 \rightarrow 0.4398$ and NIIRS falls just **0.04** (SNR $508.7 \rightarrow 481.3$). All eight named weather conditions from Arctic dry to heavy haze are **GO**; the whole 36-cell visibility $\times$ PWV grid spans 4.45–4.50 NIIRS. **No condition meets the NIIRS ≥ 5.0 goal** — the best case reaches 4.50. The baseline noise budget is 99.8 % signal_shot (502.0 e- RMS of a 502.4 e- RMS total, signal 252,041 e-). A real-MODTRAN cross-check validates both axes: measured visibility response $23 \rightarrow 5$ km gives τ $0.555 \rightarrow 0.511$ against the model's $0.552 \rightarrow 0.526$, and the PWV slope matches at -0.038 vs -0.039 per cm.

Regime. extended — one MWIR radiance field, so signal_shot carries the whole scene.

Takeaway. GSD, not weather, caps NIIRS. The GIQE-5 GSD term carries a -3.32 coefficient against SNR's $+1.559$, and at 7.5 m GSD the penalty is -8.20 before SNR is considered; reaching NIIRS 5.0 needs $\text{GSD} \leq \sim 5.4$ m, i.e. a ~ 1.7 m focal length rather than 1.2 m. MWIR is robust to aerosol because extinction scales as $\lambda^{-1.3}$, making $4.2 \mu\text{m}$ scattering $\sim 13\times$ weaker than at $0.55 \mu\text{m}$, and robust to water because the MWIR H_2O bands are already saturated.

Where to go deeper. scenarios/03_raj_mission_planner/3.2_weather_sensitivity/. A GUI baseline ships.

13.5.2 3.3 — Multi-Sensor Comparison for Procurement

Mission setup. Three vendors have bid MWIR sensors. At a common operating point — 600 km, a 300 K extended scene, 8 ms — how do they compare on SNR, NIIRS, NEDT, MTF and GSD; which meet the requirements; and where should each vendor invest to gain the most interpretability? A composition scenario: the chain plus giqe5_sensitivity, no new model.

Key inputs.

Quantity	Value
Source	three vendor spec columns transcribed to a common workbook (PDF spec-sheet parsing is out of scope)
Vendor A	3.7–4.8 μm , balanced
Vendor B	3.0–5.0 μm , fast f/3, large pixel, 77 K
Vendor C	3.7–4.8 μm , 10 μm pixel behind f/5, warmer and lower-QE
Common operating point	600 km, 300 K extended scene, 8 ms
Requirements	SNR ≥ 50 , NIIRS ≥ 4.0 , NEDT ≤ 50 mK, GSD ≤ 1.5 m, MTF at Nyquist ≥ 0.25

Headline results.

Metric	Vendor A	Vendor B	Vendor C
SNR	1160	2449	491
NIIRS	4.83	4.54	4.89
NEDT [mK]	23.7	11.4	56.0
GSD [m]	9.0	19.2	3.4
MTF at Nyquist	0.27	0.43	0.00

Compliance: A 4/5, B 4/5, C 2/5. **No proposal is fully compliant, and the binding failure is GSD** — none reaches 1.5 m from 600 km at the proposed f-numbers, which would need $\approx f/30$ or more at these pitches. Vendor C’s MTF at Nyquist of 0.00 is correct, not a bug: a 10 μm pixel behind f/5 MWIR optics puts Nyquist far above the diffraction cutoff ($Q \gg 2$). A +10 % improvement in GSD or RER is worth $\approx +0.14$ NIIRS for every vendor, against only +0.07 for SNR.

Regime. extended — the large absolute SNRs are whole-scene 300 K MWIR values, appropriate for a relative comparison and not as detection thresholds.

Takeaway. The actionable procurement finding is that **the GSD requirement is infeasible as written**: relax it, lower the orbit, or ask for longer focal lengths. And because NIIRS is logarithmic in SNR but strongly responsive to resolution, vendor investment belongs in the optics, not the detector.

Where to go deeper. `scenarios/03_raj_mission_planner/3.3_multi_sensor_comparison/`. A GUI baseline ships.

13.5.3 3.4 — Off-Nadir Performance Degradation

Mission setup. A VNIR panchromatic pushbroom on a 600 km sun-synchronous orbit. “Can I still get useful imagery at 45° off-nadir?” The trade is image quality against access area: how much NIIRS and GSD are surrendered to reach a target 527 km from the nadir track.

Key inputs.

Quantity	Value
Aperture / focal length	35 cm / 350 cm (f/10) TMA
Obscuration / WFE	20 % / 0.05 waves RMS at 633 nm
Band / pitch / QE	450–900 nm / 8 μm / 80 %
Integration / altitude	0.5 ms / 600 km, 10:30 LTAN, solar zenith 30°
Nadir GSD / Q	1.37 m / 0.844 (slightly undersampled)
Sweep	path zenith 0–45° in 5° steps

Headline results. Over $0 \rightarrow 45^\circ$: slant range 600.0 $\rightarrow$ 814.8 km, air mass 1.0000 $\rightarrow$ 1.3580, ground range $0 \rightarrow$ **527.2 km**, band-mean τ 0.6594 $\rightarrow$ 0.5581 (–15.9 %). Cross-track GSD grows 1.37 $\rightarrow$ **1.86 m (+36 %)** while along-track grows 1.37 $\rightarrow$ **2.63 m (+92 %)** — the ground sample becomes rectangular. **SNR rises, 59.8 $\rightarrow$ 71.6 (+19.8 %)**, because the growing path radiance adds more photons than the transmission loss removes — the atmospheric-veiling effect. NIIRS falls 5.43 $\rightarrow$ 4.86, a **–0.57 penalty**, smaller than the –0.69 the geometric-mean GSD term alone would give because the rising SNR partly offsets it. Nadir diagnostics: NEDT 64.0 mK, Strehl 0.9065, RER 0.5372, EE 1×1 0.3634, folded MTF at Nyquist 0.4544 with a 0.5000 alias fraction. RADIANT’s `gsd_cross_track_m` and `gsd_along_track_m` now reproduce the scenario’s independent spherical-Earth calculation on **both** axes to double-precision round-off (worst residual 1.4×10^{-13} %). (*The walkthrough’s “Access vs Quality Trade” table, gap-closure list and one refresh note carry an older vintage — NIIRS 5.35 / 5.14 / 4.78, well margin 27.6 dB, a –0.90 penalty at 45° — against the refreshed sweep table quoted here.*)

Regime. extended — one reflective radiance field, hence no separate `background_shot` term.

Takeaway. SNR going *up* off-nadir is real but misleading for detection work: *contrast* SNR falls, because veiling washes out the target-background difference. The honest cost of a 527 km access radius is -0.57 NIIRS, driven almost entirely by the along/cross GSD asymmetry.

Where to go deeper. `scenarios/03_raj_mission_planner/3.4_off_nadir_agility/`. A GUI baseline ships.

13.5.4 3.5 — Nighttime MWIR Imaging Feasibility

Mission setup. Can an airborne sensor image a warm building complex (295 K) against terrain (288 K) **at night**, and how does MWIR compare with LWIR for a 7 K thermal scene? The scenario exercises the first-class extended contrast reference, the NEDT and MRT-at-Nyquist metrics, and an analytic solar-versus-thermal comparison.

Key inputs.

Quantity	Value
Scene	295 K target against 288 K terrain ($\Delta T = 7$ K)
Platform	3 km AGL airborne, tropical atmosphere
Water column	4.1 cm precipitable water — set explicitly , not implied by the profile
Bands	MWIR 3.5–5.0 μm and LWIR 8–12 μm
Optics / pitch	f/4, 30 μm pixels
Integration / well	0.2 ms / 1×10^7 e $^-$ — sized so neither band saturates
Terrain envelope	NOAA LST strip, 287.6–288.6 K

Headline results. MWIR: SNR 375.7, contrast SNR 38.1, NEDT 72.5 mK ($\Delta T/\text{NEDT} = 97\times$), MRT at Nyquist 0.399 K. LWIR: SNR 2773.3, contrast SNR 101.0, NEDT 21.6 mK ($\Delta T/\text{NEDT} = 324\times$), MRT 0.350 K. **Both bands detect the 7 K contrast with wide margin**; LWIR wins on every figure, as the ~ 10 μm Planck peak of a 290 K scene implies. On solar independence, band-integrated thermal against a daytime reflected-solar upper bound: MWIR 1.379 vs 0.305 W/m 2 /sr ($\times 5$), LWIR 32.60 vs 0.0331 W/m 2 /sr ($\times 986$). Across the terrain envelope MWIR contrast SNR stays ≥ 34 (34.6 at the hottest background, 40.4 at the coolest) and LWIR spans 89.3–108.8.

Regime. extended thermal self-emission, with `source.contrast_reference` set to the 288 K terrain so `contrast_snr` is a true two-pixel differential with combined target-plus-reference noise.

Takeaway. Verdict: yes. And the night case is the *stronger* one for MWIR: by day the $\times 5$ thermal-to-solar margin means roughly 20 % reflected-solar contamination — the glint problem — while at night that term is exactly zero. A real-MODTRAN check notes the parametric model is $2.2\times$ too absorbing in MWIR and $0.81\times$ too transparent in LWIR on this tropical column, both pushing the same way, so the quoted MWIR-versus-LWIR margin overstates LWIR's advantage.

Where to go deeper. `scenarios/03_raj_mission_planner/3.5_nighttime_mwir_feasibility/`. A GUI baseline ships.

13.6 Persona 4 — Lisa, Detection/Targeting Analyst

Lisa produces collection guidance. Her scenarios ask at what range, at what hour, and under what counter-measure a target is actually findable — and two of the four turn on contrast rather than sensitivity.

13.6.1 4.2 — Maritime Ship Classification (Johnson DRI)

Mission setup. With an airborne MWIR sensor on a 5 km UAV, at what range can each ship class be Detected, Recognized and Identified? First consumer of `radiant.performance.johnson_criteria`, which turns the Johnson resolved-cycle criteria (1 / 4 / 6.4 cycles across the target) into ranges.

Key inputs.

Quantity	Value
IFOV	12.5 μ rad (15 μ m pitch / 1.2 m focal length)
Platform altitude	5 km
Critical dimension	$\sqrt{L \cdot H}$, the 2-D-target convention
N50 cycles	1 (detection), 4 (recognition), 6.4 (identification)
Geometric horizon	$\sqrt{2R_E h} = 252$ km

Headline results. Identification resolution range by class: small boat (3.5 m critical dimension) **22 km**, patrol craft (13.4 m) **84 km**, corvette (30.0 m) **188 km**, frigate (44.2 m) 276 km, destroyer (52.8 m) 330 km, container ship (94.9 m) 593 km. The binding limit is resolution for the first three and the **252 km horizon** for the last three. The small boat is resolution-limited at every task — detection 139 km, recognition 35 km, identification 22 km. For detection, every class except the small boat is horizon-limited.

Regime. No chain run at all: DRI ranges are pure geometry (target size, IFOV, N50). The signal chain would enter only through a contrast/MTF extension.

Takeaway. The fleet splits at the frigate: above ~44 m critical dimension, line-of-sight is the wall and resolution is ample; below it, closing range is the only option. Small, fast craft are the hard maritime ISR problem, and this quantifies why. The model is contrast-blind and therefore the optimistic bound — a low-contrast target at dusk identifies at shorter range, and reliable (>90 %) identification needs ~1.5× the N50 cycles, shrinking every range by about a third.

Where to go deeper. `scenarios/04_lisa_analyst/4.2_maritime_ship_classification/`. **No GUI baseline ships.**

13.6.2 4.3 — Camouflage Effectiveness Analysis

Mission setup. Lisa evaluates three thermal camouflage nets against an airborne LWIR FLIR at 3 km. A hot vehicle (engine deck ~380 K, oxidized steel) sits in scrub (305 K, $\epsilon = 0.96$); the nets drape it and re-emit at their own near-ambient 310 K. Their emissivity spectra arrive in three different vendor forms — a dense measured CSV, a spectrally shaped curve, and a three-point quote sheet.

Key inputs.

Quantity	Value
Sensor / band	airborne LWIR FLIR at 3 km, 8–12 μm
Bare vehicle	oxidized steel, ASTER library, $\epsilon \approx 0.80$
Net A	broadband metalized weave, mean $\epsilon \approx 0.60$
Net B	spectrally shaped — low 8–10 μm , high 10–12 μm
Net C	$\epsilon \approx 0.93$, given at only three wavelengths (8.0 / 10.5 / 14.0 μm)
Net surface temperature	310 K

Headline results.

Option	Contrast [e-]	SCNR	Well fill [%]	Signature reduction
Bare vehicle	+1,851,019	1283.5	32.3	—
Net A ($\epsilon \approx 0.60$)	−509,543	353.3	12.6	72.5 %
Net B (shaped)	−273,017	189.3	14.6	85.3 %
Net C ($\epsilon \approx 0.93$)	+82,721	57.4	17.6	95.5 %

Sub-band SCNR shows Net B’s shaping directly: 232.1 in 8–10 μm against 41.5 in 10–12 μm , so a half-band sensor would rank the nets differently than the full FLIR. Detection range is **edge-limited at 17.1 km for every option** — the 80° zenith sweep cap, not SCNR.

Regime. extended — a draped net fills the pixel. Because the extended regime carries no separate scene-background photon term, the differential is formed explicitly from two runs: $SCNR = |S_{option} - S_{scrub}| / \text{noise}_{scrub}$. Spectral emissivity enters through the tabulated-radiance path (source.target.user_radiance_path) because the chain has no spectral target-emissivity input of its own.

Takeaway. Camouflage is radiance matching, not emission lowering. The intuitive low- ϵ choice over-corrects: Net A at $\epsilon = 0.60$ reads distinctly *cold* (−510 k e-), a large negative contrast an *|contrast|* detector sees just as well as a hot one. Net C, whose high flat emissivity sits near the scrub’s 0.96, cuts the signature 95.5 %. And no net defeats detection at 3 km — they reduce signature, which is the operational metric.

Where to go deeper. scenarios/04_lisa_analyst/4.3_camouflage_effectiveness/. A GUI base-line ships.

13.6.3 4.4 — Time-of-Day (Diurnal) Thermal Detectability

Mission setup. Over a 24-hour cycle, when is a painted-metal vehicle detectable against its soil background in the LWIR, and when does it wash out? The physics is thermal crossover: twice a day the two surfaces radiate equally, contrast collapses, and the target vanishes regardless of detector sensitivity. The diurnal temperature profile is input data; no new framework model is involved.

Key inputs.

Quantity	Value
Sensor / band / altitude	LWIR, 8–12 μm , 3 km AGL
Target	painted metal, $\epsilon = 0.92$, low thermal inertia (large early swing)
Background	soil, $\epsilon = 0.95$, higher inertia (smaller lagged swing)

Quantity	Value
Profile	measured 24-hour CSV, sampled every 0.5 h
Integration time	0.1 ms — 8 ms saturates the well (>99 %) and destroys the contrast
Detectability threshold	contrast SNR ≥ 10

Headline results. Contrast SNR runs −98.3 at 00:00 ($\Delta T = -4.06$ K, cold target), +18.5 at 06:00, **+140.6 at 12:00** ($\Delta T = +10.06$ K), +17.8 at 18:00, −66.0 at 21:00. Physical-temperature crossovers ($\Delta T = 0$) fall at **04:12 and 19:48**; **radiance** crossovers (contrast = 0) at **05:12 and 18:48** — offset by about an hour. Washout windows (|contrast SNR| < 10) are $\approx 05:30$ – $06:00$ and $\approx 18:30$ – $19:00$, roughly 30 minutes each. **Median NEDT across the day is 36.7 mK, essentially constant.**

Regime. extended for each pixel run; the differential is built at the scenario level as $(S_t - S_b) / \sqrt{N_t^2 + N_b^2}$, because the chain's own contrast_e is populated only in the sub-pixel regime.

Takeaway. **The washout is a scene effect, not a sensor effect** — the detector is exactly as sensitive at crossover as at noon; there is simply nothing to detect. And because the target is *less* emissive than the background, it must run a few kelvin warmer to match it, so the radiance crossover is offset from the temperature crossover by ~ 1 hour. An analyst planning around the temperature crossover mis-times the washout by that hour. A real-MODTRAN check moves the crossings by only $\approx \pm 10$ minutes.

Where to go deeper. scenarios/04_lisa_analyst/4.4_time_of_day_analysis/. A GUI baseline ships.

13.6.4 4.5 — Microbolometer UAV Altitude Trade (NETD-Specified)

Mission setup. An uncooled microbolometer is specified only by its NETD — the way uncooled detectors are actually quoted. How high can the UAV fly and still detect a small warm ground target? Second consumer of the $D^*/NEP/NETD$ converter set (radiant.performance.detectivity, .nep_electrons, .nep_netd).

Key inputs.

Quantity	Value
Vendor NETD	50 mK
Optics	f/1, 486 μ rad IFOV
Target	1 m, $\Delta T = 4$ K against background
Frame time	16 ms (the bolometer thermal time constant)
Detection floor	$4 \times \text{NETD} = 200$ mK (recognition-grade)
Altitude sweep	1–11 km

Headline results. Converter outputs: $dP/dT = 1.515 \times 10^{-10}$ W/K, $NEP = 7.576 \times 10^{-12}$ W, $D^* = 1.254 \times 10^9$ Jones — the textbook order of magnitude for an uncooled microbolometer, about 100× below a cooled photon detector. The altitude trade: at 1 km, GSD 0.49 m, fill fraction 1.000, τ 0.921, apparent ΔT 3682 mK; at 5 km, 2.43 m / 0.170 / 0.748 / 508 mK; at 7 km, 3.40 m / 0.087 / 0.715 / **248 mK (detect)**; at 9 km, 4.37 m / 0.052 / 0.695 / **145 mK (no detect)**. The target goes sub-pixel at **2.1 km**, and the **detection ceiling is 7.5 km** — 1 km lower than this walkthrough previously reported, because the refreshed atmosphere

is more absorbing on long slant paths. Across 1–11 km the fill fraction falls $\times 28.6$ while τ falls only $\times 1.34$, so dilution outweighs attenuation about 21:1 in the $38\times$ total contrast collapse.

Regime. Sub-pixel dilution applied as an apparent-contrast threshold ($f \cdot \Delta T \cdot \tau$ against $4\times$ NETD), not a chain contrast_snr. **A false saturation warning fires:** the chain checks a photoelectron count against a charge well, but a bolometer measures resistance change over its thermal frame. The SNR the baseline reports is a photon-FPA quantity with no bolometric meaning; the scenario's actual metric is independent of the well and ADC path.

Takeaway. The ceiling is set by resolution, not sky clarity, so the fix is a longer focal length or a closer approach. An operator flying to the previously-published 8.5 km ceiling would have been 1 km above the detection limit.

Where to go deeper. scenarios/04_lisa_analyst/4.5_altitude_trade_uav/. A GUI baseline ships.

13.7 Persona 5 — Tom, Optical Designer

Tom owns the optics and the spatial budget. His scenarios are sampling, chromaticism, line-of-sight stability and stray-light trades, and each one finds that the metric a designer reaches for first — MTF at Nyquist, veiling glare, Strehl — is not the one that binds.

13.7.1 5.2 — Pixel Pitch and the Sampling Parameter Q

Mission setup. The optics are fixed: a 30 cm f/4 Ritchey-Chrétien working 3.5–5.0 μm from a 500 km orbit. Six vendors offer detectors from 8 to 30 μm pitch, each pitch carrying its own well, dark current, QE and read noise — small pixels quiet and shallow, large ones deep and noisy. Which pitch balances resolution against sensitivity while meeting every requirement?

Key inputs.

Quantity	Value
Aperture / f-number	30 cm / f/4
Band / geometry	3.5–5.0 μm / 500 km LEO, nadir
Airy disk at 4.25 μm	41.6 μm ($2.44 \lambda f / \#$)
Pitch candidates	8, 12, 15, 18, 24, 30 μm , each with matched vendor specs
Matched-spec span	QE 70 $\rightarrow$ 68 %, dark 15 $\rightarrow$ 120 e-/s, FWC 40 ke- $\rightarrow$ 2.5 Me-, read 8 $\rightarrow$ 28 e- RMS
Requirements	GSD < 10 m, MTF at Nyquist $\geq$ 0.10, EE $1\times 1 \geq$ 0.30, SNR $\geq$ 100

Headline results. Across 8 $\rightarrow$ 30 μm : $Q = \lambda (f/\#) / p$ runs 2.12 $\rightarrow$ 0.57, GSD 3.3 $\rightarrow$ 12.5 m, SNR 199.8 $\rightarrow$ 1268.8, MTF at Nyquist 0.000 $\rightarrow$ 0.409, EE 1×1 0.146 $\rightarrow$ 0.572, NEDT 149.7 $\rightarrow$ 23.6 mK. **Two candidates pass everything — 15 μm and 18 μm —** and an SNR/GSD figure of merit picks **18 μm** (94.3 against 15 μm 's 80.0) at $Q = 0.94$, SNR 706.9, GSD 7.5 m, NEDT 42.3 mK. The 8 μm pixel fails both spatial requirements; 12 μm now fails EE 1×1 at 0.269 against the 0.30 floor, having passed before the cell-area-overlap EE_box refinement lowered every EE 1×1 by 11–13 %; 24 and 30 μm fail GSD. **MTF at Nyquist of exactly 0.000 at 8 μm is physical, not a defect** — Nyquist at 62.5 cy/mm sits above the ~ 59 cy/mm diffraction cutoff, so there is genuinely no modulation there to measure.

Regime. extended — the 310 K ground scene fills every candidate pixel, so the trade is collection area against sampling with no EE_box coupling.

Takeaway. Signal scales as p^2 , so oversampling is expensive: $18 \rightarrow 8 \mu\text{m}$ throws away $5\times$ the signal to buy resolution the $f/4$ optics cannot deliver. The binding *upper* limit on pitch is GSD and the binding *lower* one is ensquared energy — MTF at Nyquist, the metric a designer reaches for first, is comfortable across the entire compliant range. And each pitch must carry its own vendor specs: sweeping pitch alone would give the wrong answer.

Where to go deeper. scenarios/05_tom_optical_designer/5.2_pixel_pitch_optimization/. A GUI baseline ships. (The refresh notes call this “an 8–12 μm scene” where the band is 3.5–5.0 μm — boil-erplate carried from an LWIR sibling; the movement they quote is this scenario’s own.)

13.7.2 5.3 — Monochromatic vs Polychromatic PSF

Mission setup. With the 18 μm pitch chosen in 5.2, is the spatial analysis that chose it even accurate? Diffraction scales linearly with wavelength, so across a 3.5–5.0 μm band — a 43 % span — the Airy diameter grows $34.2 \rightarrow 48.8 \mu\text{m}$. Does a monochromatic PSF at band center give the right spatial metrics, or is a flux-weighted polychromatic PSF required?

Key inputs.

Quantity	Value
Optics / pitch	$f/4$, 18 μm (the 5.2 selection)
Band	3.5–5.0 μm
Per- λ probes	3.5, 4.0, 4.25, 4.5, 5.0 μm in ± 50 nm narrow bands
PSF model	optics.psf_n_wavelengths = 1 (mono), 5, 11, 21
Weighting	in-band photon spectral flux of the 300 K scene

Headline results. The per-wavelength ladder is monotone in both directions: MTF at Nyquist $0.328 \rightarrow 0.210$, EE 1×1 $0.478 \rightarrow 0.358$, FWHM $19.5 \rightarrow 23.6 \mu\text{m}$ and SNR $88 \rightarrow 244$ across $3.5 \rightarrow 5.0 \mu\text{m}$ — tighter PSF and fewer photons at the short end. Full-band, mono ($N=1$) gives MTF at Nyquist 0.267, EE 1×1 0.414, EE 3×3 0.878, RER 0.610, FWHM 21.3 μm ; poly ($N=11$) gives 0.248, 0.396, 0.871, 0.596, 21.1 μm . The chromaticism error is therefore **+7.6 % on MTF at Nyquist and +4.7 % on EE 1×1** , +2.3 % on RER, and negligible on EE 3×3 (+0.8 %), FWHM (+0.8 %) and SNR (+0.0 %). $N=11$ converges to within 0.3 % of $N=21$. The 18 μm pixel still passes every 5.2 requirement under polychromatic analysis, with EE 1×1 clearing its floor by 31 % rather than 47 %. (*Takeaway 6 of the walkthrough quotes the poly values as 0.246/0.394 where its own table reads 0.248/0.396, and calls the EE 3×3 error 0.9 % against the table’s 0.8 % — stale prose against a refreshed table.*)

Regime. extended. SNR is identical across every PSF model, because in the extended regime SNR depends on total signal and noise, not PSF shape.

Takeaway. **Monochromatic analysis always overstates, never understates**, and by a knowable amount: band center is sharper than the flux-weighted average, and a thermal source’s in-band photon flux is biased toward the long-wavelength end. The error concentrates entirely in the single-pixel metrics — EE 1×1 and MTF at Nyquist — which is exactly where point-source detection sensitivity lives.

Where to go deeper. `scenarios/05_tom_optical_designer/5.3_mono_vs_poly_psf/`. A GUI baseline ships.

13.7.3 5.4 — Jitter Tolerance: Deriving the Pointing Requirement

Mission setup. A 50 cm $f/10$ VNIR panchromatic imager on a 500 km sun-synchronous orbit, GSD 0.80 m. The 5.0 m focal length that buys the fine GSD also amplifies angular jitter onto the focal plane: how much line-of-sight wander can the spacecraft allow? The sweep is 51 *full-chain* evaluations, not an analytic approximation — the jitter kernel convolves into the EffectivePSF and PerformanceStage reads RER and NIIRS back off it.

Key inputs.

Quantity	Value
Aperture / focal length	50 cm / 500 cm ($f/10$)
Obscuration / WFE	30 % / 0.05 waves RMS
Band / pitch / QE	450–700 nm / 8 μm / 85 %
Integration / FWC	0.5 ms / 100,000 e-
GSD / Q / IFOV	0.80 m / 0.72 (undersampled) / 1.6 μrad
Jitter sweep	0–5.0 μrad , 51 points
Thresholds	$\Delta\text{NIIRS} = -0.5$, $\Delta\text{NIIRS} = -1.0$, absolute NIIRS = 6.0

Headline results. Baseline signal 2109 e- (2.1 % well), noise 46.2 e- RMS of which `signal_shot` is 98.6 %, and — from the refreshed sweep — zero-jitter MTF at Nyquist 0.2186, RER 0.5724, NIIRS 6.03. **SNR is exactly 45.6 at all 51 sweep points, spread 0.0000:** jitter blurs the image without removing a photon, so NIIRS degrades entirely through the RER term. Focal-plane blur is $\sigma_{fp} = \text{jitter} \times f$, so 1 μrad is 5 μm = 0.625 pixels. Jitter MTF at Nyquist falls 1.0000 $\rightarrow$ 0.1455 (1 μrad) $\rightarrow$ 0.0072 (1.6 μrad , one full IFOV); RER 0.5724 $\rightarrow$ 0.4013 $\rightarrow$ 0.1193 at 5 μrad ; NIIRS 6.03 $\rightarrow$ 5.52 $\rightarrow$ 3.77. The budget lines are **1.0 μrad for $\Delta\text{NIIRS} = -0.5$, 1.8 μrad for -1.0 , and 0.2 μrad for the absolute NIIRS = 6.0 floor.** Past ~ 2.6 μrad RER drops below 0.20 and the NIIRS tail is extrapolated GIQE-5 output. (*The walkthrough’s “Baseline Results” box — MTF 0.2330, RER 0.5483, NIIRS 5.97 — and its claim that the NIIRS = 6.0 floor is unreachable at any jitter are of an earlier vintage than the refreshed tables quoted here.*)

Regime. extended — a reflective ground scene fills the pixel, which is why SNR is jitter-invariant. For point-source detection jitter *would* cost per-pixel SNR; GIQE-5 assumes extended targets.

Takeaway. How the requirement is *written* changes the answer by 5 $\times$: an absolute NIIRS ≥ 6.0 spec consumes essentially the whole jitter budget at 0.2 μrad , while a relative $\Delta\text{NIIRS} \leq 0.5$ allows 1.0 μrad . And the Gaussian-PSF shortcut used in the first version of this study overestimated the budget by ~ 20 %, because the real Airy-plus-IPC PSF has wider tails than a Gaussian.

Where to go deeper. `scenarios/05_tom_optical_designer/5.4_jitter_induced_blur/`. A GUI baseline ships.

13.7.4 5.5 — Stray Light and Veiling Glare

Mission setup. Tom’s FRED stray-light analysis returns three things: a 3 % veiling-glare index, 2.5 W/m² of out-of-field stray irradiance, and a 2-D stray PSF RADIANT cannot ingest. What do the first two cost in

contrast, SNR and NIIRS, and how much veiling glare can the design tolerate? The scene is a daytime VNIR pan rooftop ($\rho = 0.30$) against vegetation ($\rho = 0.15$) from 7 km.

Key inputs.

Quantity	Value
Optics / band	15 cm f/6 / 0.5–0.8 μm
Platform / illumination	airborne 7 km, solar zenith 30°
Target / background reflectance	0.30 / 0.15
Stray mode 1	veiling_glare, VGI = 3 %
Stray mode 2	absolute_irradiance, 2.5 W/m ² out-of-field
Well	3.0×10^5 e- (target pixel saturates)
Budget	$\Delta\text{NIIRS} \leq 0.2$ and contrast SNR ≥ 50

Headline results. Clean: SNR 546.7, contrast SNR 217.4, NIIRS 11.052. **3 % veiling glare** adds 2.92×10^4 stray e- $\rightarrow$ SNR 522.0, contrast SNR 208.6, $\Delta\text{NIIRS} -0.031$. **The 2.5 W/m² out-of-field term** adds 5.52×10^6 stray e- — several times the signal itself — taking SNR to 124.3, contrast SNR to 49.4 and ΔNIIRS to **-1.003**, a full NIIRS level. Tolerance: VGI can rise to **~10 %** before either budget clause breaks. The scenario also found and fixed the veiling_glare mode: it had scaled in-FOV irradiance by the *pixel IFOV* solid angle instead of the *f-cone*, under-reporting stray by $\sim 10^7$ – 10^8 so that any VGI produced $\sim$ zero stray. Fixed, the mode reproduces the identity $\text{stray}_e = \text{VGI} \cdot \text{signal}_e$ to the digit — 9.729×10^4 e- at VGI 10 %. (The absolute NIIRS values of ~ 11 sit far above the rating scale's 9-point ceiling — unflagged GIQE-5 extrapolation on a fine-GSD airborne pan scene; only the ΔNIIRS column is meaningful.)

Regime. extended. Stray light is a **pedestal, not a signal**: it is common to target and background, cancels exactly in the target–background contrast, and degrades contrast SNR purely through the shot noise it adds.

Takeaway. The number the optical designer worries about is the cheap one. Three percent of veiling glare costs 0.03 NIIRS; the out-of-field irradiance costs a full level, and that is the term the baffle design must control. What the scalar model does *not* capture is the spatial half — there is no veiling-glare MTF and no 2-D PSF/PST importer, so the radiometric hit is modeled and the contrast-modulation hit is not.

Where to go deeper. scenarios/05_tom_optical_designer/5.5_stray_light_veiling_glare/. A GUI baseline ships.

13.8 Persona 6 — Dr. Chen, Researcher

Dr. Chen validates. Every one of her scenarios checks RADIANT against something outside it — real MODTRAN output, a hand-integrated Planck calculation, ROC theory, a retrieval Jacobian — and reports the residual whether or not it flatters the tool.

13.8.1 6.2 — Atmospheric Model Intercomparison

Mission setup. How far does RADIANT's parametric SimpleAtmosphere (Beer-Lambert with tuned band fits) diverge from real MODTRAN 6 across the six standard atmosphere profiles at one fixed geometry — and what does that divergence do to SNR? The reference is the delivered MODTRAN 6 A-block, not a synthetic

stand-in; the script falls back to synthetic tape7 with a loud banner when the gitignored real set is not staged, so a bare clone still runs.

Key inputs.

Quantity	Value
Profiles	us_standard, tropical, midlat summer/winter, subarctic summer/winter
Geometry	nadir, 100 km sensor (the A-block matrix, not the catalog's 10°/500 km)
Band	3.5–5.0 μm
Scene	300 K target against 288 K background
Reference	real MODTRAN 6 A1–A6 .tp7, imported via <code>atmosphere.model = "modtran"</code>
Comparison	identical sensor config run twice per profile, isolating the atmosphere term

Headline results. Band-mean τ residuals (MODTRAN – Simple, relative to MODTRAN) span a **uniform –4.7 % to –9.8 %** across all six profiles — Simple is systematically slightly *too transparent*, and the offset no longer varies with climate. SNR residuals run **–6.6 % to –16.6 %** and now track the τ residuals (tropical worst on both, subarctic_winter best on both), with real-MODTRAN SNR nearly profile-independent at 567–579. The τ table is the acceptance evidence for the gas-band calibration: the first real-data run found residuals spanning **–43 % to +62 %**, and the gas-band recalibration that finding triggered collapsed them **6×** to the band above.

Regime. extended thermal contrast in the MWIR — both target and background are attenuated by the same column, so the SNR-relevant quantity is the contrast radiance difference rather than absolute τ .

Takeaway. SNR is now more sensitive to atmosphere-model choice than raw transmittance is, which reverses this scenario's own earlier conclusion. An atmosphere model contributes two separable things: attenuation, which largely cancels in a contrast ratio, and its own path emission, which does not. Judging a model by τ alone understates its effect on a thermal SNR prediction. (libRadtran remains absent — fabricating plausible numbers would defeat the purpose of an intercomparison the same way a fake MODTRAN would.)

Where to go deeper. `scenarios/06_dr_chen_researcher/6.2_atmospheric_intercomparison/`. A GUI baseline ships.

13.8.2 6.3 — Noise Model Verification Against Hand Calculation

Mission setup. Dr. Chen is writing a paper comparing RADIANT against analytic noise models. She needs every noise term checked against a hand calculation built from CODATA constants alone — sharing nothing with RADIANT but the constants and the solar irradiance table.

Key inputs.

Quantity	Value
Optics	30 cm f/4, 70 % transmission, 293 K
Detector	18 μm pitch, QE 70 %, dark 100 e-/s, read 20 e- RMS, FWC 2 Me-, 14-bit at 1 e-/DN
Scene / band / integration	300 K, $\epsilon = 0.95$ / 3.5–5.0 μm / 5 ms
Platform / atmosphere	8 km airborne / exo (vacuum)

Quantity	Value
Entry path	raw vendor units through <code>Sensor.set(..., unit=...)</code> , cross-checked to 1e-12

Headline results. Every noise term agrees to 0.00 %: `signal_shot` 1280.68, `dark_shot` 0.71, `read_noise` 20.00, quantization 0.29 e- RMS, RSS total 1280.83 e- RMS, SNR 1280.52. The underlying signals match to better than 0.01 % — 1,506,203 thermal + 133,931 solar = 1,640,135 e- by hand against RADIANT’s 1,640,136. That exactness required two upgrades to the *hand* model, not the code: a photon-weighted spectral integral (the band-center E_{photon} shortcut reads ~5.5 % low for a 300 K source in this band) and the Kirchhoff reflected-solar term ($\rho = 1 - \epsilon = 0.05$ of the TOA solar spectrum, ~9 % of the in-band signal). `background_shot` and `nearfield_shot` are both 0 by design. **NEDT is the one disagreement: 21.79 mK reported against 23.92 mK exact, 8.9 % apart** — a recorded limitation, because the stage uses the single-wavelength Planck-factor form and its SNR numerator includes the reflected-solar signal, which does not vary with target temperature. (*The performance table reports MTF at Nyquist 0.2668 where the MTF budget’s own system row reads 0.2688.*)

Regime. extended — the 300 K target fills the pixel IFOV, so there is no separate scene-background photon stream, and the background inputs define only the contrast scene.

Takeaway. A 0.00 % agreement across all terms simultaneously pins the Planck integral, the solar coupling, the Kirchhoff reflectance, the pixel étendue, the QE and transmission application and the shot-noise square root in one check. The two hand-model upgrades are the lesson: **a thermal-only textbook formula verifies a nighttime scene**, and a band-center photon energy is a 5.5 % error on any wide-band thermal source.

Where to go deeper. `scenarios/06_dr_chen_researcher/6.3_noise_model_verification/`. A GUI baseline ships.

13.8.3 6.4 — Synthetic Scene Generation for Algorithm Testing

Mission setup. Dr. Chen is developing an LWIR target-detection algorithm and needs raw material to test it against: pixel-level signal and noise for a multi-target scene, a simulated noisy 1-D strip, an SNR map, and a ROC curve per target. First consumer of `radiant.performance.roc`.

Key inputs.

Quantity	Value
Sensor	5 cm aperture, f/20, 25 μm pitch (25 μrad IFOV), 0.5 ms
Band / background	LWIR 8–12 μm / uniform 290 K
Targets	five, 10–200 km range, 15–40 K hotter than background
Reference target	305 K, 3 m, $\epsilon = 0.93$
Fill fraction	$ff = (\text{size}/\text{GSD})^2$, capped at 1
Detection model	equal-variance Gaussian, $P_d = Q(Q^{-1}(P_{fa}) - \text{SNR})$

Headline results. Background pixel 4.972×10^5 e- with $\sigma = 713$ e-, shot-noise-limited. **All five nominal targets are trivially detected** — contrast SNR 487.3 (10 km) down to 51.2 (200 km, the only sub-pixel one at $ff = 0.36$), every $P_d = 1.000$ at $P_{fa} = 10^{-4}$. A ROC of those five is uninformative, so the detection science lives in the range sweep, where $ff \propto 1/R^2$ walks the reference target down: contrast SNR 51.2 (200 km) $\rightarrow$

8.2 (500 km) $\rightarrow$ 3.2 (800 km) $\rightarrow$ 0.51 (2000 km). **Reliable detection** ($P_d \geq 0.9$) **holds to ≈ 535 km and the 50/50 range is ≈ 700 km.** The informative gap is at 800 km, where AUC is still 0.988 — good *separation* — but P_d at $P_{fa} = 10^{-4}$ is only 0.302.

Regime. extended for every chain run, with the scene assembled analytically from it. In the extended regime the per-pixel signal is **geometrically** range-independent — radiance $\times$ a fixed pixel solid angle — but not atmospherically so: the nominal table shows the filled-pixel target signal falling 31 % over 10 $\rightarrow$ 200 km through transmittance alone, and the sweep’s analytic ff dilution holds the atmospheric factors at each target’s own range rather than claiming an identity.

Takeaway. The gap between AUC and a strict- P_{fa} P_d is the whole design trade. At 800 km the target is well separated from the background by any integrated measure, and a detector held to one false alarm in ten thousand still finds it only one time in three. The near targets being “too easy” is the physical answer, not a modeling artifact, and the scenario moves the analysis rather than tuning the scene.

Where to go deeper. scenarios/06_dr_chen_researcher/6.4_synthetic_scene_generation/. A GUI baseline ships.

13.8.4 6.5 — Emissivity Sensitivity for Temperature Retrieval

Mission setup. An LWIR radiometer measures radiance and the analyst inverts it for surface temperature — but the inversion needs an *assumed* emissivity. How badly does an error in that assumption bias the retrieved temperature, and how does the bias compare with the sensor’s own NEDT? First consumer of radiant. performance.temperature_retrieval.

Key inputs.

Quantity	Value
True scene	T = 300 K, $\epsilon = 0.95$
Band	LWIR 8–12 μm
System NEDT	50 mK
Assumed- ϵ sweep	0.90–1.00
Method	band-averaged Planck; Brent root-find inversion plus the analytic Jacobian

Headline results. The Jacobian at the operating point gives $\partial L / \partial \epsilon = 38.50 \text{ W/m}^2/\text{sr}$ and $\partial L / \partial T = 0.598 \text{ W/m}^2/\text{sr/K}$, hence $dT/d\epsilon = -64.4 \text{ K per unit } \epsilon$, or **$-0.64 \text{ K per } 0.01 \epsilon$** . The exact inversion tracks it: assumed $\epsilon = 0.90$ retrieves 303.34 K (**+3.34 K**), $\epsilon = 0.94$ gives +0.65 K, $\epsilon = 0.96$ gives -0.64 K , $\epsilon = 1.00$ gives **-3.11 K** . A lower assumed ϵ over-estimates T, because the surface must be hotter to emit the same radiance. **The NEDT-equivalent emissivity uncertainty is ± 0.0008 (0.08 %):** to keep the retrieval bias below the sensor’s own 50 mK, emissivity must be known to better than a tenth of a percent. Over a realistic ± 0.05 uncertainty the bias reaches 3.3 K — **$67\times$ the NEDT floor.**

Regime. No chain regime applies — this is a band-radiance inversion with its Jacobian, run against a declared NEDT rather than through the signal chain.

Takeaway. Emissivity knowledge, not detector NEDT, limits absolute LWIR thermometry. Buying a lower-NEDT detector does nothing for temperature accuracy until ϵ is pinned down — and because the retrieval error is a *bias*, not noise, frame averaging (which beats down NEDT) does not touch it.

Where to go deeper. `scenarios/06_dr_chen_researcher/6.5_spectral_emissivity_sensitivity/`.
No GUI baseline ships — the only scenario in personas 5–7 without one.

13.9 Persona 7 — Karen, Test Engineer

Karen measures what the model predicts. Her scenarios are bench comparisons — a blackbody calibration, a slanted-edge MTF, a cold-stop background, a TVAC temperature sweep — and the interesting result is always the residual, and what it turns out to be made of.

13.9.1 7.2 — Radiometric Calibration Verification

Mission setup. Karen ran a NIST-traceable blackbody calibration at five set points from 280 to 360 K, recording 100-frame mean DN at each. She needs RADIANT’s as-built prediction beside the measurement **in DN**, the unit her data system actually records, plus responsivity, a linearity check and per-point uncertainty — with the lab ambient and the instrument’s own self-emission modeled, because that is what a calibration’s offset term physically is.

Key inputs.

Quantity	Value
Optics	15 cm / 30 cm (f/2.0), $\tau = 0.72$ net
Modeled train	$3 \times R = 0.98$ mirror + AR cold window $\Rightarrow$ emitting $\epsilon \approx 0.045$, not the workbook’s $\epsilon = 28\%$
Optics temperature	20 °C (bench ambient)
Detector	15 μm pitch, QE 75 %, dark 5×10^4 e-/s, read 30 e- RMS, 2 Me- well
Band / integration / gain	3.7–4.9 μm / 0.25 ms / 125 e-/DN, 14 bit
Set points	280, 300, 320, 340, 360 K (5–60 % well)

Headline results. Predicted DN runs 742.1 $\rightarrow$ 9615.1 against measured 795.5 $\rightarrow$ 9815.0, residuals **–6.72 % at the cold end narrowing to –2.04 % at the hot end**. Fitting measured = $a \cdot \text{predicted} + b$ decomposes that entirely into two physical knobs: **$a = 1.0162$** (real responsivity +1.62 % against the as-built gain spec) and **$b = +43.6$ DN** of instrument offset. Percent residuals are largest at the cold end not because the model is worse there but because a fixed offset is a larger fraction of a small signal. Responsivity dDN/dT rises 42.47 $\rightarrow$ 198.15 DN/K over the sweep as the Planck derivative steepens; radiance responsivity is 1059 DN/(W/m²/sr). Linearity holds to **0.107 % of full scale**. Calibration uncertainty on 100-frame means is 5.9 mK (280 K) and 4.4 mK (360 K). Separately, RADIANT models **73.8 DN of warm-optics near-field** from first principles — which now *exceeds* the 43.6 DN fitted residual, itself informative.

Regime. extended — the blackbody fills the aperture, so the scene-background photon term is skipped and the lab-ambient parameters feed only the contrast scene. The instrument terms that genuinely move the offset, near-field and dark, are both modeled.

Takeaway. Raw residuals of **–2 to –7 % are not a model failure; they are an uncalibrated instrument**. Two coefficients absorb them completely, which is the entire point of the exercise. That the modeled near-field now exceeds the fitted offset says either the bench train is better coated than the assumed $R = 0.98$ or its barrel is colder than 20 °C — a testable statement, which is where 7.4 picks up.

Where to go deeper. [scenarios/07_karen_test_engineer/7.2_radiometric_calibration/](#). A GUI baseline ships.

13.9.2 7.3 — MTF Measurement vs Prediction

Mission setup. Karen measured system MTF with a slanted-edge target (ISO 12233) at 650 nm. She has as-built WFE from interferometry (0.07 waves RMS at 633 nm) and knows the detector sits 5 μm from best focus. Overlay the prediction, decompose the budget, compute the residual, and find out what the residual is made of.

Key inputs.

Quantity	Value
Optics	20 cm Cassegrain, f/3.0, 25 % obscuration, 82 % transmission
Band / test wavelength	550–750 nm / 650 nm collimator
Detector	10 μm pitch, 100 % fill, IPC 1.0 %
As-built WFE / defocus	0.07 waves RMS at 633 nm / 5 μm
Sampling	f_{Nyq} 50.0 cy/mm against a 512.8 cy/mm cutoff; $Q = 0.195$
Atmosphere	exo (bench)

Headline results. Measured MTF at Nyquist **0.4441** against RADIANT's **0.4575**, with a **residual RMS of 0.0206** across all 50 measured points — slanted-edge measurement-noise level. The analytic four-term composition reads 0.5339 with a 0.0606 RMS residual, so **RADIANT now beats the analytic model**. The budget at Nyquist: optics (diffraction + obscuration + WFE + defocus-Z4, from one pupil) 0.7535, pixel aperture 0.6364, IPC 0.9602, everything else unity, system product 0.4607 against the PSF path's 0.4575. Defocus is negligible: 5 μm costs 0.9 % of MTF at Nyquist, and even 15 μm only 7.4 %. Two residual explainers were tested on a grid and **neither is supported** — TIS surface roughness (5 nm $\Rightarrow$ RMS 0.0220) makes the fit *worse*, and electronics blur (1 μm $\Rightarrow$ 0.0205) moves it by 0.0001, below the measurement noise, so both are bounded rather than detected. Other metrics: Strehl 0.9663, RER 0.7978, FWHM 10.22 μm .

Regime. A bench measurement, not a scene: GSD, NIIRS and NEDT are all reported N/A (altitude = 0, no thermal scene in the VNIR), and the noise budget is dark 0.32, read 8.00, quantization 2.31 e- with essentially no photon flux.

Takeaway. **This scenario's own diagnosis drove a model change.** At the earlier vintage the residual was 0.0917 RMS and the walkthrough diagnosed it as the *shape ambiguity of scalar WFE* — a white-noise phase screen dumps aberrated energy into a compact halo and drops low frequencies toward the Strehl plateau, where a real optic's smooth aberrations hold them near 1. The fix replaced the screen with a deterministic low-order Zernike expansion and the residual fell **4.5×**. The secondary lesson is the rejection workflow itself: a hypothesis that does not reduce the residual is reported as rejected, not quietly fitted. And at $Q = 0.195$ the pixel sinc, not the optics, is the dominant contributor — improving WFE buys almost nothing.

Where to go deeper. [scenarios/07_karen_test_engineer/7.3_mtf_measurement_vs_prediction/](#). A GUI baseline ships.

13.9.3 7.4 — Cold-Stop Undersizing Sweep

Mission setup. In a TVAC campaign on an MWIR imager, the cold stop *is* the aperture stop: a ~77 K cryogenic aperture deliberately built a little smaller than the primary so that alignment and thermal tolerances can never let the focal plane see past it to warm structure. Undersizing buys that certainty and pays in photons. How much margin can the camera afford, does it help the background, and what do six shuttered-background lab readings mean?

Key inputs.

Quantity	Value
Optics	25 cm / 1.0 m (f/4.0), net $\tau = 0.68$
Modeled train	$3 \times R = 0.98$ mirror ($\epsilon = 0.02$ each) + AR cold window ($\epsilon = 0$) $\Rightarrow \epsilon_{\text{emit}} 0.06$, not 0.32
Detector	dark 499,376 e-/s (from 80 fA/pixel), read 25 e- RMS
Band / integration	3.70–4.80 μm / 8 ms
Sweep	undersizing $u = 0 \rightarrow 10\%$
Requirement	shuttered background < 40,000 e-

Headline results. One effective pupil drives everything: $D_{\text{eff}} = (1 - u)D$, and from it A_{collect} , N_{eff} , the complex pupil (hence PSF and MTF alike) and the étendue cone Ω_{cone} . At $u = 0$: signal 2,994,945 e-, near-field 106,631 e-, SNR 1699.3, NEDT 17.00 mK, MTF at Nyquist 0.3017. Over $0 \rightarrow 10\%$: A_{collect} **-19.0 %**, Ω_{cone} **-18.8 %**, near-field **-18.8 %**, signal **-19.0 %**, SNR **-10.0 %**, MTF at Nyquist **-11.6 %** — about 1 % of SNR and 1.2 % of resolution per 1 % of pupil diameter. The 40,000 e- requirement **fails at both ends** (106,631 $\rightarrow$ 86,561 e-). Karen’s six lab readings, 35,500 $\rightarrow$ 55,750 e-, all sit *below* the model, inverting to an implied per-mirror reflectance of $R \approx 0.990\text{--}0.993$ against the assumed 0.98.

Regime. extended — the blackbody fills the FOV, so background_e = 0 by design and the only background terms are warm-optics near-field and dark. The bench runs as exo with a placeholder sensor_altitude_m — a recorded limitation.

Takeaway. Undersizing is not a background-control knob. Signal and near-field fall together because both come from the same effective pupil — the physics the deleted optics.nearfield_fraction “leakage” knob concealed by letting a cold stop attenuate in-cone emission it cannot touch. The levers that actually scale the near-field are optics temperature and coating emissivity. And the 57 % spread across cold-stop *positions* is what the new rules cannot explain and should not fit: with a true aperture stop, position cannot move the background, so a monotone rise says the stop has stopped being the aperture stop — a finding for the mechanical team, recorded as an accepted limitation of the near-field model.

Where to go deeper. scenarios/07_karen_test_engineer/7.4_cold_stop_sweep/. A GUI baseline ships.

13.9.4 7.5 — Performance at Temperature Extremes

Mission setup. A TVAC sweep of the FPA operating temperature from 70 to 95 K, imaging the 300 K chamber shroud. Karen measured dark current at each point — it goes *super-Arrhenius* above ~85 K — and QE at three. She needs SNR and NEDT versus temperature, the noise budget as dark grows, the QE(T)-versus-dark split, and an operating point with margin against the acceptance spec.

Key inputs.

Quantity	Value
Optics	f/2.0, net $\tau = 0.74$; train modeled as $3 \times R = 0.98 + \text{AR cold window } (\epsilon_{\text{emit}} 0.06)$
Scene	300 K chamber shroud, fills the aperture
Sweep	FPA temperature 70–95 K, measured $J(T)$ driving <code>detector.dark_rate_e_per_s</code>
QE	interpolated from three measured points; every run repeated with QE frozen at 77 K
Integration	0.6 ms
Spec	$\text{SNR} \geq 750$ and $\text{NEDT} \leq 35$ mK

Headline results. The measured dark curve matches an Arrhenius fit ($E_a = 0.240$ eV) exactly through 82 K and then diverges catastrophically: **+10 % at 85 K, +90 % at 88 K, +201.8 % at 90 K, +735.8 % at 95 K** (395,723,319 against a fitted 47,345,161 e-/s). Across the sweep SNR falls $814.7 \rightarrow 674.6$ and NEDT rises $34.01 \rightarrow 41.07$ mK, driven by dark shot noise climbing **0.9 $\rightarrow$ 466.5 e- RMS** from negligible to the second-largest term. Over $70 \rightarrow 95$ K **QE falls 9 % while dark current rises 294,612 $\times$** , and NEDT with QE(T) versus QE frozen differs by only a few percent. Spec holds through **82 K** (NEDT 34.88 mK, 0.1 mK of margin); 85 K fails on NEDT. The correctly-modeled warm-optics near-field (the étendue-cone model raised it $7,496 \rightarrow 31,840$ e-) costs ~ 6 K of the compliant range the $\epsilon = 1 - \tau$ fallacy had appeared to buy. (*Key Results recommends operating at 79 K with a 3 K guard band; the next-steps list says 85 K, the first temperature the same table fails on.*)

Regime. extended — the 300 K shroud fills the aperture, so the background photon term is skipped and the budget is signal-shot + dark-shot + read + near-field + quantization.

Takeaway. **Extrapolating a cold-side activation energy is the failure mode this test exists to catch:** the Arrhenius line would call 95 K eight times better than it is, because above the knee a different physical mechanism (defect-assisted tunneling) dominates and no single E_a fits both regimes. The guard band protects against cooler drift *and* against the NEDT slope steepening from ~ 0.25 mK/K near the compliance edge to ~ 1.4 mK/K past 92 K.

Where to go deeper. `scenarios/07_karen_test_engineer/7.5_environmental_temp_extremes/`. A GUI baseline ships.

13.10 Persona 8 — Interpolation Demonstrations (a tool demonstration, not a persona)

Folder 08 sits deliberately outside the closed thirty-five-scenario catalog. It has no persona and no mission owner — it exists to demonstrate the per-family MODTRAN atmosphere interpolator (`scripts/synth_modtran/family_interpolate.py`) with runnable examples, and it follows the same scenario-trio convention as everything else.

13.10.1 8.1 — Off-Nadir Angle Interpolation

Mission setup. A customer requirement lands at 37.5° off-nadir, which is not one of the MODTRAN run matrix's zenith-fan points (0° , 30° , 45° , 60° , all us_standard, 100 km sensor, nadir target). Does interpolating between the two bracketing runs actually beat grabbing the nearest one?

Key inputs.

Quantity	Value
Family	zenith_fan_us_standard — real MODTRAN 6 runs A1/B1/B2/B3
Query	37.5° path zenith
Bracketing runs	B1 at 30° , B2 at 45°
Method	linear in $\log \tau$, on the airmass axis $\sec \theta$
Band	3.5–5.0 μm

Headline results. A holdout test predicts the real 45° run from its 30° and 60° neighbors and compares against ground truth (in-band $\tau = 0.4988$): interpolating in $\log \tau$ **linear in angle** gives 0.4785, a **−4.07 %** error; **linear in airmass** gives 0.4983, **−0.10 %**; nearest-neighbor (30°) gives 0.5329, **+6.84 %**. At the actual 37.5° query, interpolated τ is 0.5185 with chain SNR 552.1, against nearest-neighbor τ 0.4988 and SNR 541.6 — a nearest-neighbor error of **−3.8 % in transmittance and −1.9 % in SNR**.

Regime. Not a regime study — one atmosphere-data-source comparison fed through an identical chain config evaluated at the true 37.5° geometry, isolating the data-source effect.

Takeaway. The method beats nearest-neighbor by about $1.7\times$ against real ground truth, and the residual that remained had a knowable cause with a $40\times$ fix: optical depth scales with airmass, not angle. The axis correction was a coordinate transform, not new data, and it now governs the shipped atmosphere library too.

Where to go deeper. scenarios/08_interpolation_demonstrations/8.1_off_nadir_angle_interpolation/. A GUI baseline ships.

13.10.2 8.2 — Target-Altitude Interpolation

Mission setup. A stratospheric-sensor mission needs atmosphere data for a target at 15 km — not one of the altitude ladder's points (0, 1, 5, 10, 20, 29 km, all midlat_summer, 35 km sensor, nadir). Same method as 8.1, a different axis type, to show the tool generalizes.

Key inputs.

Quantity	Value
Family	altitude_ladder_stratospheric — synthetic tape7 data
Query	15 km target altitude
Ladder spacing	1, 4, 5, 10, 9 km gaps — deliberately non-uniform
Band	8–12 μm

Headline results. The ladder reads in-band τ 0.5774 (0 km), 0.7039 (1 km), 0.8822 (5 km), 0.9151 (10 km), **0.9291 interpolated at 15 km**, 0.9447 (20 km), 0.9790 (29 km). Nearest-neighbor selection (20 km) is **+1.7**

% in transmittance and **+0.7 % in full-chain SNR** — the SNR error is smaller because the extended-scene target/background contrast partially cancels atmosphere effects common to both terms. *The walkthrough is explicit that the +0.7 % SNR figure was carried forward, not re-measured, in the last refresh sweep: the runner’s chain half needs a generated synthetic MODTRAN set that is absent on a clean checkout.*

Regime. extended. Reproducing the scenario requires `python scripts/generate_synthetic_tape7.py` first; the runner now names that prerequisite rather than failing with a bare `FileNotFoundError`.

Takeaway. The 15 km query lands in the ladder’s widest gap, which is exactly where nearest-neighbor is weakest and interpolation earns its keep. The method does not care whether the free axis is an angle or an altitude — it only needs monotone behavior in optical depth along that axis. Note the recurring friction: full-well saturation silently zeroed the atmosphere effect on the first attempt, the third scenario to hit that failure mode.

Where to go deeper. `scenarios/08_interpolation_demonstrations/8.2_target_altitude_interpolation/`. A GUI baseline ships (it uses an inline builder and does not need the synthetic set).

13.10.3 8.3 — Boost-Phase Target-Altitude Sweep (skeleton)

Mission setup. A space-based MWIR sensor in a 500 km orbit tracks a booster continuously from launch to burnout. As the booster climbs, the absorbing column between it and the sensor shortens. Can one config sweep geometry.target_altitude_m from 0 to 300 km against the shipped interpolated library and produce physically sensible τ and SNR?

Key inputs.

Quantity	Value
Sensor / band	500 km LEO, nadir; 3–5 μm
Target	900 K plume, 4 m^2
Atmosphere	interpolated, midlat_summer_ladders (real MODTRAN, 5 cm^{-1} FWHM slit)
Interpolation axes	sensor_altitude_m, target_altitude_m
Sweep	target altitude 0–300 km

Headline results. Three regimes. **Interpolated (0–29 km):** τ_{up} rises 0.4143 $\rightarrow$ 0.9425 and SNR 304.33 $\rightarrow$ 485.23. **Pending (29–100 km):** above the ladder’s 29 km ceiling but below the 100 km atmosphere top, so the interpolator refuses the query and the script marks the rungs **PENDING** rather than inventing numbers — this band is the acceptance driver for the MODTRAN boost-ladder expansion. **Vacuum (≥ 100 km):** $\tau_{\text{up}} \equiv 1.0000$ by the exo-altitude leg, with SNR 590.18 (100 km) $\rightarrow$ 1183.67 (300 km).

Regime. `sub_pixel`, locked by `source.regime_override`. A 4 m^2 plume at LEO slant range subtends ≈ 4 μrad against a ~ 16.7 μrad IFOV — 0.22 $\times$ the PSF FWHM, too large for the point-source approximation and far too small to fill a pixel.

Takeaway. SNR keeps rising across the vacuum leg even with τ pinned at 1, and the scenario separates the two causes explicitly so the climb is not misread as residual absorption: the slant range is *closing*, from 400 km to 200 km, so the plume’s fill fraction grows. The engineering lesson is in the friction note — for a sweep, check the saturation envelope at **both** ends, because here the brightest case is the last rung, not the first.

Where to go deeper. `scenarios/08_interpolation_demonstrations/8.3_boost_phase_target_altitude_sweep/`. **No GUI baseline ships.**

13.11 Direction-General Scenarios (folder 10)

Folder 10 is not a persona either. It is the acceptance suite for the direction-general geometry work and the Geometry-Flexibility work: one scenario per cell of the observer $\times$ target matrix that RADIANT could not previously *express*, because `core/viewing_triangle.py` rejected a sensor below its target and the canonical target-side path zenith θ_o was bounded to $[0, \pi/2)$. Each scenario is a Category-D validation with independent cross-checks, and each reports the dual-path consistency residual explicitly. The fourth member of the folder, 10.2, is covered at full depth as a case study.

13.11.1 10.1 — Ground-to-Air MWIR Detection

Mission setup. A range test engineer has a ground-based MWIR search-and-track camera on a tripod at a sea-level desert range; the trial target is a small turbojet UAS cruising at 10 km, at night. The test plan sweeps the camera from straight up down to 30° elevation and asks the ordinary questions: what does pointing elevation cost, how far can the target be held, and does any of it agree with MODTRAN?

Key inputs.

Quantity	Value
Optics	100 mm / 200 mm (f/2.0), $\tau = 0.75$
Modeled train	$3 \times R = 0.98$ mirror + AR cold window $\Rightarrow$ emitting $\epsilon = 0.06$, not the datasheet's 25 %
Detector	15 μm pitch, QE 75 %, dark 5×10^4 e-/s, read 250 e- RMS, 11 Me- well
Band / integration	3.0–5.0 μm / 0.5 ms
Target	550 K nozzle, $\epsilon = 0.90$, $2.827 \times 10^{-3} \text{ m}^2$ at 10 km
Atmosphere / illumination	simple, midlat summer, 23 km visibility, rural aerosol / night
Sweep	sensor-side zenith $\zeta_{low} = 0 \rightarrow 60^\circ$ (elevation $90 \rightarrow 30^\circ$)

Headline results. At the nominal 60°-elevation point: slant range 11,543.99 m, band-mean τ 0.4520, signal 8.062×10^4 e- against a sky background of 1.507×10^5 e-, **SNR 144.64, NEDT 610.8 mK**. Over the whole sweep ($90^\circ \rightarrow 30^\circ$ elevation) τ falls $0.4995 \rightarrow 0.2668$, signal falls **7.21 $\times$** and SNR $204.52 \rightarrow 29.24$. The extra factor over the 3.98 $\times$ inverse-square loss is air mass; meanwhile **up-path radiance rises 64.1 %** ($0.4388 \rightarrow 0.7203 \text{ W/m}^2/\text{sr}$), which is Kirchhoff, not a sign error — the same extra absorbing column is an extra emitting one. Detection range (SNR = 5), walked with the full chain along the real ray, runs **66.88 km at the zenith down to 44.20 km at 60°**. RADIANT's own `detection_range_m` metric is deliberately **absent**, with a named result-typed failure: the continuation past a 10 km target is still inside the atmosphere and the metric layer has no altitude-resolved extinction profile to integrate. (*The near-field retune paragraph in the scenario's own walkthrough reports post-retune values of SNR 142.23 and NEDT 621.2 mK that the sweep table below it does not carry.*)

Regime. `point_source` — the nozzle's angular extent is 0.0602 of the PSF FWHM, inside the ≤ 0.10 bound RADIANT enforces, and `EE_box` (0.5941) is applied once to the target term only. Eleven ground-projection

metrics are off by scene class, and GSD stays absent even when the group is force-enabled: the ground-plane cosine projection is undefined at an incidence angle of 150° .

Takeaway. The MODTRAN anchor is reported as a **characterization, not an agreement claim**: τ is systematically too transparent in the MWIR (+29.5 % on a 1 km column, converging to +9.3 % by 20 km) and up-path radiance is too low (−45.3 % at the scenario’s own geometry), both pushing the same way, so **the quoted SNR should be read as $\approx 40\text{--}50\%$ optimistic in absolute terms with the trends reliable** — which is all a pointing-elevation trade needs.

Where to go deeper. scenarios/10_direction_general/10.1_ground_to_air_mwir_detection/. A GUI baseline ships.

13.11.2 10.3 — Ground-to-Space SST in the Visible Band

Mission setup. A space-surveillance site runs a 1 m visible tracking telescope on a small LEO object at 700 km during the terminator window: the sun is 12° below the *site*’s horizon, so the sky at the telescope is dark while the object overhead is still in full sunlight. Before the direction-general geometry work the scene was doubly inexpressible — the sensor had to be above the target, and the sun was hard-bounded above the horizon.

Key inputs.

Quantity	Value
Site / telescope	900 m MSL / 1.000 m aperture, $f/10$, 60 % transmission
Band / detector	400–900 nm / 15 μm pitch, QE 80 %, read 5 e [−] RMS, 400 ke [−] well
Exposure	5 ms (set by tracking accuracy, not the well)
Target	700 km, $\rho = 0.25$, $A_{proj} = 1.00\text{ m}^2$, solar phase 35°
Target door	source.target.user_intensity_path — a measured $I(\lambda)$ CSV
Pointing / sun	$\zeta_{low} = 20^\circ$ / solar depression 12° , relative azimuth 45°
Turbulence / visibility	HV-5/7 C_n^2 profile / 100 km

Headline results. At the nominal tasking: band-mean τ_{up} 0.7652, Fried parameter $r_0 = \mathbf{19.820\text{ cm}}$, **EE_box 0.12075**, signal 49,220 e[−], **SNR 221.78**, detection range (SNR = 3) 24,678 km. The measurement is decisively **seeing-limited**: seeing FWHM 3.214 μrad (0.663'') against diffraction’s 0.793 μrad , a ratio of 4.05, and turbulence takes MTF at Nyquist from 0.46250 to **0.00862** while RER falls $0.7704 \rightarrow 0.3317$. Across the pass ($\zeta_{low} 0 \rightarrow 75^\circ$) SNR falls 6.6 $\times$, with transmittance contributing only 1.56 $\times$ and r_0 shrinking as $\sec \zeta^{-3/5}$ — and with the Rayleigh term correct, **seeing plus inverse-square, not extinction, is most of the decay**. Four anchors: the closed-form point-source identity reproduces the chain signal to 0.000 %, the apparent magnitude lands at 5.54 mag (inside the 4–8 naked-eye-satellite band), transmittance reciprocity holds to 1.1×10^{-16} , and **published astronomical extinction FAILS** — 0.261 mag/airmass against a published 0.12–0.20 band. (*Two stale figures ride along: the deliberate-non-anchor paragraph still calls that failure 1.4 $\times$ the published top where the refreshed anchor reads 1.3 $\times$, and the closing to-do list quotes a daylight sky pedestal of 3.2774 W/m²/sr/ μm where the scenario walkthrough’s §9 computes 3.8794.*)

Regime. point_source, entered through the intensity door because RADIANT has no reflective point-source door: the reflective path multiplies by $\max(\cos \theta_s, 0)$, identically zero in exactly the terminator window this scenario exists to model. That door also *strips* the sun, so τ_{sun} never multiplies the target term and the analyst owns the illumination gate.

Takeaway. The 1 m aperture buys photons, not resolution — doubling D would leave the long-exposure blur diameter unchanged and only double D/r_0 . The seeing itself should be read as optimistic by roughly 2×: the Hufnagel-Valley ground term is conventionally above *ground* level but the model evaluates it against MSL, so a 900 m site silently loses its own boundary layer. And the extinction anchor’s failure is instructive rather than fatal: the MODTRAN-scored comparison on the same band *improved* while this one worsened, which localizes the residual to aerosol attribution rather than the gas fit.

Where to go deeper. scenarios/10_direction_general/10.3_ground_to_space_sst_visible/. A GUI baseline ships.

13.11.3 10.4 — LEO-to-GEO Space-to-Space SDA

Mission setup. A 500 km LEO host carries a 35 cm MWIR staring telescope pointed *up* at the geostationary belt. Three numbers size the tasking plan: can it detect a reference GEO communications bus in eclipse and at what single-frame SNR, how far past GEO does that capability extend, and does the relative angular rate force rate-tracking or permit a stare? The physics was never the blocker — both endpoints sit above h_{atm_top} , so the path is vacuum by construction — only the altitude-ordering gate was.

Key inputs.

Quantity	Value
Telescope	350 mm / 2100 mm (f/6.0), 25 % obscuration, 60 % transmission, 0.05 waves WFE
Detector	18 μm pitch (8.571 μrad IFOV), QE 75 %, dark 1000 e-/s, read 25 e- RMS, 100 ke- well
Band / integration	3.5–5.0 μm / 500 ms rate-tracked stare
Geometry	500 km LEO $\rightarrow$ 35,786 km GEO, $\zeta_{low} = 0^\circ$, night (eclipse)
Target	280 K gray body, $\epsilon = 0.85$, 20 m ² projected area
Rate-track residual	1 % of the open-loop LOS rate

Headline results. EE_box 0.245670, in-pixel signal 1295.78 e-, total noise 207.947 e- RMS, **SNR 6.23, detection range (SNR = 5) 39,451 km** — 1.12× the LEO $\rightarrow$ GEO range, so the belt sits only just inside the single-frame horizon. *Scene* background shot noise is **exactly zero** — the up-looking LOS exits into deep space and selects ColdSpaceBackground — but that is precisely why the telescope’s **own** emission dominates: nearfield_shot is 202.034 e- RMS, **94 % of the noise power** (CU-380, 2026-10-03; the six declared surfaces at the datasheet’s 180 K bench temperature). A point source against cold space has no other background, so this scenario measures the instrument against itself, and it is the case in the whole suite where warm optics matter most. The kinematics are the design driver: LEO 1108.508 $\mu\text{rad/s}$ against GEO 72.940 $\mu\text{rad/s}$ gives an open-loop LOS rate of **128.709 $\mu\text{rad/s}$** , which drags the point source across **7.5 pixels** in a 500 ms inertially-fixed stare, collapsing EE_box to 0.0597 and SNR to 1.53 — below the detection threshold, so the open-loop arm publishes **no detection range at all** (the solver declines rather than extrapolating, Rule 17’s metric-layer carve-out). **Open-loop SNR peaks at 100 ms** and then falls — past that the smear kernel grows faster than $\sqrt{t}$ — and its peak of 2.28 never reaches threshold either. The rate-tracked curve keeps rising with t , but no longer as $\sqrt{t}$: since CU-380 the dominant term is warm-optics shot noise, which accumulates with integration time exactly as the signal’s own does, so the tracked arm is **warm-optics limited** rather than background-free. Every vacuum transport identity is checked bitwise, not toleranced.

Regime. `point_source`, finalized in `OpticsStage`. A 20 m² bus at 35,286 km subtends 0.1267 μ rad — 68 $\times$ smaller than the detector IFOV and far inside the 14.8 μ rad diffraction core — so all the flux lands in one PSF and the in-pixel signal follows inverse-square, which is what makes a detection range meaningful at all. The reported NEDT of 1005 mK is *not* a figure of merit here: it is a radiance-contrast sensitivity defined against a scene that fills the pixel, and this target fills 0.02 % of one.

Takeaway. **The tracking loop, not the aperture or the signature, is the design driver.** A 1 % rate residual holds smear to 0.075 px and the full 500 ms is usable; an untracked stare does not merely lose SNR, it **never reaches the detection threshold at any swept integration time**. Rate tracking here is the difference between a sensor and no sensor. The scenario also retired a framework defect found here and in 10.2 independently: the detection-range solver froze the noise at the reference range, which made R_{det} conservative by 15.2 % and dependent on where the chain was evaluated, back when signal shot noise carried 51 % of the noise power. CU-380 has since taken that share to 3 %, so the two noise models now agree to 0.2 % — the correction *vanishes for a background-limited chain*, and this chain is now background-limited by its own optics, which demonstrates the limit rather than describing it.

Where to go deeper. `scenarios/10_direction_general/10.4_leo_to_geo_exo/`. A GUI baseline ships.

Flagship-Mission Validation

Everything before this chapter shows RADIANT *being used*. This chapter asks the prior question: **when RADIANT prints a number, is the number right?**

The honest answer has three parts, and they carry different weights.

1. **Four flight instruments.** Sentinel-2 MSI, Landsat 8/9 TIRS, Aqua MODIS, and Landsat 9 OLI-2 all publish performance figures — per-band SNR at a stated reference radiance, or per-band NEdT at a stated scene temperature — together with enough instrument description to build the model that should reproduce them. Scenarios 9.1–9.4 do exactly that. This is the strongest evidence the model has, because the comparison is against hardware in orbit rather than against another model.
2. **A MODTRAN 6 run set.** The atmosphere backends are measured against 130 delivered tape7 runs spanning six profiles, five geometry classes, and the full 0.3–14 μm span. That is a model-to-model comparison — MODTRAN is not the sky — but MODTRAN is the community reference, and the parity record states the residuals rather than claiming agreement.
3. **A hand calculation.** One MWIR case is worked from CODATA constants to SNR by hand, step by step, and asserted by an integration test to a relative tolerance of 10^{-4} . It validates the arithmetic of the chain, not its physics choices, and it is the anchor everything else is ultimately checked against.

The chapter closes with what none of that covers, which is a longer list than the first three sections might suggest.

14.1 How to read a flagship comparison

Every one of the four flagship scenarios is built the same way, and the construction is what makes the comparison meaningful:

- **The anchor is quoted at-aperture.** Published SNR requirements are stated *at* a reference radiance L_{ref} or L_{typ} in $\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$; published NEdT is measured against an onboard blackbody. So each scenario drives the chain with that radiance (or that blackbody) through a **vacuum path** (`atmosphere.model:exo`). The atmosphere and the scene model are deliberately taken out of the loop. What is under test is the aperture-to-electrons radiometry: solid angle, collecting area, throughput, quantum efficiency, integration time, and the noise stack.
- **Every instrument input carries a provenance class.** The four source-data dossiers maintained with the validation records (Sentinel-2 MSI, Landsat TIRS, MODIS TEB, Landsat OLI-2) tag each number as *published*, *model value*, or *assumption*, with the citation. Aperture and focal length are published; band-average quantum efficiency usually is not.
- **The assumption envelope is declared before the run.** Where an input is an assumption, the dossier states a range as well as a central value, and the comparison reports the SNR envelope that range produces. A prediction band, not a point, is what an assumption-carrying model is entitled to claim.

- **The verdict vocabulary is fixed.** **CONSISTENT** means the published value is reachable inside the declared envelope, or the residual maps onto a plausible value of a named unpublished parameter. **INCONCLUSIVE** means the residual is explainable but the explanation is not yet sourced. There is no “PASS”.
- **The implied-parameter inversion is the diagnostic.** In the shot-noise limit $\text{SNR} \propto \sqrt{\eta \tau t_{\text{int}} N_{\text{TDL}}}$, so a measured SNR can be inverted for the throughput product it implies. A residual that implies a physically plausible $\eta \tau$ is a modeling assumption being wrong in a known direction; one that implies an impossible $\eta \tau$ would be a defect.

Provenance of the numbers below. Every table in the four flagship sections is quoted from the scenario’s committed `walkthrough.md`. Because the four scenarios run in under a second each on a vacuum path with no fenced data, they were re-run while this chapter was written: all four reproduce their committed tables. Two numbers of extra precision have moved since the walkthroughs were written and are flagged where they appear. Envelope and implied-throughput columns that the walkthroughs summarize in prose are quoted from a fresh run of `scripts/run_external_validation.py`, which is the script the dossiers are executable through, and are labeled as such.

14.2 Sentinel-2 MSI — per-band SNR at the reference radiance

14.2.1 Mission context

Copernicus Sentinel-2 carries the MultiSpectral Instrument: a 150 mm-pupil three-mirror-anastigmat push-broom imager in a 786 km sun-synchronous orbit, imaging 13 bands from 443 nm to 2190 nm at 10 m, 20 m, and 60 m ground sample distance across a 290 km swath. It is the most heavily used optical Earth-observation instrument in civil service, and ESA publishes both its **radiometric requirement** and its **on-orbit measured performance** band by band.

14.2.2 The published anchor

ESA’s SentiWiki mission pages ([S2-SW], <https://sentiwiki.copernicus.eu/web/s2-mission>) give, in Table 3, each band’s reference radiance L_{ref} [W/m²/sr/μm] and the SNR required at it; Table 4 gives the SNR measured on Sentinel-2C. Five bands are modeled — four VNIR silicon bands and one SWIR MCT band:

Band	λ center [nm]	FWHM [nm]	L_{ref} [W/m ² /sr/μm]	SNR required [-]	SNR measured, S2C [-]
B2	492.7	64	128.00	154	162
B3	559.8	35	128.00	168	—
B4	664.6	30	108.00	142	175
B8	832.8	118	103.00	174	—
B11	1613.7	88	4.00	100	133

Source: the Sentinel-2 MSI source-data dossier, which carries the full instrument table with per-row confidence classes.

14.2.3 How the scenario models it

Each band is one config file in `scenarios/09_flagship_missions/9.1_sentinel2_msi_snr/`. The scene is a flat two-point user-radiance spectrum equal to that band’s L_{ref} (`data/s2_msi*_lref.csv`), propagated through `atmosphere.model: exo`. The regime is **extended** — a uniform scene fills the pixel footprint, so the ensquared-energy factor is unity and never applied. TDI is modeled as a 2-line charge sum, which is what the published “1 TDI stage for 2 lines” means radiometrically: signal doubles, one read. The line time follows from GSD and ground-track speed, ≈ 1.50 ms for the 10 m bands. B11 differs in pixel pitch (15 μm) and in its QE assumption; nothing else changes between the five files.

14.2.4 The comparison

RADIANT central prediction against both published figures (walkthrough table, verbatim, with the ratios computed from it):

Band	RADIANT SNR [-]	ESA required [-]	RADIANT / required [-]	S2C measured [-]	RADIANT / measured [-]
B2	253	≥ 154	1.64	162	1.56
B3	208	≥ 168	1.24	—	—
B4	193	≥ 142	1.36	175	1.10
B8	337	≥ 174	1.94	—	—
B11	331	≥ 100	3.31	133	2.49

The assumption envelope and the shot-limit inversion, from `scripts/run_external_validation.py`:

Band	Envelope, SNR [-]	Collected signal [e-]	Implied $\eta\tau$ from measured [-]
B2	188.0–308.6	65,615	0.153
B3	157.2–251.5	44,847	—
B4	145.0–232.6	38,506	0.341
B8	253.7–430.2	115,183	—
B11	263.6–375.3	110,810	0.091

14.2.5 What the residuals mean

Every band clears its requirement, several by a wide margin, and the prediction sits *above* the measured value wherever a measured value exists. That direction is expected and is the honest reading of the model: RADIANT is computing a photon-limited SNR with generous throughput assumptions, and it models no pixel-response non-uniformity, no detector striping, no calibration-transfer noise, and no quantization of the on-ground processing chain — all of which are present in the flight number.

Of the three bands with a measured anchor, only **B4** has its measurement inside the declared envelope (175 sits comfortably in 145.0–232.6). B2’s 162 and B11’s 133 fall *below* their envelopes, and the dossier resolves both by inversion rather than by widening the envelope after the fact:

- **B2** implies $\eta\tau = 0.153$. At a band-average transmission near 0.6 that is a blue-end quantum efficiency of roughly 0.20 — low against the 0.50 assumed, but entirely ordinary for a front-illuminated silicon CMOS detector at 493 nm. The scenario’s verdict is **CONSISTENT** with the QE assumption named as the responsible input.

- **B11** implies $\eta\tau = 0.091$, which no plausible MCT quantum efficiency reaches. The unpublished parameter here is the SWIR integration time: the SWIR focal plane runs on its own timing, and a substantially shorter dwell than the 20 m-band line time would close the gap on its own. Because no source for that dwell was found, the scenario records B11 as **INCONCLUSIVE pending a sourced SWIR t_{int}** , not as agreement.

That distinction — CONSISTENT for the four bands whose residual maps onto a plausible value of a named parameter, INCONCLUSIVE for the one that does not — is the whole discipline of this chapter in one table.

14.3 Landsat 8 TIRS — thermal NEdT

14.3.1 Mission context

The Thermal Infrared Sensor on Landsat 8 and 9 is the canonical spaceborne LWIR noise-equivalent-temperature-difference benchmark: an f/1.64 four-element refractive telescope (178 mm focal length) at 705 km, imaging two thermal bands onto three 640×512 GaAs QWIP sensor chip assemblies at roughly 39 K, with an onboard variable- temperature blackbody for calibration. Its on-orbit NEdT was measured by the instrument team against that blackbody and published.

14.3.2 The published anchor

[Montanaro 2014b] — Montanaro, Levy & Markham, *On-Orbit Radiometric Performance of the Landsat 8 Thermal Infrared Sensor*, Remote Sensing 6(12) — Table 2:

Band	Scene T [K]	NEdT spec [K]	NEdT measured [K]
B10 (10.6–11.2 μm)	300	0.400	0.049
B11 (11.5–12.5 μm)	300	0.400	0.052

Provenance and the full instrument table, including the [Reuter 2015] optical prescription and the [Jhabvala 2011] focal-plane parameters: the Landsat TIRS source-data dossier.

14.3.3 How the scenario models it

A 300 K unit-emissivity blackbody through a vacuum path — the same view the published measurement is taken against. The optics are the published prescription (f/1.64, 178 mm, total transmission 0.49); the detector is the published 25 μm QWIP with the as-flown 3.49 ms integration time, 4×10^7 e-/s dark rate, and 260 e- RMS read noise.

One input is not published in a usable form and is the scenario's one inversion. The published conversion efficiency $CE = g\eta = 0.8\%$, taken literally, does not fill the well anywhere near the published saturation temperatures. Inverting instead on those saturation points — full well at 400 K for B10 and 370 K for B11 — gives $CE_{\text{eff}} = 1.64 \times 10^{-2}$ (B10, $\times 2.05$ the published value) and 1.36×10^{-2} (B11, $\times 1.70$). The two CE values bracket the answer, and the scenario reports the bracket rather than picking one silently.

14.3.4 The comparison

Walkthrough table, verbatim:

Band	RADIANT NEdT, this config [mK]	Bracket [mK]	Spec [mK]	Measured [mK]
B10	58	58–124	≤ 400	49
B11	52	52–89	≤ 400	52

The re-run reproduces these to the digit the walkthrough rounds to: 58.07 mK (B10) and 52.02 mK (B11), against collected signals of 1.62×10^6 e- and 2.90×10^6 e- respectively. The saturation-CE and raw-CE brackets at three scene temperatures, from the same run of `scripts/run_external_validation.py`:

Band	Scene T [K]	Raw-CE prediction [mK]	Saturation-CE prediction [mK]	Measured [mK]	Well fill [-]
B10	270	102.5–153.1	63.6–84.3	57	0.23
B10	300	90.3–124.1	58.0–71.3	49	0.35
B10	320	85.7–112.6	56.0–66.3	45	0.46
B11	270	76.3–103.4	55.0–68.6	60	0.32
B11	300	70.8–89.4	52.0–61.1	52	0.49
B11	320	68.9–83.9	51.0–58.3	51	0.62

(The inner width of each bracket is the read-noise envelope: 260 e- RMS for the ROIC typical value, up to 1033 e- RMS when the 1000 e- electronics spec ceiling is RSS'd in.)

14.3.5 What the residuals mean

B11's prediction lands on the flight value exactly: 52 mK predicted, 52 mK measured, at the same 300 K scene. B10 predicts 58 mK against 49 mK measured, an 18 % over-prediction of the noise — the model is *conservative*, which is the safe direction for a noise floor and the direction the missing conversion-efficiency detail would push.

Both bands sit at roughly one-eighth of the 400 mK specification, which is the more important structural check: a model that could not distinguish a 400 mK requirement from a 50 mK instrument would be useless for the trade studies the rest of this volume performs.

The scenario also cross-checks the raw signal against the instrument team's own published radiometric budget ([Jhabvala 2011]) and agrees within about 25 %. The remaining difference is the physically correct one: the published budget is dominated by calibration-**stability** terms — blackbody and optics temperature knowledge — which are not detector noise and are deliberately absent from a detector-noise floor. Verdict: CONSISTENT.

14.4 Aqua MODIS — thermal emissive bands

14.4.1 Mission context

MODIS is a 705 km whiskbroom scanner with a 17.78 cm aperture, a double-sided scan mirror completing a 1.4771 s revolution, 1354 frames per Earth scan, and separate focal planes at roughly 83 K for the short- and long-wave infrared. Aqua MODIS has a twenty-year on-orbit calibration record, which makes it the best-characterized thermal emissive band set in existence.

14.4.2 The published anchors

Two independent anchors, which is why this scenario validates two different claims.

NASA's MODIS specification pages ([SPEC]) publish each thermal band's **typical scene radiance** L_{typ} — itself the 300 K band-averaged Planck radiance — and its NEdT specification; [XIONG-2023] Table 5 publishes the Aqua measured NEdT:

Band	λ range [μm]	L_{typ} [$\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$]	NEdT spec [mK]	NEdT measured [mK]
20	3.660–3.840	0.45	50	20
29	8.400–8.700	9.58	50	20
31	10.780–11.280	9.55	50	20
32	11.770–12.270	8.94	50	30

Provenance: the MODIS TEB source-data dossier.

14.4.3 How the scenario models it

A 300 K blackbody through a vacuum path, with the published effective focal lengths (380.9 mm short-wave / 282.1 mm long-wave), the published detector sizes (540 μm and 400 μm), and the frame-derived integration times (323.3 μs ; band 29 is four on-board-averaged 73.3 μs samples). Optical transmission (0.40) and quantum efficiency (0.70) are assumptions carrying envelopes; detector noise is the *named unknown* the scenario exists to bound.

14.4.4 Claim 1 — the spectral chain against four published Planck anchors

Published L_{typ} against RADIANT's 300 K band-averaged Planck radiance (from a run of `scripts/run_external_validation.py`):

Band	Published [$\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$]	RADIANT [$\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$]	Difference [%]
20	0.45	0.450	−0.0
29	9.58	9.583	+0.0
31	9.55	9.555	+0.1
32	8.94	8.946	+0.1

Four independent bands spanning 3.7 μm to 12 μm agree to 0.1 % or better — which is the printed precision of the published column. This is the cleanest validation in the chapter, because it is a direct comparison

against published numbers with **no** assumption class involved: the Planck integral, the band edges, and the spectral grid are all that participate.

14.4.5 Claim 2 — the NEdT floor, and why it is a bound

Band	RADIANT photon/read floor [mK]	Spec [mK]	Measured [mK]	Implied detector noise σ_{det} [e-]
20	8.10–10.56	50	20	7.4×10^3
29	2.12–2.73	50	20	2.4×10^5
31	1.76–2.27	50	20	4.4×10^5
32	1.89–2.44	50	30	6.2×10^5

The band-20 chain value at the scenario’s central assumptions is ≈ 10 mK, reproduced on re-run as 10.27 mK. The floors for bands 29, 31, and 32 are computed pre-readout by `scripts/run_external_validation.py`, because those bands well-clip in the full chain at 300 K — a modeling limitation recorded in the scenario’s own `gaps.md`, not a physical statement about MODIS.

The required ordering holds in every band: **floor < measured < spec**. That is the signature of a detector-noise-limited instrument, and it is exactly what MODIS is — photoconductive HgCdTe with generation-recombination and 1/f noise in the short-wave bands, photovoltaic crosstalk in the long-wave ones. The measured 20–30 mK sits 10–40× above the photon floor.

The floor is therefore a **bound, not a prediction**, and the chapter says so rather than claiming a 2 mK model matches a 20 mK instrument. The useful output is the last column: the implied detector noise a detector model would have to reproduce to close the gap. That column is a specification for future work, written in the units the work would be done in.

14.5 Landsat 9 OLI-2 — nine-band SNR with a modeled coating train

14.5.1 Mission context

The Operational Land Imager 2 is a 135 mm four-mirror off-axis anastigmat pushbroom imager at 705 km, covering nine reflective bands from the 443 nm coastal-aerosol band to the 2.2 μm SWIR, with silicon PIN detectors for the VNIR and HgCdTe for the SWIR on a single 210 K focal plane, digitized to 14 bits. Requirements are published per band at a typical radiance L_{typ} , and on-orbit SNR has been characterized for both OLI and OLI-2.

14.5.2 The published anchor

Requirements from [Irons 2012] (*The next Landsat satellite: The Landsat Data Continuity Mission*, Remote Sensing of Environment 122); Landsat 8 measured values from [Morfitt 2015], recalled at $\pm 15\%$ and flagged as such in the scenario’s own gap list. Provenance: the Landsat OLI-2 source-data dossier.

14.5.3 How the scenario models it

This scenario is the one that exercises the most machinery, and it is worth reading for that reason alone even where the validation verdict is unsurprising.

The optical train is **not** a lumped `transmission_scalar`. It is six `optical_elements` rows: four protected-silver mirrors carrying a reflectance curve with $\varepsilon = 1 - R$ derived per Kirchhoff's law, an anti-reflection-coated focal-plane window, and the band's interference filter. Every curve is **synthetic** — no per-surface curves are published for OLI — generated by `scripts/gen_oli2_coatings.py` from vendor-typical anchors with the filter 50 % points pinned to the published band edges. The end-to-end band-average throughput this produces, for the red band, is $0.975^4 \times 0.985 \times 0.90 \approx 0.80$.

All nine bands live in **one** configuration set (`oli2_all_bands_study.yaml`). The shared instrument is stated once; what is configured is the band filter (as a configured element row — one complete element entry per configuration), the L_{typ} spectrum, the band edges, the quantum efficiency, the dark rate, and — for the panchromatic band alone — an 18 μm pitch and a 1.8 ms line time against 36 μm and 3.6 ms for the 30 m bands.

14.5.4 The comparison

Walkthrough table, verbatim:

Band	RADIANT SNR		L8 measured [-] (recalled ± 15 %)	Prediction / measured [-]
	[-]	Required [-]		
B1 Coastal aerosol	177	≥ 130	~230	0.77
B2 Blue	496	≥ 130	~360	1.38
B3 Green	472	≥ 100	~300	1.57
B4 Red	341	≥ 90	~225	1.51
B5 NIR	238	≥ 90	~200	1.19
B6 SWIR1	328	≥ 100	~265	1.24
B7 SWIR2	367	≥ 100	~180	2.04
B8 Pan	227	≥ 80	~145	1.56
B9 Cirrus	139	≥ 50	~165	0.84

Collected signals run from 3.95×10^4 e- (B9) to 2.82×10^5 e- (B2). Every band is shot-noise dominated: the 200 e- RMS read noise contributes ≤ 6 % of the total noise, and dark charge stays under 180 e- per frame even in the SWIR.

14.5.5 What the residuals mean

Verdict: CONSISTENT. Every prediction clears its requirement, and every prediction-to-flight ratio lands in the 0.8–2.0 $\times$ envelope that scenario 9.1 established for assumption-class $\eta\tau$ — with B7 at 2.04 sitting on the boundary.

The two bands that land *below* flight are the informative ones. B1 (16 nm wide) and B9 (21 nm wide) are the two narrow bands, where the band-edge roll-off of the synthetic super-Gaussian filter eats a larger fraction of the in-band integral than it does in a wide band, and where the assumed quantum efficiency (0.70 at 443 nm; 0.80 for 1.37 μm MCT) was deliberately set conservative. The flight instrument evidently does better than

the synthetic coating model. The recalled measured values themselves carry $\pm 15\%$, which covers a good part of both gaps.

14.5.6 The structural check — configuration set against standalone files

The nine per-band standalone config files remain in the folder as cross-checks, and `scripts/check_all_bands_parity.py` asserts that the configuration set reproduces each one. Walkthrough table, verbatim:

Band	Study SNR [-]	Standalone SNR [-]	Relative difference [-]
B1_CA	176.566166598	176.566166598	0.00e+00
B2_Blue	496.146430356	496.146430356	0.00e+00
B3_Green	472.394526650	472.394526650	0.00e+00
B4_Red	340.629354646	340.629354646	0.00e+00
B5_NIR	237.635101135	237.635101135	0.00e+00
B6_SWIR1	327.616523306	327.616523306	0.00e+00
B7_SWIR2	367.385730198	367.385730198	0.00e+00
B8_Pan	226.565673500	226.565673500	0.00e+00
B9_Cirrus	138.894479928	138.894479928	0.00e+00

The re-run reproduces the study-versus-standalone verdict exactly — all nine bands bit-identical, every relative difference 0.00e+00, against an acceptance bar of 10^{-9} . Two of the *absolute* values have moved in their sixth significant figure since the walkthrough was written (B6_SWIR1 327.616523306 → 327.616524064, a relative move of 2×10^{-9} ; B7_SWIR2 367.385730198 → 367.387242489, a relative move of 4.1×10^{-6}), which is a re-baselining of the two SWIR bands somewhere upstream, far below the precision of any published comparison in the table above. The committed walkthrough is quoted as it stands, and the drift is recorded rather than absorbed.

This is not a physics result — the equality is expected *by construction*, because materializing configuration N reproduces standalone N’s effective document field for field, so both sides drive the same deterministic chain. It is a regression guard on the configuration-set machinery, and it is the kind of check that catches a configured-row bug the physics tests would not. It has caught one already: the study once shared a single composite filter element whose union carried both strips’ roll-off across the 1 nm B1/B2 seam, giving $\leq 0.5\%$ differences in those two bands. Per-band filter entries removed the artifact.

14.6 What the four flagships share

Read together, the four comparisons establish a narrower claim than “RADIANT is validated”, and it is worth stating the narrow claim precisely:

- **Extended-scene radiometry, aperture to electrons, is right to within the assumption envelope of the unpublished instrument parameters.** Four instruments, twenty bands, spanning 443 nm to 12.3 μm , silicon and MCT and QWIP, pushbroom and whiskbroom, SNR and NEdT.
- **The Planck integral and the band-averaging are right to 0.1 %** against four independently published radiance anchors (the MODIS section, claim 1).

- **The noise stack is right to roughly 20 % or better where the instrument is photon-and-read-limited** (the TIRS section: 52 mK against 52 mK, 58 mK against 49 mK) and correctly identifies itself as a *bound* where the instrument is detector-limited (the MODIS section, claim 2).
- **The direction of every residual is explained**, and the explanations are all of one kind: RADIANT computes an idealized photon-limited performance, so it over-predicts flight SNR wherever the flight instrument carries non-uniformity, striping, or calibration-transfer noise that the model does not represent.

Every one of these comparisons runs on a **vacuum path**. None of them validates the atmosphere, the scene model, the spatial path, or the calibration model. That is what the next two sections and *What validation does not yet cover* are for.

14.7 Atmosphere — parity against a MODTRAN 6 run set

The atmosphere backends are measured against a MODTRAN 6 run matrix rather than against field radiometry. The full record — every table, every test that pins it, and a seventeen-entry limitations register — is the MODTRAN-parity validation record. This section quotes its headline numbers; it does not replace it.

The run set. 132 authored rows, of which **130 tape7 runs are delivered**, plus four ground-level flux sidecars = 134 artifacts. They span seventeen blocks: the six standard profiles at nadir, a zenith fan at 30°/45°/60°, partial columns, a visibility and water-vapor ladder, sky-irradiance runs, airborne and space sensor geometries, thermal downwelling at the 48.2° diffusivity angle, a 5 × 5 horizontal-path grid from 5 km to 100 km range, and near-horizon probes out to 89.5°. Two rows — the refraction on/off calibration pair — remain unrun.

Thermal path radiance. Band-mean model/MODTRAN ratios, before and after the escape-resolved emission-temperature model landed:

Anchor	Geometry	MWIR 3–5 μm , before → after [-]	LWIR 8–12 μm , before → after [-]
O3	ground ↔ 10 km, zenith 48.2°, down	2.013 → 1.141	1.326 → 1.055
O5	ground ↔ 100 km, zenith 48.2°, down	2.422 → 1.217	1.430 → 1.093
K1	ground ↔ 1 km, nadir, up	0.379 → 0.358	0.530 → 0.515
H5	ground ↔ 100 km, zenith 48.2°, up	1.041 → 0.721	1.263 → 1.074

Aggregate over the fourteen enforced anchors, LWIR: $\text{RMS } |\ln r|$ **0.3342 → 0.2611**, a 22 % improvement. The tall down-looking columns went from over-reading the MWIR by a factor of two to within about 20 %.

Effective emission temperature. Because MODTRAN reports τ and thermal path radiance separately, its own effective emission temperature is recoverable exactly, which isolates the temperature error from the opacity error:

Metric	Band	One-temperature [K]	Height-resolved [K]
$\max \Delta T $ over the anchor set	MWIR	25.2	10.4
$\max \Delta T $ over the anchor set	LWIR	23.2	9.5
$\text{RMS } \Delta T $	MWIR	9.5	4.3

Metric	Band	One-temperature [K]	Height-resolved [K]
RMS $ \Delta T $	LWIR	10.4	3.2

Transmittance. The gas and water calibration was fitted on one profile (US standard); the five non-calibration profiles — tropical, mid-latitude summer and winter, subarctic summer and winter — hold band-mean τ within ± 0.012 of MODTRAN in the water-relevant windows. Up-looking partial columns converge to within 2 % in the LWIR by 10–20 km, and are systematically too transparent for shallow columns (at 1 km, model 0.889 against MODTRAN 0.781 in the 8–12 μm band) because the calibration was fitted to full columns. Constant-altitude horizontal arms hold to ± 3.5 % in the LWIR across the 3–15 km altitude sweep.

Where the simple backend stops working. The analytic arm’s exponential law $\tau(2L) = \tau(L)^2$ is exactly where a correlated- k band model disagrees, and the disagreement is monotone with range. Down the 3 km row of the horizontal grid, model/MODTRAN:

Range	5 km	10 km	25 km	50 km	100 km
LWIR 8–12 μm [-]	1.03	1.01	0.95	0.87	0.82
MWIR 3–5 μm [-]	1.09	0.88	0.43	0.11	0.01

This is the documented reason long-range MWIR horizontal work needs the MODTRAN or interpolated backend rather than the simple one — and it is a good example of the register’s general posture: the number is published, with its trend, rather than the regime being quietly excluded.

The limitations register names its own dominant residual. The largest known atmosphere error is the *region-flat spectral shape* of the calibrated opacity model: there is no line structure inside the seventeen calibrated regions, which under-reads up-looking MWIR thermal radiance by 25–40 % on columns deeper than 5 km. Alongside it: the visible and near-infrared sky is a provisional single-scatter model that under-predicts the daytime visible sky by roughly $2\times$ near the horizon (and warns at runtime below 3 μm); refraction is unmodeled, worth about 0.5° of lift near the horizon; and the visible band’s opacity is right in total but mis-attributed between gas and aerosol, which matters to any product that separates the two.

14.8 MWIR single-wavelength ground truth

The oldest anchor in the codebase is also the simplest: one MWIR case worked by hand from CODATA 2018 constants to an SNR, with every intermediate value written down. The single-wave ground-truth record is the full derivation; `examples/ground_truth_mwir.yaml` is the configuration, and `tests/integration/test_ground_truth_mwir.py` asserts the chain against it.

The configuration is deliberately stripped of everything that could hide an error: a 300 K, $\varepsilon = 1.0$ target at 4.0 μm on a three-point grid [3.999, 4.000, 4.001] μm , a vacuum atmosphere, a 0.30 m aperture at $f/5$, 0.75 optical transmission, a 15 μm pixel, 0.70 quantum efficiency, **zero** dark current, 5.0 ms integration, 30 e-RMS read noise, unity gain, 16-bit conversion.

The chain, step by step:

Step	Quantity	Hand value	Unit
1	Planck radiance $B(4.0 \mu m, 300 K)$	7.2197642257e-01	W/m ² /sr/μm
2	At-aperture radiance (vacuum)	7.2197642257e-01	W/m ² /sr/μm
3	After optics, $\times 0.75$	5.4148231693e-01	W/m ² /sr/μm
4	Collecting area $\frac{\pi}{4} D^2$	7.0685834706e-02	m ²
5	Pixel solid angle p^2/f^2	1.0000e-10	sr
6	Photon rate	7.7073e+07	photon/s/pixel/μm
7	Electron rate, $\times \eta$	5.3951e+07	e-/s/pixel/μm
8	Signal over the band, $\times t_{int}$	539.508	e-
9	Shot noise $\sqrt{S}$	23.23	e- RMS
10–12	Dark / read / quantization noise	0.0 / 30.0 / 0.2887	e- RMS
13	Total noise (RSS)	37.94	e- RMS
14	SNR	14.22	–

The chain reproduces step 8 as 539.50846835 e- against the constant-radiance hand value 539.50809863 e-, a relative difference of 6.9×10^{-7} . That single number is the entire error budget: it is the trapezoid rule capturing the curvature of the Planck function across a 0.002 μm band, and it is three orders of magnitude inside the test's 10^{-4} tolerance. Every other step is exact to float64 — the constants are CODATA 2018 exact values, the vacuum atmosphere is the identity, and the geometry is two closed-form expressions.

What this anchor validates is the **arithmetic and unit handling** of the radiometric core: solid angle, collecting area, the photon-energy conversion λ/hc , the spectral integral, and the noise RSS. What it does not validate is any *modeling choice* — there is no atmosphere, no background, no spatial path, no real detector physics in it. It is the floor of the validation stack, not its ceiling.

14.9 What validation does not yet cover

RADIANT's validated core is **extended-scene, vacuum-path radiometry from aperture to electrons, plus the photon-and-read noise stack**: four flight instruments, twenty bands, 443 nm to 12.3 μm, agreeing within the declared envelopes of their unpublished parameters, on top of a hand-calculated arithmetic anchor good to 7×10^{-7} .

Outside that core the evidence thins in five specific ways, and none of them is closed by the chapter above.

1. **Nothing here validates the spatial path against measurement.** No flight or laboratory MTF, ensquared-energy, or NIIRS figure is anchored anywhere in the repository. The PSF/MTF dual-path consistency check proves the two spatial paths agree with *each other* to 2e-2, which is an internal-consistency guarantee and not an external one, and the one scenario that compares predicted against “measured” MTF uses a synthetic measurement.
2. **The atmosphere is validated against MODTRAN, not against the sky.** It is a model-to-model comparison over one deck set. Its residuals are large where the register says they are large — 25–40 % on deep up-looking MWIR columns, a factor of two on the near-horizon daytime visible sky, an order of magnitude on 100 km MWIR horizontal arms — and no geometry outside the delivered run matrix is measured at all.

3. **The flagship comparisons deliberately bypass the atmosphere and the scene model.** All four run `atmosphere.model: exo` against a stated at-aperture radiance, so there is no end-to-end validated case in which a scene model, an atmosphere, and a sensor model are simultaneously under test against a published number.
4. **The point-source and sub-pixel regimes have no external anchor.** Every flagship is extended-scene; detection range, the Johnson criteria, and the ensquared-energy coupling that governs sub-pixel targets are exercised by scenarios but validated against no published instrument.
5. **The detector and calibration models are datasheet-anchored, not measurement-anchored.** 1/f and generation-recombination noise, inter-pixel capacitance, persistence, well saturation, and the whole post-NUC calibration residual model rest on vendor datasheet values and internal consistency. The MODIS section above quantifies exactly how much that matters: MODIS's measured NEdT sits 10–40× above the photon floor RADIANT can currently compute, and the detector-noise term that accounts for the difference is named as an unknown rather than modeled.

A reader deciding whether to trust a RADIANT number should ask which of those five categories it falls in before asking how many digits it printed.

Three further boundaries are worth stating in the same spirit, because they are properties of the *evidence*, not of the model:

- **Recalled and assumption-class inputs are present in every flagship.** Landsat 8's measured OLI SNRs are recalled at $\pm 15\%$. Quantum efficiencies, optical transmissions, and one integration time are assumptions with envelopes rather than citations. A comparison that agrees inside such an envelope is a consistency test, not a calibration.
- **The comparisons are single-point, not distributional.** Each band is compared at one scene radiance or one scene temperature (TIRS at three temperatures is the exception). Nothing here tests how the residual behaves across the full dynamic range of any instrument.
- **Scenario numbers carry a vintage.** RADIANT's physics has been corrected repeatedly, each correction re-baselining affected walkthroughs. The four flagship walkthroughs were re-run for this chapter and reproduce; the rest of the volume quotes committed values at their stated vintage, and the scenario's own `walkthrough.md` is always the authority.

None of this argues against using the tool. It argues for reading the validation register before quoting a number outside the validated core — and, where a decision turns on a number outside it, for adding the anchor that would bring it inside.

Trade-Study Cookbook

A trade study is a question of the form “*how does this number move when I change that one?*” — and RADIANT answers six distinct shapes of that question with six distinct tools. Choosing the wrong one is the most common way to spend an afternoon producing a plot nobody can act on.

This chapter is the recipe card for all six. Each entry says **when to reach for it**, gives a short scripted recipe with its **real output**, names the **GUI route** where one exists, and points at the case study earlier in this volume that uses it in anger. It is deliberately not the reference: the project’s Trade Studies Guide holds the full API surface, the configuration-set model, and the result-interpretation notes, and this chapter does not repeat them.

15.1 The worked configuration, and three habits

Every recipe below runs against `examples/mwir_leo_minimal.yaml` — a 0.30 m f/4 telescope at 8 km altitude looking straight down at a 300 K, $\varepsilon = 0.95$ extended scene through a mid-latitude-summer atmosphere, on an 18 μm -pitch focal plane with a 2 Me- well. Every output block is what the recipe actually printed, run from the repository root as `PYTHONPATH=src python <script>.py`.

Three habits apply to all six, and each of them has cost someone a re-run:

- **Declare the envelope before you run.** A tolerance, a threshold, or a pass band chosen *after* the numbers are on screen is not an analysis result. Put `TOLERANCE_PCT`, the SNR floor, or the acceptable NEDT in the script as a named constant at the top.
- **Read the warnings.** Every trade tool in RADIANT evaluates the full chain, and the chain warns rather than silently clipping. A sweep whose curve flattens may be showing you diminishing returns — or a saturated well, which is a different conclusion entirely. Recipe 1 walks straight into that case on purpose.
- **Nothing mutates your session.** `sweep`, `sweep_2d`, `solve_for`, `monte_carlo` and `sensitivity` all evaluate against copies of the parameter set, and the GUI runs its dialogs against a Sensor clone. You can trade freely from a configured session without disturbing it.

15.2 Recipe 1 — The 1-D sweep

15.2.1 When to reach for it

One continuous axis, one metric, and you want the **shape** of the relationship: where the knee is, where the requirement is crossed, whether the curve is linear or square- root. If you find yourself naming individual

sweep points, you want a configuration set instead (see the Trade Studies Guide); if you want the crossing point rather than the curve, skip to Recipe 3.

15.2.2 The recipe

```
import numpy as np
from radiant.api import Sensor

sensor = Sensor.from_yaml("examples/mwir_leo_minimal.yaml")

sweep = sensor.sweep(
    "optics.aperture_diameter_m",      # dot-path, canonical unit (m)
    np.linspace(0.10, 0.60, 6),        # the axis values [m]
    metric="snr",                      # a metrics key, or a callable
)

for diameter_m, snr in zip(sweep.values, sweep.metric_values, strict=True):
    print(f" D = {diameter_m:.2f} m   SNR = {snr:8.2f} [-]")

hit = sweep.at_metric_threshold(1000.0) # first point at or above the threshold
print(f"first point with SNR >= 1000: {hit}")
sweep.to_csv("aperture_sweep.csv")      # tidy per-point export

D = 0.10 m   SNR =   374.54 [-]
D = 0.20 m   SNR =   749.31 [-]
D = 0.30 m   SNR =  1124.03 [-]
D = 0.40 m   SNR =  1414.17 [-]
D = 0.50 m   SNR =  1414.17 [-]
D = 0.60 m   SNR =  1414.17 [-]
first point with SNR >= 1000: (0.30000000000000004, 1124.0273378815184)
```

Read the curve before believing it. The first three points scale exactly linearly with diameter — which is the shot-noise limit’s signature, since signal grows as D^2 and shot noise as D . The last three are identical to the digit, and *that* is not physics: the run warns

```
pixel saturated: signal 5.054e+06 e- clipped to 2.000e+06 e- – the well
capacity remaining after the non-signal pedestal
```

The flat top is the 2 Me- well filling, not an aperture knee. A sweep that crosses a saturation boundary is answering a different question on either side of it; either shorten the integration time along with the aperture, or stop the axis before the clip.

Useful extras on SweepResult: param_name and metric_name (what was swept, what was measured), results (the full ChainResult per point, unless you passed keep_results=False), sweep[metric_key] to pull a *second* metric out of the retained results without re-running, and n_workers=N on the call for parallel evaluation.

15.2.3 From the GUI

Run → Run Sweep... is the same call behind a dialog. Pick the parameter from the schema-driven list (every float ParameterDef), type Start / Stop / Points **in the parameter’s own input unit** — the chip beside the field shows that unit and the session’s current value — optionally tick *Log spacing*, pick the metric, and Run.

The range seeds itself at $\times 0.5 / \times 1.5$ of the current value, clamped to the schema bounds, so an opened dialog is already bracketing where you are. The curve plots in the units you typed; a progress bar tracks done/total and Cancel aborts cleanly (there are no partial results on cancel, by API contract, and the dialog says so). **Copy as script** puts a runnable reproduction of the sweep on the clipboard — the canonical values that actually ran — which is the intended graduation path from dialog to script. **File → Export Sweep CSV...** writes the retained result.

15.2.4 Seen in anger

The **MWIR maritime surveillance** case study sweeps aperture from 0.15 m to 0.45 m at fixed f/2.5 and reports the result that matters: SNR does not move at all while NIIRS climbs from 3.53 to 5.13. Aperture bought resolution, not signal — the kind of finding that only a swept curve makes obvious. The **wavefront-error budget** case study sweeps $0 \rightarrow 0.25$ waves RMS over thirteen points and reads the first grid point past each degradation threshold, with a note on the difference between sweep resolution and interpolated crossing.

15.3 Recipe 2 — The 2-D sweep

15.3.1 When to reach for it

Two continuous axes whose **interaction** is the question. Use it when you suspect the optimum in one parameter depends on the other — aperture against integration time, pitch against focal length — and a pair of 1-D sweeps would mislead by holding the wrong thing fixed. Two coarse axes beat one fine axis: an 8×4 grid is 32 evaluations and shows you the whole surface.

15.3.2 The recipe

```
from radiant.api import Sensor

sensor = Sensor.from_yaml("examples/mwir_leo_minimal.yaml")

grid = sensor.sweep_2d(
    "optics.aperture_diameter_m", [0.20, 0.30, 0.40],      # [m]
    "spectral_integration.integration_time_s", [0.001, 0.005], # [s]
    metric="snr",
)

print(grid.grid.shape)      # (3, 2) — axis 1 varies down the rows
print(grid.grid.round(1))
grid.to_csv("aperture_dwell_grid.csv")

(3, 2)
[[ 335.   749.3]
 [ 502.6 1124. ]
 [ 670.2 1414.2]]
```

The grid is indexed [i1, i2] — first parameter down the rows, second across the columns — and to_csv writes it in long form with both axis values per cell, which is the shape a plotting library or a spreadsheet wants.

Note the bottom-right cell: 0.40 m at 5 ms is the saturated corner from Recipe 1, so the surface's apparent plateau is a well limit rather than an optimum. On a real 2-D study, plot the saturation mask alongside the metric.

15.3.3 From the GUI

The same **Run** → **Run Sweep...** dialog. Tick **Second parameter (2-D grid)** and a second parameter/Start/Stop/Points block appears, each with its own unit chip and log-spacing option. The result renders as a heatmap with both axes and the color bar unit-suffixed in the units you typed. Progress, cancel, **Copy as script** and CSV export behave exactly as in the 1-D case.

15.3.4 Seen in anger

No tier-1 case study runs a `sweep_2d` — which is itself informative. When the two axes are *discrete and labeled* (three atmospheres × six targets) the right tool is Recipe 6's batch matrix, and that is what the **target detection matrix** case study uses. The 2-D sweep earns its place when both axes are continuous and the answer is a surface, not a table of named cells.

15.4 Recipe 3 — Solve for a parameter

15.4.1 When to reach for it

You already know the answer you need — SNR = 50, NEDT = 30 mK, NIIRS = 4.5 — and want the parameter value that produces it. A sweep can only tell you which of its grid points crossed; a solve gives you the crossing itself, usually in fewer evaluations than a coarse sweep would take.

15.4.2 The recipe

```
from radiant.api import Sensor
from radiant.api.solve import SolveBracketError

sensor = Sensor.from_yaml("examples/mwir_leo_minimal.yaml")

try:
    solution = sensor.solve_for(
        "optics.aperture_diameter_m",  # the free parameter
        250.0,                        # the target metric value [-]
        bounds=(0.05, 1.0),           # bracket, in the parameter's input unit [m]
        metric="snr",
    )
except SolveBracketError as exc:
    print(f"target not bracketed: {exc}")
else:
    print(f"{solution.param_name} = {solution.solution:.5f} m")
    print(f"achieved {solution.metric_name} = {solution.achieved:.3f} [-] "
          f"in {solution.n_evaluations} evaluations")
```

```
optics.aperture_diameter_m = 0.06678 m
achieved snr = 250.000 [-] in 9 evaluations
```

Three things to know. The solve is **Brent root-finding on the forward model**, so the metric must be monotone-ish and *actually vary* over the bracket: a plateaued metric cannot be bracketed, and the error says so. `SolveBracketError` reports **both endpoint metric values**, which is usually enough to see whether you bracketed the wrong side or the wrong parameter. And `solution.result` carries the full `ChainResult` at the solution, so you can check the rest of the chain — saturation, MTF, noise split — at the point the solver landed on, rather than trusting one metric in isolation.

15.4.3 From the GUI

Tools → Solve for Parameter.... Pick the free parameter, the target metric and its value, and the bracket in the parameter’s input unit. The Brent iteration runs on a worker thread against a **clone**, so the session is untouched until you decide otherwise; success reports the solution with its unit, the achieved metric, and the evaluation count, and offers **Apply solution** — one `sensor.set` on the live sensor. A non-bracketing target surfaces the API’s actionable error, both endpoint values included, inline in the dialog.

15.4.4 Seen in anger

No tier-1 case study drives the solver — the closest is scenario 7.4’s cold-stop study, whose walkthrough records that the inverse-solver gap it was written against was closed by `Sensor.solve_for`, and whose own inversion parameter has since been removed. The live demonstrations are scenario 7.2 (radiometric calibration verification) and the solver’s own unit tests. In practice the solver’s most valuable use is exactly the one this recipe shows: converting a requirement into a design value without building a sweep around it first.

15.5 Recipe 4 — Monte Carlo tolerance analysis

15.5.1 When to reach for it

Your inputs are not single values but distributions — a coating transmission with a $\pm 5\%$ spread, a QE that varies across a lot, an aperture with a manufacturing tolerance — and the question is not “what is the SNR” but “**what fraction of built units meet the requirement**”. Reach for this when a specification has to survive manufacturing, not just design.

15.5.2 The recipe

```
from radiant.api import Sensor

REQUIRED_SNR = 1050.0          # declare the bar BEFORE the run

sensor = Sensor.from_yaml("examples/mwir_leo_minimal.yaml")
sensor.set_tolerance("optics.aperture_diameter_m", "gaussian", std_fraction=0.02)
sensor.set_tolerance("detector.qe_value", "gaussian", std=0.03)

mc = sensor.monte_carlo(n_trials=64, seed=42)
```

```

print(f"mean    {mc.mean('snr'):.2f} [-]")
print(f"std     {mc.std('snr'):.2f} [-]")
print(f"5th percentile {mc.percentile('snr', 5):.2f} [-]")
p_ok = mc.probability_of_exceeding("snr", REQUIRED_SNR)
print(f"P(SNR >= {REQUIRED_SNR:.0f}) = {p_ok:.2f}")
corr = {k: round(v, 2) for k, v in mc.correlation("snr").items()}
print("correlation with inputs:", corr)
mc.to_csv("mc_trials.csv")      # one row per trial

mean    1122.04 [-]
std     27.99 [-]
5th percentile  1071.41 [-]
P(SNR >= 1050) = 0.98
correlation with inputs: {'detector.qe_value': 0.75, 'optics.aperture_diameter_m': 0.76}

```

Four distributions are available — gaussian, uniform, truncated_gaussian, log_normal — and the keyword arguments are the distribution's own (std, std_fraction, mean, and so on). **The seed is part of the result:** the same seed and trial count reproduce the run exactly, which is what makes an MC number quotable in a review. Sixty-four trials is a demonstration size; a percentile you intend to defend wants several hundred, and the cost is linear.

The correlation column is the part people under-use. Here the two tolerated inputs land at 0.75 and 0.76 — i.e. neither dominates, and tightening only one of them will not move the distribution much. When one input correlates at 0.9 and the rest at 0.2, you have found the tolerance worth paying to tighten.

15.5.3 From the GUI

There is no Monte Carlo dialog. **Run → Monte Carlo...** opens the scripting window with a **prefilled scaffold**: the tolerances already set on the session appear as comments at the top (or, if none are set, a one-line example of how to set one), followed by the monte_carlo call, the summary statistics, and a commented to_csv export. That is a deliberate design choice — a tolerance study has too many defensible shapes to put behind a form, so the GUI hands you the script and gets out of the way. Tolerances set through the GUI's ± badges persist into the config file's `_radiant.tolerances` block, so a scaffolded script picks up the session's declared spreads.

15.5.4 Seen in anger

examples/scripts/tolerance_analysis.py, walked in the **Scripting Examples** chapter of this volume: three tolerated parameters, 50 trials, and the statistics printed with units. No tier-1 case study runs a Monte Carlo — the eight were chosen for breadth of persona, and tolerancing is a specialist's tool that a systems engineer reaches for once a program rather than weekly.

15.6 Recipe 5 — Sensitivity ranking

15.6.1 When to reach for it

Before any of the above. You have twenty parameters and no idea which three matter, and you want a defensible ordering to spend the rest of the study on. A one-at-a-time sensitivity is cheap — two evaluations per parameter — and it produces exactly that ordering.

15.6.2 The recipe

```
from radiant.api import Sensor

sensor = Sensor.from_yaml("examples/mwir_leo_minimal.yaml")

ranking = sensor.sensitivity(
    metric="snr",
    param_names=[
        "optics.aperture_diameter_m",
        "detector.qe_value",
        "spectral_integration.integration_time_s",
        "optics.transmission_scalar",
    ],
    delta_fraction=0.01,      # ±1 % perturbation
)

for entry in sorted(ranking.entries, key=lambda e: abs(e.sensitivity), reverse=True):
    print(f"  {entry.param_name:45s} {entry.sensitivity:+.3f}")
print(f"nominal {ranking.metric_name} = {ranking.entries[0].metric_nominal:.2f} [-]")

    optics.aperture_diameter_m          +1.000
    detector.qe_value                    +0.500
    optics.transmission_scalar            +0.500
    spectral_integration.integration_time_s +0.500
nominal snr = 1124.03 [-]
```

The sensitivity is normalized: a value of 2.0 means a 1 % change in the parameter produces a 2 % change in the metric, and a negative value means the metric falls as the parameter rises. It is therefore directly comparable across parameters with wildly different units, which is the whole point.

This particular ranking is also a validation of the chain in miniature. In the shot-noise limit $\text{SNR} \propto D\sqrt{\eta\tau t_{\text{int}}}$ — aperture with exponent 1, and quantum efficiency, transmission and integration time each with exponent $\frac{1}{2}$. The numerical ranking recovers 1.000, 0.500, 0.500, 0.500 exactly. When a sensitivity comes back at an exponent you cannot explain from the physics, that is worth a second look before it is worth a design decision.

Each SensitivityEntry also carries nominal_value, metric_nominal, metric_plus and metric_minus, so the two-sided behavior is inspectable — useful where a parameter is near a boundary and the response is asymmetric. Omit param_names entirely and the analysis runs over the tolerated parameters, or over all float parameters if none are tolerated.

15.6.3 From the GUI

None. Sensitivity is a scripting-only surface; run it from the scripting window (**Tools → Scripting Window**, Ctrl+Shift+P, or Command+Shift+P on macOS), where sensor is already bound to the loaded session.

15.6.4 Seen in anger

Scenario 6.3 (noise-model verification) closes by ranking parametric sensitivities against hand calculations. Among the tier-1 case studies, the **InSb versus HgCdTe** shootout does the same job structurally — it establishes which noise term dominates before arguing about detector choice, which is the sensitivity question asked in the noise budget’s own language.

15.7 Recipe 6 — The batch matrix

15.7.1 When to reach for it

Two or more **discrete, labeled** axes whose cells you want in a table with the labels attached: three sensors × four targets × two atmospheres. Unlike a sweep, each cell can change several parameters at once, the axis labels survive into the output, and a cell that fails is *recorded* rather than aborting the run — which matters when a corner of the matrix is physically invalid.

15.7.2 The recipe

```
from radiant.api import Sensor
from radiant.api.batch import BatchRunner

CFG = "examples/mwir_leo_minimal.yaml"

axes = [
    ("aperture", {"25 cm": {"optics.aperture_diameter_m": 0.25},
                  "35 cm": {"optics.aperture_diameter_m": 0.35}}),
    ("dwell", {"2 ms": {"spectral_integration.integration_time_s": 0.002},
               "5 ms": {"spectral_integration.integration_time_s": 0.005}}),
]

def evaluate(sensor, labels):
    """One cell: the sensor arrives with this cell's overrides applied."""
    result = sensor.evaluate()
    return {"snr": result.metrics["snr"], "nedt_mK": 1e3 * result.metrics["nedt_K"]}

runner = BatchRunner(
    base_config={},  # unused here – the factory loads the YAML
    axes=axes,
    sensor_factory=lambda cfg: Sensor.from_yaml(CFG),
)
batch = runner.run(evaluate)

print("failed cells:", batch.n_failed)
```

```
for row in batch.rows:
    print(f" {row['aperture']:>6} x {row['dwell']:>5}: "
          f"SNR {row['snr']:.1f} [-]   NEdT {row['nedt_mK']:.1f} [mK]")
```

failed cells: 0

```
25 cm x 2 ms: SNR    592.3 [-]   NEdT    47.4 [mK]
25 cm x 5 ms: SNR    936.7 [-]   NEdT    30.0 [mK]
35 cm x 2 ms: SNR    829.3 [-]   NEdT    33.8 [mK]
35 cm x 5 ms: SNR   1311.4 [-]   NEdT    21.4 [mK]
```

`batch.pivot("snr", rows="aperture", cols="dwell")` turns the same rows into a briefing table — `{row_label: {col_label: value}}`, with failed cells as `None` — which is one call away from a rendered matrix. The last axis varies fastest; axis names become result columns, so "error" is reserved.

The two structural properties worth designing around: the `evaluate` callback returns **whatever columns you want** (a derived metric, a detection range, a pass/fail flag — not just chain metrics), and a `RadiantError` inside a cell is captured into that row's error column so the batch completes. A matrix that silently dropped its invalid cells would misreport coverage; this one reports `n_failed` and keeps the row.

15.7.3 From the GUI

Run → Batch Run..., like Monte Carlo, opens the scripting window with a prefilled `BatchRunner` skeleton — two labeled axes, an `evaluate` function, the `run`, a `pivot` and the failed-cell count — for the same reason: the shape of a batch is the analysis, and it belongs in a script.

15.7.4 Seen in anger

The **target detection matrix** case study is this recipe at full scale: twelve targets × four atmospheres × three sensors — 144 cells — with a bisection search for detection range inside each cell's `evaluate`, early exits at nadir and at the swath edge, and `pivot` used to build the briefing tables directly. Read it for how much analysis can live inside the cell callback.

15.8 Choosing between them

Your question	Reach for	Chapter
"What does the curve look like?"	1-D sweep	Recipe 1
"Do these two parameters interact?"	2-D sweep	Recipe 2
"What value hits my requirement?"	<code>solve_for</code>	Recipe 3
"What fraction of units will pass?"	Monte Carlo	Recipe 4
"Which parameters matter at all?"	Sensitivity ranking	Recipe 5
"Fill in this table of named cases."	<code>BatchRunner</code>	Recipe 6
"Compare a handful of named designs."	Configuration set	Trade Studies Guide

Two boundaries are worth restating because they are the ones people cross by accident. A **configuration set** is for up to twelve *named* designs that differ in several parameters each, save as one document, and appear side by side in the GUI's comparison view — if you are naming sweep points, that is what you want. A

batch matrix is for the Cartesian product of labeled axes evaluated into a tidy table — if you are adding a thirteenth configuration to trace a trend, you want a sweep instead.

And whichever you reach for: the numbers it prints are only as good as the configuration underneath. The flagship-validation chapter says exactly how far that underneath has been checked.

Scenario Index

Fifty-one scenario folders live under `scenarios/` in the RADIANT repository. This appendix indexes every one of them, in catalog order, with the place in this volume where it is covered.

Coverage is total, at tiered depth: eight scenarios are walked at full depth as case studies, four are the subject of the flagship-validation chapter, and the remaining thirty-nine appear as digests in the compendium. Nothing in the repository is unrepresented here.

A.1 How to read the table

- **Scenario** — the folder name’s numeric identifier. Two folders share the identifier 2.7; they are distinguished in the Title column and in the digest compendium, and the identifier collision is a property of the tree, not a typo.
- **Persona** — the catalog persona the folder belongs to. Folders 8, 9 and 10 are not personas: 8 is a set of tool demonstrations, 9 is the external-validation suite, and 10 is the direction-general acceptance suite for the geometry work.
- **GUI** — *yes* when the folder ships a GUI-openable baseline. For personas 1–8 and 10 that is an `inputs/<slug>.gui.yaml` opened through **File → Open YAML**; for the flagship folders it is the per-band config files at the folder root, which open the same way (marked *configs*).
- **Runner** — *yes* when the folder ships an executable program under `scripts/` (`PYTHONPATH=src python scenarios/<...>/scripts/<name>.py`). A folder without one is driven from the GUI or from `radiant run`.
- **Covered in** — where in this volume to read about it. *Case study* names the full-depth chapter; *Validation* names the mission’s section of the flagship-mission validation chapter; *Digest* is the entry in the scenario digest compendium, filed under the same identifier.

Every folder, without exception, carries a `walkthrough.md` (the narrative and the committed results), a `gui_workflow.md` (the GUI route and its requirements), and a `gaps.md` (what the scenario could not do, and why). Those three files are the authority for their scenario; this volume’s chapters quote them.

Scenario	Title	Persona	GUI	Runner	Covered in
1.1	MWIR Maritime Surveillance Trade Study	Sarah, systems engineer	yes	yes	Case study — MWIR Maritime Surveillance
1.2	VNIR Pan Imager: GSD vs Aperture vs Altitude	Sarah, systems engineer	yes	yes	Digest 1.2
1.3	Dual-Band MWIR/LWIR Wildfire Detection	Sarah, systems engineer	yes	yes	Digest 1.3
1.4	TDI Pushbroom Optimization: Line Rate vs SNR	Sarah, systems engineer	yes	yes	Digest 1.4

Scenario	Title	Persona	GUI	Runner	Covered in
1.5	Obscured Aperture and Spider Vanes	Sarah, systems engineer	yes	yes	Digest 1.5
1.6	MWIR Point-Source Space Domain Awareness	Sarah, systems engineer	no	yes	Digest 1.6
2.1	InSb vs HgCdTe Noise Budget Shootout at 77 K	Mike, detector engineer	yes	yes	Case study — InSb versus HgCdTe on One Bench
2.2	1/f Noise Corner Frequency, LWIR Staring Array	Mike, detector engineer	yes	yes	Digest 2.2
2.3	IPC Impact on MTF	Mike, detector engineer	yes	yes	Digest 2.3
2.4	Persistence Characterization: Bright-Source Recovery	Mike, detector engineer	no	yes	Digest 2.4
2.5	Well Capacity Optimization: Integration Time vs Dynamic Range	Mike, detector engineer	yes	yes	Digest 2.5
2.6	DROIC vs Analog ROIC: Single-Frame HDR	Mike, detector engineer	no	yes	Digest 2.6
2.7	Calibration-Limited NEDT: Two-Point NUC Floor	Mike, detector engineer	no	yes	Digest 2.7
2.8	Real-Part Quick Start: GeoSnap-18 by Name	Mike, detector engineer	no	yes	Digest 2.8
2.9	Up/Down Counting: In-Pixel Background Subtraction	Mike, detector engineer	no	yes	Digest 2.9
3.1	ISR Orbit Geometry and Pass Planning	Raj, mission planner	yes	yes	Case study — ISR Pass Planning
3.2	Weather Sensitivity: How Bad Can It Get?	Raj, mission planner	yes	yes	Digest 3.2
3.3	Multi-Sensor Comparison for Procurement	Raj, mission planner	yes	yes	Digest 3.3
3.4	Off-Nadir Performance Degradation	Raj, mission planner	yes	yes	Digest 3.4
3.5	Nighttime MWIR Imaging Feasibility	Raj, mission planner	yes	yes	Digest 3.5
4.1	Target Detection Matrix	Lisa, detection/targeting analyst	yes	yes	Case study — A Target Detection Matrix
4.2	Maritime Ship Classification (Johnson DRI)	Lisa, detection/targeting analyst	no	yes	Digest 4.2
4.3	Camouflage Effectiveness Analysis	Lisa, detection/targeting analyst	yes	yes	Digest 4.3
4.4	Time-of-Day (Diurnal) Thermal Detectability	Lisa, detection/targeting analyst	yes	yes	Digest 4.4
4.5	Microbolometer UAV Altitude Trade (NETD-Specified)	Lisa, detection/targeting analyst	yes	yes	Digest 4.5
5.1	WFE Budget Allocation	Tom, optical designer	yes	yes	Case study — Allocating a Wavefront-Error Budget
5.2	Pixel Pitch and the Sampling Parameter Q	Tom, optical designer	yes	yes	Digest 5.2
5.3	Monochromatic vs Polychromatic PSF	Tom, optical designer	yes	yes	Digest 5.3
5.4	Jitter Tolerance: Line-of-Sight Stability Requirements	Tom, optical designer	yes	yes	Digest 5.4

Scenario	Title	Persona	GUI	Runner	Covered in
5.5	Stray Light and Veiling Glare	Tom, optical designer	yes	yes	Digest 5.5
6.1	Published-Datasheet Benchmark (D*/NETD)	Dr. Chen, researcher	yes	yes	Case study — Benchmarking Against a Published Datasheet
6.2	Atmospheric Model Intercomparison	Dr. Chen, researcher	yes	yes	
6.3	Noise Model Verification Against Hand Calculation	Dr. Chen, researcher	yes	yes	Digest 6.3
6.4	Synthetic Scene Generation for Algorithm Testing	Dr. Chen, researcher	yes	yes	Digest 6.4
6.5	Emissivity Sensitivity for Temperature Retrieval	Dr. Chen, researcher	no	yes	Digest 6.5
7.1	Predicted vs Measured NEDT Reconciliation	Karen, test engineer	yes	yes	Case study — Reconciling Predicted and Measured NEDT
7.2	Radiometric Calibration Verification	Karen, test engineer	yes	yes	
7.3	MTF Measurement vs Prediction	Karen, test engineer	yes	yes	Digest 7.3
7.4	Cold-Stop Undersizing Sweep	Karen, test engineer	yes	yes	Digest 7.4
7.5	Performance at Temperature Extremes	Karen, test engineer	yes	yes	Digest 7.5
8.1	Off-Nadir Angle Interpolation	Tool demonstration	yes	yes	Digest 8.1
8.2	Target-Altitude Interpolation	Tool demonstration	yes	yes	Digest 8.2
8.3	Boost-Phase Target-Altitude Sweep (skeleton)	Tool demonstration	no	yes	Digest 8.3
9.1	Sentinel-2 MSI — SNR at Reference Radiance	Flagship validation	yes, <i>configs</i>	no	Validation — Sentinel-2 MSI
9.2	Landsat 8 TIRS — Thermal NEdT	Flagship validation	yes, <i>configs</i>	no	Validation — Landsat 8 TIRS
9.3	MODIS (Aqua) Thermal Emissive Bands	Flagship validation	yes, <i>configs</i>	no	Validation — Aqua MODIS
9.4	Landsat 9 OLI-2 — Nine-Band SNR, Per-Element Coated Optics	Flagship validation	yes, <i>configs</i>	yes	Validation — Landsat 9 OLI-2
10.1	Ground-to-Air MWIR Detection	Direction-general	yes	yes	Digest 10.1
10.2	Air-to-Air Level-Arm MWIRIRST	Direction-general	yes	yes	Case study — Air-to-AirIRST on a Level Arm
10.3	Ground-to-Space SST, Visible Band	Direction-general	yes	yes	
10.4	LEO-to-GEO Up-Looking Space-to-Space SDA	Direction-general	yes	yes	Digest 10.4

A.2 Counts

Folder	Scenarios	Case studies	Validation	Digests
1 — Sarah, systems engineer	6	1	0	5
2 — Mike, detector engineer	9	1	0	8
3 — Raj, mission planner	5	1	0	4
4 — Lisa, detection/targeting analyst	5	1	0	4
5 — Tom, optical designer	5	1	0	4

Folder	Scenarios	Case studies	Validation	Digests
6 — Dr. Chen, researcher	5	1	0	4
7 — Karen, test engineer	5	1	0	4
8 — Tool demonstrations	3	0	0	3
9 — Flagship missions	4	0	4	0
10 — Direction-general	4	1	0	3
Total	51	8	4	39

Thirty-eight of the fifty-one ship an `inputs/*.gui.yaml` baseline and the four flagship folders ship GUI-openable per-band configs — forty-two GUI entry points in all. Forty-eight ship an executable runner; the three without one — the three flagship folders that are pure configuration comparisons — are driven by `radiant run <config>` or by the repository-level `scripts/run_external_validation.py`, which is the canonical executable form of the external-validation dossier.

The catalog itself — one paragraph per scenario, with the persona’s motivating question — is the repository’s scenario catalog. It is the front door for “which scenario is closest to my problem”; this appendix is the front door for “where does this volume discuss it”.